

BiPo Event Classification in OSIRIS Using Machine Learning

Master's Thesis

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Abstract

The reliable classification of BiPo events is crucial for radiopurity measurements in the OSIRIS detector, a pre-detector system for the Jiangmen Underground Neutrino Observatory (JUNO). Traditionally, such events have been identified using a sequence of manually optimized selection cuts. This thesis investigates whether machine learning (ML) methods offer a superior alternative for BiPo classification.

A simulated toy Monte Carlo dataset with an equal distribution of signal and background classes was used to benchmark classical cuts, optimized cuts via Optuna, artificial neural networks, decision trees, and recurrent neural network (RNN) architectures, including LSTM and BiLSTM models enhanced with attention mechanisms. Among these approaches, the BiLSTM with attention and pairwise post-processing achieved the best performance, significantly outperforming both cut-based methods and other machine learning models in distinguishing BiPo events from background.

Despite these encouraging results, limitations arise from the use of simplified simulations, artificial class balances, and the absence of real detector effects. Future work must incorporate the full OSIRIS simulation, address the severe class imbalance expected in real data, and explore more sophisticated model architectures and uncertainty estimation methods.

This study highlights the promise of machine learning in advancing neutrino detection techniques, a small step closer toward unveiling the invisible threads of the universe.

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Chapter 1

Introduction

Neutrino physics stands at the frontier of our understanding of fundamental particles and their interactions. First postulated by Wolfgang Pauli in 1930 to explain the apparent non-conservation of energy in beta decay, neutrinos were later experimentally confirmed by Clyde Cowan and Frederick Reines in 1956 through their detection in a nuclear reactor experiment [1]. As electrically neutral, nearly massless fermions, neutrinos are key to answering open questions in particle physics and cosmology, including the nature of mass generation, Charge Parity (CP) violation in the lepton sector, and the structure of the Standard Model itself. Experimental efforts across the globe are focused on probing neutrino oscillations, mass hierarchy, and other properties that remain elusive.

One such experiment is the JUNO, a multi-purpose neutrino experiment currently in the Liquid Scintillator (LS) filling phase in southern China. It features a 20 kt LS detector and is primarily designed to determine the neutrino mass hierarchy by performing a high-precision measurement of the oscillated energy spectrum of electron antineutrinos from nearby nuclear power plants. JUNO aims for an unprecedented energy resolution of $3\%/\sqrt{E(\text{MeV})}$, enabling a determination of the mass hierarchy with a statistical significance of approximately 3σ after six years of data taking. Beyond its primary objective, JUNO also aims to detect neutrinos from various sources such as the atmosphere, the sun, the Earth (geo-neutrinos), supernovae, and even potential dark matter annihilation signatures.

Online Scintillator Internal Radioactivity Investigation System (OSIRIS), the pre-detector for JUNO, is designed to perform radiopurity measurements on liquid scintillator batches prior to their use in the main detector. A key part of this effort is the reliable identification of BiPo coincidences. BiPo events are fast-decay coincidences in uranium and thorium chains, and they are a crucial signature for identifying and reducing radioactive backgrounds. Their accurate classification is thus essential to the sensitivity of experiments like JUNO. Traditionally, this classification has relied on a series of well-defined selection cuts applied to physical variables such as energy, time, and spatial separation. While effective to a degree, these cuts are inherently rigid and often struggle to capture the complexity of multidimensional feature correlations or overlapping event signatures.

To address this limitation, the present work explores Machine Learning (ML) techniques as a flexible and Monte Carlo (MC) data-driven alternative for BiPo classification. By learning patterns directly from the underlying distributions, ML methods can potentially uncover subtle features that remain faint to conventional cut-based strategies. The central question explored in this thesis is whether machine learning models can outperform classical cut-based methods in the classification of BiPo events under OSIRIS conditions.

This thesis does not aim to develop a real-time event classifier for OSIRIS, nor does it incorporate low-level detector simulation data. Instead, it focuses on evaluating model performance on a preprocessed, structured dataset representing simplified Monte Carlo events. Beyond this specific application, the methods explored here reflect a growing trend in particle physics toward data-driven, algorithmic approaches, offering potential insights for other experiments facing complex classification challenges.

The structure of this thesis is as follows: Chapter 2 lays out the theoretical foundations of neutrino physics, the JUNO and OSIRIS detectors, BiPo event characteristics, and the motivation for applying machine learning. Chapter 3 describes the dataset used, including event labeling and features. Chapter 4 presents the classical and optimized cut-based classification methods, which serve as benchmarks for the ML-based approaches. Chapters 5 and 6 explore artificial neural networks, decision trees, and recurrent neural network architectures, including Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (BiLSTM) models with attention mechanisms. A comparison of these methods is provided in Chapter 7, including limitations and outlook, followed by a summary in Chapter 8.

Through this progression, the thesis aims to evaluate the applicability of machine learning to improve BiPo classification in OSIRIS, setting the stage for future applications in JUNO and beyond.

Chapter 2

Theory and Context for Neutrino Physics in JUNO and OSIRIS

This chapter outlines the theoretical and experimental context underpinning this work. It begins with an overview of neutrino oscillations and the importance of determining the neutrino mass hierarchy which is one of the key open questions in particle physics. The JUNO experiment is introduced as a leading effort to resolve this question, with OSIRIS serving as its liquid scintillator test facility.

A particular focus is placed on BiPo events, fast decay sequences that form a critical background in low-energy neutrino detection. Their identification is essential for the success of both OSIRIS and JUNO. To address this, the final section motivates the use of machine learning as a tool to improve classification performance in this complex experimental environment.

2.1 Neutrino Physics, Oscillations, and Mass Hierarchy

Neutrinos are fundamental particles in the Standard Model of particle physics, notable for their extremely small mass, lack of electric charge, and exclusive interaction via the weak force [2, 3]. They exist in three flavors: electron neutrino (ν_e), muon neutrino (ν_μ), and tau neutrino (ν_τ), each paired with a corresponding charged lepton. Each flavor also has an associated antiparticle, known as an antineutrino (e.g., $\bar{\nu}_e$). In the Standard Model, neutrinos were initially considered massless, but experimental evidence from solar, atmospheric, reactor, and accelerator neutrino experiments has since confirmed that they possess non-zero mass and undergo flavor transitions, a phenomenon known as neutrino oscillation [4, 5, 6].

2.1.1 Neutrino Oscillations

A key property of neutrinos is their ability to oscillate between flavors, known as neutrino oscillations. These describe periodic transitions from one neutrino flavor to another. The phenomenon was first hinted at by the solar neutrino problem, where the observed solar neutrino flux was significantly lower than predicted by the Standard Solar Model [4]. This discrepancy was later confirmed by experiments such as GALLEX [7], GNO [8], and SAGE [9], all of which were sensitive only to the electron neutrino flux component. The missing flux was explained by the flavor-oscillating nature of neutrinos.

Direct experimental evidence for flavor transformation came from the Sudbury Neutrino Observatory (SNO), which could detect both charged-current (electron neutrino specific) and neutral-current (flavor-blind) interactions, thus confirming the presence of non-electron neutrino flavors in the solar neutrino flux [5]. Neutrino oscillations imply that the mass and weak interaction eigenstates are not identical and that neutrinos must have non-zero mass. These features were not originally included in the Standard Model [3]. Neutrino oscillations reveal physics beyond the Standard Model but make it harder to measure fundamental properties, blessed with a curse, the phenomenon both solves and creates problems.

In the three-flavor formalism of neutrino oscillations in vacuum, the flavor eigenstates ν_α (with $\alpha \in \{e, \mu, \tau\}$) are expressed as linear combinations of the mass eigenstates ν_i (with $i \in \{1, 2, 3\}$) via the unitary Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix U [3]:

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i} |\nu_i\rangle \quad \text{and} \quad |\bar{\nu}_\alpha\rangle = \sum_i U_{\alpha i}^* |\bar{\nu}_i\rangle \quad (2.1)$$

Here, $U_{\alpha i}$ are the elements of the PMNS matrix U , which connects the flavor and mass eigenstates:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} \quad (2.2)$$

In the case of three active neutrino flavors, the PMNS matrix is a 3×3 unitary matrix. Its standard parameterization involves three mixing angles ($\theta_{12}, \theta_{13}, \theta_{23}$) and one CP-violating phase δ :

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (2.3)$$

Multiplying these matrices gives the explicit form of U :

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{13}s_{23}e^{i\delta} & c_{12}c_{23} - s_{12}s_{13}s_{23}e^{i\delta} & c_{13}s_{23} \\ s_{12}s_{23} - c_{12}s_{13}c_{23}e^{i\delta} & -c_{12}s_{23} - s_{12}s_{13}c_{23}e^{i\delta} & c_{13}c_{23} \end{pmatrix} \quad (2.4)$$

with $s_{ij} = \sin \theta_{ij}$ and $c_{ij} = \cos \theta_{ij}$.

If neutrinos are Majorana particles, the full PMNS matrix includes two additional phases via the diagonal matrix:

$$P_\nu = \text{diag}(e^{i\alpha_1/2}, e^{i\alpha_2/2}, 1)$$

which do not affect oscillation probabilities and are therefore omitted in this treatment [6].

The survival probability of an electron antineutrino of energy E after traveling a distance L is then given by:

$$P(\bar{\nu}_e \rightarrow \bar{\nu}_e) = 1 - \sin^2(2\theta_{12}) \cos^4(\theta_{13}) \sin^2\left(\frac{\Delta m_{21}^2 L}{4E}\right) - \sin^2(2\theta_{13}) \left[\cos^2(\theta_{12}) \sin^2\left(\frac{\Delta m_{31}^2 L}{4E}\right) + \sin^2(\theta_{12}) \sin^2\left(\frac{\Delta m_{32}^2 L}{4E}\right) \right] \quad (2.5)$$

with $\Delta m_{ij}^2 = m_i^2 - m_j^2$ being the squared mass differences [6]. Neutrino oscillations in the three-flavor model are fully described by six independent parameters: the three mixing angles, two mass splittings, and the CP-violating phase. The oscillation amplitudes and frequencies are governed by the mixing angles and squared mass differences, respectively. These parameters have been measured with increasing precision across numerous experiments, including (but not limited to) Super-Kamiokande, Daya Bay, and NO ν A, as highlighted in [10].

A compilation of current global best-fit values for the oscillation parameters can be found in Table 2.1. The sign of the larger mass difference Δm_{31}^2 is still unknown and determines whether the neutrino mass hierarchy follows a normal or inverted ordering, this open question remains central in modern neutrino physics [11].

Table 2.1: Global best-fit values of neutrino oscillation parameters from the ICI24 analysis including Super-Kamiokande atmospheric data. Here, *bfp* refers to the best fit point obtained from a global χ^2 minimization, with the quoted uncertainties corresponding to the 1σ confidence level. The parameter $\Delta m_{3\ell}^2$ is defined as Δm_{31}^2 for normal ordering and Δm_{32}^2 for inverted ordering, providing a unified notation for the atmospheric mass-squared splitting across both mass hierarchies [11].

Parameter	Normal Ordering (best fit)		Inverted Ordering ($\Delta\chi^2 = 6.1$)	
	bfp $\pm 1\sigma$	3σ range	bfp $\pm 1\sigma$	3σ range
$\sin^2 \theta_{12}$	$0.308^{+0.012}_{-0.011}$	$0.275 \rightarrow 0.345$	$0.308^{+0.012}_{-0.011}$	$0.275 \rightarrow 0.345$
$\theta_{12}/^\circ$	$33.68^{+0.73}_{-0.70}$	$31.63 \rightarrow 35.95$	$33.68^{+0.73}_{-0.70}$	$31.63 \rightarrow 35.95$
$\sin^2 \theta_{23}$	$0.470^{+0.017}_{-0.021}$	$0.435 \rightarrow 0.585$	$0.550^{+0.012}_{-0.017}$	$0.440 \rightarrow 0.584$
$\theta_{23}/^\circ$	$43.3^{+1.1}_{-0.9}$	$41.3 \rightarrow 49.9$	$47.9^{+0.7}_{-0.9}$	$41.5 \rightarrow 49.8$
$\sin^2 \theta_{13}$	$0.02215^{+0.00056}_{-0.00058}$	$0.02030 \rightarrow 0.02388$	$0.02231^{+0.00056}_{-0.00056}$	$0.02060 \rightarrow 0.02409$
$\theta_{13}/^\circ$	$8.56^{+0.11}_{-0.11}$	$8.19 \rightarrow 8.89$	$8.59^{+0.11}_{-0.11}$	$8.19 \rightarrow 8.89$
$\delta_{\text{CP}}/^\circ$	212^{+26}_{-41}	$124 \rightarrow 364$	274^{+22}_{-25}	$201 \rightarrow 335$
Δm_{21}^2 [10^{-5} eV 2]	$7.49^{+0.19}_{-0.19}$	$6.92 \rightarrow 8.05$	$7.49^{+0.19}_{-0.19}$	$6.92 \rightarrow 8.05$
$\Delta m_{3\ell}^2$ [10^{-3} eV 2]	$+2.513^{+0.021}_{-0.019}$	$+2.451 \rightarrow +2.578$	$-2.484^{+0.020}_{-0.020}$	$-2.547 \rightarrow -2.421$

2.1.2 Neutrino Mass Hierarchy

One of the primary goals of experiments like JUNO is to resolve one of the major open questions in neutrino physics: the ordering of neutrino mass eigenstates, also referred to as the Neutrino Mass Hierarchy (NMH). There are two possible scenarios for the arrangement of the three known mass eigenstates (ν_1, ν_2, ν_3):

- **Normal Ordering (NO)**

In this scenario, the masses are sorted as

$$m_1 < m_2 < m_3 \quad (2.6)$$

and the hierarchy is associated with

$$|U_{e1}|^2 > |U_{e2}|^2 > |U_{e3}|^2 \quad (2.7)$$

implying that m_1 corresponds to the eigenstate with the largest electron neutrino component.

- **Inverted Ordering (IO)**

Here, the lightest mass eigenstate is m_3 , and the ordering becomes:

$$m_3 < m_1 < m_2 \quad (2.8)$$

Although the absolute value of the larger mass-squared splitting Δm_{31}^2 is well known, the sign of this quantity remains undetermined, leaving the true mass ordering ambiguous. This uncertainty persists because the experimental precision of past measurements was not sufficient to distinguish between the two orderings. However, the smaller splitting Δm_{21}^2 is known to be positive, as confirmed by solar neutrino observations and matter effects through the MSW mechanism.

Figure 2.1 illustrates the two possible orderings, including the relative flavor contributions to each mass eigenstate. The figure clearly shows how the mass levels and flavor contents differ between NO and IO.

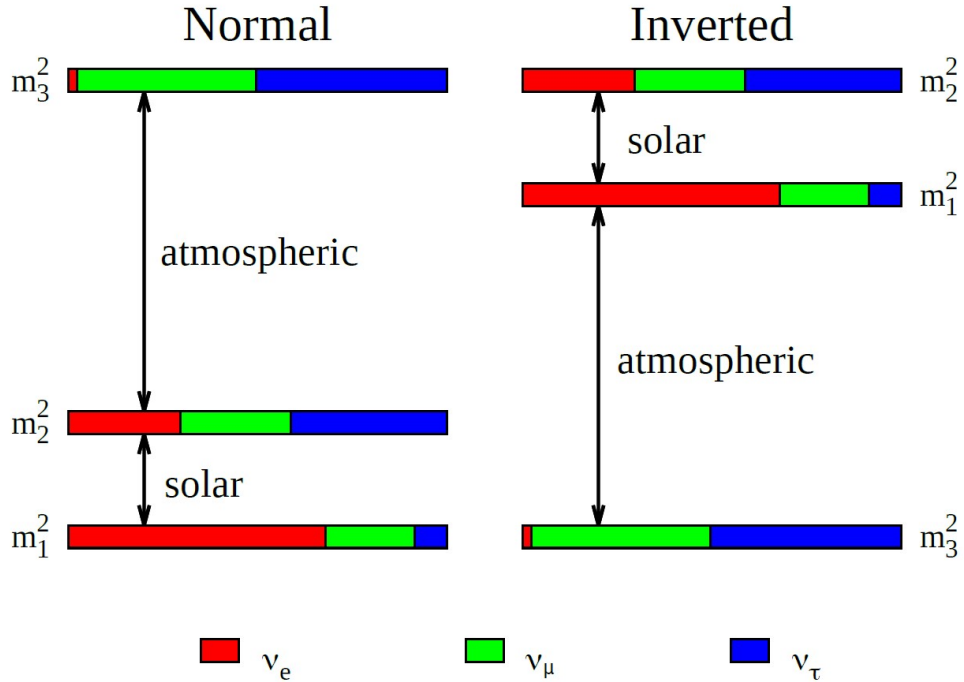


Figure 2.1: Schematic illustration of the two possible neutrino mass orderings. In the NO (left), the lightest state is m_1 , and the heaviest is m_3 , while in the IO (right), m_3 is the lightest. The bars represent the flavor composition of each mass eigenstate, with red, green, and blue segments corresponding to ν_e , ν_μ , and ν_τ , respectively. The diagram highlights how flavor contributions and relative mass splittings differ depending on the ordering. Adapted from [12]

Recent global fits from T2K [13], NOvA [14], and Super-Kamiokande [15] slightly favor the NO. Nevertheless, discrepancies remain, and the final resolution requires even more precise measurements.

In addition to identifying the ordering itself, determining the mass hierarchy has broader implications. For example:

- It affects the interpretation of long-baseline oscillation experiments and the analysis of δ_{CP} .
- It plays a crucial role in the prediction and search for neutrinoless double beta decay ($0\nu\beta\beta$).
- It impacts our understanding of the origin of neutrino mass and mixing.

Experiments like JUNO aim to resolve the mass hierarchy by detecting fine features in the oscillation spectrum with high energy resolution. Other facilities such as NOvA, DUNE, and Hyper-K [14, 16, 17] are also pursuing this goal using complementary methods like long-baseline accelerator beams or atmospheric neutrinos. Additionally, the MSW effect provides a terrestrial tool to probe the sign of Δm_{31}^2 by analyzing how neutrinos traverse dense matter.

Resolving the neutrino mass hierarchy is therefore not only essential for completing our understanding of the Standard Model's lepton sector, but also pivotal in guiding the direction of future experimental and theoretical work in neutrino physics.

2.2 The JUNO Experiment

The Jiangmen Underground Neutrino Observatory (JUNO) is a next-generation, multi-purpose liquid scintillator detector currently under construction in China. Designed with a target mass of 20 kt and located approximately 700 m underground, JUNO's primary objective is the determination of the neutrino mass hierarchy through the precise measurement of the energy spectrum of reactor antineutrinos. To this end, JUNO achieves an unprecedented energy resolution of $3\%/\sqrt{E[\text{MeV}]}$ for liquid scintillator detectors, a critical feature that allows for the resolution of fine oscillatory features in the antineutrino spectrum.

Two nearby nuclear power plants, Taishan and Yangjiang, located 53 km from the experimental site, provide the required flux of electron antineutrinos. This distance has been carefully selected to optimize sensitivity to oscillation patterns that are indicative of the neutrino mass ordering. With a high event rate of approximately 60 antineutrino interactions per day and an energy resolution surpassing all previous large-scale detectors, JUNO is uniquely positioned to extract precise measurements of oscillation parameters such as $\sin^2 \theta_{12}$, Δm_{21}^2 , and $|\Delta m_{ee}^2|^1$ to within 1%.

Beyond its primary mission, JUNO's large volume and superb resolution open the door to a wide array of additional physics goals, including the study of solar, atmospheric, geo, and supernova neutrinos, the search for sterile neutrinos, and tests of non-standard interactions and symmetries. Its ability to observe time-correlated signals from radioactive decay chains also provides a unique environment for background studies, which are crucial for next-generation neutrino and dark matter experiments.

2.2.1 Neutrino Mass Hierarchy with JUNO

The precise determination of the neutrino mass hierarchy is the primary physics goal of JUNO. By detecting antineutrinos from nearby nuclear reactors via the Inverse Beta Decay (IBD) reaction:

$$\bar{\nu}_e + p \rightarrow e^+ + n, \quad (2.9)$$

the experiment reconstructs the neutrino energy spectrum with high accuracy. The positron provides a prompt signal via scintillation light, while the neutron thermalizes and is captured after a short delay, producing a delayed signal. The energy of the incoming antineutrino is related to the visible prompt energy by $E_{\bar{\nu}_e} \approx E_{\text{prompt}} + 0.8 \text{ MeV}$.

The survival probability of reactor antineutrinos, $P(\bar{\nu}_e \rightarrow \bar{\nu}_e)$, is sensitive to the mass ordering of the neutrino eigenstates. As such, JUNO aims to determine the mass hierarchy by observing the interference pattern in the energy spectrum at baseline of 53km [19]. With six years of data taking, JUNO expects to reach a median sensitivity of $\sim 3\sigma$ for distinguishing between the normal and inverted mass hierarchies.

¹ $\Delta m_{ee}^2 \equiv \cos^2 \theta_{12} \Delta m_{31}^2 + \sin^2 \theta_{12} \Delta m_{32}^2$ is the effective mass-squared difference governing $\bar{\nu}_e$ disappearance in reactor experiments [18].

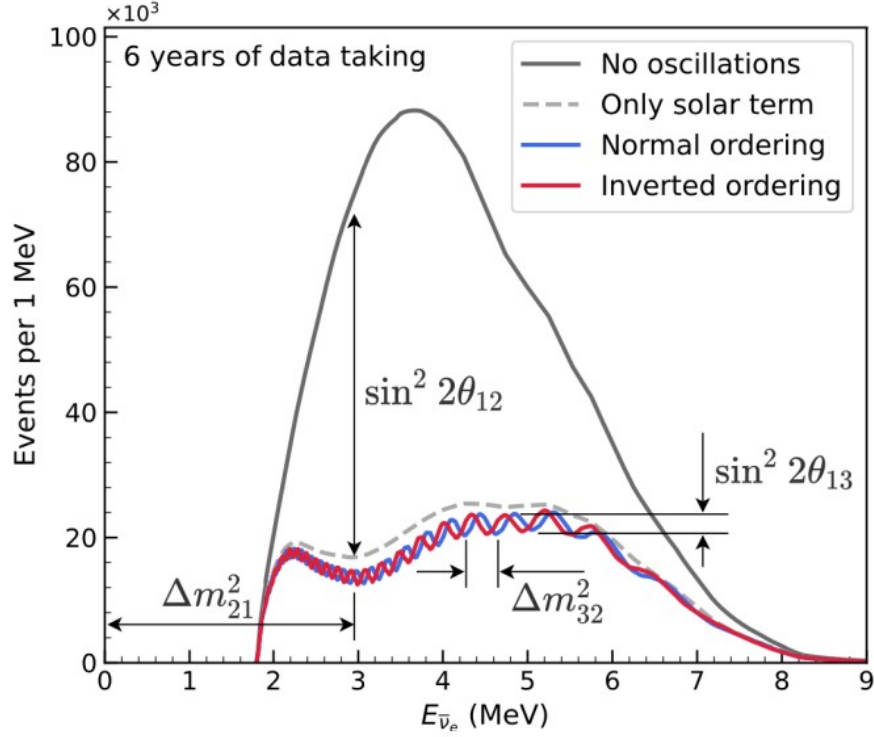


Figure 2.2: Calculated energy spectrum of electron antineutrinos expected at JUNO illustrating the impact of neutrino oscillations. JUNO’s high energy resolution enables it to distinguish fine oscillation structures arising from both solar-scale (Δm_{21}^2) and atmospheric-scale (Δm_{31}^2 , Δm_{32}^2) mass splittings. The figure shows the expected spectra under different scenarios: no oscillations, solar term only, and full oscillation with normal and inverted mass orderings. Adapted from [19].

2.2.2 Detector Overview

The Central Detector (CD) of JUNO consists of a spherical Acrylic Vessel (AV) with a diameter of 35.4 m and a wall thickness of 12 cm, filled with 20 kt of linear alkylbenzene (LAB) based liquid scintillator. The vessel is supported by a stainless-steel truss structure and submerged in a water pool that acts as both shielding and a Cherenkov veto system [6].

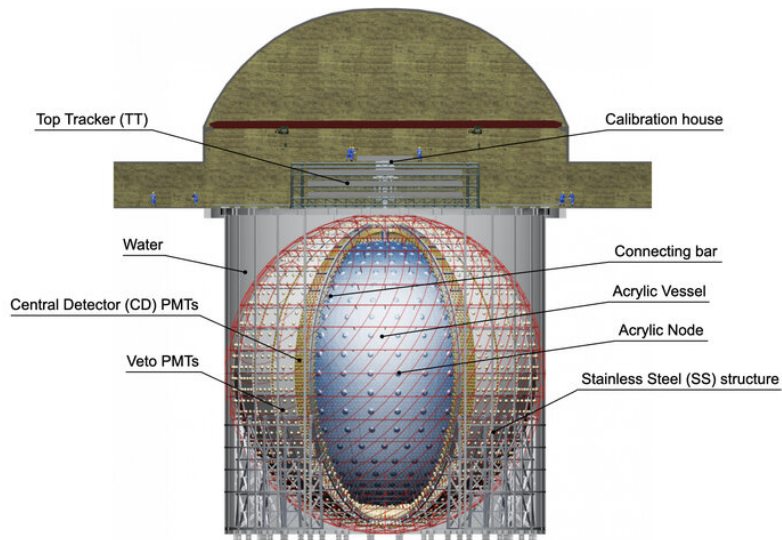


Figure 2.3: Schematic of JUNO’s CD. The LS-filled AV is surrounded by a steel support structure and a water Cherenkov veto. Figure from [6].

A total of 17,600 20-inch Photomultiplier Tube (PMT)s and 25,600 3-inch PMTs are mounted on the stainless steel frame of the JUNO CD. The 20-inch PMTs include approximately 5,000 dynode-type

tubes from Hamamatsu and 13,000 Microchannel Plate (MCP) PMTs from NNVT. The PMT system is optimized to achieve a charge resolution of better than 15% and a timing resolution of a few nanoseconds. The total photocathode coverage reaches over 75%, while the photon detection efficiency, determined by the PMT quantum efficiency, exceeds 35% [6].

2.2.3 Liquid Scintillator Composition

The LS mixture consists of:

- **Solvent:** Linear alkylbenzene (LAB), chosen for its chemical stability and high light yield when combined with the fluor described below.
- **Fluor:** 2,5-diphenyloxazole (PPO), at a concentration of 2.5 g/L. In combination with LAB, the mixture yields approximately 10^4 photons per MeV of deposited energy [12].
- **Wavelength Shifter:** bis-MSB (1,4-bis(2-methylstyryl)benzene), at 15 mg/L, to shift the emission peak to 420 nm, optimizing detection by the PMTs.

The LAB-based LS is prepared on-site, purified via water extraction and nitrogen stripping, and continuously monitored using the OSIRIS detector to ensure radiopurity [20].

2.2.4 Additional Systems

Muon Veto and Water Cherenkov Detector: The water pool surrounding the AV serves as a muon veto and shields against external radiation. It contains 2400 additional 20-inch PMTs and achieves a tracking efficiency of 98% for single muons [21].

Top Tracker: A top tracker system based on plastic scintillator panels inherited from the OPERA experiment provides further muon rejection and tracking capabilities [22].

JUNO-TAO Near Detector: Located 30 m from the Taishan reactor, JUNO-TAO is a 2.6 t liquid scintillator detector featuring gadolinium loading and Silicon Photomultiplier (SiPM). It provides high-statistics measurements of the reactor spectrum with an outstanding energy resolution of $1.5\%/\sqrt{E}$ [MeV] [23]. This helps constrain reactor-related uncertainties in the main JUNO analysis.

2.2.5 Energy Resolution and Sensitivity

The exceptional energy resolution goal of $3\%/\sqrt{E}$ [MeV] is central to JUNO’s design. Achieving this requires minimizing noise in the PMTs, maximizing light yield from the scintillator, and carefully controlling systematics. The combination of high photon statistics (about 1200 p.e./MeV) and low noise electronics enables JUNO to resolve oscillation features at the sub-percent level [23].

2.3 The OSIRIS Detector

The Online Scintillator Internal Radioactivity Investigation System (OSIRIS) is a dedicated subsystem of the JUNO Liquid Handling System (LHS), built to monitor the radiopurity of the LS before it is filled into the CD. It consists of a cylindrical AV with a capacity of 18 t of LS, surrounded by an array of 64 PMTs identical to the MCP PMTs from NNVT used in the JUNO CD [20]. To suppress background from muons and ambient radioactivity, OSIRIS is further equipped with a surrounding water Cherenkov veto system containing additional 12 PMTs.

Developed as a precision tool born of OSIRIS’ unique design requirements, the system operates in two modes: a *batch mode*, in which the vessel is filled and measured over several weeks, and a *continuous mode*, where LS flows through the vessel continuously and is monitored in real-time [20]. The system aims to reach a sensitivity level of 10^{-16} g/g for both ^{238}U and ^{232}Th , ensuring that any impurities in the scintillator that might have evaded upstream purification processes are identified and excluded before filling JUNO’s detector [21].

2.3.1 Motivation

The main objective of OSIRIS is to ensure that the LS used in JUNO achieves ultra-low background levels necessary for precision measurements of reactor and solar neutrinos. Even trace amounts of long-lived radioactive impurities, particularly from the ^{238}U and ^{232}Th decay chains, can mimic IBD-like signals or degrade the energy resolution (of ^{14}C), thereby obscuring true neutrino interactions. Reducing background is crucial for detecting low-energy signals, such as those from solar neutrinos. It is especially important for solar neutrinos, which relies on a clean and precise energy spectrum [23]. The corresponding radiopurity requirements for various isotopes are summarized in Table 2.2, alongside comparison values achieved by KamLAND and Borexino [20].

Table 2.2: Radiopurity requirements for the main physics modes of JUNO (IBD and solar) along with values achieved in KamLAND and Borexino. Adapted from [20].

Chain/isotope	JUNO IBD	JUNO solar	KamLAND	Borexino
^{238}U	1×10^{-15} g/g	1×10^{-16} g/g	$(5.0 \pm 0.2) \times 10^{-18}$ g/g	$< 1 \times 10^{-18}$ g/g
^{232}Th	1×10^{-15} g/g	1×10^{-16} g/g	$(1.3 \pm 0.1) \times 10^{-17}$ g/g	$< 1 \times 10^{-18}$ g/g
^{210}Po	–	5×10^{-24} g/g	$\sim 2 \times 10^{-23}$ g/g	$< 1 \times 10^{-25}$ g/g
^{40}K	1×10^{-16} g/g	1×10^{-17} g/g	$(7.3 \pm 1.2) \times 10^{-17}$ g/g	$< 1 \times 10^{-19}$ g/g
^{14}C	1×10^{-17} g/g	1×10^{-17} g/g	$(3.98 \pm 0.94) \times 10^{-18}$ g/g	$(2.7 \pm 0.1) \times 10^{-18}$ g/g
^{85}Kr	–	$500 \text{ (d}\cdot\text{kt)}^{-1}$	$858 \pm 59 \text{ (d}\cdot\text{kt)}^{-1}$	$68 \pm 18 \text{ (d}\cdot\text{kt)}^{-1}$

BiPo Coincidence Events

OSIRIS achieves its sensitivity primarily by exploiting BiPo coincidence signals, which are short-lived β - α cascade decays from bismuth to lead isotopes. In particular, two such coincidence chains are targeted: $^{214}\text{Bi} \rightarrow ^{214}\text{Po}$ (from the ^{238}U chain) and $^{212}\text{Bi} \rightarrow ^{212}\text{Po}$ (from the ^{232}Th chain). These decays occur within narrow time windows governed by the exponential lifetimes of the daughter isotopes: $\tau = 237 \mu\text{s}$ for ^{214}Po and $\tau = 0.431 \mu\text{s}$ for ^{212}Po . These are the shortest lifetimes among all isotopes in their respective chains, making them ideal for tagging via coincidence detection. Their clean, prompt signatures provide a powerful tool for real-time radiopurity monitoring [22].

The decay chains of ^{238}U and ^{232}Th are shown in Figures 2.4 and 2.5, respectively. The isotopes relevant for the BiPo coincidence tagging, $^{212}\text{Bi}/^{212}\text{Po}$ and $^{214}\text{Bi}/^{214}\text{Po}$, are highlighted and form the basis of OSIRIS’s monitoring capability. While the rates of these decays provide valuable information about contamination, they do not directly reflect the full decay chain activity due to potential breaks in secular equilibrium, such as the introduction of radon from air.

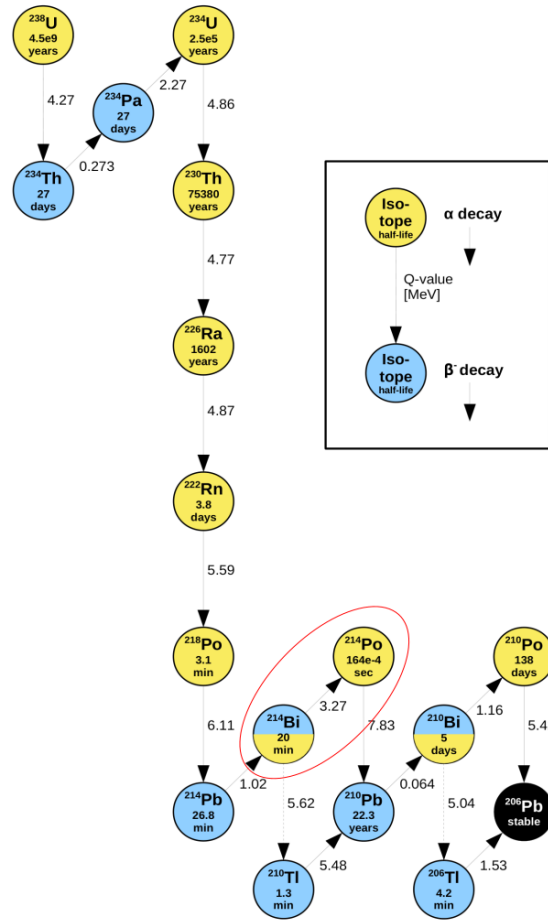


Figure 2.4: Full decay chain of ^{238}U . The isotopes decaying via β are marked in blue, and those via α are yellow. Q-values and half-lives are included. Figure adapted from [24].

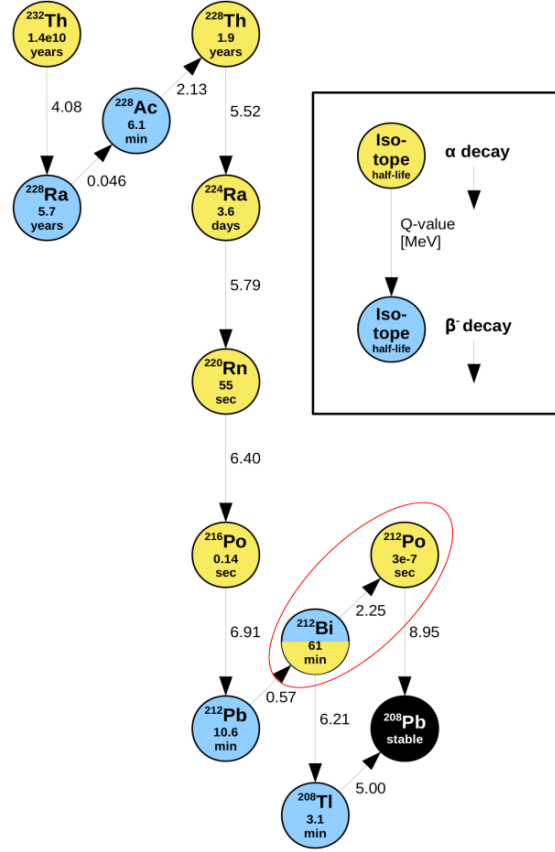


Figure 2.5: Full decay chain of ^{232}Th . BiPo coincidence is derived from ^{212}Bi and ^{212}Po . Figure adapted from [24].

These diagrams also underline the tagging method's effectiveness. By correlating prompt and delayed signals and measuring their rates, OSIRIS can detect contaminations as low as 10^{-16} g/g for ^{238}U and ^{232}Th [20]. For the identification of the decays, the lifetimes and branching ratios of relevant isotopes are listed in 2.3.

Table 2.3: Overview of BiPo isotope decays relevant for the detection of ^{232}Th and ^{238}U contamination. Data adapted from [24], originally based on the GEANT4 RadioactiveDecay database.

Isotope	Mean Life	Decay	Branching Ratio [%]	Q-value [keV]
^{212}Bi	87.4 min	β^-	64.06	2254
		α	35.94	6207
^{212}Po	$0.431 \mu\text{s}$	α	100	8954
^{214}Bi	28.7 min	β^-	99.98	3272
		α	0.02	5617
^{214}Po	$237 \mu\text{s}$	α	100	7833

In conclusion, OSIRIS not only ensures that JUNO maintains the ultra-low background levels necessary for its primary and solar neutrino measurements, but also provides a real-time verification system during the detector's filling phase. The BiPo coincidence mechanism and corresponding decay chain analysis are powerful methods for identifying and quantifying radioactive impurities.

2.3.2 Detector Design

The OSIRIS detector is designed to ensure high-precision monitoring of the radiopurity of the LS used in JUNO. Its architecture includes both inner and outer detector components, carefully integrated for

optimal sensitivity and background suppression [20].

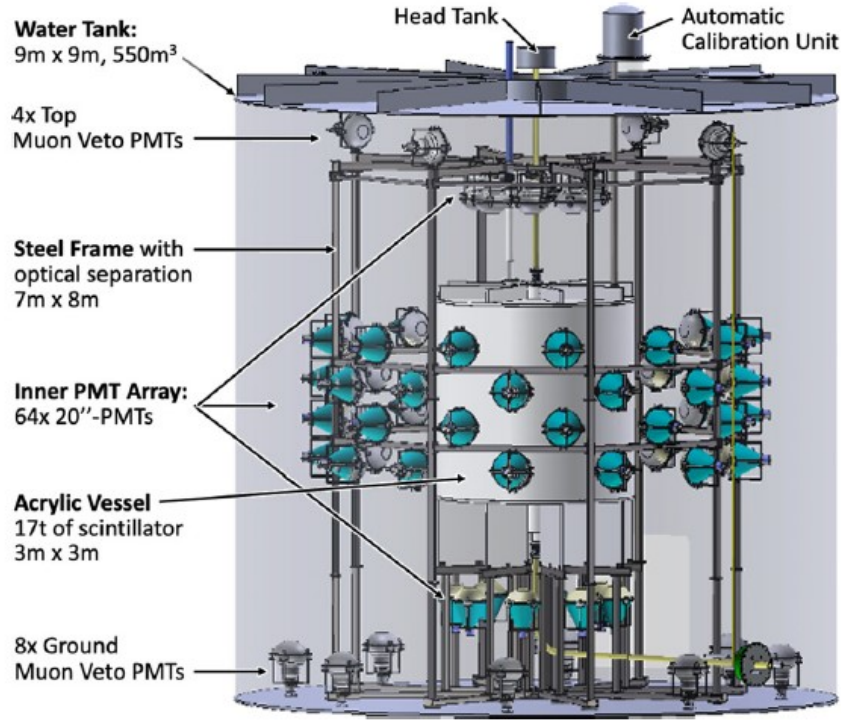


Figure 2.6: Illustration of the OSIRIS detector showing the water tank, AV, PMT arrangement, and auxiliary systems. The design supports both batch and continuous LS monitoring with integrated calibration and veto mechanisms [20].

- Acrylic Vessel:** A cylindrical vessel with a diameter and height of 3 m, and wall thicknesses of 3 cm on the sides and 5 cm at the top (following reparation after mechanical failure). It holds approximately 17–18 t of liquid scintillator. Support fins at the top and bottom enhance mechanical stability. Fixed calibration and monitoring instruments (e.g., ACU, thermometer, and level meter) are installed on the vessel, while radioactive calibration sources can be inserted via dedicated pipes.
- Inner PMT Array:** Comprises 64 20-inch MCP PMTs sourced from JUNO’s main detector. Originally, iPMTs were intended, but due to failures in the High Voltage (HV) boards, discovered in some iPMTs after shipping, standard MCP-based LPMTs were used instead. These are arranged in rings around the vessel to provide about 9% optical coverage, maintaining a distance of 1.3 m from the vessel to optimize light yield and reduce background from PMT glass [25].
- PMT Steel Frame:** The octagonal frame (7 m wide, 8 m high) supports the AV and houses the PMTs, temperature sensors, and calibration systems. It also hosts PET sheets that provide optical separation between inner and outer detectors. The black interior side absorbs photons to prevent timing distortions by reflections, while the white exterior enhances light reflection in the outer volume.
- Water Tank:** Enclosing the entire setup, the cylindrical water tank is 9 m × 9 m in size and holds 550 t of ultra-pure water. It serves as a 4π shield against external gamma radiation. The inner surface is lined with HDPE and Tyvek to enhance water tightness and reflectivity.
- Muon Veto:** To suppress backgrounds from cosmic muons and spallation products, 12 veto PMTs are installed, 4 on top of the steel frame and 8 around the bottom. These detect Cherenkov light in the water tank and achieve veto efficiencies of 90-97% depending on the trigger model.
- Liquid Handling System:** Enables both batch and continuous operation of OSIRIS. It regulates LS temperature and stratification, manages inlets/outlets, and prevents contamination via a buffer tank and nitrogen over-pressure system.
- Calibration Systems:**

- *Automated Calibration Unit (ACU)*: Inherited from Daya Bay, uses a combination of LED, potassium, and multi-gamma sources (e.g., ^{137}Cs , ^{60}Co , ^{65}Zn). Mounted off-axis in the top cleanroom at a distance of 1.21 m from the center, it provides vertex and energy calibration.
- *Laser Calibration System (LCU)*: Includes 24 laser injection points with picosecond timing precision. The laser light is diffused by PTFE balls to create a uniform illumination pattern for calibrating PMT charge and timing.
- **Data Acquisition (DAQ)**: OSIRIS uses an EventBuilder software for data acquisition. The identification of BiPo coincidences and monitoring of real-time variations in event rates are performed by the RootSorter software. DAQ electronics (GCUs, BECs) are inherited from JUNO’s infrastructure and handle PMT signal processing.

2.3.3 Operation Modes and Sensitivity

OSIRIS supports two operational modes: *batch* and *continuous*, each suited for different stages of LS deployment and with distinct sensitivity profiles.

Batch Mode

In batch mode, the AV is filled with a fixed sample of LS, which is monitored over several days or weeks. This mode is ideal during the commissioning phase and when validating the purification chain [20]. Its static nature allows investigation of short-lived radon isotopes (^{222}Rn for ^{238}U and ^{220}Rn for ^{232}Th), which produce additional BiPo decays that can mimic ^{212}Bi – ^{212}Po and ^{214}Bi – ^{214}Po signals. Over time, the radon-induced decay rate decreases exponentially, while the background from ^{238}U and ^{232}Th remains constant. This decay pattern allows for accurate separation and quantification. For example, with an initial ^{222}Rn activity of 130 decays/day, a sensitivity limit at the IBD level can be achieved in 7 days, while a solar-level limit requires 22 days.

Continuous Mode

The default mode for regular operation is continuous mode, where LS flows at a controlled rate of approximately $1\text{ m}^3/\text{h}$ from top to bottom of the AV. This design enables real-time monitoring during JUNO’s CD filling. In this mode, temperature gradients and diffusers prevent mixing of incoming and resident scintillator, preserving a stratified structure. Since radon decays during LS transit, a vertical activity gradient is formed. The radon background can thus be spatially resolved and statistically separated from the uniform U/Th-induced BiPo signals. OSIRIS divides the Fiducial Volume (FV) into horizontal layers (e.g., 14 layers of 20 cm each), enabling position-dependent rate analysis. Depending on the ^{222}Rn rate, sensitivities as low as $6 \times 10^{-15}\text{ g/g}$ for ^{232}Th are achievable [20].

Overall, OSIRIS is optimized to detect U/Th-induced BiPo events with high sensitivity in both modes, enabling it to ensure LS radiopurity at the stringent levels required for JUNO’s physics program.

2.3.4 Signal and Background

The OSIRIS detector identifies ^{232}Th and ^{238}U contamination in the LS via detection of BiPo coincidence signals, where a β -decay of a bismuth isotope is followed by a delayed α -decay of the subsequent polonium isotope within a short time window. These decays occur at the end of the thorium and uranium decay chains and are ideal markers for residual radioactivity in the LS. The identification of true BiPo events requires distinguishing these time- and space-correlated decays from a variety of background sources.

External Gamma Background The dominant background in OSIRIS arises from uncorrelated gamma photons that can mimic BiPo signals by randomly satisfying the coincidence criteria. These gammas primarily originate from the decay of ^{40}K and from isotopes in the ^{232}Th and ^{238}U chains, located in the surrounding rock and the PMT glass [12]. Additionally, radon isotopes such as ^{222}Rn and ^{220}Rn can spread into the water shield, where convection may carry them closer to the AV, allowing their progeny’s high-energy gammas to reach the LS [24]. In scenarios of high radon contamination, ^{222}Rn contributions can become comparable to those from the rock and PMT glass, forming one of the most significant external background sources. A simulated spectrum showing this effect is illustrated in Figure 2.7, adapted from [24] and also referenced in [20].

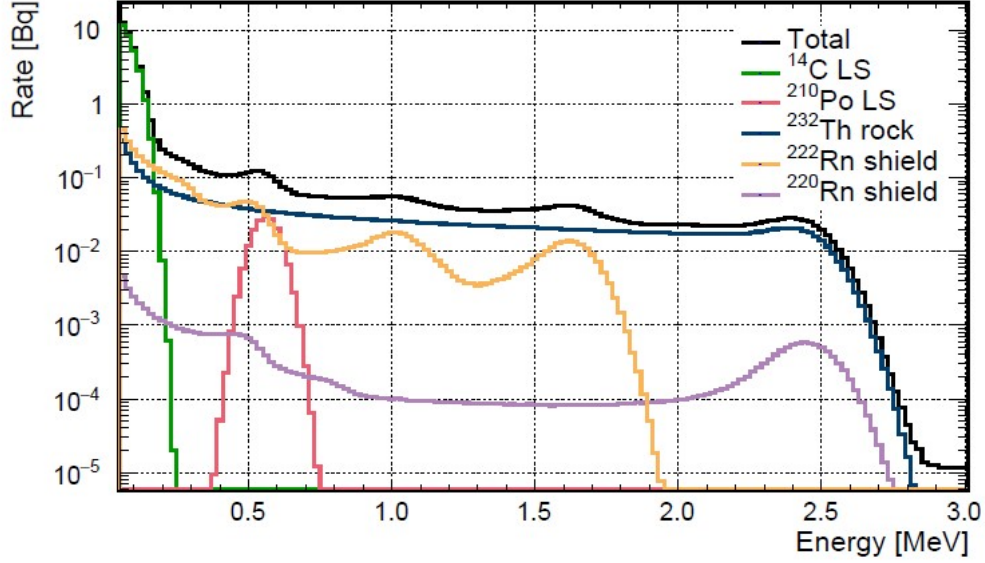


Figure 2.7: Simulated energy spectrum of the total background in the OSIRIS LS volume, including contributions from the surrounding rock, PMT glass, and radon contamination in the water shield. The peaks from ^{222}Rn are visible around 1 MeV and 1.6 MeV, while the overall spectrum is dominated by gammas from ^{232}Th decay in the rock. Adapted from [24].

Internal Radioactivity While external gamma backgrounds dominate, residual internal radioactivity in the LS also contributes. This includes long-lived isotopes such as ^{14}C , as well as ^{40}K , ^{85}Kr , and all daughters of the U/Th decay chains. These internal contributions can deposit visible energy and, when occurring in close succession, may mimic BiPo signals [24, 12].

Cosmogenic Background Despite its underground location with ~ 700 meters of overburden, OSIRIS remains susceptible to cosmic muons and their secondary products. Muon-induced spallation can generate fast neutrons and cosmogenic isotopes such as ^8He and ^9Li , which can imitate BiPo decays by successively emitting an electron and a neutron. These neutrons, when captured by hydrogen, produce 2.2 MeV gamma photons that may form accidental coincidences [12]. The muon veto system significantly reduces this background, with suppression efficiencies up to 96% depending on the veto configuration [12]. Fast neutrons originating from muons in the rock, undetectable by the veto system, were shown to contribute negligibly based on simulation studies [20].

Accidental Coincidences Accidental backgrounds arise from pairs of uncorrelated single events that coincidentally satisfy the BiPo criteria. Their likelihood depends on the rate of singles and the coincidence time window. Given the low expected true BiPo rate (~ 1 count per day at solar levels), even rare accidental pairs can introduce significant background if not effectively suppressed [24, 12].

Signal Selection Strategy To extract BiPo signals, OSIRIS applies four consecutive selection criteria: a FV cut, energy cuts for the β and α components, and constraints on both the time and spatial separation between the two signals [12]. This selection suppresses a large portion of the accidental coincidences while preserving the true correlated decays from BiPo events.

2.3.5 BiPo Event Selection

The identification of BiPo events in OSIRIS is crucial for assessing the radiopurity of the LS. To discriminate true BiPo decays from accidental coincidences or other background signals, a set of four sequential cuts is applied. These cuts exploit the spatial, energetic, and temporal correlation between the prompt β -decay of a bismuth isotope and the delayed α -decay of a polonium daughter. The cut parameters were optimized based on simulations of signal and background within the detector geometry [24].

1. Fiducial Volume Cut To suppress external gamma backgrounds and surface-related contamination, events are selected from a cylindrical FV within the LS. The FV cut removes events near the AV

boundaries, where background rates are elevated due to reduced shielding or surface effects. In OSIRIS, this corresponds to selecting events within a central cylinder of 1.4 meters in both height and diameter [20]. This spatial restriction ensures a high purity selection region while maintaining sufficient signal efficiency.

2. Energy Cuts BiPo decays present a distinct energy profile: the β -decay of bismuth produces a broad energy spectrum (up to a few MeV), while the α -decay of polonium results in a mono-energetic signature. Due to quenching in the LS, the visible energy of the α signal is significantly reduced, appearing in the range of 0.5–1 MeV. Separate energy windows are defined for Bi and Po candidates to reject background while capturing the signal with high efficiency [20, 24].

3. Timing Cut The BiPo coincidence is defined by a characteristic delay time Δt between the prompt Bi decay and the delayed Po decay. The lifetimes of the relevant isotopes, ^{212}Po (431 ns) and ^{214}Po (237 μs), enable strong discrimination against random coincidences. A minimum delay of 200 ns is conservatively assumed to allow temporal separation by the readout electronics. The upper bound of the timing window is optimized for each decay chain to balance signal retention and background suppression [20].

4. Distance Cut Given the short time between decays and low recoil energy, true BiPo decays occur nearly at the same location. By enforcing a maximum allowed spatial separation Δr between the reconstructed positions of the Bi and Po candidates, OSIRIS efficiently rejects spatially uncorrelated accidental events. The optimal Δr value is derived considering the detector’s spatial resolution [24].

The final cut values and their efficiencies are summarized in Table 2.4, adapted from [20].

Table 2.4: BiPo event selection cuts and their corresponding signal efficiencies for $^{212}\text{BiPo}$ and $^{214}\text{BiPo}$ events. The FV is defined as a cylinder of 2.8 m height and diameter. $E_{\text{Bi, Po}}$ are the energy windows selected for prompt and delayed decays, respectively. Δt and Δr refer to the applied timing and distance cuts. For Δr , values in parentheses indicate the rejection efficiency for accidental coincidences. Adapted from [20].

Cut	$^{212}\text{Bi-Po}$		$^{214}\text{Bi-Po}$	
	Range	Efficiencies (%)	Range	Efficiencies (%)
FV	< 140 cm	–	< 140 cm	–
E_{Bi}	0.14–2.5 MeV	98	1.4–3.5 MeV	83
E_{Po}	0.9–1.5 MeV	100	0.7–1.1 MeV	99
Δt	0.2–2 μs	62	0.2–711 μs	95
Δr	< 72.5 cm	98 (94)	< 45 cm	88 (98)
Total	–	60	–	69

2.4 Motivation for Using Machine Learning

Identifying BiPo events in OSIRIS is a challenging task, as it requires separating true decay coincidences from a dominant and diverse background. While classical selection cuts can isolate certain signal regions, they are inherently limited by their rigidity and do not scale well to multi-dimensional correlations or overlapping feature spaces. Events that fall in between well-separated classes are often misclassified.

Machine learning (ML) techniques offer a powerful alternative. In high-dimensional feature spaces, where manual thresholds tend to drown in noise and parameter tuning becomes increasingly ineffective, ML models can learn non-linear decision boundaries and internal representations of patterns within energy, time, and spatial distributions, without relying on explicit, feature-by-feature cuts. This enables the models to exploit subtle correlations and interdependencies that are not easily accessible through manual optimization of cut parameters.

In particular, Artificial Neural Network (ANN) and Recurrent Neural Network (RNN) can be trained to distinguish between different types of events by leveraging the full structure of the input data. When trained on simulated datasets labeled with known event types, these models can generalize to unseen data and provide probabilistic classification, facilitating not just binary decisions but uncertainty-aware tagging.

Given the ultimate goal of identifying rare signal processes such as $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ coincidences amidst a large background, machine learning enables a data-driven approach that improves both sensitivity and signal retention. Additionally, the flexibility of ML models makes them suitable for adapting to future improvements in detector simulations, data volume, and analysis scope, offering a scalable path forward as OSIRIS and JUNO continue to develop.

2.5 Summary

This chapter introduced the theoretical and experimental framework underlying this thesis. The first section reviewed the fundamentals of neutrino physics, including the phenomenon of neutrino oscillations, the mass hierarchy problem, and the importance of precision measurements in resolving outstanding questions in particle physics.

The second part presented the JUNO, a next-generation LS detector designed to determine the neutrino mass ordering and perform high-precision measurements of reactor and solar neutrino fluxes. JUNO's design, featuring a 20 kt target volume and excellent energy resolution, enables sensitivity to oscillation parameters at an unprecedented scale.

In the final section, OSIRIS was described. As a subsystem of JUNO's LHS, OSIRIS plays a critical role in verifying the radiopurity of the LS before it is filled into the CD. Its design, including a dedicated AV, PMT array, calibration systems, and muon veto, enables precise measurement of trace contaminants, especially from the ^{238}U and ^{232}Th decay chains. The BiPo coincidence method was highlighted as the primary signal for detecting these impurities, and the chapter outlined both the background sources and event selection techniques used in this analysis.

Together, these components form the foundation for the data analysis and event classification studies that follow in this thesis.

Chapter 3

Dataset Description

This chapter provides an overview of the dataset used throughout this thesis. A detailed understanding of the input data is essential, as it forms the basis for both the cut-based and machine learning classification approaches developed later.

Section 3.1 introduces the structure of the dataset, including its origin, features, truth labels, and overall composition. This is followed by Section 3.2, which explores the distributions of key features such as energy, spatial coordinates, and time differences, supported by statistical fits.

3.1 Description of the Data

The data used in this work was generated using a toy Monte Carlo simulation designed to model BiPo events and their detection characteristics. It has the following features: Event Time (s), total number of collected Photoelectrons (p.e.) from all PMTs, spatial coordinates (X, Y, Z) in mm, and truth labels that encode the event type. The dataset consists of 6,443,032 simulated events, classified into six categories. These include internal and external background events, as well as BiPo decays from the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains. The distribution of events across these categories is summarized in Table 3.1.

The internal background is simulated as a flat energy distribution between 0 and 10 MeV, with a mix of internal gammas and neutron recoils. The external background is modeled as purely gamma radiation. An energy resolution of 4.7% (independent of energy) is applied to all events in the simulation.

Table 3.1: Event distribution in the dataset. The dataset contains a total of 6,443,032 simulated events. The classification consists of six categories: internal and external backgrounds, as well as BiPo events from the ^{238}U and ^{232}Th decay chains. The percentages indicate the fraction of events belonging to each class.

Truth Label	Event Type	Percentage of Events
0	Internal Background	16.73%
1	^{214}Bi	16.81%
2	^{214}Po	16.81%
3	External Background	16.73%
4	^{212}Bi	16.46%
5	^{212}Po	16.46%

It is important to note that this class distribution does not reflect a real experimental scenario. In practice, background events (labels 0 and 3) would dominate the dataset, potentially making up more than 99% of all recorded events, while BiPo coincidences (labels 1, 2, 4, 5) are rare. Moreover, among the signal events, ^{214}Bi and ^{214}Po (labels 1 and 2) are generally expected to be more prevalent than ^{212}Bi and ^{212}Po (labels 4 and 5), due to differences in half-lives and decay chain abundances in typical detector environments. Exact rates depend on the detector geometry, background suppression techniques, and levels of radioactive contamination. However, since these values are highly experiment-specific and not available for this toy dataset, a uniform class distribution is intentionally used to facilitate training and evaluation of the machine learning models.

The first preprocessing step includes converting the p.e. to energy using the following relation:

$$E \text{ [MeV]} = \frac{\text{p.e.}}{280}$$

This conversion factor reflects the simulated photoelectron yield for 1 MeV electrons at the center of the OSIRIS detector, which is approximately 280 p.e./MeV. It results from the optical coverage of the PMT array (9%) and detector-specific features such as shielding geometry and light collection efficiency [20]. The resulting energy values in MeV are used for all subsequent analysis steps.

3.2 Data Exploration and Fitting

The dataset was analyzed in terms of its key feature distributions to verify that the toy Monte Carlo simulation reproduces the expected physical behavior of radioactive decay.

3.2.1 Decay Curve Fits for the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ Events

Figs. 3.1 and 3.2 show the decay curves associated with the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ coincidences, respectively. These curves are constructed by calculating the time difference between the Bi and Po events in each identified BiPo pair. In this case, the pairs are identified using the truth labels provided by the simulation: each ^{214}Bi or ^{212}Bi event (label 1 or 4) is paired with the immediately following ^{214}Po or ^{212}Po event (label 2 or 5), which always appears consecutively in the dataset. Once a Bi–Po pair is identified, the time difference between the two events is computed. Since the Po isotope is the daughter nucleus in both decay chains and decays after the Bi event, the resulting time difference distribution reflects the decay timing of the Po isotopes.

The observed time distributions are fitted with the exponential decay law:

$$N(t) = N_0 e^{-\frac{t}{\tau}} \quad (3.1)$$

where $N(t)$ is the number of events as a function of time t , N_0 is the number of events at $t = 0$, and τ is the mean lifetime of the decaying isotope. The mean lifetime is related to the half-life by the relation $t_{1/2} = \ln(2) \cdot \tau$.

For the $^{214}\text{BiPo}$ events, the fit yields a mean lifetime of $\tau = 2.37 \cdot 10^{-4}$ s, corresponding to a half-life of approximately 164 μs . This is consistent with the known half-life of ^{214}Po [26], the daughter of ^{214}Bi in the ^{238}U decay chain.

Similarly, for the $^{212}\text{BiPo}$ events, the fit gives $\tau = 4.288 \cdot 10^{-7}$ s, which corresponds to a half-life of approximately 297 ns. This value agrees with the known half-life of ^{212}Po [26], the daughter of ^{212}Bi in the ^{232}Th decay chain.

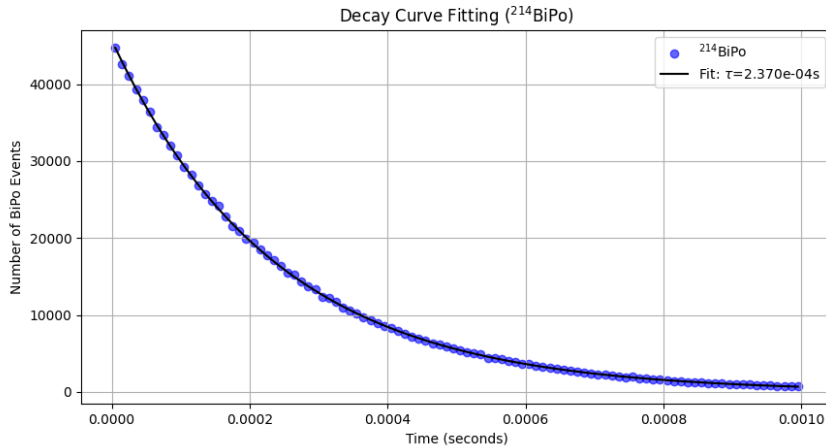


Figure 3.1: Decay curve for $^{214}\text{BiPo}$ events, derived from the time difference between ^{214}Bi and ^{214}Po decays. The fitted mean lifetime is $\tau = 2.37 \times 10^{-4}$ s, corresponding to a half-life of 164 μs , consistent with the known lifetime of ^{214}Po .

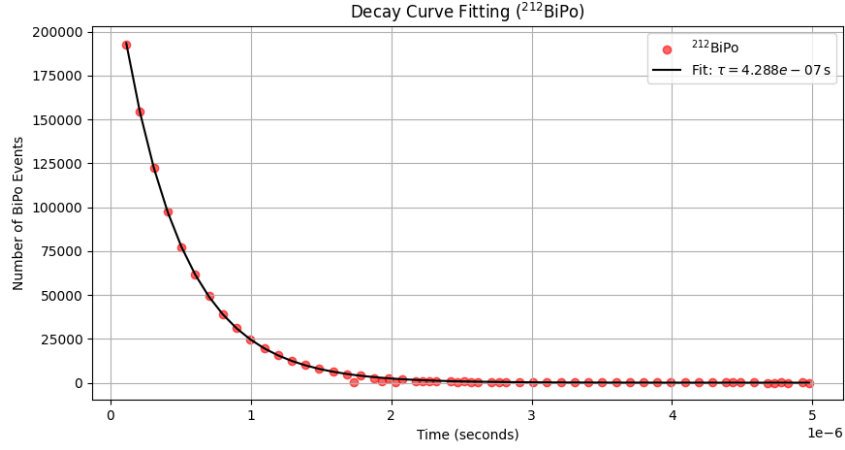


Figure 3.2: Decay curve for $^{212}\text{BiPo}$ events, calculated from the time delay between ^{212}Bi and ^{212}Po events. The extracted mean lifetime is $\tau = 4.288 \times 10^{-7}$ s, giving a half-life of 297 ns, consistent with that of ^{212}Po .

3.2.2 Energy Distributions of ^{214}Bi , ^{214}Po , ^{212}Bi , and ^{212}Po

Energy distributions for the four isotopes of interest, ^{214}Bi , ^{214}Po , ^{212}Bi , and ^{212}Po , are shown in Figs. 3.3 through 3.6.

For the Bi isotopes, which undergo beta decay, the energy spectra are broad and continuous due to the distribution of electron energies. As such, no functional fit is applied to these distributions, since a Gaussian function would not be a physically accurate representation of the beta decay spectrum. The histograms for ^{214}Bi and ^{212}Bi are instead shown directly to illustrate their overall shapes in Figures 3.3 and 3.4.

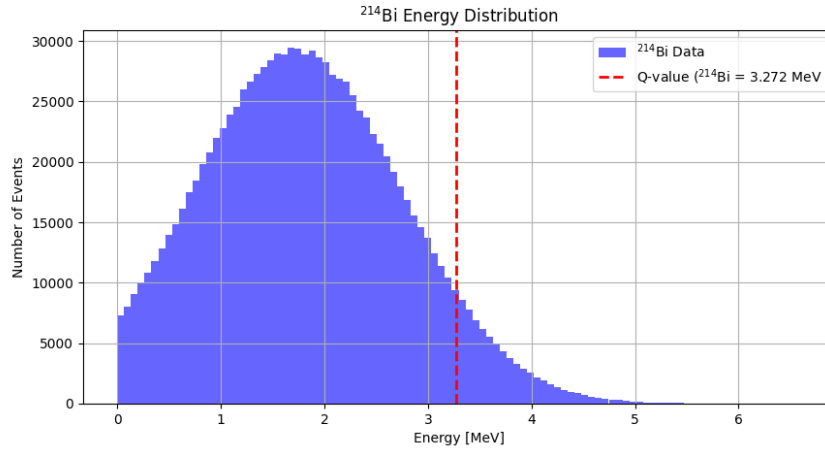


Figure 3.3: Energy distribution of ^{214}Bi , exhibiting a broad spectrum characteristic of beta decay

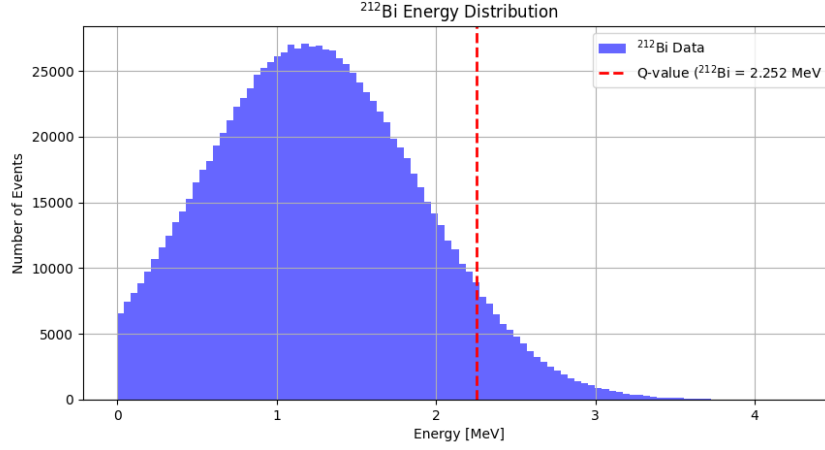


Figure 3.4: Energy distribution of ^{212}Bi , also showing a wide beta spectrum without a sharp peak

In contrast, the Po isotopes decay via alpha emission, which is monoenergetic for a given isotope. As a result, their energy distributions are sharply peaked and well-approximated by Gaussian functions. These fits provide a simplified but effective description of the energy resolution observed in the detector. The Gaussian function used is given by:

$$f(E) = A \cdot \exp\left(-\frac{(E - \mu)^2}{2\sigma^2}\right) \quad (3.2)$$

where A is the amplitude (peak height), μ is the mean energy (peak position), and σ is the standard deviation representing the detector's energy resolution.

For ^{214}Po (Fig. 3.5), the peak occurs at 0.765 MeV with a standard deviation of $\sigma = 0.036$ MeV. For ^{212}Po (Fig. 3.6), the peak is located at 0.877 MeV with $\sigma = 0.041$ MeV.

This clear difference between the broad beta decay spectra of Bi isotopes and the narrow alpha peaks of Po isotopes, though overlapping in energy, remains essential for event classification. The distinct spectral shapes allow statistical separation of BiPo coincidences and support background suppression.

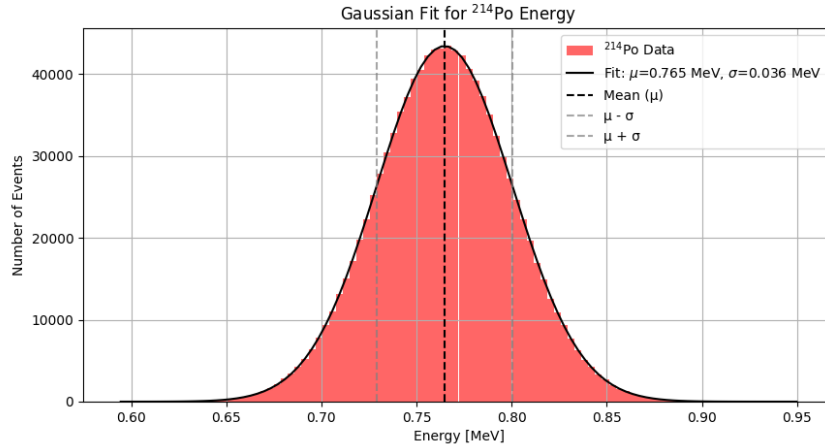


Figure 3.5: Gaussian fit of the ^{214}Po energy distribution, peaking at 0.765 MeV with $\sigma = 0.036$ MeV.

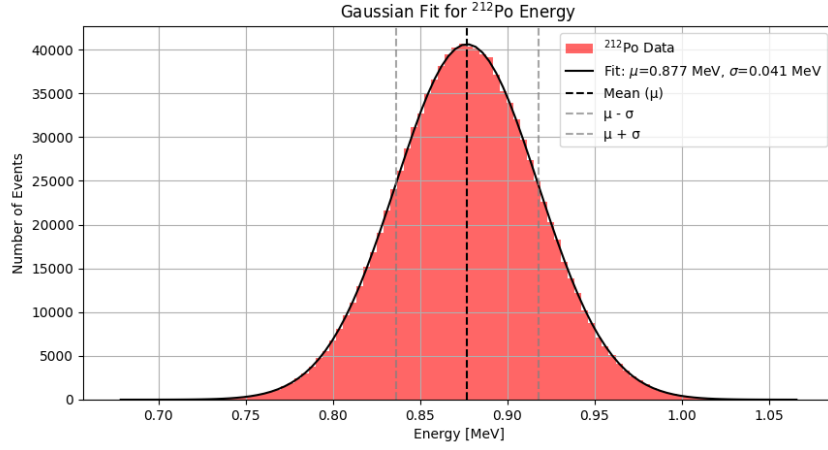


Figure 3.6: Gaussian fit of the ^{212}Po energy distribution, peaking at 0.877 MeV with $\sigma = 0.041$ MeV.

3.2.3 Comparison of the Energy distributions

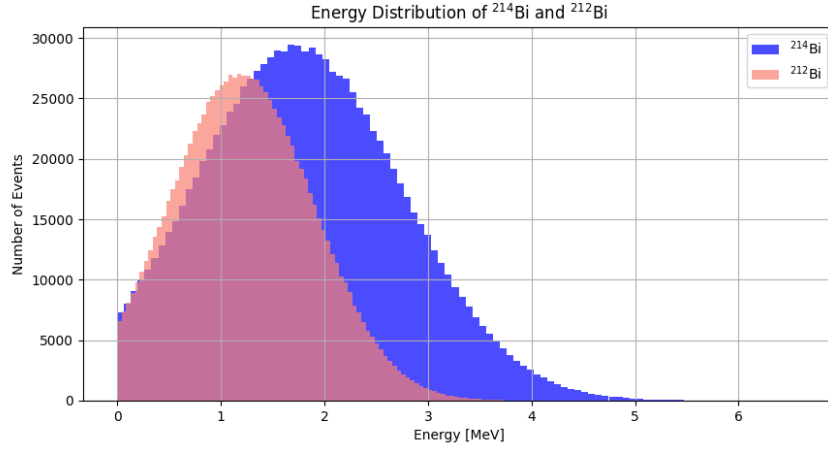


Figure 3.7: Energy distributions of ^{214}Bi and ^{212}Bi . The two distributions exhibit significant overlap, making it challenging to distinguish between them.

Fig. 3.7 shows the energy distribution of events corresponding to ^{214}Bi and ^{212}Bi . The two distributions overlap significantly, with the ^{214}Bi spectrum (blue) extending to higher energies compared to the ^{212}Bi spectrum (salmon). This significant overlap becomes the main reason for developing separate models for the two chains, as will be seen in Section 5.2.3.

Similarly, Figure 3.8 shows the energy distributions for ^{214}Po and ^{212}Po . In this case, the overlap is much smaller, and the ^{212}Po spectrum (salmon) extends to slightly higher energies than the ^{214}Po spectrum (blue). As will be seen in the next chapter, this overlap leads to some misclassifications in this energy range.

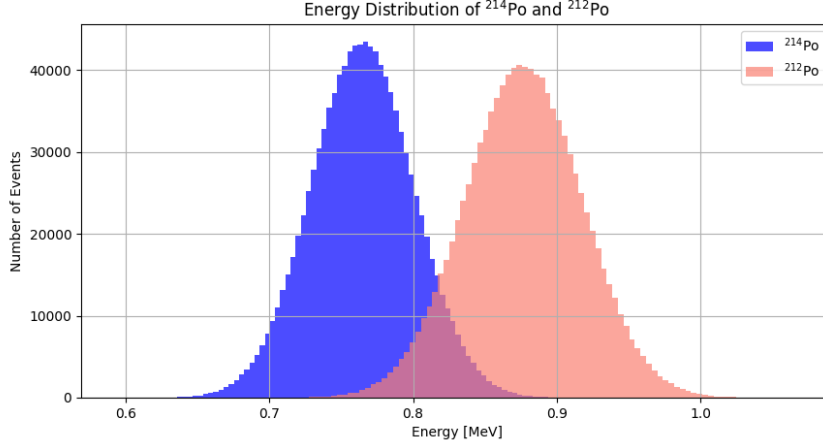


Figure 3.8: Energy distributions of ^{214}Po and ^{212}Po . Unlike the Bi isotopes, the overlap between these distributions is minimal.

Fig. 3.9 shows a combined energy spectrum in the region of interest, allowing for better visualization of individual contributions. The full energy spectrum can be found in Appendix B.1. Colored overlays highlight the contributions from different signal truth categories. The energy distributions discussed above can be seen in this plot as well. The contributions in the gray areas are from the internal and external backgrounds, along with the signals.

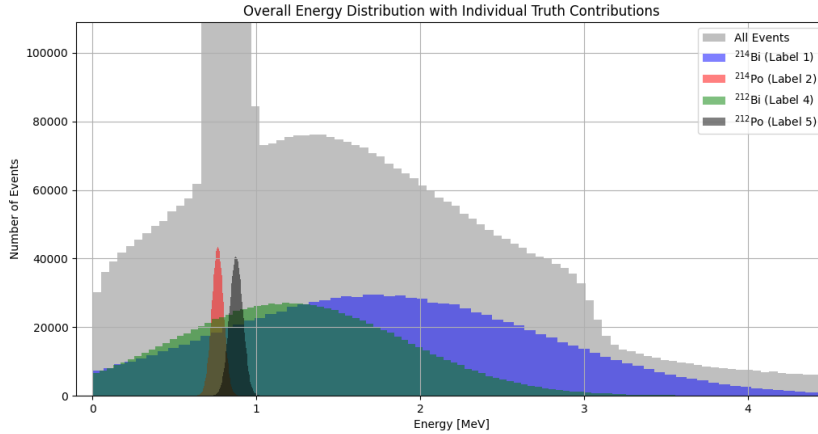


Figure 3.9: Zoomed-in view of all energy distributions in the dataset. Colored overlays highlight contributions from different signal truth categories, including BiPo events, and gray areas represent background components along with signals. The full-range energy spectrum is available in Appendix B.1.

3.2.4 Spatial Distribution of the External Background

Figs. 3.10 and 3.11 show the spatial distribution of the external background (label 3), which originates from external gamma sources. These interactions contribute significantly to the detector background and exhibit a clear spatial pattern.

The radial distribution (Fig. 3.10) shows the highest number of events near the detector's physical outer edge. This trend aligns with the physical expectation that external gammas interact more frequently near the detector's boundaries before decaying deeper inside. The detector has a radius of 1500 mm, but the histogram extends beyond this limit, which might initially seem unphysical. However, this effect arises from the spatial reconstruction process, which can predict vertices beyond $r = 1500$ mm.

The z-coordinate distribution (Fig. 3.11) shows the accumulation of events near ± 1500 mm, which corresponds to the upper and lower boundaries of the cylindrical detector. This pattern is expected because external gammas are more likely to occur near the detector surfaces, similar to the radial distribution. Between ± 1000 mm, the event density is relatively uniform, suggesting that once the gammas penetrate the detector, their distribution becomes more homogeneous along the central axis. Again,

some event counts appear slightly outside the expected detector range, which is an artifact of the spatial reconstruction.

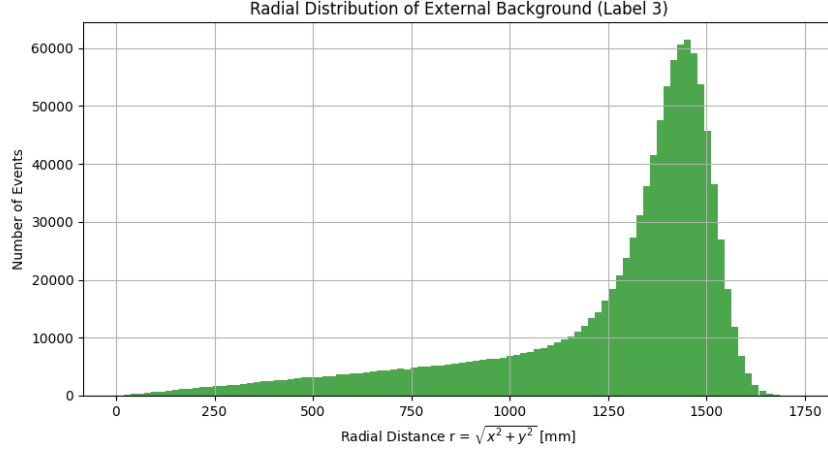


Figure 3.10: Radial distribution of the external background (Label 3). Events are concentrated near the detector's outer edge, consistent with the expectation that external gamma interactions occur more frequently near boundaries. Due to the simulated reconstruction effects, the histogram extends beyond the physical detector radius (1500 mm).

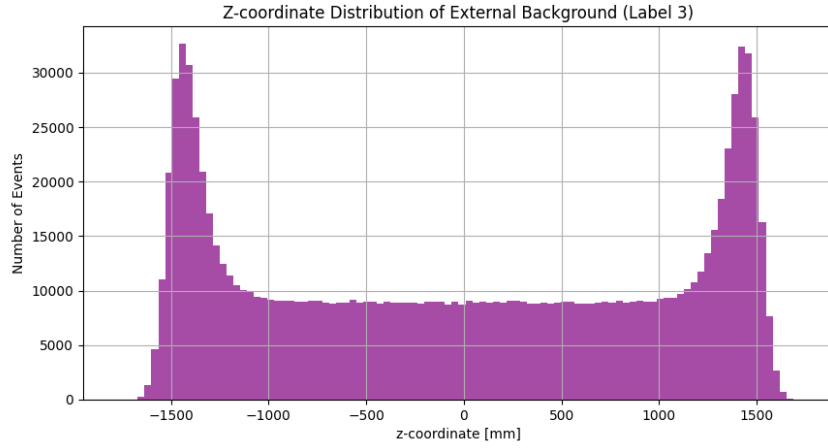


Figure 3.11: Z-coordinate distribution of the external background (Label 3). Events accumulate near ± 1500 mm, corresponding to the top and bottom boundaries of the cylindrical detector. The relatively uniform distribution between ± 1000 mm suggests homogeneous gamma interactions in the detector's central region. Some events extend beyond the expected range due to simulation reconstruction artifacts.

Chapter 4

Classical Event Selection and Optimization

Before introducing machine learning-based approaches for BiPo event classification, it is important to establish a baseline using classical, cut based selection methods. These techniques rely on well-defined thresholds applied to physical observables such as energy, timing, and spatial separation, derived from prior studies. While limited in flexibility, such cuts provide an interpretable and physics motivated way to isolate signal-like events and suppress background.

The first section in this chapter (Section 4.1) presents the classical selection strategy, including cut values adapted from literature [20] and their effect on the signal efficiency and background suppression. This is followed by an optimization step (Section 4.2), where the cut thresholds are treated as hyperparameters and optimized using the Optuna framework [27] to improve classification performance.

Finally in Section 4.3, a detailed comparison is made between the results of the classical and optimized selections in terms of signal efficiency, background rejection, and spectral features. These results serve as a benchmark for evaluating the gains provided by machine learning approaches in the following chapters.

4.1 Classical Cuts based Selection

As mentioned in Section 2.3.5, a series of five consecutive cuts is used to select BiPo coincidence events: FV, Energy (For Bi and Po), Time Difference, and Distance cuts. These cuts are designed to filter out background events while maximizing the classification efficiency of true BiPo events.

The cut values are based on results from a prior optimization study in [24]. The selection criteria were optimized using a grid search algorithm over the multi-dimensional parameter space defined by the five main cut parameters: FV, E_{Bi} , E_{Po} , Δr , and Δt . The lower time threshold of $\Delta t = 0.2 \mu\text{s}$ was fixed based on the detector's timing resolution, while the upper limit was varied during the grid search. The optimization strategy and final selection values are fully detailed in [24]. The grid search procedure iteratively evaluates combinations of cut values to maximize the BiPo tagging sensitivity using the Feldman-Cousins (FC) method. The FC sensitivity refers to the expected 90% confidence level upper limit on the signal strength under the background-only hypothesis and serves as a performance metric to optimize BiPo detection efficiency while maintaining statistical rigor [28].

The obtained selection cuts from [20], which are also reproduced in Table 2.4, are summarized below for both $^{212}\text{BiPo}$ and $^{214}\text{BiPo}$.

Table 4.1: Selection cuts from [20], same as Table 2.4. FV is the fiducial volume of the detector, E_{Bi} , E_{Po} are the energy windows selected for prompt and delayed decays, and Δt and Δr are time and distance cuts.¹

Cut	$^{212}\text{BiPo}$	$^{214}\text{BiPo}$
FV	$< 140 \text{ cm}$	$< 140 \text{ cm}$
E_{Bi}	$0.14 - 2.5 \text{ MeV}$	$1.4 - 3.5 \text{ MeV}$
E_{Po}	$0.9 - 1.5 \text{ MeV}$	$0.7 - 1.1 \text{ MeV}$
Δt	$0.2 - 2 \mu\text{s}$	$0.2 - 711 \mu\text{s}$
Δr	$< 72.5 \text{ cm}$	$< 45 \text{ cm}$

4.1.1 Efficiency Calculation

To quantify the effectiveness of these cuts and compare each classification technique, two key metrics are evaluated after applying the selection cuts:

- True Positive Rate (TPR) : The percentage of correctly classified BiPo events among all selected events:

$$\text{TPR} = \frac{\text{True BiPo Events}}{\text{Total Selected Events}} \cdot 100\% \quad (4.1)$$

- Efficiency: The percentage of true BiPo events in the dataset that are successfully classified:

$$\text{Efficiency} = \frac{\text{True BiPo Events}}{\text{Total BiPo Events in the Dataset}} \cdot 100\% \quad (4.2)$$

These values provide insight into the performance of the selection criteria and indicate how well the cuts separate BiPo events from background noise.

4.1.2 Application of the Classical Cuts

The selection process was performed separately for the ^{238}U and ^{232}Th decay chains. The order in which these cuts are implemented does not affect the results. Here the cuts were applied in the following sequence:

- **Energy Cuts:** The energy selection was applied separately to Bi and Po, restricting events to the energy ranges defined in Table 4.1. Only events within these specified ranges were retained.
- **FV Cut:** Events occurring outside a radial distance of 1400 mm were rejected. After this cut, all remaining events lie within the FV of the detector and the defined energy ranges.
- **Time cut:** The most stringent and effective cut is the time cut, as Bi decays into Po within a well-defined time window characterized by the Po mean lifetimes ($\tau = 237 \mu\text{s}$ for ^{214}Po and $\tau = 0.431 \mu\text{s}$ for ^{212}Po). The BiPo pairs are identified by matching each Bi event with the nearest Po event occurring within a predefined time window after the Bi decay. Only Po events that fall within this time range are considered valid pairs, ensuring that the time difference is consistent with the expected exponential decay of the Po isotope. Once a valid pair is found, both events are removed from consideration in subsequent pairings, ensuring that no event is reused. This step reduces background events by selecting only pairs within the expected decay timeframe. After applying this cut, the number of events classified as Bi and Po match.
- **Distance cut:** The distance cut is applied by identifying BiPo event pairs that, in addition to falling within the expected time window, also occur within a specified spatial separation. Only pairs whose reconstructed positions are closer than the defined distance threshold are retained, ensuring that they originate from spatially correlated decays. As with the time cut, once a valid spatially and temporally consistent pair is found, both events are excluded from further consideration to avoid reuse. The events passing this final selection are classified as BiPo coincidences.

4.1.3 Results and Visualization

Efficiency of the Cuts

To evaluate the effectiveness of the applied cuts, two key metrics, TPR and Efficiency (Eqs. 4.1 and 4.2), are analyzed. These metrics are computed after applying the cut values summarized in Table 4.2, which were obtained through the grid search procedure described earlier. The computed values for both the chains are summarized in Table 4.2.

From Table 4.2, it is evident that for the $^{214}\text{BiPo}$ chain, approximately 47.34% of true signal events remain after the cuts, while a significant portion (over 50%) is removed. To better understand the origin of selected events, the class-wise composition is summarized in Table 4.3. For $^{214}\text{BiPo}$, 68.66% of the selected events originate from the correct signal class, with only 31.33% originating from $^{212}\text{BiPo}$.

In contrast, for the $^{212}\text{BiPo}$ chain, although the TPR is nearly perfect (99.99%), the efficiency drops to 14.54%. The cause of this efficiency drop will be discussed in the following subsections. Among the selected events, 98.9% belong to the correct class, while 1.1% are misclassified events, originating from the $^{214}\text{BiPo}$ chain.

Table 4.2: Calculated TPR and efficiencies for both the chains after application of the classical cuts. Here, TPR measures the fraction of selected events that are true BiPo signals and is therefore sensitive to the dataset composition, especially the relative amount of background. Efficiency, on the other hand, quantifies the fraction of all true BiPo events retained after the cuts and remains independent of background abundance.

Metrics	$^{214}\text{BiPo}$	$^{212}\text{BiPo}$
TPR	68.66%	99.99%
Efficiency	47.34%	14.54%

Table 4.3: Class-wise composition of selected events after classical cuts, showing the proportion of true signal and misclassified events for each BiPo chain.

Event Source	$^{214}\text{BiPo}$ Selected Events	$^{212}\text{BiPo}$ Selected Events
True signal events	68.66%	98.9%
Cross-chain contamination	31.33%	1.1%
Background contamination	0%	0%

Decay Curves of $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ Events

Figures 4.1 and 4.2 show the distributions of time differences between Bi and Po events on a logarithmic scale, which reflect the exponential decay behavior of the daughter isotopes ^{214}Po and ^{212}Po , respectively. These distributions follow the characteristic lifetime of the Po nuclei.

In Figure 4.1, the exponential decay of ^{214}Po is clearly visible. A high initial event count is observed, largely due to contamination from misclassified $^{212}\text{BiPo}$ events, which decay on a much shorter timescale, as discussed in Section 3.2.1.

For $^{212}\text{BiPo}$ (Fig. 4.2), the extracted decay curve remains limited in quality despite adjusting the bin size. The logarithmic scale makes the rapid drop in event count due to the short half-life of ^{212}Po more apparent. While the early part of the distribution still reflects the expected exponential shape, the tail lacks sufficient counts to yield a smooth and statistically meaningful curve.

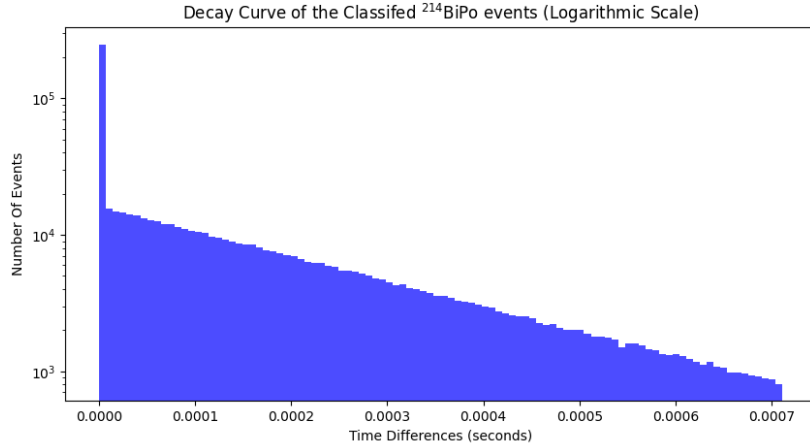


Figure 4.1: Time difference distribution of classified $^{214}\text{BiPo}$ events, corresponding to the decay of ^{214}Po with a high initial count of events.

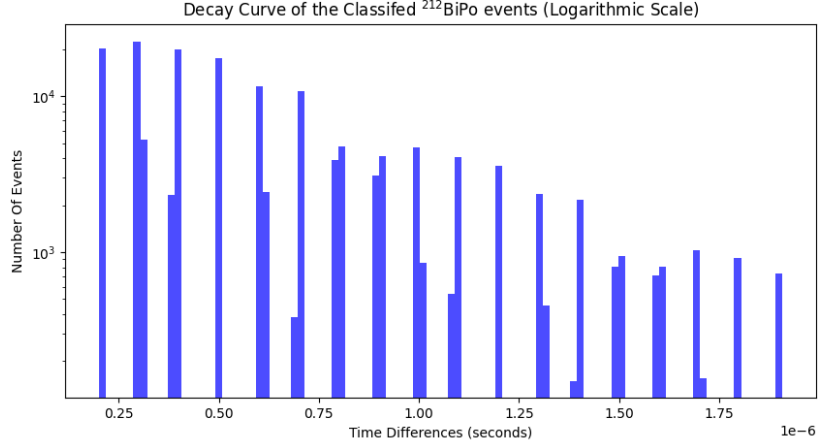


Figure 4.2: Time difference distribution of classified $^{212}\text{BiPo}$ events. The short-lived exponential decay of ^{212}Po is slightly visible despite the low statistics.

Energy Spectra of ^{214}Bi , ^{214}Po , ^{212}Bi , and ^{212}Po

The energy distribution of ^{214}Bi after classification via the classical cuts is shown in Fig. 4.3). Comparing it with the original distribution in Figure 3.4, it is evident that the lower half of the energy spectrum is entirely removed. This is expected, as the peak of the distribution is at 1.733 MeV, while the cut is applied at 1.4 MeV, effectively eliminating lower-energy events.

The energy distribution of ^{214}Po (Fig. 4.4) shows an additional peak along with the characteristic monoenergetic peak. This trailing component, extending beyond the expected distribution, is due to misclassified events from the other decay chain. A comparison with Figure 3.8 shows that this excess originates in the overlapping energy region of the two chains and extends into the energy range characteristic of the other chain

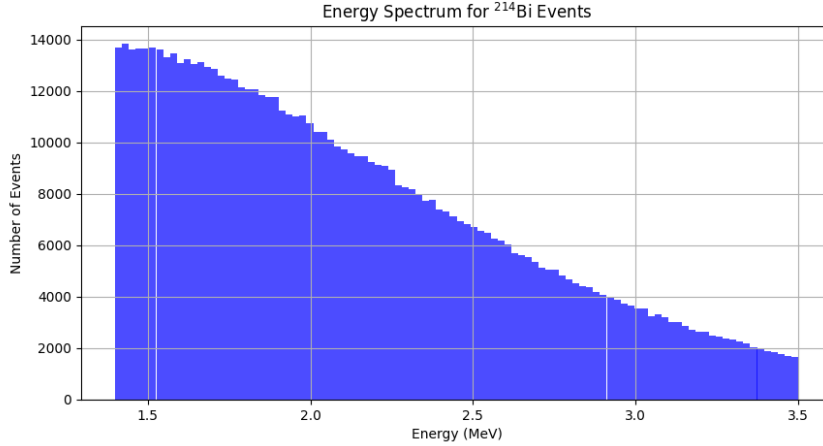


Figure 4.3: Energy distribution of classified ^{214}Bi events after applying classical cuts. The lower half of the spectrum is removed due to the 1.4 MeV threshold, cutting off much of the low-energy tail.

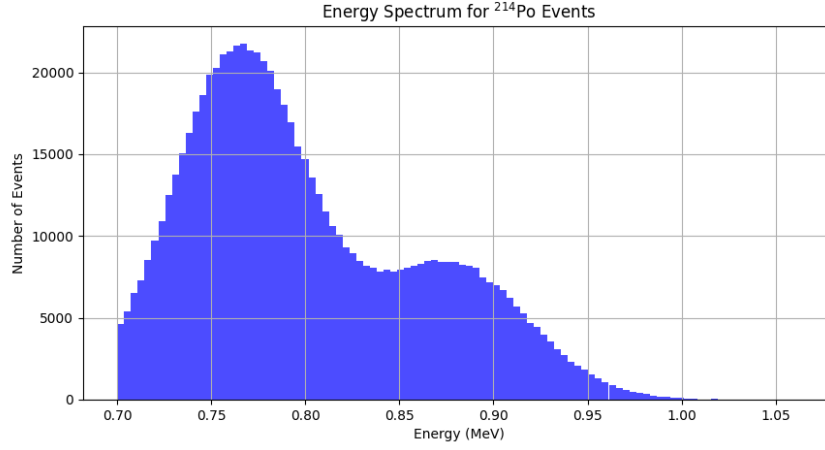


Figure 4.4: Energy distribution of classified ^{214}Po events. Alongside the expected monoenergetic peak, a trailing excess appears due to misclassified events from the $^{212}\text{BiPo}$ chain.

For the second chain, despite the low statistics, the shape of ^{212}Bi energy distribution (Figure 4.5) remains largely consistent with its original distribution in Figure 3.4. However, the monoenergetic peak of ^{212}Po is completely absent in Figure 4.6. This is primarily due to the energy cut at 0.9 MeV, which eliminates most of the events, including the peak at 0.877 MeV. As a result, very few ^{212}Po events are retained, leading to low statistics for this chain. The energy cut is thus the primary factor responsible for this effect and the selection criteria disproportionately impact this chain.

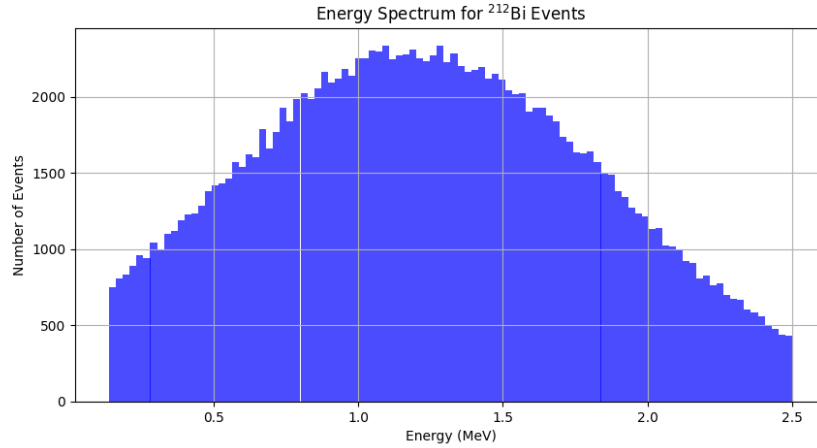


Figure 4.5: Energy distribution of classified ^{212}Bi events. Despite lower statistics, the overall spectral shape is consistent with expectations from beta decay.

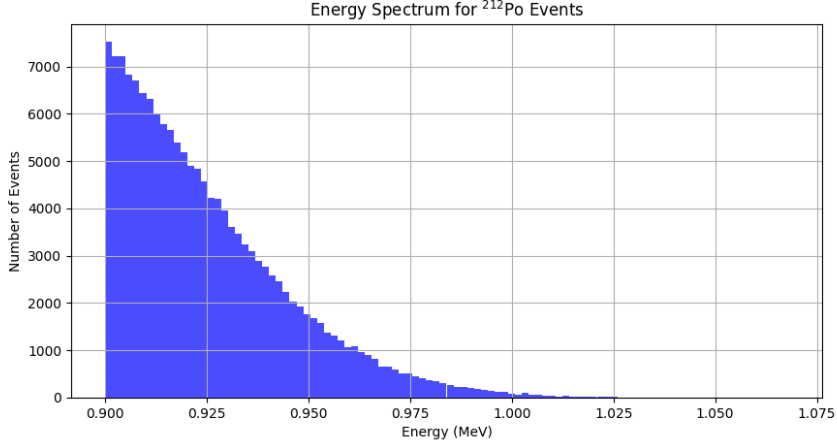


Figure 4.6: Energy distribution of classified ^{212}Po events. The monoenergetic peak at 0.877 MeV is missing due to the 0.9 MeV cut, which removes the majority of true ^{212}Po events.

2D Energy Correlations of Bi-Po Events

The 2D histograms (Figs. 4.7 and 4.8) display the energy distributions of classified $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ coincidences. The color intensity represents event density, with darker regions indicating higher concentrations. These plots show how the energy cuts influence the selection process and highlight potential contamination between the two chains.

The histogram for the $^{214}\text{BiPo}$ events (Fig. 4.7) indicates a high concentration of events around $E_{\text{Po}} \approx 0.75 - 0.85$ MeV, which corresponds to the ^{214}Po monoenergetic peak that occurs at 0.785 MeV. The ^{214}Bi energy spectrum, which originally peaked at 1.733 MeV, is visibly truncated below 1.4 MeV due to the applied energy cut, effectively removing lower energy events. However, there is a trailing component in E_{Po} extending into the energy range characteristic of ^{212}Po , suggesting some degree of misclassification.

The $^{212}\text{BiPo}$ histogram (Fig. 4.8), on the other hand, is much sparser, with most events concentrated in the lower energy region of ^{212}Bi around 1 MeV, while ^{212}Po appears almost entirely absent. The expected monoenergetic peak at 0.877 MeV is missing, which can be directly attributed to the energy cut at 0.9 MeV that removes most events in this range.

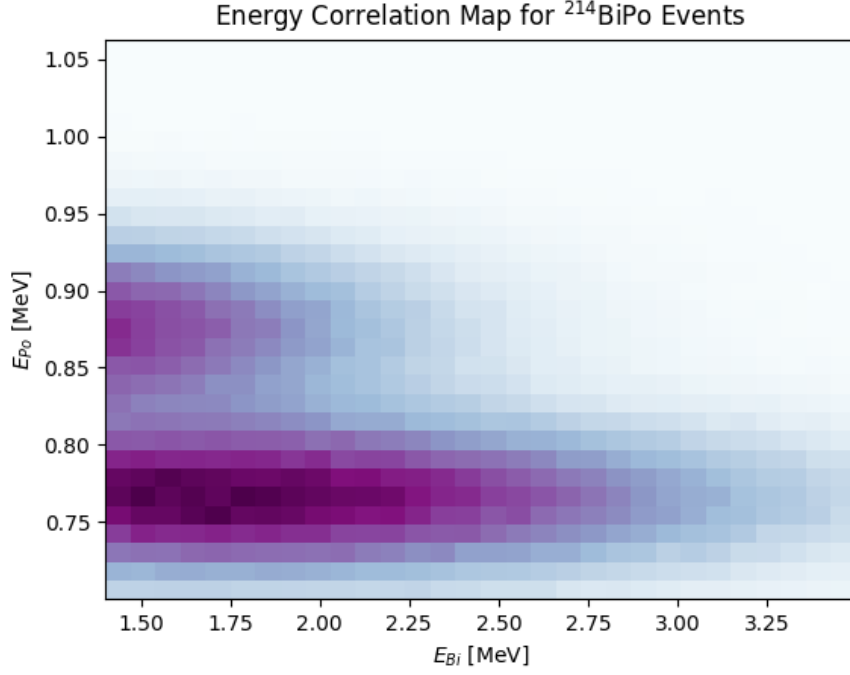


Figure 4.7: 2D energy distribution of classified $^{214}\text{BiPo}$ events after applying classical cuts. A dense cluster is visible near the ^{214}Po monoenergetic peak at ~ 0.785 MeV, while the ^{214}Bi energy distribution is truncated below 1.4 MeV. A faint trail in E_{Po} suggests misclassification from the $^{212}\text{BiPo}$ chain.

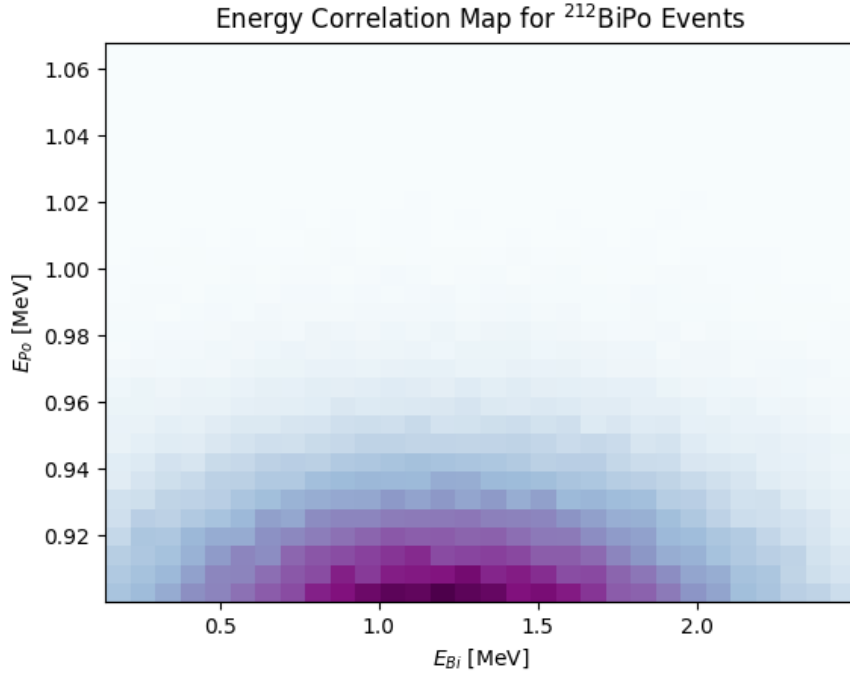


Figure 4.8: 2D energy distribution of classified $^{212}\text{BiPo}$ events after applying classical cuts. The distribution is sparse, and the expected ^{212}Po peak at 0.877 MeV is absent due to the 0.9 MeV lower cut. Some ^{212}Bi events are retained, but their Po counterparts are mostly excluded.

4.2 Optimizing Cuts Using Optuna

Hyperparameter optimization is essential for tuning selection criteria in classification tasks. In the context of BiPo event tagging using classical cuts, the cut parameters, such as FV, energy windows, time

difference, and spatial distance thresholds, are treated as hyperparameters. This is because they are not learned directly from the data but must be set manually or optimized externally to achieve the best classification performance. Traditionally, this is done using grid search, which exhaustively evaluates all possible parameter combinations but is computationally expensive and memory-intensive. In contrast, Optuna is a next-generation define-by-run hyperparameter optimization framework designed to dynamically explore and refine the search space while incorporating efficient pruning strategies.

4.2.1 How Optuna Works

The general optimization problem can be expressed as:

$$\theta^* = \arg \max_{\theta \in \Theta} f(\theta) \quad (4.3)$$

where:

- $f(\theta)$ is the objective function that evaluates the performance of a given set of parameters θ
- Θ is the search space of hyperparameters
- θ^* represents the optimal set of parameters

Optuna formulates the hyperparameter optimization process as a black box optimization problem, where an objective function is minimized (or maximized) over a given parameter space. Optuna introduces several novel design criteria aimed at enhancing flexibility, efficiency, and ease of use [27]:

- **Define-by-Run API:** Optuna constructs the search space dynamically during execution using standard Python syntax. This enables conditional and adaptive search strategies, as hyperparameters can be defined within control structures based on earlier suggestions.
- **Efficient Search and Pruning:** It integrates the Tree-structured Parzen Estimator (TPE) for intelligent sampling and Asynchronous Successive Halving Algorithm (ASHA) for pruning underperforming trials, reducing computational cost.
- **Versatile Architecture:** Optuna is designed for scalability, supporting distributed computing while remaining lightweight enough for interactive optimization tasks.

Optuna employs the TPE [27], a Bayesian optimization approach that models the parameter space using two probability densities:

$$l(\theta) = P(\theta \mid f(\theta) < \gamma), \quad g(\theta) = P(\theta \mid f(\theta) \geq \gamma) \quad (4.4)$$

where:

- $l(\theta)$ represents the distribution of promising hyperparameters (those yielding better-than-threshold performance).
- $g(\theta)$ represents the distribution of less effective hyperparameters.
- γ is an adaptively chosen threshold defining the top-performing trials.

Here, $P(\theta \mid f(\theta))$ does not represent a true analytical likelihood but a probability density function estimated non-parametrically from past trials. TPE models these densities using kernel density estimation (KDE), constructing $l(\theta)$ and $g(\theta)$ from the observed hyperparameter configurations and their corresponding objective function values. These approximations guide the selection of new hyperparameters by maximizing the expected improvement over the current best performance.

New candidate hyperparameters are sampled by maximizing the expected improvement:

$$\theta_{\text{new}} = \arg \max_{\theta} \frac{l(\theta)}{g(\theta)} \quad (4.5)$$

This method efficiently focuses exploration on high-potential areas of the parameter space while reducing unnecessary evaluations, making it significantly more efficient than traditional grid or random search methods.

Optuna integrates an asynchronous pruning mechanism to stop underperforming trials early, improving computational efficiency. Using the ASHA, the optimizer evaluates intermediate results and terminates less promising trials, allowing resources to be allocated more efficiently to better-performing configurations [27].

ASHA works by:

1. Running multiple trials in parallel and evaluating them periodically.
2. Comparing intermediate performance metrics to pre-defined thresholds.
3. Pruning (stopping) trials that fall below the performance threshold.

This pruning strategy significantly speeds up the optimization process by eliminating poor hyperparameter choices early, reducing overall computation time while maintaining high optimization accuracy.

4.2.2 Applying Optuna to Cut Optimization

Traditional cut selection relies on values from previous studies, which may not be optimal for the current dataset. By contrast, Optuna dynamically adjusts the cut values based on statistical performance, leading to a more efficient classification process, searching for optimal cut thresholds that maximize classification performance based on a selected metric. The optimization process is structured as follows:

Defining the Parameter Space

For both $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$, the cut values for energy selection, time difference, spatial separation, and FV (radial and z-coordinate) were treated as tunable hyperparameters.

$$\theta = \{r_{fv}, h_{fv}, E_{Bi_{\min}}, E_{Bi_{\max}}, E_{Po_{\min}}, E_{Po_{\max}}, t_{\max}, d_{\max}\} \quad (4.6)$$

where:

- r_{fv}, h_{fv} are the FV cuts (radial and z coordinate or height)
- $E_{\min}, E_{\max}$ are the lower and upper bounds for energy selection
- $t_{\max}$ is the maximum allowed time difference between Bi and Po events
- $d_{\max}$ is the maximum allowed spatial separation between Bi and Po events

The search space for each chain was defined separately, considering their distinct energy ranges and decay characteristics. The ranges set for Optuna's optimization process can be seen in table 4.4

Table 4.4: Search Ranges for Hyperparameter Optimization Using Optuna for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$

Cut Parameter	$^{214}\text{BiPo}$ Search Range	$^{212}\text{BiPo}$ Search Range
r_{fv} (mm)	800 – 1500	800 – 1500
h_{fv} (mm)	1000 – 1500	1000 – 1500
$E_{Bi_{\min}}$ (MeV)	0.1 – 3	0.05 – 2.5
$E_{Bi_{\max}}$ (MeV)	1 – 5	1 – 4.5
$E_{Po_{\min}}$ (MeV)	0.1 – 2	0.1 – 1.5
$E_{Po_{\max}}$ (MeV)	0.5 – 3	0.8 – 5
$t_{\max}$ (μs)	0.2 – 7110	0.2 – 30
$d_{\max}$ (mm)	250 – 700	450 – 1200

The selection of search ranges for hyperparameter optimization in $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ was based on a combination of physical constraints, experimental considerations, and exploratory testing:

FV Cuts (r_{fv}, h_{fv})

The upper bound for the FV cut was set at 1500 mm, corresponding to the detector's maximum radius. This ensures that no events are selected beyond the physical boundaries of the detector. Unlike classical cuts, where only the radial FV r_{fv} was considered, an additional height parameter h_{fv} was introduced to allow for a more flexible selection. The lower limit for these parameters (r_{fv}, h_{fv}) was set at 800 mm and 1000 mm, respectively. This lower bound was chosen as part of an exploratory approach to investigate whether restricting events to a more central region could improve classification.

Time Difference (t_{max})

The lower bound of $0.2 \mu\text{s}$ was imposed by the hardware limit, as events occurring within shorter time intervals cannot be reliably resolved. The upper limit was chosen based on the expected decay time of the isotopes while allowing for flexibility. Specifically, the upper bound for $^{214}\text{BiPo}$ was set at $7110 \mu\text{s}$, which corresponds to 10 times the original cut values, ensuring that potential long-lived decays were not prematurely excluded. For $^{212}\text{BiPo}$, the limit was kept at $30 \mu\text{s}$, reflecting its significantly shorter decay time.

Spatial Separation (d_{max})

The range for maximum allowed spatial separation between Bi and Po events was chosen to symmetrically encompass the values used in the classical cuts, allowing for variations both above and below the standard cut values. For $^{214}\text{BiPo}$, the search range of 250 mm to 700 mm ensures that proximity events are prioritized while still allowing for some spatial separation. In the case of $^{212}\text{BiPo}$, a slightly broader range was set between 450 mm and 1200 mm to investigate whether extending the spatial allowance could improve classification.

Energy Selection ($E_{\text{Bi}}, E_{\text{Po}}$)

The energy ranges were selected to ensure that the entire decay spectrum was included, avoiding the excessive truncation observed in classical cuts. The lower bounds were set to capture as much of the energy spectrum as possible, while the upper bounds were chosen to cover the full expected range of the Bi and Po decays without including unnecessary background regions. The ranges were designed to ensure that no significant portion of the spectrum was arbitrarily removed, as was the case in the classical selection criteria.

Objective Function Definition

The optimization goal is to maximize the Signal-to-Noise Ratio (SNR), defined as:

$$SNR = \frac{S}{\sqrt{S+B}} \quad (4.7)$$

where:

- S is the number of correctly classified signal (Bi + Po) events (True Positives)
- B is the number of misclassified background events (False Positives)

The choice of SNR as the optimization metric is motivated by its direct relevance to event classification performance. The original cuts were calculated using the FC approach with a grid search for cut optimization, which is well-suited for setting statistical confidence limits, particularly in low-statistics scenarios. However, for this work, SNR is preferred due to the following reasons:

- **Direct Impact on Classification Performance:** SNR provides a straightforward measure of how well the BiPo events are distinguished from the background. A higher SNR indicates a better separation between signal and background, making it a more natural choice for machine learning-based optimization.
- **Relevance to High Event Rates:** The FC method is particularly valuable in low-rate scenarios where statistical fluctuations significantly affect the confidence intervals. In this case, the toy MC dataset reflects high expected signal and background rates, where such statistical corrections are minimal. Therefore, SNR serves as a practical and sufficiently accurate metric for optimizing the selection cuts.
- **Computational Efficiency:** FC-based optimization requires recalculating confidence intervals for each trial, which is computationally expensive, especially in an iterative framework like Optuna. SNR, on the other hand, allows for faster convergence, making it more suitable for hyperparameter tuning.

If needed, an FC-based approach can be tested separately to compare its effectiveness against SNR-based optimization. However, for this thesis, SNR was chosen as the optimization metric due to its computational efficiency, direct alignment with classification goals, and suitability for large datasets.

It is important to note that SNR is sensitive to the underlying composition of the dataset, particularly the ratio of signal to background events. In scenarios with a high background fraction, the SNR naturally decreases, even for a fixed set of cuts.

Optimization Process

The optimization process follows these steps:

1. **Sampling Cut Values:** Optuna dynamically samples different cut values for FV, energy, time, and spatial separation within the predefined search range.
2. **Evaluating Performance:** Each sampled set of cut values is evaluated using the SNR, which serves as the objective function.
3. **Refining the Search Space:** The TPE guides the search process by prioritizing regions in the hyperparameter space that are more likely to yield high SNR values.
4. **Pruning Inefficient Trials:** Optuna employs an early stopping mechanism to terminate unpromising trials, thereby reducing computational time and focusing resources on better-performing hyperparameter configurations.
5. **Running Multiple Trials:** The optimization process was conducted with 100 trials for the $^{214}\text{BiPo}$ chain and 120 trials for the $^{212}\text{BiPo}$ chain. The first chain converged more quickly, while the second required additional iterations to reach stable cut values. While Optuna includes a pruning mechanism to stop underperforming trials early, pruning was applied at the trial level based on overall objective value (SNR), rather than individual event evaluations. As intermediate results are not reported in this optimization setup, the pruning decision is deferred until each trial completes.

The evolution of the optimization process can be seen in Appendix C.2, where training curves illustrate how Optuna explores different points in the hyperparameter space. Initially, the objective function fluctuates as the algorithm explores various cut values, but over time, the best-performing region is refined. Since the optimization goal is to maximize SNR, the loss curve appears to increase, indicating the discovery of improved cut parameters that yield a higher SNR value.

Final Selection of Optimized Cuts

After running multiple trials, the optimal cut values θ^* are selected based on the highest observed SNR, ensuring the best trade-off between signal retention and background suppression.

4.2.3 Results and Visualization

The final optimized selection cuts, obtained using the Optuna hyperparameter optimization framework, are presented in Table 4.5. Compared to the classical selection, the optimized cuts result in a wider energy selection range, particularly for $^{212}\text{BiPo}$, and an improved event retention rate.

Cut	$^{212}\text{BiPo}$	$^{214}\text{BiPo}$
FV (r)	< 1483 mm	< 1144 mm
FV (h)	< 1422 mm	< 1220 mm
E_{Bi}	0.47 – 2.99 MeV	0.45 – 4.60 MeV
E_{Po}	0.29 – 0.97 MeV	0.50 – 1.03 MeV
Δt	0.015 – 1.50 μs	0.0014 – 711 μs
Δr	< 898 mm	< 498 mm
SNR	1.22	6.0

Table 4.5: Optimized selection cuts obtained through Optuna. FV represents the fiducial volume of the detector in radial (r) and height (h) dimensions, $E_{Bi, Po}$ are the energy windows for prompt and delayed decays, and Δt and Δr are the time and spatial separation constraints.

These optimized cuts will be used in the following sections to evaluate event classification performance, decay characteristics, and energy distributions. The impact of these adjustments is examined through various visualizations comparing classical and optimized selections.

Efficiency of the Optimized Cuts

To evaluate the effectiveness of the optimized cuts on the toy MC dataset, the same key metrics, TPR and Efficiency (Eqs. 4.1 and 4.2), are analyzed. The computed values for both chains are summarized in Table 4.6.

The table shows that for the $^{214}\text{BiPo}$ chain, the efficiency remains nearly the same at 46.81%, compared to 47.34% with the classical cuts. However, the TPR has slightly decreased from 68.66% to 61.95%, indicating that the optimized selection criteria are now removing a larger fraction of true signal events. Most of the misclassified signals still originate from the other chain.

For the $^{212}\text{BiPo}$ chain, the optimized cuts result in a significant improvement in efficiency, increasing from 14.54% to 51.52%. This means that a much larger proportion of true BiPo events is retained compared to the classical selection. However, this comes at the cost of a slight decrease in TPR, from 99.99% to 91.86%, indicating that a small fraction of misclassified events passes the selection. Most of these misclassified events come from the other chain, confirming that inter-chain misclassification is a major contributor to background contamination.

The increased efficiency for the $^{212}\text{BiPo}$ chain demonstrates the effectiveness of the optimized cuts in preserving more true signal events, while the small reduction in TPR suggests a balance between background rejection and signal retention. The implications of these results, along with a direct comparison with the classical cuts, are further discussed in Section 4.3.

Metrics	$^{214}\text{BiPo}$	$^{212}\text{BiPo}$
TPR	61.95%	91.86%
Efficiency	46.81%	51.52%

Table 4.6: Calculated TPR and efficiencies for both the chains after application of the optimized cuts.

Decay Curves of $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ Events

Figures 4.9 and 4.10 show the time difference distributions of BiPo events after applying the optimized selection criteria. These distributions correspond to the decay times of the daughter isotopes ^{214}Po and ^{212}Po , respectively.

In Figure 4.9, the expected exponential decay of ^{214}Po is clearly observed across a broad time range. As in the case of classical cuts, an initial excess of events is seen, caused by misclassified $^{212}\text{BiPo}$ events with much shorter decay times (see Section 3.2.1).

For $^{212}\text{BiPo}$ (Figure 4.10), the optimized cuts now yield a much clearer exponential decay curve, in contrast to the classical selection where low statistics hindered visibility. A small number of additional events appear beyond the expected lifetime window, due to contamination from misclassified $^{214}\text{BiPo}$ events. Despite this, the distribution follows the characteristic decay shape and demonstrates significantly improved signal retention.

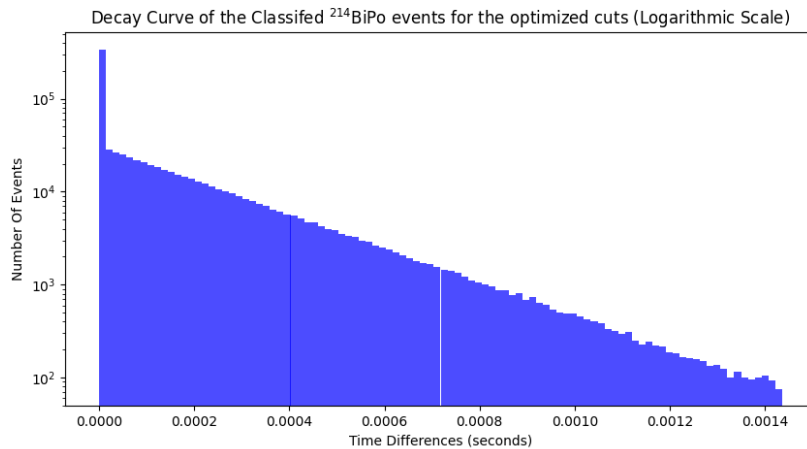


Figure 4.9: Time difference distribution of classified $^{214}\text{BiPo}$ events after applying optimized cuts. The exponential decay behavior of ^{214}Po is visible across a wide time range.

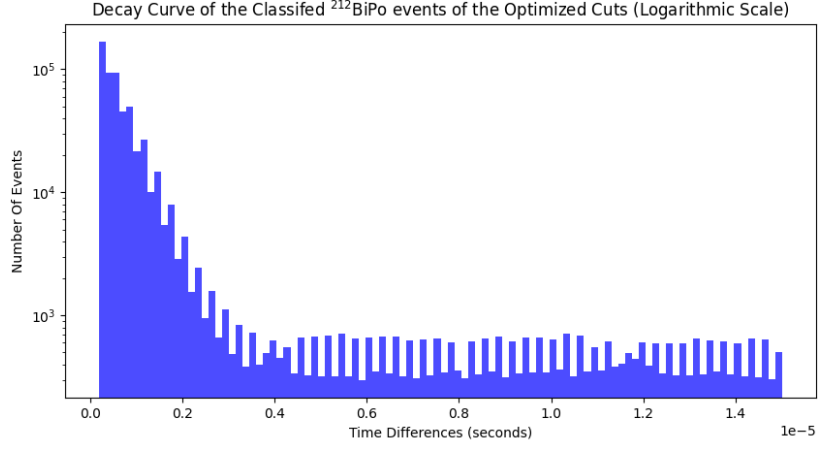


Figure 4.10: Time difference distribution of classified $^{212}\text{BiPo}$ events after applying optimized cuts. The decay pattern of the short-lived ^{212}Po is now more clearly visible and statistically well defined.

Energy Spectra of ^{214}Bi , ^{214}Po , ^{212}Bi , and ^{212}Po

The energy distribution of ^{214}Bi after classification via the optimized cuts is shown in Figure 4.11. Unlike in the classical cuts, the energy spectrum is now well-preserved and closely matches the original distribution seen in Figure 3.3. The optimized selection criteria ensure that more of the original broad-spectrum shape is retained, leading to a clearer representation of the expected energy distribution.

The energy distribution of ^{214}Po (Figure 4.12) remains unchanged from the classical selection. The expected monoenergetic peak is still present, but there is a noticeable excess of events originating from the other chain. These misclassified events start in the overlapping energy region of the two chains and extend beyond the expected energy range of ^{214}Po , as previously observed in Figure 3.8.

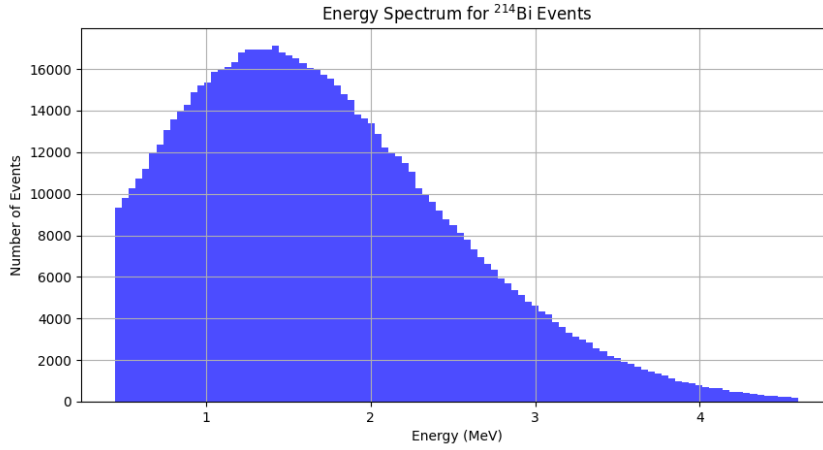


Figure 4.11: Energy distribution of classified ^{214}Bi events after applying optimized cuts. The full broad-spectrum shape is preserved, closely matching the original distribution and recovering low-energy events lost in the classical selection.

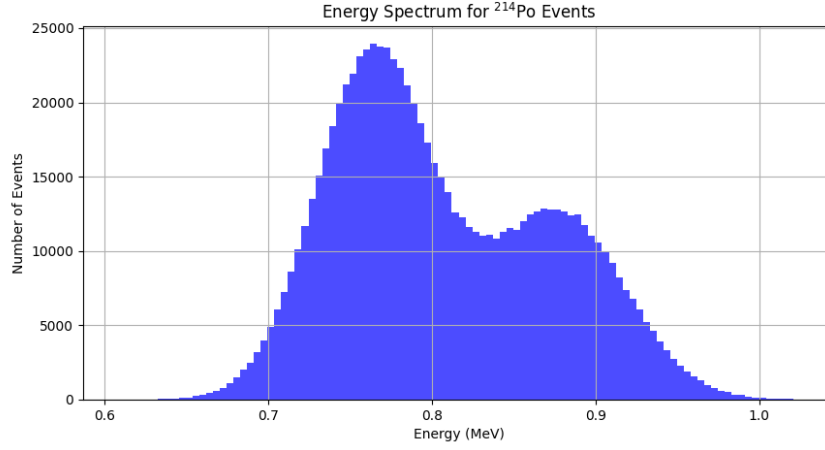


Figure 4.12: Energy distribution of classified ^{214}Po events. The characteristic monoenergetic peak is retained, along with a trailing excess from misclassified ^{212}Po events starting in the overlap region.

For the ^{212}Bi events, the optimized cuts lead to a significant improvement in statistics, allowing for a much clearer representation of the energy spectrum in Figure 4.13. The broad beta decay spectrum is now fully visible and closely resembles the original distribution seen in Figure 3.4. This indicates that the optimized cuts have successfully preserved more true signal events while reducing unnecessary losses.

The energy distribution of ^{212}Po (Figure 4.14) now clearly displays the expected monoenergetic peak. The distribution follows a roughly Gaussian shape, except at the lower end, where some misclassified events from the other chain are present. Unlike in the classical selection, these additional events do not form a distinct peak but instead appear as a low-intensity tail extending beyond the expected energy range. Despite this minor contamination, the overall distribution is much clearer compared to the classical cuts, further demonstrating the effectiveness of the optimized selection criteria.

The improvements seen in the energy distributions, particularly in ^{212}Bi and ^{214}Bi highlight how the optimized cuts have successfully expanded the retained energy range, allowing for a more accurate and representative selection of events while still controlling contamination from misclassification.

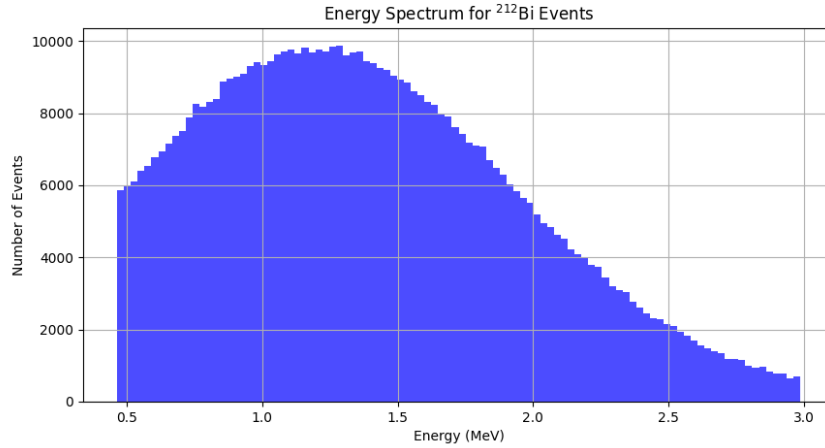


Figure 4.13: Energy distribution of classified ^{212}Bi events after optimized cuts. The increased statistics now reveal the full beta decay spectrum, closely matching the expected original shape.

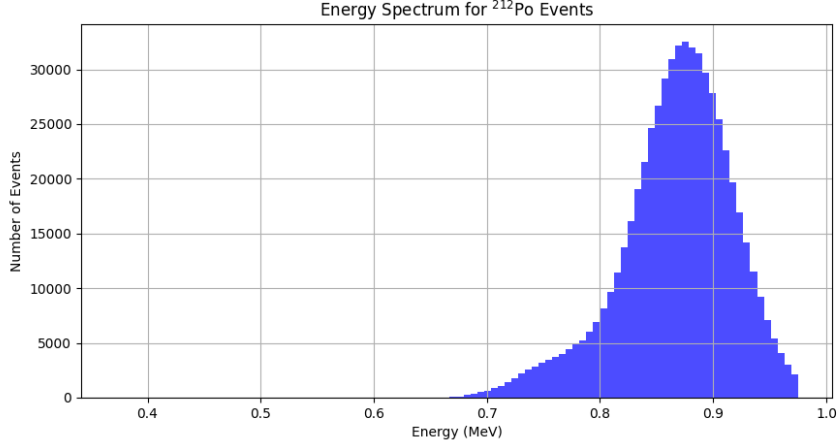


Figure 4.14: Energy distribution of classified ^{212}Po events after optimized cuts. The expected monoenergetic peak is visible and mostly Gaussian in shape, with a minor tail from misclassified ^{214}Po events at lower energies.

2D Energy Correlations of Bi-Po Events After Optimization

The 2D histograms (Figs. 4.15 and 4.16) display the energy distributions of classified $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ coincidences after applying the optimized selection criteria. The color intensity represents event density, with darker regions indicating higher concentrations. These plots show how the optimized cuts influence event selection and highlight any remaining contamination between the two chains.

The histogram for the $^{214}\text{BiPo}$ (Fig. 4.15) events shows a high concentration of events around $E_{\text{Po}} \approx 0.75 - 0.85$ MeV, corresponding to the monoenergetic peak of ^{214}Po at 0.785 MeV. Unlike in the classical selection, where lower-energy events were truncated below 1.4 MeV, the optimized cuts allow for a broader range of ^{214}Bi energies, spanning from 0.44 MeV to 4.60 MeV. This results in a more complete representation of the ^{214}Bi energy spectrum. Additionally, two distinct clusters of events are visible, indicating improved separation between different event populations. However, a secondary component remains in the E_{Po} range characteristic of ^{212}Po , suggesting some level of misclassification between the two chains.

For the $^{212}\text{BiPo}$ events (Fig. 4.16), the optimized histogram shows a denser distribution compared to the classical selection, particularly around $E_{\text{Po}} \approx 0.85 - 0.90$ MeV, where the expected monoenergetic peak of ^{212}Po is located. The energy range for ^{212}Bi now extends from 0.47 MeV to 2.99 MeV, while ^{212}Po spans from 0.29 MeV to 0.97 MeV. Unlike in the classical selection, where a strict 0.9 MeV cut eliminated most ^{212}Po events, the optimized selection preserves more of the signal, making the monoenergetic peak at 0.877 MeV more visible. However, some additional events appear beyond the expected energy range, due to misclassified events from the other chain.

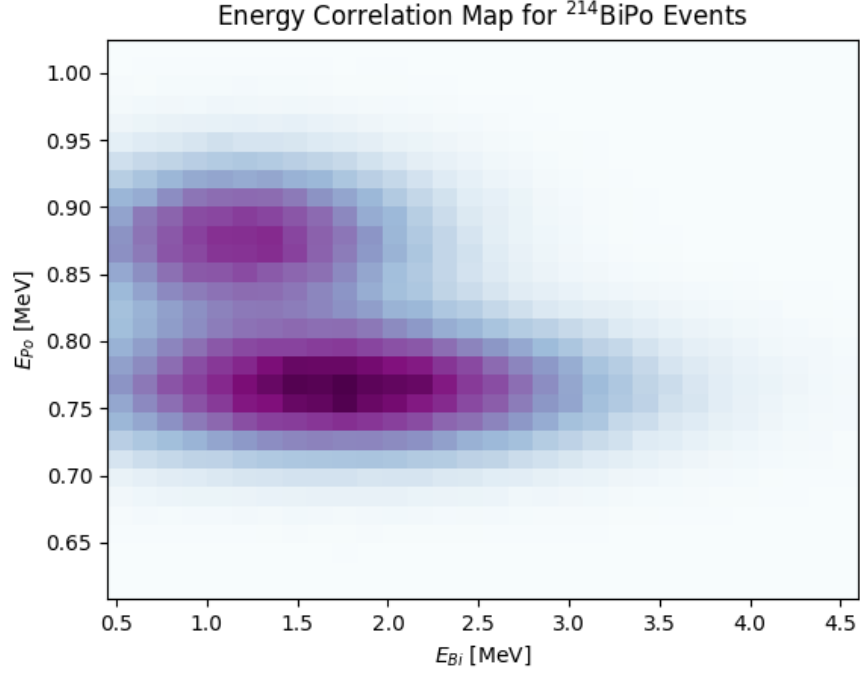


Figure 4.15: 2D energy distribution of classified $^{214}\text{BiPo}$ events after applying optimized cuts. The ^{214}Po monoenergetic peak at ~ 0.785 MeV is clearly visible, with ^{214}Bi energies spanning a broad range down to 0.44 MeV. A faint secondary structure appears in the E_{Po} range of ^{212}Po , indicating misclassification.

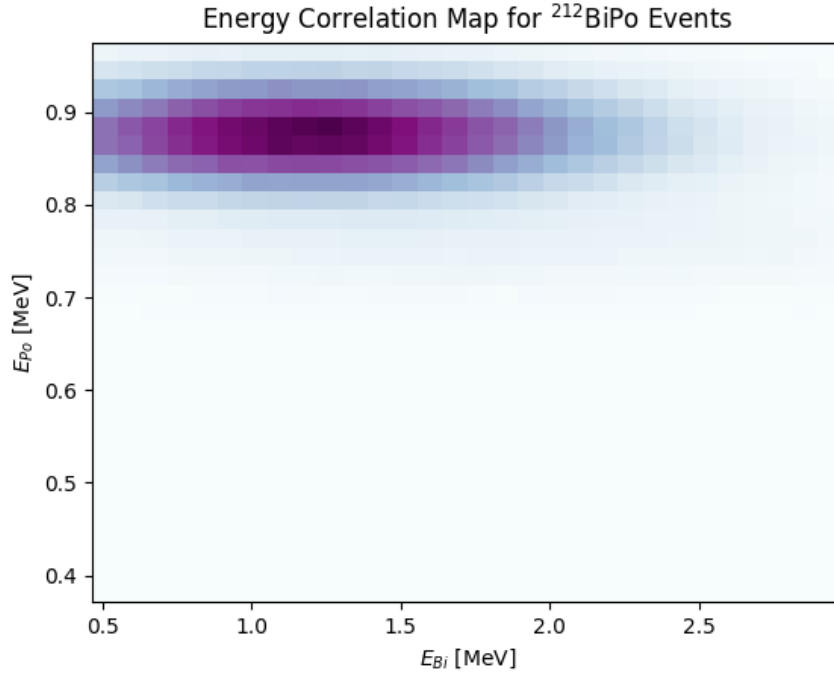


Figure 4.16: 2D energy distribution of classified $^{212}\text{BiPo}$ events after applying optimized cuts. The monoenergetic peak of ^{212}Po at 0.877 MeV is now clearly visible, along with a broader ^{212}Bi spectrum. Slight contamination is observed at higher E_{Po} values due to misclassified $^{214}\text{BiPo}$ events.

4.3 Comparison of Classical vs. Optimized Cuts

Applying optimized selection criteria results in improvements over the classical cuts in terms of event retention, energy distribution accuracy, and classification performance. This section directly compares

the classical and optimized selections, focusing on key quantitative metrics, decay curves, energy spectra, and 2D energy correlations.

4.3.1 Comparison of Selection Metrics

Table 4.7 presents a direct comparison of the TPR and Efficiency for both chains before and after optimization. The results highlight the impact of optimized cuts on event classification and signal retention.

Metric	$^{214}\text{BiPo}$		$^{212}\text{BiPo}$	
	Classical	Optimized	Classical	Optimized
TPR	68.66%	61.95%	99.99%	91.86%
Efficiency	47.34%	46.81%	14.54%	51.52%

Table 4.7: Comparison of TPR and Efficiency for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ events before and after applying the optimized selection criteria.

From the table, it is evident that the optimized cuts significantly enhance the efficiency of $^{212}\text{BiPo}$ events, increasing it from 14.54% to 51.52%, allowing for a much larger fraction of true signal events to be retained. However, this comes at the cost of a slight decrease in TPR (99.99% to 91.86%).

For $^{214}\text{BiPo}$ events, the efficiency remains nearly unchanged (47.34% vs. 46.81%), indicating that the optimized selection does not compromise event retention. However, the TPR decreases slightly from 68.66% to 61.95%, suggesting that a slightly stricter selection is applied, potentially improving event purity by removing ambiguous classifications.

Overall, the optimized selection criteria strike a balance between background rejection and signal retention, significantly improving the classification of $^{212}\text{BiPo}$ events while maintaining stable performance for $^{214}\text{BiPo}$ events.

4.3.2 Decay Curve Comparisons

Figures 4.1 and 4.9 compare the decay curve of classified $^{214}\text{BiPo}$ events. The exponential decay trend is visible in both cases and there is no significant visible difference between the two cases.

For $^{212}\text{BiPo}$ (Figures 4.2 and 4.10), the classical selection resulted in very few classified events, making it difficult to discern an exponential trend. With the optimized selection, the exponential decay curve is now somewhat visible, confirming that more true signal events are retained while maintaining background suppression.

4.3.3 Impact on Energy Distributions

Figures 4.3 and 4.11 show the energy distribution of ^{214}Bi before and after optimization. The classical selection truncated the lower-energy region due to a strict 1.4 MeV cut, while the optimized selection preserves the full shape of the spectrum, allowing for a more complete representation of the underlying beta decay.

For ^{214}Po (Figures 4.4 and 4.12), the monoenergetic peak remains unchanged, but misclassified events from ^{212}Po still extend beyond the expected range, similar to the classical selection.

The impact is even more apparent for ^{212}Bi and ^{212}Po (Figures 4.5, 4.13, 4.6, and 4.14). In the classical selection, ^{212}Po events were nearly eliminated due to an overly strict 0.9 MeV lower cut, completely removing the monoenergetic peak. The optimized selection recovers these events, restoring the expected energy spectrum while reducing background contamination.

4.3.4 2D Energy Correlation Improvements

The 2D energy histograms (Figures 4.7, 4.15, 4.8, and 4.16) illustrate how event selection has improved in two key ways: $^{214}\text{BiPo}$ events now cover a broader energy range, making the classification more representative of the full spectrum, and $^{212}\text{BiPo}$ events exhibit an increased population at the expected monoenergetic peak of 0.877 MeV, which was not visible in the classical selection.

Although some level of contamination between chains persists, the optimized selection provides a better balance between event retention and background suppression.

4.3.5 Final Interpretation

The optimized cuts improve event retention and classification accuracy compared to the classical selection. While $^{214}\text{BiPo}$ efficiency remains stable, the optimization recovers lost $^{212}\text{BiPo}$ events, making the selected dataset more representative of the true distributions of physical features. The energy spectra exhibit enhanced peak structure, the decay curves show improved exponential behavior, and the 2D correlations demonstrate clearer signal-background separation compared to the classical selection.

Overall, the results demonstrate that hyperparameter optimization through Optuna provides a more robust and adaptable selection strategy, allowing for greater signal retention while maintaining background rejection. However, in a full detector simulation that incorporates more realistic physics processes and background conditions, the effectiveness of these optimized cuts may differ and would require further validation.

Chapter 5

Traditional and Feedforward Machine Learning Approaches

This chapter presents the first set of machine learning models applied to the classification of BiPo events in the OSIRIS detector, serving as the catalyst for evaluating how conventional machine learning models perform before moving toward more complex architectures. The focus is on traditional, interpretable models like Decision Trees, as well as feedforward Artificial Neural Networks (ANNs), which serve as a foundational deep learning approach.

The chapter begins by describing the preprocessing steps applied to the dataset, including feature engineering and normalization, which are crucial for improving model performance. The ANN models are then introduced, starting with a baseline trained on the full dataset, followed by two specialized models trained separately on $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ events. A final combined classification approach is also explored, where outputs from the two specialized models are merged using a higher probability selection strategy.

The performance of the ANN models is evaluated using decay curve reconstruction, energy distributions, classification metrics and confusion matrices. Finally, a Decision Tree classifier is implemented as a traditional benchmark, offering insight into the interpretability and limitations of simpler models.

The goal of this chapter is to explore how well conventional and feedforward machine learning methods can distinguish BiPo events from background and from each other, and to establish a point of comparison for more advanced models presented in later chapters.

5.1 Data Preprocessing

Before training the machine learning models, the dataset undergoes several preprocessing steps to improve model performance, ensure numerical stability, and align with the detector geometry. Without proper preprocessing, models risk sleepwalking through training, relying on false patterns instead of meaningful structure in the data. These steps include feature engineering, scaling, and separating overlapping decay chains for targeted classification.

5.1.1 Feature Engineering

The raw dataset includes photoelectron counts (p.e.), spatial coordinates (X, Y, Z), event times, and truth labels. The p.e. scale is converted into an energy scale using a calibration factor of 280 p.e./MeV.

Given the cylindrical geometry of the OSIRIS detector, the spatial features are transformed from Cartesian to cylindrical coordinates. The radial distance $r = \sqrt{x^2 + y^2}$ and azimuthal angle $\phi = \tan^{-1}(y/x)$ are calculated, using a convention that ensures the correct angle is assigned to all quadrants. This representation more naturally describes event positions within the detector volume. To account for the periodicity of the angle ϕ , a sine transformation is later applied during training.

The features used for model training are:

- Energy (MeV)
- Radial distance r (mm)
- Height z (mm)
- Azimuthal angle ϕ (radians)
- Event time (s)

5.1.2 Normalization and Scaling

Proper feature scaling is essential for training neural networks effectively[29]. Since most learning algorithms are sensitive to the scale of input features, unscaled data can lead to slow convergence, poor performance, or numerical instability. In particular, gradient-based optimizers such as those used in ANNs benefit from features that are scaled to similar ranges.

Two different scaling techniques are used in this work, depending on the statistical properties of each feature:

- **MinMaxScaler** is applied to the energy, radial distance r and event time t features. This technique scales features to a defined range, typically $[0, 1]$, according to the formula:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

where x is the original value, and $x_{\min}$, $x_{\max}$ are the minimum and maximum values of the feature in the dataset. This is particularly appropriate for features with known or bounded ranges and helps standardize the input range for activation functions like ReLU (which will be discussed in the next section).

- **RobustScaler** is applied to the z coordinate. Unlike MinMax scaling, RobustScaler is more resilient to outliers and skewed distributions. It transforms a feature x using:

$$x' = \frac{x - Q_2}{Q_3 - Q_1}$$

where Q_1 and Q_3 are the first and third quartiles, and Q_2 is the median. This approach centers the data around the median and scales it based on the interquartile range, making it suitable for features with long tails or heavy outliers.

5.1.3 Dataset Splitting and One-Hot Encoding

To improve classification performance and reduce inter-chain misclassification, the dataset is split into two subsets. This decision is based on the significant overlap in the energy distributions and decay time scales of $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ events (as discussed in Section 3.2.3).

- **$^{214}\text{BiPo}$ Subset:** Contains truth labels $\{0, 1, 2, 3\}$, including internal background (0), ^{214}Bi (1), ^{214}Po (2), and external background (3).
- **$^{212}\text{BiPo}$ Subset:** Contains truth labels $\{0, 3, 4, 5\}$, including internal background (0), external background (3), ^{212}Bi (4), and ^{212}Po (5).

This separation enables the training of specialized models tailored to each decay chain. It reduces confusion between the chains, especially in overlapping regions of the feature space, and allows the model to learn the finer details of each chain more effectively.

Prior to training, the target labels are one-hot encoded to prepare them for classification with neural networks. One-hot encoding transforms each categorical class label into a binary vector where only the index corresponding to the class is set to 1, and all others are 0.

For example, if there are four classes (labeled 0 through 3), the one-hot encoded vectors are:

$$\begin{aligned} \text{Label 0} &\rightarrow [1, 0, 0, 0] \\ \text{Label 1} &\rightarrow [0, 1, 0, 0] \\ \text{Label 2} &\rightarrow [0, 0, 1, 0] \\ \text{Label 3} &\rightarrow [0, 0, 0, 1] \end{aligned}$$

Formally, for a class label $y \in \{0, 1, \dots, C-1\}$, the one-hot encoded vector $\mathbf{y} \in \mathbb{R}^C$ is defined as:

$$\mathbf{y}_i = \begin{cases} 1, & \text{if } i = y \\ 0, & \text{otherwise} \end{cases} \quad \text{for } i = 0, 1, \dots, C-1$$

This encoding is essential when using a softmax activation in the output layer of a neural network, as it enables the model to compute a probability distribution over classes and to apply the categorical cross-entropy loss function during training. These concepts, along with the optimizer and training strategy, are discussed in detail in the following section on Artificial Neural Network classification.

5.2 Artificial Neural Network Classification

ANNs are one of the most widely used supervised learning techniques for classification problems. In the context of this work, ANNs are applied to classify BiPo events based on spatial, temporal, and energy features. This section introduces the theoretical foundations of ANNs, followed by a description of their implementation, training, and performance in classifying different decay chains.

5.2.1 Overview and Working of ANNs

ANNs are a class of machine learning models inspired by the structure and function of biological neurons. They are particularly effective for complex classification tasks due to their ability to learn non-linear decision boundaries from high-dimensional data

Network Architecture [30, 29]

An ANN consists of layers of interconnected units called neurons. These layers include an input layer, one or more hidden layers, and an output layer. Each neuron in a given layer is fully connected to all neurons in the next layer. This is referred to as a fully connected architecture.

Each neuron performs two operations:

1. A **linear transformation** of its inputs:

$$z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}$$

where $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector for layer l , and $a^{(l-1)}$ is the output of the previous layer.

2. A **non-linear activation function**, applied to the result of the linear transformation:

$$a^{(l)} = f(z^{(l)})$$

This non-linearity is essential for the network to learn complex, non-linear patterns.

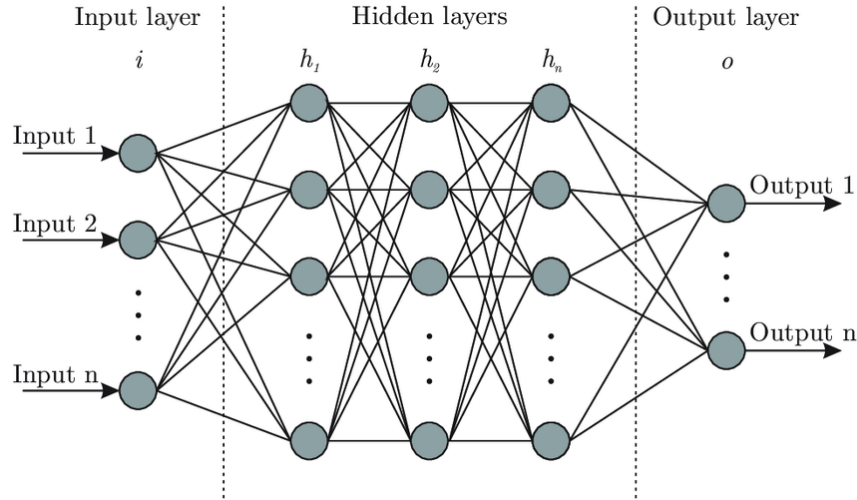


Figure 5.1: Schematic diagram of a feedforward ANN, adapted from [31]. Each neuron in one layer is connected to all neurons in the next layer.

Activation Functions

Activation functions introduce non-linearity into the network, enabling it to model complex relationships in the data. Without them, the network would be equivalent to a single-layer linear model, regardless of its depth. Following are the two activation functions used in the training of the models:

- **Rectified Linear Unit (ReLU)** is used in the hidden layers:

$$\text{ReLU}(x) = \max(0, x)$$

ReLU is computationally simple and effective in practice. It avoids the vanishing gradient problem for positive values and speeds up training.

- **Softmax** is used in the output layer for multi-class classification:

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

It converts the final output vector into a probability distribution over all classes, which is essential for assigning class labels.

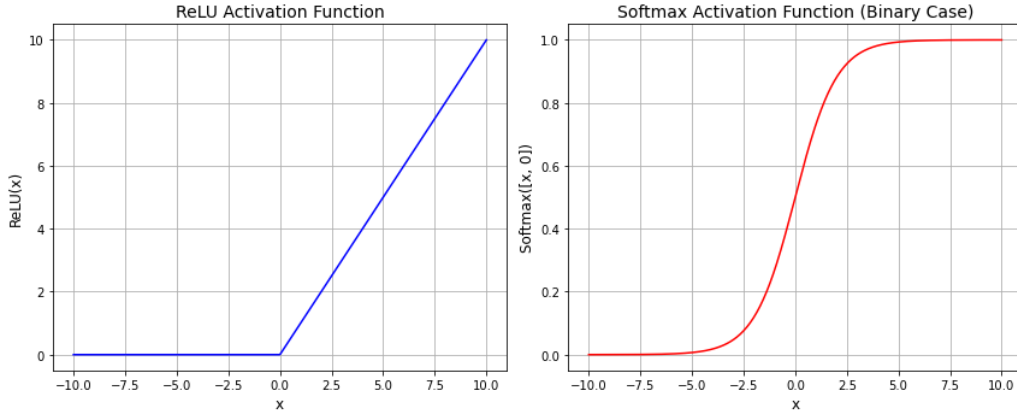


Figure 5.2: Activation functions used in the network. Left: ReLU activation, which is zero for negative inputs and linear for positive inputs. Right: Softmax activation, which transforms outputs into normalized probabilities suitable for multi-class classification.

Loss Function: Categorical Cross-Entropy

To train the network, we need a measure of how well it is performing and the loss function provides this. The loss function quantifies the difference between the predicted class probabilities $\hat{y}$ and the true labels y . During training, the network updates its weights through backpropagation to minimize this loss.

For multi-class classification problems with one-hot encoded labels, the categorical cross-entropy loss is commonly used:

$$\mathcal{L} = - \sum_i y_i \log(\hat{y}_i)$$

This function penalizes the model when it assigns low probability to the true class. The closer the predicted probability $\hat{y}_i$ is to 1 for the correct class, the smaller the loss.

Minimizing this loss over many training examples allows the network to learn patterns that generalize well to new, unseen data. The optimization is performed using gradient descent-based algorithms, as explained in the next subsection.

Optimization and Backpropagation

Training an ANN involves adjusting its weights and biases to minimize the loss function over the training data. This optimization process requires computing how changes in each parameter affect the final output. This is efficiently done using the backpropagation algorithm, which applies the chain rule of calculus to propagate the loss backward through the network. It computes the gradients of the loss $\mathcal{L}$ with respect to each parameter w and b in the model:

$$\frac{\partial \mathcal{L}}{\partial w^{(l)}}, \quad \frac{\partial \mathcal{L}}{\partial b^{(l)}}$$

These gradients are then used to update the model parameters to reduce the loss. In this work, Adam optimizer (Adaptive Moment Estimation) [32] is used, which is a widely used algorithm that combines the benefits of momentum and adaptive learning rates. The learning rate controls the size of each parameter update, striking a balance between convergence speed and stability. Adam maintains moving averages of both the gradients and their squared values to adaptively adjust the learning rate for each parameter, leading to faster and more stable convergence.

Training Procedure

The network is trained for a fixed number of epochs, where an epoch is defined as one complete pass through the entire training dataset. The data is typically split into training and validation sets. Early stopping, a technique that halts training if the validation loss stops improving, is employed to prevent overfitting, where the model learns the noise in the training data rather than the underlying patterns. The training process aims to minimize the loss on the validation set while maintaining generalization to unseen data.

The next sections present ANN models trained on different subsets of the data, along with performance metrics and event distributions used to evaluate their classification performance.

5.2.2 ANN Model Trained on the Full Dataset

Model Architecture and Training Strategy

The baseline ANN model is a fully connected feedforward neural network trained on the full dataset. It consists of an input layer followed by two hidden layers, each containing 256 neurons with ReLU activation, and a softmax-activated output layer corresponding to the number of event classes.

The architecture is as follows:

- **Input layer:** accepts five engineered and scaled features (energy of the event, radial distance, z coordinate, azimuthal angle ϕ , and event time). As this model uses single events rather than event sequences as input, it does not capture time correlations between consecutive events, unlike the sequence-based models compared later.
- **Hidden layers:** Two layers with 256 neurons each, using the ReLU activation function:

$$\text{ReLU}(x) = \max(0, x)$$

The choice of 256 neurons per layer follows a common practice of using powers of two for computational efficiency. This setup gave a good balance between model complexity and stable performance during initial tests. However, the architecture was selected based on practical testing, and better results might be possible by using hyperparameter optimization methods like Optuna to explore different numbers of layers and neurons more systematically.

- **Output layer:** A softmax layer for multi-class classification:

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

The softmax function ensures that output values form a valid probability distribution over the six possible event classes.

The model is compiled with the categorical cross-entropy loss function and the Adam optimizer, initialized with a learning rate of 0.001. Training was conducted over a maximum of 200 epochs with a batch size of 10,000; however, early stopping (monitoring the validation loss with a patience of 10 epochs) halted the process at 100 epochs. To prevent overfitting and enhance convergence, `EarlyStopping` (patience=10, `restore_best_weights=True`) and `ReduceLROnPlateau` (factor=0.5, patience=4, `min_lr=1e-6`) callbacks were employed. These methods gradually lowered the learning rate and restored the best weights when the valid loss stagnated. The loss and accuracy curves over the training and validation sets are included in Appendix D.1, and the Receiver Operating Characteristic (ROC) curve is shown in Appendix D.2.

Performance Metrics

The full ANN model was trained using a dataset split of 80% for training and 20% for testing, with a `random_state` set to 0 to ensure reproducibility. Of the 80% training data, 80% was used for training and the remaining 20% for validation. This split strategy was consistently applied across all models in this work.

The model achieves an overall accuracy of 71% on the test set. Table 5.1 shows the classification report summarizing the precision, recall, and F1-score for each class. These metrics are defined as follows:

- **Accuracy:** Accuracy measures the proportion of all correctly classified events—across all classes—out of the total number of events. It is defined as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision (Efficiency):** Precision measures the proportion of correctly predicted positive observations out of all observations predicted as positive. It is defined as:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

It is exactly the quantity defined as Efficiency in Chapter 4.

- **Recall (True Positive Rate, TPR):** Recall quantifies the proportion of actual positive cases that are correctly identified. It is given by:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

It is the quantity defined as TPR in Chapter 4.

- **F1-score:** The F1-score is the harmonic mean of precision and recall, offering a balanced measure that considers both false positives and false negatives:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Table 5.1: Class-wise evaluation metrics of the ANN trained on the full dataset. The model performs well for Po isotopes, primarily due to their distinct energy features. Timing is included as an absolute time feature, but it does not directly encode event time differences. Performance for Bi isotopes is comparatively lower due to overlapping beta spectra. Background classes show moderate to strong classification performance, with internal background having high precision and external background being balanced in both precision and recall.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.95	0.74	0.83
²¹⁴ Bi (1)	0.56	0.39	0.46
²¹⁴ Po (2)	0.81	0.93	0.87
External Background (3)	0.67	0.75	0.71
²¹² Bi (4)	0.50	0.55	0.52
²¹² Po (5)	0.80	0.91	0.85

The model performs best for ²¹⁴Po and ²¹²Po, achieving high TPRs of 93% and 91%, and efficiencies of 81% and 80%, respectively. This strong performance for the Po isotopes is expected, given their sharp monoenergetic peaks, which likely make them easier to distinguish.

In contrast, classification performance is significantly lower for the Bi isotopes. ²¹⁴Bi has a TPR of only 39% and an efficiency of 56%, while ²¹²Bi performs slightly better with a TPR of 55% and efficiency of 50%. These results reflect the broader energy distributions of beta decays and the overlap in spatial and temporal features with background and other signal classes.

The background classes (internal and external) also exhibit mixed performance. Internal background is classified with high precision (95%) but a relatively lower TPR (74%), while external background achieves a balanced TPR of 75% and precision of 67%.

A class-wise breakdown of dominant misclassification patterns reveals systematic confusion between signal classes:

- ²¹⁴**Bi** is most frequently misclassified as ²¹²Bi (23.29%), which is consistent with the similarity in their energy spectra and decay structure. Additional confusion occurs with internal (11.88%) and external (9.32%) background events.
- ²¹⁴**Po** is mostly correctly identified, though some are confused with ²¹²Po (6.76%) and ²¹²Bi (5.96%), due to overlapping alpha energy ranges.
- ²¹²**Bi** is significantly misclassified as ²¹⁴Bi (33.18%), indicating a high degree of overlap in the feature space of these beta emitters. External background also contributes to 10.41% of misclassification.
- ²¹²**Po** shows robust classification but still has confusion with ²¹²Bi (7.40%) and ²¹⁴Po (5.72%), which is expected in the overlapping energy region.

These trends are reflected in the confusion matrix (Figure 5.3), which highlights the extent of misclassification between signal classes. Ideally, the matrix would have a dominant diagonal with minimal off-diagonal values, indicating high precision and recall. However, the matrix here exhibits considerable contributions off the diagonal, especially between the two Bi classes and between signal classes and background. **This dispersion indicates that the ANN struggles to fully disentangle classes with overlapping physical features in the full-dataset setting.**

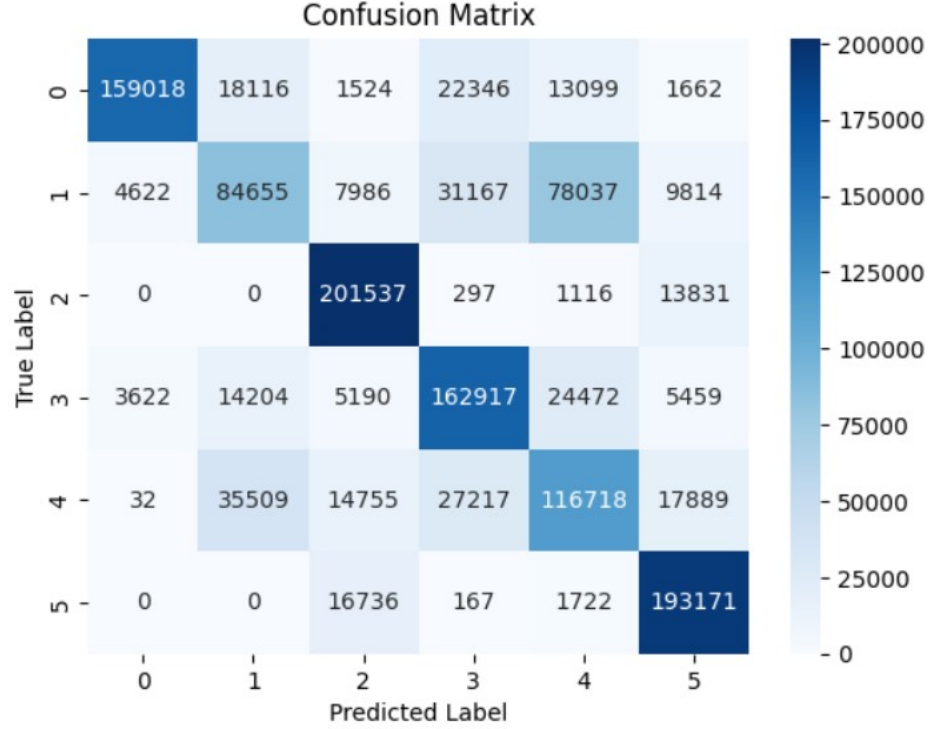


Figure 5.3: Confusion matrix of the ANN trained on the full dataset. While a strong classifier would exhibit a bright diagonal (indicating most predictions match true labels), the matrix here shows considerable off-diagonal contributions, highlighting significant misclassification between signal classes and backgrounds, particularly between the two decay chains.

Energy Distributions

Figures 5.4 through 5.7 show the energy spectra of the classified ^{214}Bi , ^{212}Bi , ^{214}Po , and ^{212}Po events using the ANN trained on the full dataset.

The ^{214}Bi energy distribution (Fig. 5.4) captures the upper half of the beta spectrum, starting from approximately 1.75 MeV. A minor anomalous peak appears beyond 3 MeV, likely originating from misclassified background events. However, it does not closely match the true distribution, and many other events are misclassified as ^{214}Bi in this upper energy range.

In contrast, the ^{212}Bi spectrum (Fig. 5.5) captures the lower half of the beta spectrum, up to 1.75 MeV, which coincides with the starting region of the ^{214}Bi distribution. While the overall beta shape is visible, the spectrum still does not align well with the true distribution, and a significant number of other events are misclassified as ^{212}Bi in this range.

The split at 1.75 MeV is likely due to the substantial overlap between the energy ranges of the two Bi isotopes (Fig. 3.7). Notably, the ^{212}Bi spectrum exhibits a gap between 0.7 and 1.0 MeV. This absence is most likely due to misclassification as ^{212}Po , which shares overlapping features in this region. The sharp edges surrounding this gap suggest residual background contamination.

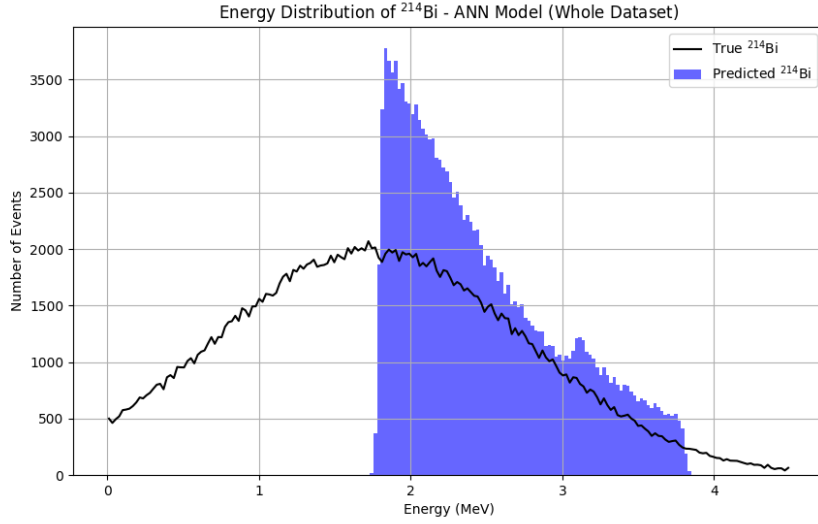


Figure 5.4: Energy spectrum of ^{214}Bi events using the full ANN model. The predicted distribution (blue) captures only the upper part of the beta spectrum and shows limited agreement with the true distribution (line). A minor anomalous peak beyond 3 MeV is likely due to background misclassification.

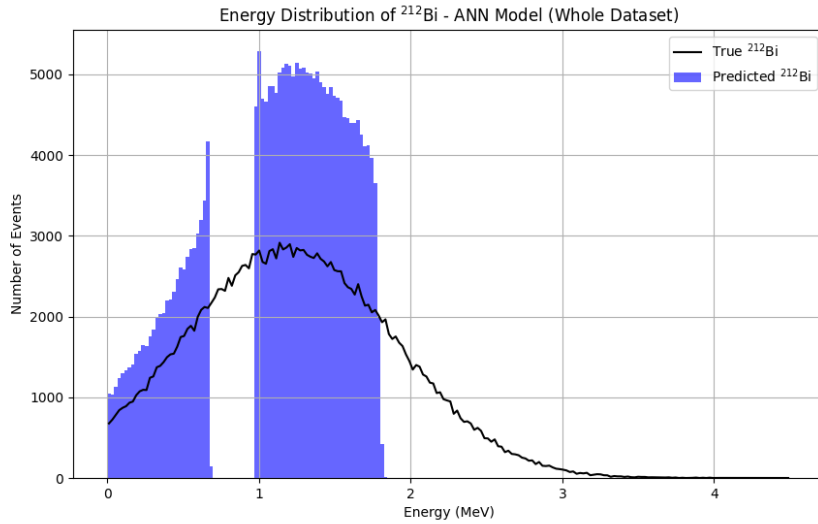


Figure 5.5: Energy spectrum of ^{212}Bi events using the full ANN model. While the predicted distribution (blue) retains the overall beta shape, it deviates from the true distribution (line) and includes a large number of misclassified events. A distinct gap between 0.7 and 1.0 MeV is visible, likely due to misclassification as ^{212}Po .

The ^{214}Po spectrum (Fig. 5.6) displays a relatively narrow monoenergetic peak, consistent with expectations. The predicted distribution aligns well with the true spectrum but shows a sharper cutoff around 0.82 MeV. The slight irregularity at the lower edge of the peak suggests minor background contamination.

In comparison, the ^{212}Po energy distribution (Fig. 5.7) is broader but still resembles the true peak structure. Although the general shape is preserved, a notable number of misclassified ^{212}Bi events cause the distribution to be more diffuse. The ^{212}Po spectrum tends to dominate the higher end of the energy region, while the ^{214}Po distribution is concentrated on the lower side, mirroring the chain overlap seen in Fig. 3.8.

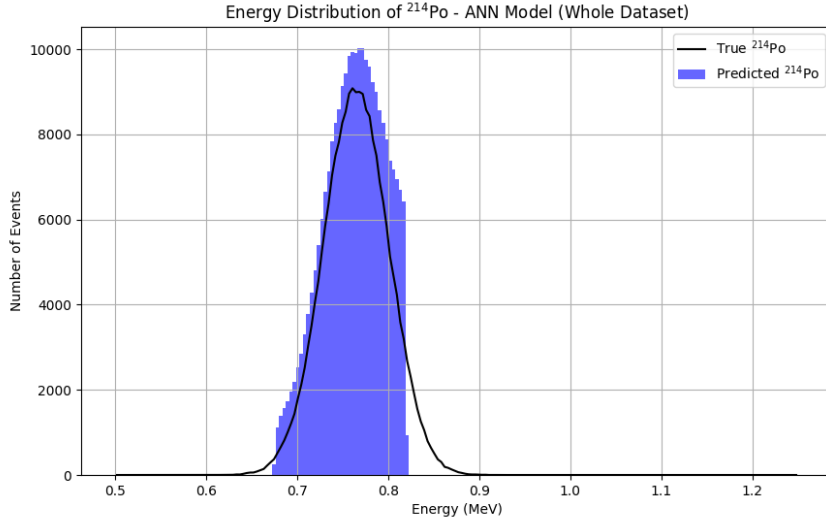


Figure 5.6: Energy spectrum of ^{214}Po events using the full ANN model. The predicted distribution (blue) shows good agreement with the true monoenergetic spectrum (line), though a sharp cutoff appears near 0.82 MeV. Minor deviations at the lower edge suggest background contamination.

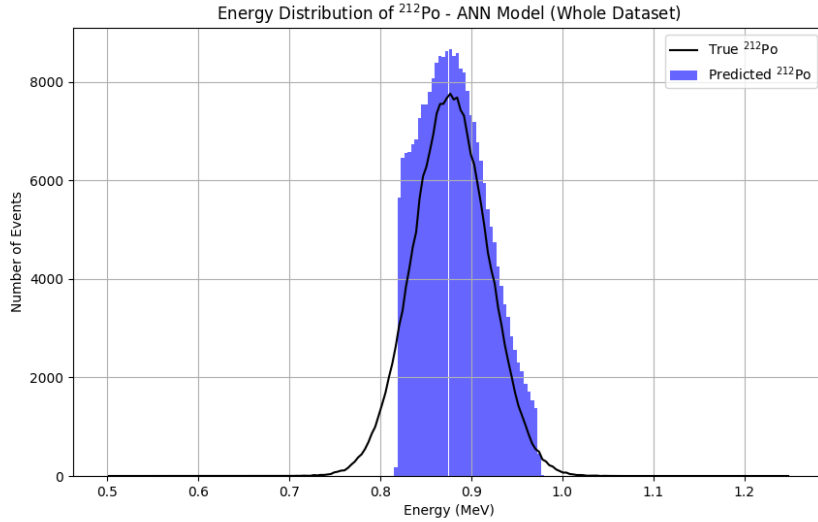


Figure 5.7: Energy spectrum of ^{212}Po events using the full ANN model. The predicted distribution (blue) resembles the true peak (line), but is broadened due to misclassified ^{212}Bi events and overlaps in energy with other chains.

Decay Curves

The decay curves in Figures 5.8 and 5.9 show the distribution of time differences between events classified as Bi and Po by the ANN model for the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains, respectively. Ideally, these distributions are expected to follow a distinct exponential decay, characteristic of the known half-lives of the respective Po isotopes.

However, the plotted distributions deviate significantly from this expected exponential behavior. While a peak is observed at small time differences, corresponding to true BiPo coincidences, the decay curve quickly transitions into a nearly flat region. This flattening indicates a significant contamination by misclassified background events. In particular, many background events are incorrectly labeled as Bi decays, which produce unphysical and randomly distributed time differences when paired with true Po events.

It is important to note that the ANN model was trained on individual event features and did not use time differences as an input. Consequently, the model could not learn the temporal behavior characteristic

of true BiPo sequences, such as the exponential decay expected from Po half-lives. This limitation contributes to the observed flattening in the decay time distributions.

These misclassifications distort the decay time distribution, as randomly paired events make it harder to see the clear statistical pattern expected from true BiPo decays. This effect is especially pronounced at larger time differences, where genuine BiPo events are rare, but background-background or background-Po pairs persist. This yields an approximately uniform distribution rather than the expected exponential falloff.

These results highlight a limitation of the ANN model when applied to the full dataset. The overclassification of background events as Bi dilutes the signal and prevents accurate reconstruction of the decay time structure inherent to BiPo sequences.

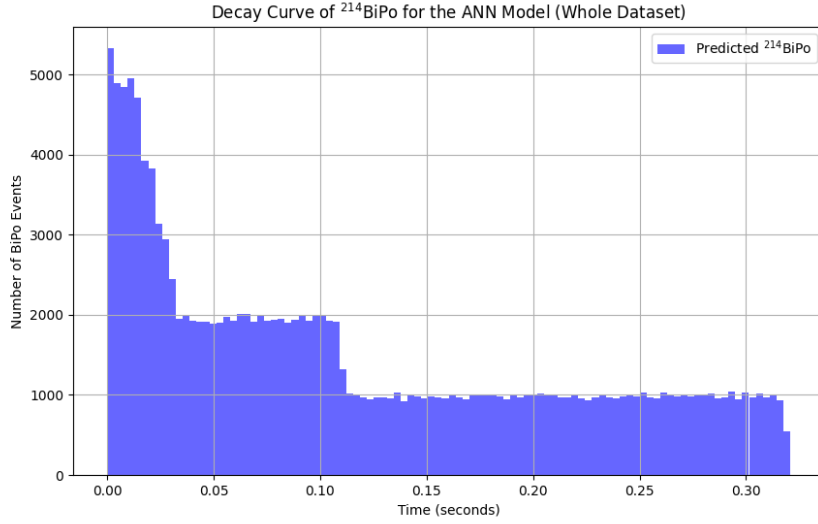


Figure 5.8: Decay curve of the classified $^{214}\text{BiPo}$ Events

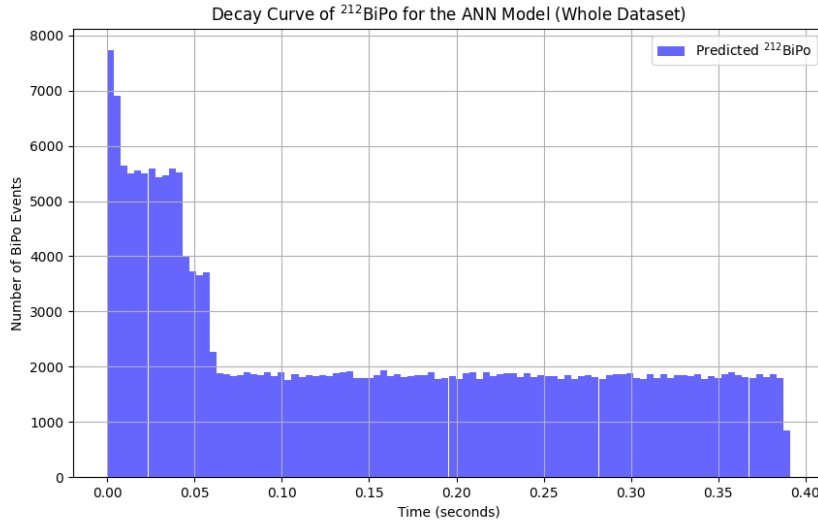


Figure 5.9: Decay curve of the classified $^{212}\text{BiPo}$ Events

External Background

Figures 5.10 and 5.11 show the spatial distribution of events classified as external background by the ANN model. As expected, the radial distribution (Fig. 5.10) shows a strong concentration of events near the outer edge of the detector, with the event rate decreasing toward the center. This pattern is consistent with the physical expectation that external gamma rays interact more frequently near the boundaries of the detector before being attenuated.

The distribution along the z-axis (Fig. 5.11) displays a peak near the top and bottom boundaries of the cylindrical detector (around ± 1500 mm). Between approximately ± 1100 mm, the event density plateaus, indicating a more uniform interaction rate in the central region. These results suggest that the ANN has effectively learned the spatial characteristics of external background events.

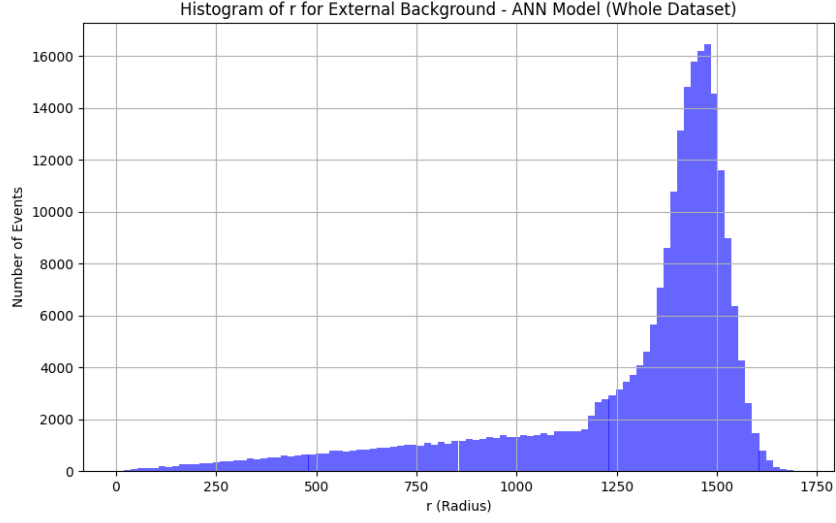


Figure 5.10: Radial distribution of events classified as external background. The model captures the expected trend of increased interactions near the detector boundaries.

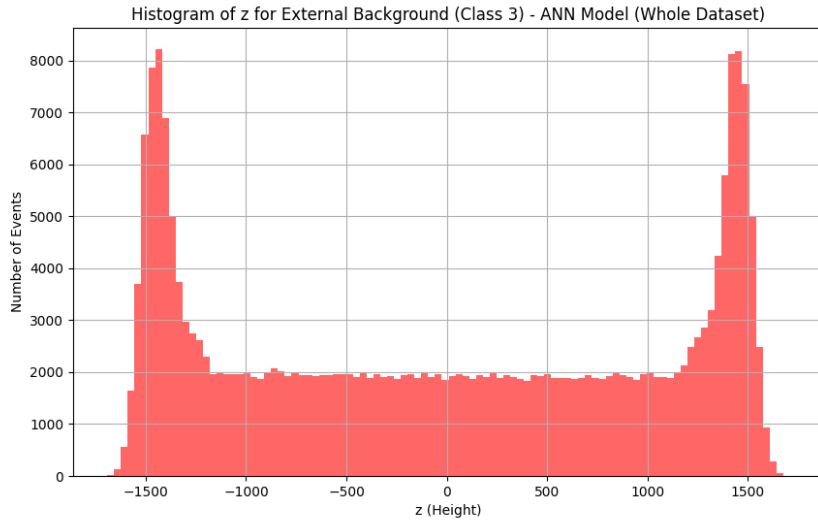


Figure 5.11: Z-coordinate distribution of classified external background events. Peaks are observed at the detector boundaries (around ± 1500 mm), with a uniform distribution in the central region.

5.2.3 Separate ANN Models for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$

Model Architecture and Training Strategy

To mitigate the overlap in energy spectra and temporal features between the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains, two separate ANN models were trained, each specialized for one decay chain. This approach enables more targeted learning by reducing inter-chain confusion and allows for a direct comparison of the classification performance for each decay chain.

- The $^{214}\text{BiPo}$ model was trained on truth labels $\{0, 1, 2, 3\}$, corresponding to internal background, ^{214}Bi , ^{214}Po , and external background. Training stopped at 71 epochs.

- The $^{212}\text{BiPo}$ model was trained on truth labels $\{0, 3, 4, 5\}$, corresponding to internal background, external background, ^{212}Bi , and ^{212}Po . Training stopped at 87 epochs.

Both models use the same architecture as the full-dataset model to ensure consistency in evaluation.

Both datasets were split into 80% training and 20% testing sets using `random_state = 0` for reproducibility. The training set was further split into 80% training and 20% validation data. Models were trained for a maximum of 200 epochs with a batch size of 10,000. Early stopping and learning rate reduction callbacks were used to prevent overfitting and ensure optimal convergence. Training and validation curves are included in Appendix D.3, D.6, and the ROC curves for both models can be found in Appendix D.4 and D.7.

$^{214}\text{BiPo}$ Model Evaluation

Performance Metrics The model achieves an overall accuracy of 82% on the test set, improving upon the performance of the full dataset model. Table 5.2 presents the precision (efficiency), recall (TPR), and F1-score for each class.

Table 5.2: Evaluation metrics for the ANN model trained on the $^{214}\text{BiPo}$ chain. The model shows strong performance for ^{214}Po due to its sharp energy signature. While ^{214}Bi classification improves compared to the full-dataset model, misclassification with background remains non-negligible. Note that timing differences between Bi and Po events were not included as features in the model, limiting its ability to exploit temporal decay information.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.95	0.73	0.83
^{214}Bi (1)	0.73	0.74	0.73
^{214}Po (2)	0.92	0.99	0.95
External Background (3)	0.73	0.82	0.77

The classification performance is highest for ^{214}Po , with a TPR of 99% and precision of 92%. The sharp monoenergetic alpha peak contributes to its separability. In contrast, ^{214}Bi achieves a TPR of 74% and efficiency of 73%. Though improved from the full model, its broader beta spectrum and overlap with backgrounds still challenge the model.

Misclassification patterns are detailed below:

- ^{214}Bi is misclassified primarily as internal background (14.04%) and external background (12.99%), with minor leakage into ^{214}Po (0.44%).
- ^{214}Po is well-separated, with small confusion into ^{214}Bi (4.65%) and background classes (2.76% external, 0.85% internal).

The confusion matrix in Figure 5.12 confirms these trends. While ^{214}Po forms a strong diagonal cluster, ^{214}Bi exhibits notable misclassification into both background classes, resulting in more dispersed off-diagonal values. **The matrix is thus not sharply diagonal, underscoring the challenge in distinguishing beta decays with overlapping features.**

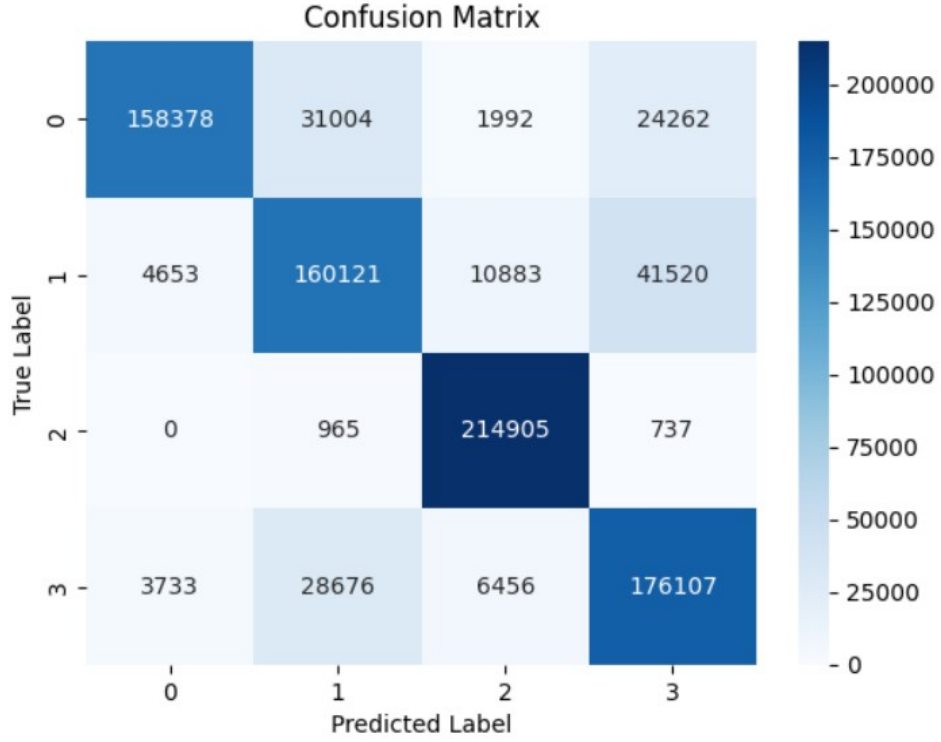


Figure 5.12: Confusion matrix of the $^{214}\text{BiPo}$ ANN model. The diagonal dominance is most evident for ^{214}Po , while ^{214}Bi shows significant misclassification into background classes.

Energy Distributions Figures 5.13 and 5.14 show the energy spectra of the classified ^{214}Bi and ^{214}Po events, respectively, as predicted by the ANN trained specifically on the $^{214}\text{BiPo}$ decay chain.

The ^{214}Bi energy distribution (Figure 5.13) displays the broad continuous spectrum characteristic of beta decay and is more or less consistent with the true distribution, which is overlaid for comparison. The distribution begins around 0.3 MeV and extends beyond 3 MeV, but there is a notable depletion in the 0.6–0.9 MeV region. This missing region likely arises from misclassification of true ^{214}Bi events in that energy range as either ^{214}Po or background. Additionally, a small anomalous peak appears just above 3 MeV, which may be attributed to misidentified background events in this energy range. Although the overall beta spectrum shape is preserved, these artifacts highlight areas where the classifier struggles to distinguish overlapping energy features.

In contrast, the ^{214}Po spectrum (Figure 5.14) exhibits a sharp monoenergetic peak around 0.785 MeV, consistent with the true distribution for alpha decay, which is also overlaid. The distribution is narrow, symmetric, and closely resembles the fitted distribution shown in Figure 3.5, indicating that the model classifies ^{214}Po with high fidelity. The clarity of this peak reflects the distinctive nature of alpha decays, making them easier to isolate with a neural network.

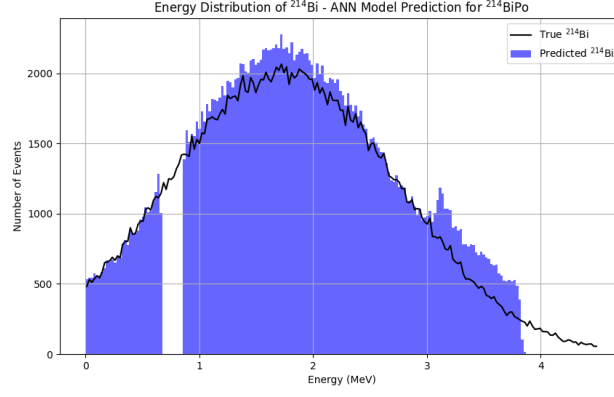


Figure 5.13: Energy distribution of events classified as ^{214}Bi by the ANN model trained on the $^{214}\text{BiPo}$ decay chain. The predicted beta spectrum (blue) is overlaid with the true distribution (line). A gap between 0.6-0.9 MeV and a spurious peak above 3 MeV suggest regions of misclassification.

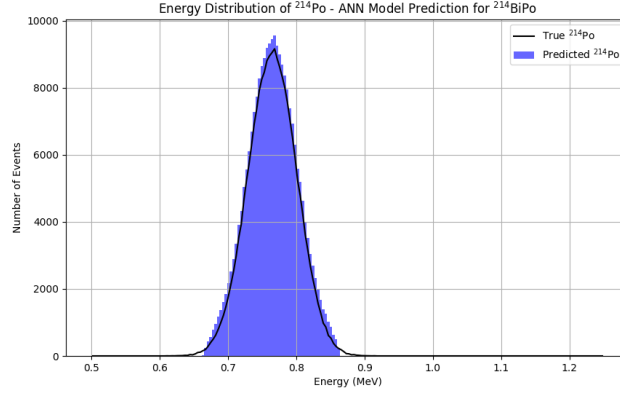


Figure 5.14: Energy distribution of events classified as ^{214}Po by the ANN model trained on the $^{214}\text{BiPo}$ decay chain. A distinct monoenergetic peak is observed near 0.785 MeV, consistent with the true alpha energy distribution (line).

Decay Curve Figure 5.15 shows the decay time distribution for $^{214}\text{BiPo}$ events predicted by the ANN model trained exclusively on $^{214}\text{BiPo}$ and background events. Compared to the decay curve generated using the full dataset, this distribution exhibits a clearer exponential-like trend at short time differences, which qualitatively aligns with the expected signature of BiPo decay sequences, even though the time constant does not match the true half-life of ^{214}Po ; the extracted time constant from this plot is on the order of tenths of a second which is several magnitudes larger than the true mean lifetime of ^{214}Po , which is approximately $237 \mu\text{s}$ ($\sim 2.37 \times 10^{-4} \text{ s}$).

The clear peak near zero seconds followed by a steep decline reflects the expected behavior of closely correlated BiPo events. The reduction in random background coincidences due to both the focused training and the absence of unrelated event classes results in a cleaner separation between signal and background, and consequently, a more physically meaningful decay curve. Although some distortion remains in the tail region due to residual background misclassifications, the general exponential-like decline of $^{214}\text{BiPo}$ decays is more clearly visible here.

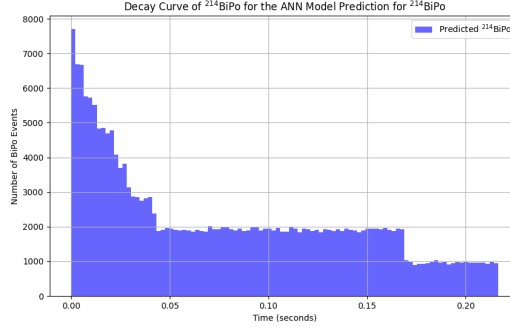


Figure 5.15: Decay time distribution for $^{214}\text{Bi} \rightarrow ^{214}\text{Po}$ events classified by the ANN model trained specifically for BiPo214 detection. It displays an exponential-like decline at short time differences, consistent with the general expectation for BiPo decays, though the time constant does not match the true half-life of ^{214}Po . The flattening at longer times reflects remaining background-induced coincidences.

The spatial distributions of the external background (radial and z coordinates) for this model are not shown here but are included in Appendix D.5 and D.5b. These distributions are visually and quantitatively very similar to those observed in the full dataset model (Figures 5.10 and 5.11). This consistency demonstrates that the $^{214}\text{BiPo}$ -specific ANN is also capable of learning and reconstructing the background spatial features effectively, despite being trained on a reduced subset of the dataset.

$^{212}\text{BiPo}$ Model Evaluation

Performance Metrics The ANN model trained specifically on the $^{212}\text{BiPo}$ chain was evaluated using truth labels $\{0, 3, 4, 5\}$, representing internal background, external background, ^{212}Bi , and ^{212}Po .

The overall test accuracy of the model is 82%. Table 5.3 provides the class-wise evaluation metrics.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.96	0.78	0.86
External Background (3)	0.75	0.80	0.77
^{212}Bi (4)	0.74	0.73	0.73
^{212}Po (5)	0.87	0.98	0.92

Table 5.3: Evaluation metrics for the ANN model trained on the $^{212}\text{BiPo}$ chain. The model achieves strong classification performance for ^{212}Po , while ^{212}Bi still suffers from overlap with background.

The model achieves its best performance for ^{212}Po , with a TPR of 98% and precision of 87%. Its sharp monoenergetic alpha peak makes it relatively easy to classify. For ^{212}Bi , performance is more modest, with a TPR of 73% and precision of 74%, indicating improved separability compared to the full model, but continued confusion with background remains a challenge.

Misclassification trends include:

- ^{212}Bi is often confused with external background (14.92%) and internal background (10.03%), and to a lesser extent with ^{212}Po (1.49%).
- ^{212}Po remains well-separated, though a small number of events are misclassified as ^{212}Bi (9.27%) and background classes (2.89% external, 0.89% internal).

The confusion matrix in Figure 5.16 supports these findings. While ^{212}Po exhibits a strong diagonal presence, ^{212}Bi shows significant off-diagonal activity, particularly into background classes. In this matrix, label 1 corresponds to external background, 2 to ^{212}Bi , and 3 to ^{212}Po . **As in the other models, the matrix does not exhibit a sharply defined diagonal, primarily due to the misclassification of ^{212}Bi events, which are confused with both internal and external backgrounds. In contrast, ^{212}Po is classified with much higher accuracy.**

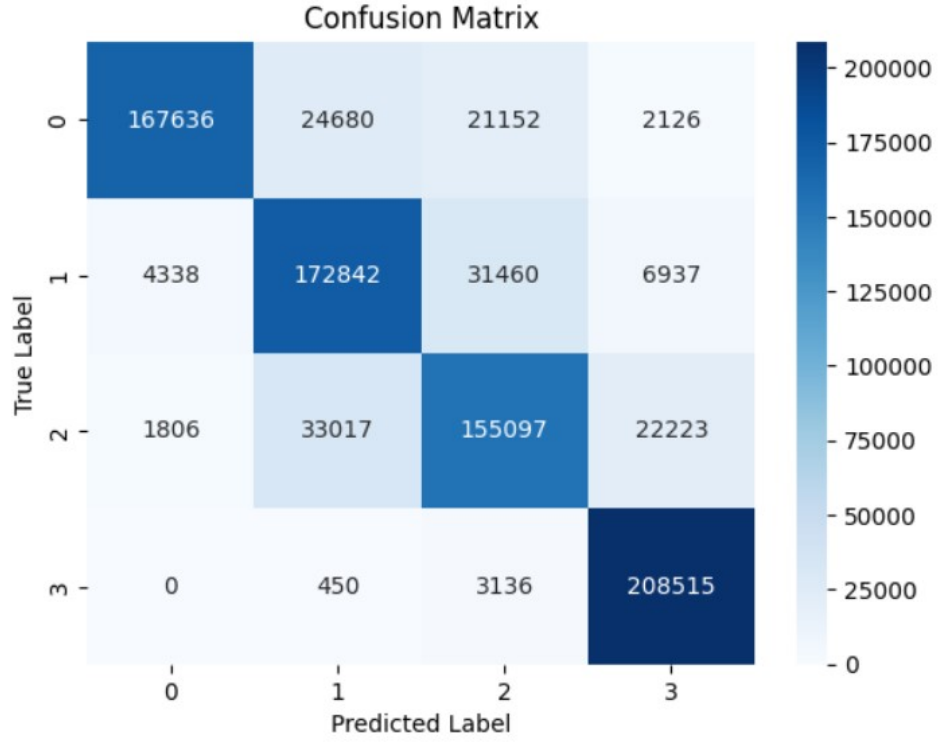


Figure 5.16: Confusion matrix of the $^{212}\text{BiPo}$ ANN model. Label 1 represents external background, 2 is ^{212}Bi , and 3 is ^{212}Po . The diagonal structure is well-defined for ^{212}Po , while ^{212}Bi shows considerable misclassification, particularly into background classes.

Energy Distributions The energy distribution of classified ^{212}Bi events, as shown in Figure 5.17, generally follows the expected continuous shape of a beta spectrum, with the true distribution overlaid for reference. Some background events are classified as ^{212}Bi , contributing to discrepancies between the predicted and true distributions. A notable gap appears between approximately 0.75 MeV and 1.0 MeV. This region is largely absent from the distribution, consistent with the behavior seen in the full dataset model (Figure 5.5). The missing events are likely due to misclassification as ^{212}Po or background, which dominate this energy region. In addition, sharp spikes are observed at the edges of the missing interval. These features suggest that background events in those ranges are being incorrectly assigned to ^{212}Bi .

For ^{212}Po , the classified energy spectrum (Figure 5.18) exhibits a narrow monoenergetic peak characteristic of alpha decays, clearly visible in the overlaid true distribution. The peak closely resembles the expected distribution observed in the Gaussian fit of the true ^{212}Po spectrum (Figure 3.6). This strong performance indicates that the model successfully identifies ^{212}Po events based on their well-defined energy signature.

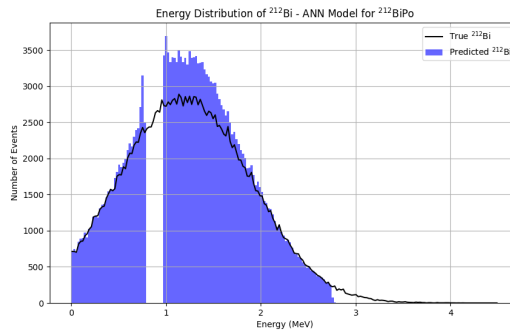


Figure 5.17: Energy spectrum of the classified ^{212}Bi events using the ANN model for the $^{212}\text{BiPo}$ chain. The predicted distribution (blue) is compared with the true distribution (line). A gap between 0.75 and 1.0 MeV is likely due to misclassification as ^{212}Po or background. Spikes near the edges of this region are attributed to background leakage.

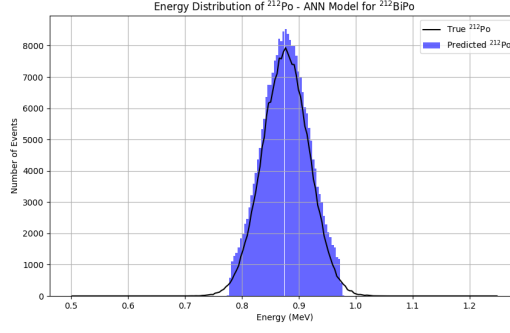


Figure 5.18: Energy spectrum of the classified ^{212}Po events using the ANN model for the $^{212}\text{BiPo}$ chain. The predicted distribution (blue) shows a sharp monoenergetic peak that closely matches the true alpha decay energy (line).

Decay Curves Figure 5.19 shows the decay curve for $^{212}\text{BiPo}$ events predicted by the ANN model trained solely on BiPo212 and background events. While the curve exhibits an initial peak followed by a general decline, the expected sharp exponential drop-off characteristic of ^{212}Po 's very short half-life ($\sim 0.3 \mu\text{s}$) is not observed. Instead, the decay appears stretched over a much longer timescale, with events extending well beyond 0.3 seconds.

This mismatch indicates that many background events are still misclassified as Bi decays, and falsely paired with Po events, resulting in unphysical time differences. The relatively flat tail and the absence of a rapid fall-off reinforce the notion that the model fails to isolate genuine BiPo212 coincidences with sufficient temporal precision. Even with a simplified dataset, the model has difficulty separating true BiPo212 events from background because ^{212}Po decays very quickly, requiring high timing precision and accurate classification.

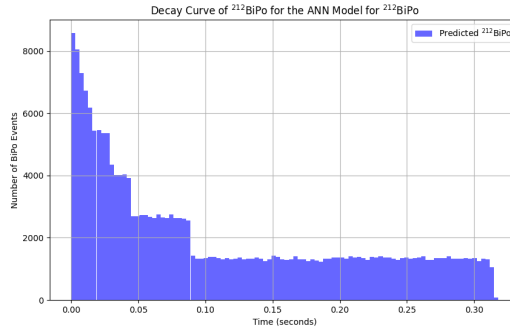


Figure 5.19: Decay time distribution for $^{212}\text{BiPo}$ events predicted by the ANN model trained specifically for BiPo212 classification. The decay curve does not exhibit the rapid exponential decline expected from the short half-life of ^{212}Po . The extended time scale and flatter shape suggest that background events are still frequently misclassified and incorrectly paired with Po candidates, distorting the physical decay signature.

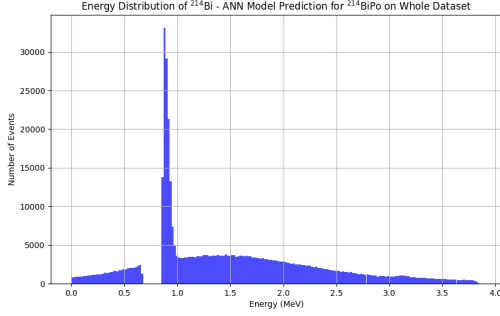
Applying Specialized Models to the Full Dataset

Each specialized model was applied to the full toy MC dataset containing all six classes to evaluate how well the chain-specific classifiers generalize beyond their training distributions. While these models were originally trained to distinguish only background and a single decay chain, applying them to the full dataset provides insight into how the networks internally separate and misclassify overlapping physical features.

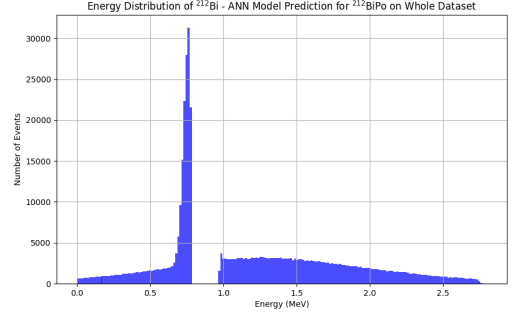
The resulting energy distributions show that both specialized models strongly favor their respective target isotopes, even when applied to the full dataset. The $^{214}\text{BiPo}$ model tends to classify a substantial number of both ^{214}Bi and ^{212}Bi events as ^{214}Bi , the $^{212}\text{BiPo}$ does the same.

Both distributions exhibit a sharp peak and a broad extension into the beta spectrum, but neither spectrum covers the full expected energy range. Instead, there is a gap between approximately 0.75 and 1.0 MeV, which is consistently missing in both models. Interestingly, this gap is flanked by different misclassification patterns: the ^{214}Bi spectrum (Fig. 5.20a) features a prominent peak just below 1.0 MeV,

while the ^{212}Bi spectrum (Fig. 5.20b) shows a sharp excess just below 0.75 MeV, corresponding to misclassified events. These edge peaks suggest that the classifiers, when restricted to only one BiPo chain during training, tend to map all BiPo-like events to the two known classes, e.g., a model trained on $^{212}\text{BiPo}$ will misclassify ^{214}Bi and ^{214}Po events as ^{212}Bi and ^{212}Po , respectively, and vice versa.



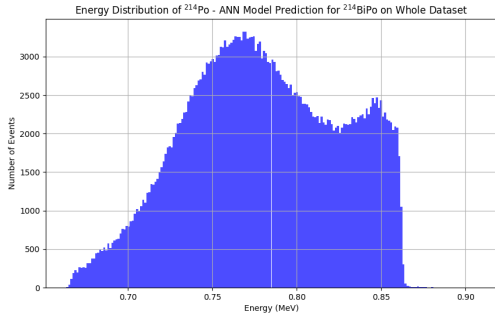
(a) Energy spectrum of ^{214}Bi events classified by the $^{214}\text{BiPo}$ model. The peak near 1 MeV reflects misclassified background events on one side of the gap.



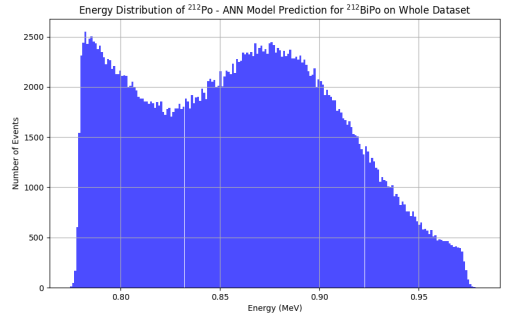
(b) Energy spectrum of ^{212}Bi events classified by the $^{212}\text{BiPo}$ model. A strong background-induced peak appears just below 0.75 MeV on the other side of the gap.

Figure 5.20: Energy distributions of Bi isotopes when the specialized models are applied to the full dataset. A consistent gap between 0.75 and 1.0 MeV is observed in both distributions, with misclassified peaks at either edge: ^{214}Bi model accumulates background near 1 MeV, while the ^{212}Bi model misclassifies background events just below 0.75 MeV. This reflects ambiguity in this region and model bias toward their respective decay chains.

For the Po isotopes, both models produce relatively clean energy peaks in the expected monoenergetic regions. However, their shapes are distorted, with broadened or multi-peaked distributions (Figs. 5.21a and 5.21b). These distortions are likely due to cross-chain contamination and classification uncertainty near the overlapping energy region around 0.83 MeV. Nonetheless, both peaks remain centered near the expected alpha decay energies, showing that the models still retain some discriminative power for Po classification.



(a) Energy spectrum of ^{214}Po events classified by the $^{214}\text{BiPo}$ model. The peak is centered near the expected energy but is broadened and asymmetric due to feature overlap.



(b) Energy spectrum of ^{212}Po events classified by the $^{212}\text{BiPo}$ model. The peak near 0.89 MeV is visible, but shape distortions indicate possible contamination from misclassified events.

Figure 5.21: Energy distributions of Po isotopes when the specialized models are applied to the full dataset. Despite showing peaks near expected alpha energies, both distributions exhibit shape distortions due to cross-chain classification.

Although the confusion matrices in this setting are not directly interpretable due to label mismatches and missing classes (resulting in zero entries), they are included in Appendix D.9 for completeness.

Overall, this evaluation highlights the limitations of using either model in isolation on mixed-chain datasets. The strong bias toward their respective decay chains and significant feature overlap reinforce the need for a combined classification approach, discussed in the following section.

5.2.4 Refined Classification with Higher Probability Selection

Model Combination Strategy

A combined classification strategy is required to enable the application of the two specialized models, each trained on a specific BiPo decay chain, to a dataset containing both chains. The final combined output can then be directly compared to the baseline ANN trained on the full dataset. Hence, a combined classification approach is employed using a maximum likelihood-based strategy. Each event is passed through both models, and the final predicted class is selected based on the following logic:

- If both models classify the event as background (truth label 0 or 3), that label is retained.
- If one model classifies the event as $^{214}\text{BiPo}$ (1 or 2) and the other as $^{212}\text{BiPo}$ (4 or 5), the model with the higher prediction confidence is selected.
- In case of no ambiguity (e.g., only one model predicts a signal label), that prediction is used.

This approach helps resolve inter-chain misclassifications while improving classification precision for true BiPo events. The results of this combined model, including decay curves and energy spectra, are discussed in the following subsections.

Performance Metrics

The refined model, after applying the post-processing strategy, was evaluated on the full test dataset. Although this approach aimed to leverage chain-specific specialization, the resulting performance metrics indicate a degradation in classification accuracy.

The model achieves an overall accuracy of 65%, significantly lower than the 71% achieved by the full ANN model. Table 5.4 shows the classification report summarizing the performance for each class.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.83	0.78	0.80
^{214}Bi (1)	0.48	0.24	0.32
^{214}Po (2)	0.80	0.83	0.81
External Background (3)	0.50	0.86	0.63
^{212}Bi (4)	0.51	0.48	0.49
^{212}Po (5)	0.83	0.69	0.75

Table 5.4: Class-wise evaluation metrics of the combined ANN model. Despite utilizing specialized networks, classification performance decreases, particularly for the Bi isotopes and background separation.

Among the signal classes, ^{214}Po performs best with a recall of 83% and efficiency of 80%, followed by ^{212}Po with a recall of 69% and efficiency of 83%. However, ^{214}Bi performs poorly with only 24% recall and 48% efficiency. ^{212}Bi also performs weakly, with both values around 50%. Background classes remain moderately classified, with external background achieving high recall (86%) but poor precision (50%), suggesting frequent misclassification of signal events into this class.

A more detailed class-wise breakdown of the dominant misclassification trends is shown below:

- **^{214}Bi :** Among events classified as this class, only 47.5% are true positives. The majority of actual ^{214}Bi events are misclassified as ^{212}Bi (32.2%), ^{212}Po (7.5%), external background (6.0%), or internal background (6.8%).
- **^{214}Po :** 99% of ^{214}Po events are correctly identified, with minor leakage into ^{212}Po (9.7%), ^{212}Bi (5.4%), and other classes.
- **^{212}Bi :** Misclassified primarily as ^{214}Bi (34.5%), as well as external background (6.7%) and internal background (5.6%). Only 48% of actual events are correctly recovered.
- **^{212}Po :** Despite relatively high efficiency (83%), this class suffers from misclassification into ^{212}Bi (6.5%), ^{214}Po (5.5%), and ^{214}Bi (3.5%).

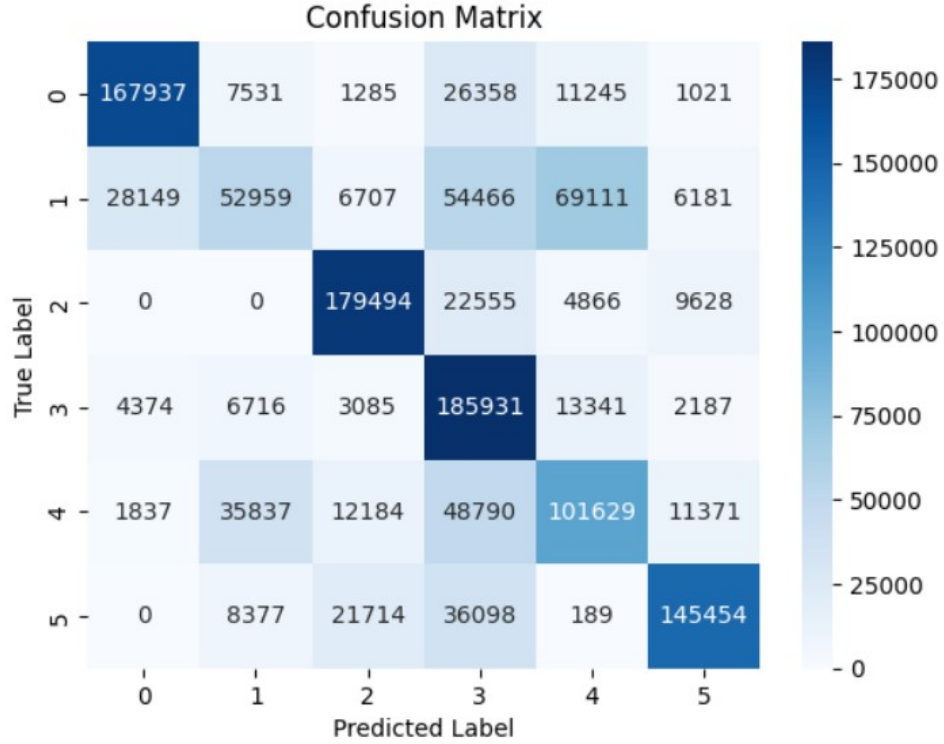


Figure 5.22: Confusion matrix of the combined ANN model. The absence of a bright diagonal highlights widespread confusion, particularly among signal classes. ^{214}Bi and ^{212}Bi suffer from substantial misclassification into each other and background classes, while ^{214}Po and ^{212}Po retain moderate separability.

The confusion matrix (Figure 5.22) reveals widespread off-diagonal activity, particularly between ^{214}Bi and ^{212}Bi and between signal classes and background. Unlike a well-performing classifier, the diagonal entries are not clearly dominant, underscoring the limitations of the combined classification strategy.

For completeness, the ROC curve for the combined model is provided in Appendix D.10, complementing this evaluation with a threshold-dependent performance overview.

Energy Distributions

The energy distributions of classified ^{212}Bi and ^{214}Bi events from the combined ANN model are shown in Figures 5.23 and 5.24. This model integrates the predictions of the individual BiPo214 and BiPo212 classifiers and assigns each event to the class with the highest probability. As a result, it attempts to resolve classification conflicts by leveraging the relative prediction strengths of both models.

The ^{214}Bi energy distribution (Figure 5.23) reveals the upper portion of the beta spectrum, extending beyond 2.5 MeV, as expected. A particularly striking feature is the sharp peak around 1.0 MeV. This feature was previously observed in the ^{212}Bi spectrum of the specialized $^{212}\text{BiPo}$ model. However, its clear appearance in the ^{214}Bi distribution within the combined model suggests that it likely originates from ^{214}Bi itself. When compared to the overlaid true distribution, it is evident that the predicted spectrum deviates significantly from the physical expectation, with many misclassified events.

In the ^{212}Bi energy distribution (Figure 5.24), the lower half of the beta spectrum is well represented, beginning near zero and extending up to approximately 1.75 MeV. Notably, the previously observed gap between 0.75 and 1.0 MeV, visible in both the full model and the BiPo212-only model, is partially recovered here. Additionally, a small excess around 0.7 MeV remains, possibly due to lingering background contamination or classification spillover from ^{214}Bi . However, a clear mismatch between the predicted and true distributions persists, indicating extensive misclassification.

Together, the two distributions reflect the expected separation between the Bi isotopes: ^{212}Bi populating the lower to mid-energy range, and ^{214}Bi occupying the higher end of the beta spectrum.

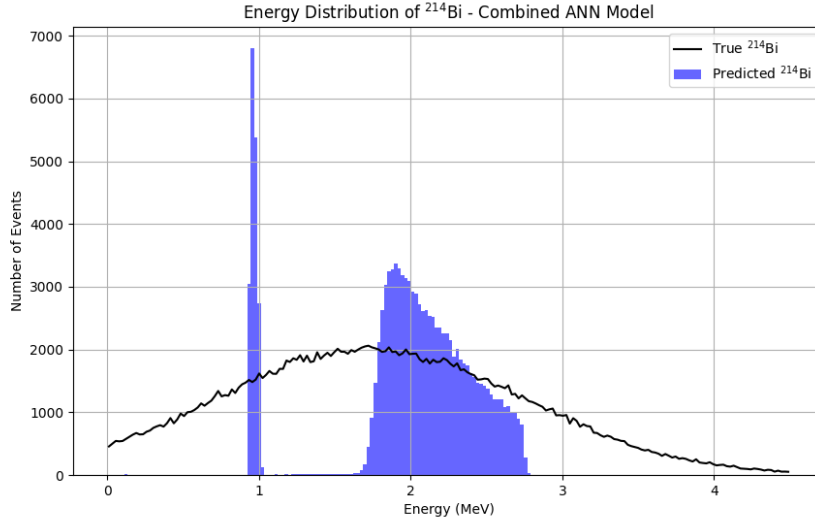


Figure 5.23: Energy distribution of ^{214}Bi events classified by the refined ANN model. The predicted distribution (blue) is compared with the true spectrum (line). The shape extends into higher energies with a sharp non-physical peak around 1 MeV, indicating significant misclassification.

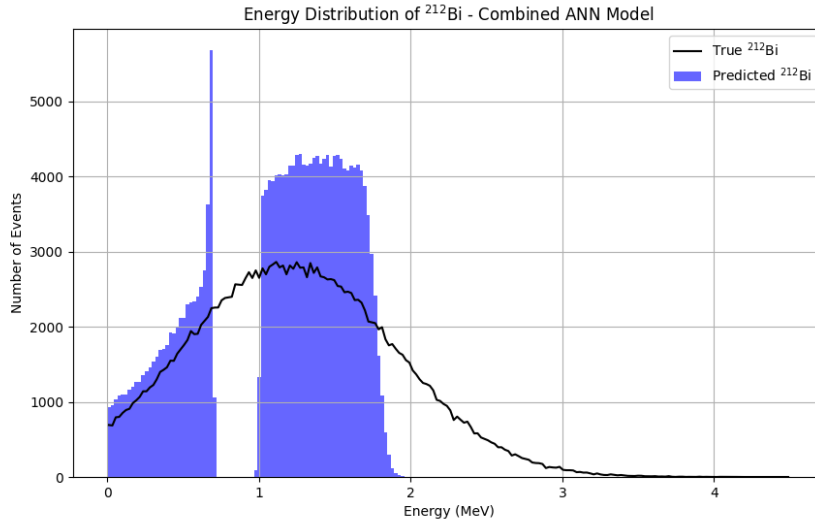


Figure 5.24: Energy distribution of events classified as ^{212}Bi using the combined ANN model. The predicted spectrum (blue) covers the lower half of the beta distribution and partially recovers the previously missing 0.75–1.0 MeV range. The overlaid true distribution (line) shows a noticeable deviation, highlighting classification inaccuracies.

Figure 5.25 shows the energy distribution of ^{214}Po events classified by the refined ANN model. While a peak is visible around 0.78 MeV, close to the expected value for alpha particles from ^{214}Po decay, the shape of the distribution is not sharply defined. The rise is gradual, the peak is broad, and the decline lacks the symmetry seen in ideal alpha spectra. Additionally, the distribution exhibits a noticeable cutoff near 0.83 MeV, which marks the onset of the energy region predominantly associated with ^{212}Po . The overlaid true distribution highlights that while the model captures the general location of the alpha energy, its classification fidelity is reduced near the overlap region, likely due to confusion between the two chains and residual background interference.

Similarly, the ^{212}Po energy distribution (Figure 5.26) presents a peak slightly above 0.85 MeV, which is consistent with the expected alpha energy for this isotope. However, the spectrum again lacks the narrow, clean shape typically associated with monoenergetic alpha decays. The peak is broad, and the spectrum drops off toward lower energies near the overlap region around 0.83 MeV, where ^{214}Po decays dominate. The comparison with the overlaid true distribution reveals incomplete separation between the

two Po classes in the transition region, as well as possible contamination from misclassified Bi events or background.

Overall, while the peaks for both isotopes are identifiable and approximately match the expected energies from alpha decay, the lack of smoothness and the abrupt cutoffs at the region of overlap limit the interpretability of the spectra. This highlights the challenge of classifying closely spaced monoenergetic signatures in the presence of overlapping features and model uncertainties, even for well-defined alpha-emitting isotopes.

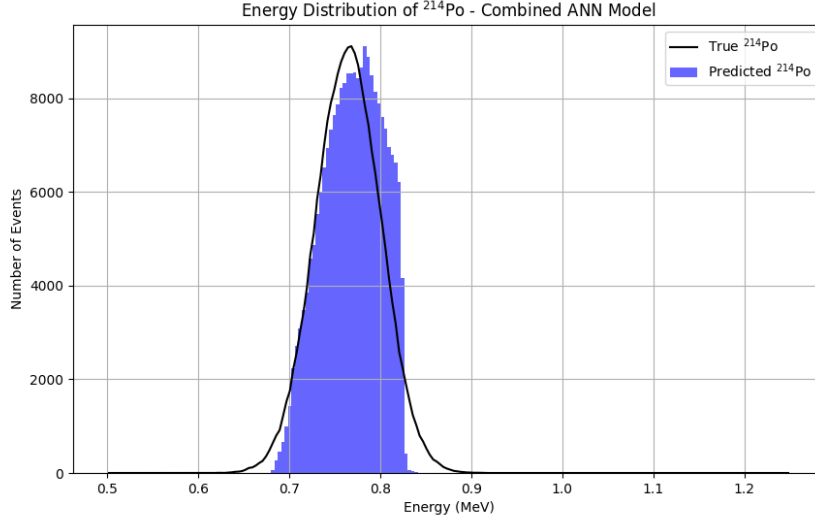


Figure 5.25: Energy distribution of ^{214}Po events classified by the combined ANN model. The predicted distribution (blue) is compared with the true alpha decay spectrum (line). A broad peak is visible around 0.78 MeV, but the lack of symmetry and the cutoff near 0.83 MeV indicate challenges in cleanly separating the two Po isotopes.

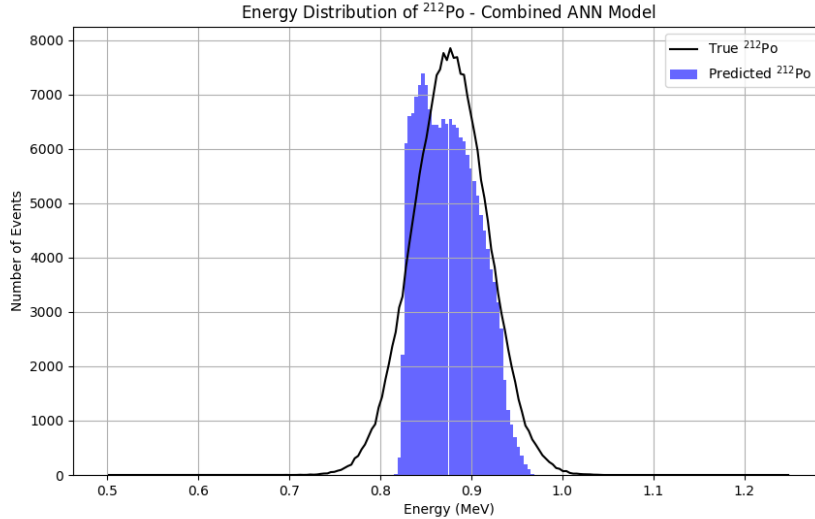


Figure 5.26: Energy distribution of ^{212}Po events classified by the combined ANN model. The predicted spectrum (blue) overlays the true distribution (line). A peak appears slightly above 0.85 MeV but lacks the narrow, clean profile of a monoenergetic alpha decay, with a sharp decline near the ^{214}Po transition region at 0.83 MeV.

Decay Curves Figures 5.27 and 5.28 show the decay time distributions for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ event pairs classified using the refined ANN model. As in previous approaches, the time difference is computed between predicted Bi and Po events, with true BiPo decays expected to follow an exponential distribution governed by the Po half-lives.

In the case of $^{214}\text{BiPo}$ (Figure 5.27), the decay curve begins with a peak at short time differences, followed by a decline. However, the distribution quickly flattens and remains relatively constant over time, indicating a large contribution from uncorrelated or misclassified background events. The lack of a clean exponential tail suggests that the model continues to misidentify a significant number of background events as Bi, despite using the combined classification strategy.

For the $^{212}\text{BiPo}$ decay (Figure 5.28), a stronger exponential-like component is initially visible, with a pronounced peak and a more gradual decline. However, as with the ^{214}Bi case, the curve becomes flat after a short time window. This flattening again points to misclassified background events paired randomly with Po candidates, overshadowing the true decay signature at longer time differences.

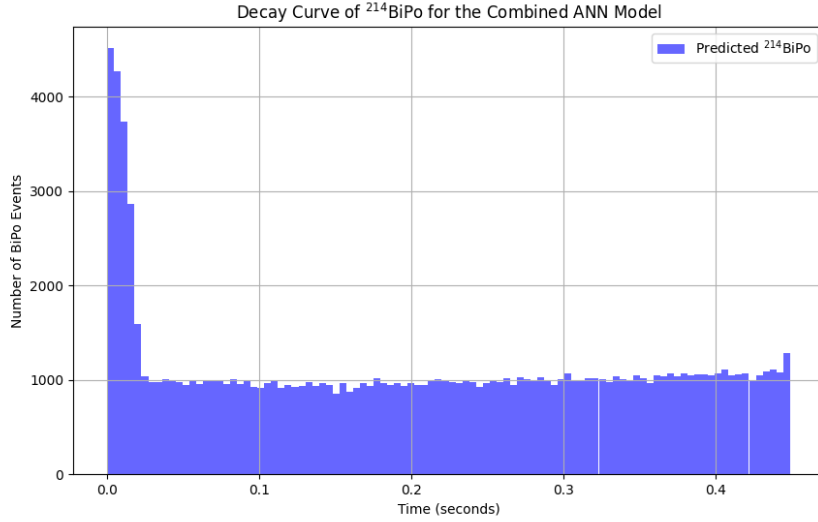


Figure 5.27: Decay curve of the classified BiPo 214 events using the combined ANN model

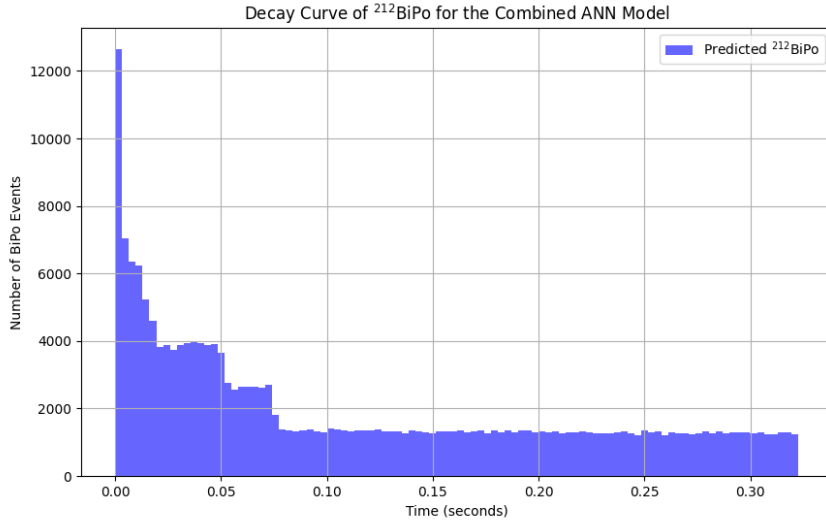


Figure 5.28: Decay curve of the classified BiPo 212 events using the combined ANN model

Despite the intention to improve classification by combining the strengths of chain-specific models, the combined ANN model ultimately underperforms in nearly every aspect when compared to the model trained on the full dataset. The decay curves for both $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ fail to exhibit clear exponential signatures, instead displaying rapid flattening indicative of severe background contamination. This issue is more pronounced than in the full-dataset model, where at least partial exponential structure was preserved. Similarly, the energy distributions of both Bi and Po isotopes in the combined model show broader, less defined peaks, with sharper cutoffs in overlap regions, reflecting reduced separation performance. Notably, some spectral features correctly attributed to background in the full model are

now misclassified as signal in the combined model, leading to misinterpretation of the energy domain. These results show that combining separately trained models does not improve overall performance and may even worsen classification by adding confusion instead of reducing it.

5.3 Traditional Classification with Decision Tree

Decision Trees are a classical and interpretable machine learning technique commonly used for classification tasks. They operate by recursively partitioning the feature space based on decision rules, making them well-suited for problems where explainability and fast inference are desired. In the context of this work, Decision Trees are applied to classify BiPo events using the same engineered features as in the neural network approach. This section begins with a brief explanation of how Decision Trees work, followed by the implementation details, model performance, and evaluation of both combined and separately trained datasets for the two decay chains.

5.3.1 Overview and Working of Decision Trees

Decision Trees are a widely used supervised learning algorithm for classification and regression tasks. They are intuitive, interpretable, and capable of handling both numerical and categorical data. In this work, Decision Trees are used to classify BiPo events based on spatial, temporal, and energy features.

A Decision Tree splits the data step by step based on feature values, creating branches that group similar classes together. At each split, the algorithm chooses the feature and threshold that best separate the classes to make the resulting groups as pure as possible.

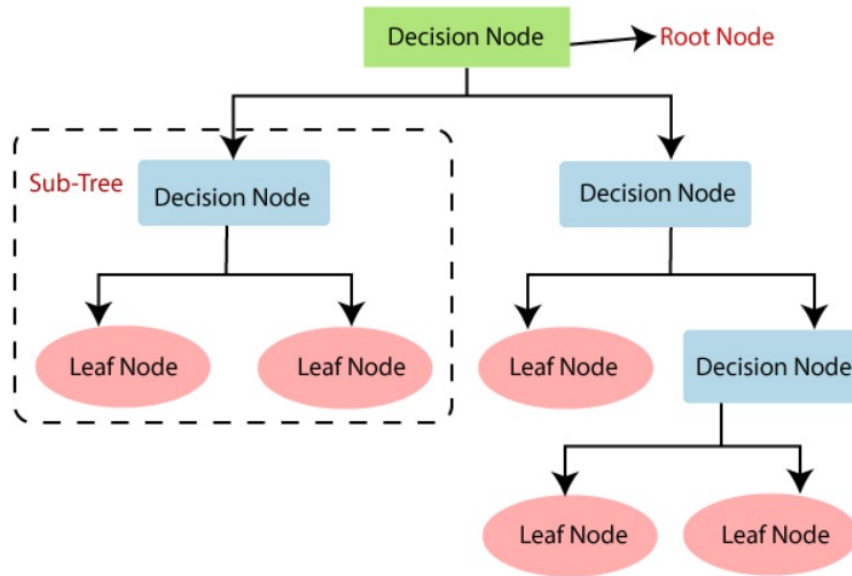


Figure 5.29: A schematic representation of a decision tree used in classification problems adapted from [33]. The tree begins at the root node (green), which is a decision node that splits the data based on a feature. Intermediate decision nodes (blue) further divide the data, while leaf nodes (pink) represent final output classes or values. The dashed box highlights a sub-tree, which is itself a smaller decision tree. This hierarchical structure allows the model to learn a sequence of decisions for classification.

Splitting Criterion: Gini Impurity

In this work, the Decision Tree uses *Gini impurity* as the criterion for selecting the best splits. Gini impurity measures the probability of misclassifying a randomly chosen element from the dataset if it were labeled according to the class distribution in the node.

For a node with K classes, the Gini impurity is defined as:

$$G = 1 - \sum_{k=1}^K p_k^2$$

where p_k is the proportion of samples of class k in the node.

The algorithm selects the feature and threshold that minimize the weighted sum of the Gini impurities of the resulting child nodes. This results in increasingly homogeneous subsets as the tree grows deeper. The top features act as points of authority in the decision process, guiding how the tree partitions the feature space.

Tree Growth and Stopping Criteria

The tree continues to grow until a stopping condition is met. Common stopping criteria include:

- Maximum tree depth
- Minimum number of samples per leaf
- No further impurity reduction possible

In practice, trees are often pruned either during the training process (as the tree is being built) or after the tree has been fully constructed to prevent overfitting. In this work, grid search was used to optimize key hyperparameters such as maximum depth and minimum samples per leaf.

Advantages and Limitations

Decision Trees are attractive due to their interpretability, minimal data preprocessing requirements, and ability to model non-linear decision boundaries. However, they can be sensitive to small variations in the data and prone to overfitting, especially with deep trees.

To mitigate this, Decision Tree models in this study are optimized using grid search, as discussed in the following sections.

5.3.2 Decision Tree Trained on the Full Dataset

Model Architecture and Training Strategy

The Decision Tree model used in this study was trained on the full dataset, with a split strategy of 80% for training and 20% for testing, ensuring reproducibility by setting the `random_state` to 0. The training data was further split into training (80%) and validation (20%) subsets. Unlike the ANN model, the target variable was not one-hot encoded, as Decision Trees inherently handle categorical data.

Initial Model and Evaluation The initial Decision Tree model was constructed without any hyperparameter optimization to establish a baseline. The model was trained using the `DecisionTreeClassifier` from Scikit-learn, with a default configuration except for setting the `random_state` to 0. After training, the model was evaluated on both validation and test sets. The validation accuracy was found to be approximately 61%, while the test accuracy also yielded a similar value of 61%. The classification report for the test set is shown in Table 5.6.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.73	0.75	0.74
^{214}Bi (1)	0.39	0.39	0.39
^{214}Po (2)	0.79	0.77	0.78
External Background (3)	0.60	0.60	0.60
^{212}Bi (4)	0.37	0.38	0.37
^{212}Po (5)	0.76	0.75	0.76

Table 5.5: Class-wise evaluation metrics of the Decision Tree model trained on the full dataset. The model performs moderately well for Po isotopes but shows significantly lower performance for Bi isotopes due to the overlap in energy spectra and beta decay characteristics. Background classification is balanced, with internal background having slightly higher precision compared to external background.

Hyperparameter Optimization with Grid Search To improve the model’s performance, hyperparameter optimization was conducted using Grid Search with Cross-Validation (`GridSearchCV`). Cross-validation is a technique used to assess how the model generalizes to an independent dataset. It involves splitting the training data into multiple subsets (folds), training the model on some of these subsets, and validating it on the remaining subsets. This process is repeated several times, and the results are averaged to provide a robust estimate of model performance.

Unlike Optuna, which employs more sophisticated sampling techniques, Grid Search was chosen because Decision Trees are computationally less intensive, allowing for a more exhaustive search without excessive resource consumption. Moreover, the relatively small number of key hyperparameters made Grid Search an appropriate and interpretable choice.

The following hyperparameters were optimized:

- **max_depth:** Controls the maximum depth of the tree. Values of {None, 5, 10, 15} were tested to explore both shallow and deeper trees. Setting `None` allows the tree to grow until all leaves are pure or contain less than the minimum samples required to split.
- **min_samples_split:** Determines the minimum number of samples required to split an internal node. Values of {2, 5, 10} were chosen to balance model complexity and generalization.
- **min_samples_leaf:** Specifies the minimum number of samples required to be at a leaf node. Values of {1, 2, 5} were considered to avoid overfitting while maintaining model flexibility.

Grid Search was performed with 3-fold cross-validation, aiming to maximize accuracy. In 3-fold cross-validation, the training data is split into three equal parts. The model is trained on two parts and validated on the third. This process is repeated three times, with each part being used as the validation set once. The final performance metric is averaged over the three iterations.

The optimal hyperparameters obtained from the search were:

$$\text{max_depth} = 10, \quad \text{min_samples_split} = 10, \quad \text{min_samples_leaf} = 2$$

These settings yielded a best cross-validation accuracy of 71.1%, indicating a significant improvement over the baseline model. In the next subsections, we will examine the model’s performance metrics in detail, analyze the energy spectrum, and study the decay curves.

The resulting Decision Tree was visualized to inspect the structure and the splitting criteria. The visualization, presented in Appendix E.1, demonstrates how the optimized model leverages input features to classify events.

Performance Metrics

The Decision Tree model achieves an overall accuracy of 71% on the test set, comparable to the ANN model trained on the full dataset. Table 5.6 summarizes the class-wise precision, recall (TPR), and F1-score.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.95	0.74	0.83
^{214}Bi (1)	0.56	0.37	0.45
^{214}Po (2)	0.82	0.92	0.87
External Background (3)	0.67	0.75	0.71
^{212}Bi (4)	0.49	0.57	0.53
^{212}Po (5)	0.80	0.91	0.85

Table 5.6: Class-wise evaluation metrics of the Decision Tree model trained on the full dataset. The model demonstrates strong performance for Po isotopes, similar to the ANN, but the classification of Bi isotopes shows limitations due to overlapping energy spectra and decay characteristics. Background classes exhibit consistent performance, with internal background achieving high precision and external background showing balanced precision and recall.

The model performs best for ^{214}Po and ^{212}Po , achieving high TPRs of 92% and 91%, and efficiencies of 82% and 80%, respectively. Similar to the ANN model, this performance is attributed to the clear monoenergetic peaks of alpha decay, which help the Decision Tree to distinguish these classes more effectively.

In contrast, the Bi isotopes are more challenging to classify accurately. ^{214}Bi has a TPR of 37% and an efficiency of 56%, while ^{212}Bi achieves a TPR of 57% and an efficiency of 49%. These values are comparable to the ANN performance and highlight the difficulty in discriminating beta decay events due to their broader spectra and overlap with background signals.

The background classes show moderate to strong classification performance. Internal background achieves high precision (95%) but a lower TPR (74%), indicating that some true internal background events are misclassified. External background is comparatively balanced with a TPR of 75% and precision of 67%.

A closer look at the class-wise misclassification patterns reveals the following:

- ^{214}Bi is most frequently misclassified as ^{212}Bi (22.76%), consistent with the overlap in energy spectra. Additional confusion occurs with internal background (12.25%) and external background (8.98%).
- ^{214}Po is predominantly correctly identified, but some events are misclassified as ^{212}Po (6.56%) and ^{212}Bi (5.88%), reflecting the energy overlap between chains.
- ^{212}Bi shows substantial misclassification as ^{214}Bi (33.17%), with additional errors involving external background (10.91%). This pattern mirrors the challenge seen in the ANN model.
- ^{212}Po shows a robust classification but is occasionally confused with ^{212}Bi (7.27%) and ^{214}Po (6.00%).

These patterns are clearly illustrated in the confusion matrix (Figure 5.30). Ideally, a highly accurate model would exhibit a dominant diagonal with minimal off-diagonal elements, indicating few misclassifications. However, similar to the ANN, the Decision Tree model shows considerable off-diagonal contributions, particularly between the two Bi classes and between signal classes and background. This lack of a sharply defined diagonal highlights the inherent difficulty in accurately separating overlapping beta decay events and background noise using a single decision tree.

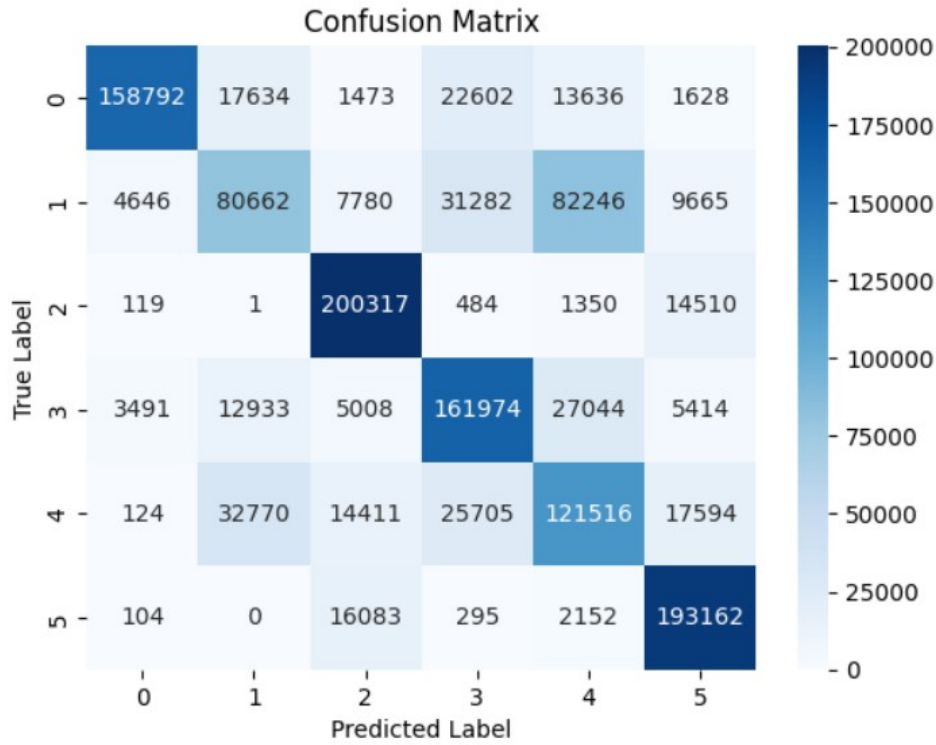


Figure 5.30: Confusion matrix of the Decision Tree model trained on the full dataset. **While the model exhibits reasonable accuracy for Po isotopes, significant misclassification is observed between Bi classes and background, reflecting the challenge of discriminating beta decays from background events.**

Energy Distributions

The energy distributions of ^{214}Bi and ^{212}Bi events predicted by the Decision Tree model on the whole dataset are shown in Figures 5.31 and 5.32, respectively.

The ^{214}Bi energy spectrum (Figure 5.31) captures the upper portion of the beta spectrum, starting from approximately 1.75 MeV and extending up to around 4 MeV. However, it does not resemble the full shape of the true distribution, which spans a broader energy range. The peak around 2 MeV reflects a high density of events, likely due to the Decision Tree's bias toward assigning events in this region to the ^{214}Bi class. The lack of lower-energy structure, where ^{212}Bi events are typically found, suggests that these events have been effectively excluded or misclassified. The overlaid true distribution highlights the disparity and the limited physical accuracy of the predicted spectrum.

In contrast, the ^{212}Bi spectrum (Figure 5.32) captures the lower half of the beta spectrum, from 0 to approximately 1.75 MeV. The distribution shows a rise toward higher energies, with a spike around 1 MeV. This feature likely results from contamination by background or ^{214}Bi events, due to overlapping energy ranges. The persistent gap between 0.7 and 1.0 MeV mirrors the behavior seen in the ANN model and suggests systematic misclassification as Po or background. Comparison with the overlaid true distribution reveals significant deviation from the expected shape, despite partially retaining the beta spectrum profile.

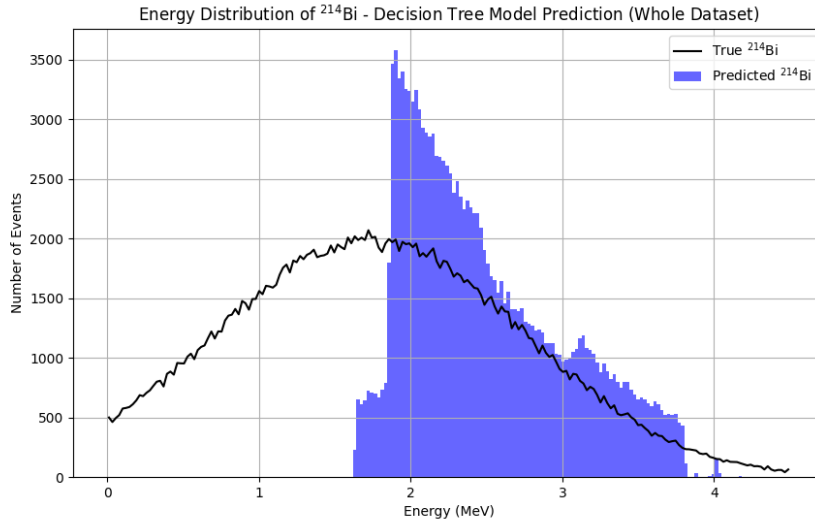


Figure 5.31: Energy spectrum of ^{214}Bi events classified by the Decision Tree model. The predicted distribution (blue) captures only the higher part of the beta spectrum, while the overlaid true distribution (line) shows a broader expected shape. The peak around 2 MeV and the absence of lower-energy events suggest strong misclassification.

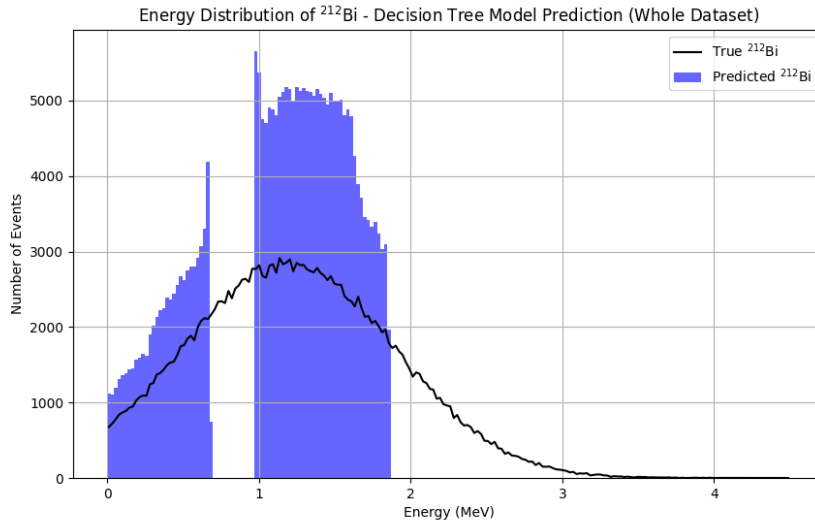


Figure 5.32: Energy spectrum of ^{212}Bi events classified by the Decision Tree model. The predicted spectrum (blue) roughly matches the lower beta energy range, while the overlaid true distribution (line) shows the expected shape. A gap between 0.7 and 1.0 MeV and a sharp feature near 1 MeV reflect background contamination and classification errors.

The energy distributions of ^{214}Po and ^{212}Po events predicted by the Decision Tree model on the whole dataset are shown in Figures 5.33 and 5.34, respectively. Unlike the ambiguous overlap seen in the Bi isotopes, the Po distributions exhibit more distinct monoenergetic peaks characteristic of alpha decay.

The ^{214}Po spectrum (Figure 5.33) displays a well-defined peak centered around 0.78 MeV, consistent with the expected alpha decay energy. The distribution is relatively narrow and agrees somewhat well with the overlaid true distribution, suggesting that the Decision Tree model can classify most ^{214}Po events with high precision. However, the spectrum exhibits a sharp drop-off just beyond 0.82 MeV, which marks a threshold beyond which the model predominantly classifies events as ^{212}Po . This cutoff corresponds to the overlapping region between the two Po isotopes discussed in Section 3.8, and likely serves as a learned decision boundary within the model.

The ^{212}Po spectrum (Figure 5.34) shows a similar monoenergetic peak around 0.89 MeV, also somewhat consistent with the overlaid true distribution. The shape is narrow, indicating accurate classification of alpha decays. The spectrum begins near 0.82 MeV—directly where the ^{214}Po distribution ends, suggesting a consistent classification threshold. This behavior reflects the Decision Tree’s reliance on distinct energy boundaries to separate isotopes, particularly within the overlap region.

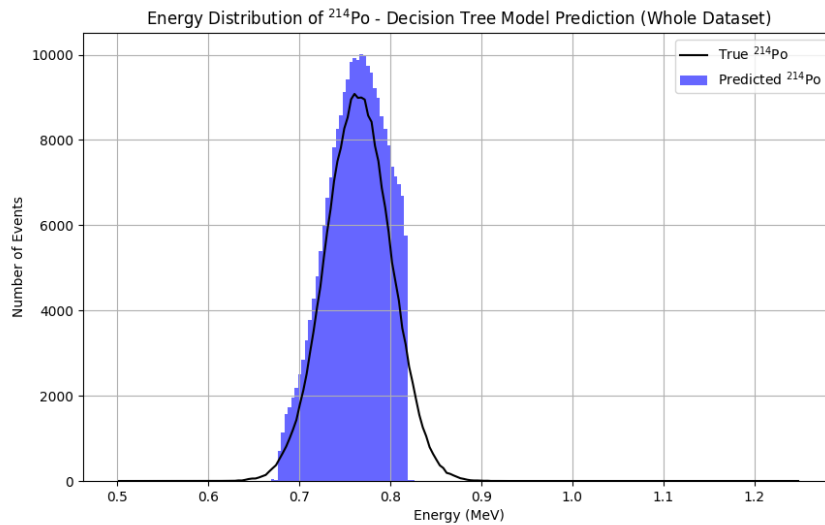


Figure 5.33: Energy spectrum of ^{214}Po events classified by the Decision Tree model using the full dataset. The narrow monoenergetic peak near 0.78 MeV matches the true distribution (line), with a sharp cutoff at 0.82 MeV indicating the transition to ^{212}Po classification.

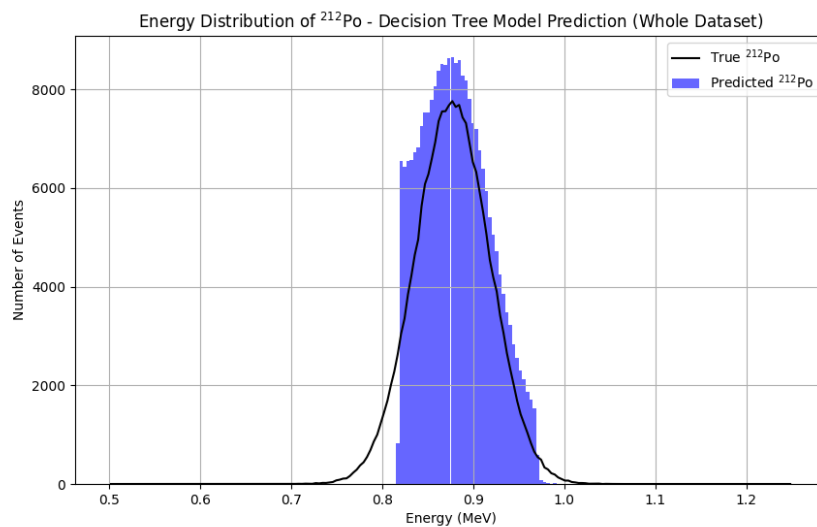


Figure 5.34: Energy spectrum of ^{212}Po events classified by the Decision Tree model using the full dataset. A clear monoenergetic peak is centered around 0.89 MeV, beginning near 0.82 MeV where ^{214}Po classification ends. The predicted shape aligns well with the true distribution (line).

Decay Curves

The decay curves of $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ predicted by the Decision Tree model on the whole dataset are shown in Figures 5.35 and 5.36, respectively. In both cases, the resulting distributions significantly deviate from the expected physical exponential decay pattern. This deviation highlights a critical limitation of the Decision Tree model when classifying decay chain events. It is important to note that the Decision Tree model, like the ANN model, does not take time differences between events as input features. Consequently, the model cannot learn the temporal decay behavior characteristic of BiPo sequences, leading to the observed flattening of the decay curves.

For the $^{214}\text{BiPo}$ chain (Figure 5.35), the decay curve exhibits a steep initial drop, consistent with the rapid decay of ^{214}Po , but then flattens out, showing a roughly uniform distribution afterward. This pattern suggests that many true $^{214}\text{BiPo}$ events are not correctly identified, while a substantial number of background or non- $^{214}\text{BiPo}$ events are incorrectly classified as $^{214}\text{BiPo}$. As a result, the expected exponential decrease is lost, and the distribution becomes largely uniform after the initial decline.

The $^{212}\text{BiPo}$ chain (Figure 5.36) shows a similar misclassification pattern but is even less physically accurate. The characteristic short decay time of ^{212}Po is not properly reflected in the curve. The distribution also flattens out rather than showing a steep decline.

Both decay curves reflect the fundamental issue that the Decision Tree model, while effective in classification tasks with more distinct features, struggles to accurately differentiate between short-lived and background events in the context of BiPo decay chains. This limitation is especially evident for the ^{212}Po decay, where the model fails to represent the rapid decrease in event count characteristic of its short half-life.

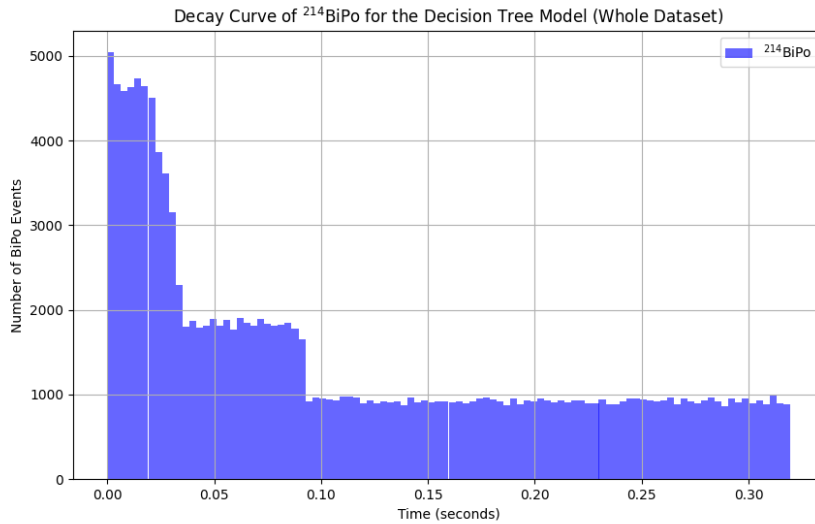


Figure 5.35: Decay Curve of the classified BiPo214 events predicted by the Decision Tree model for the whole dataset. The curve shows an initial rapid decline but then flattens out, indicating that many background events are misclassified as $^{214}\text{BiPo}$, resulting in a non-exponential pattern.

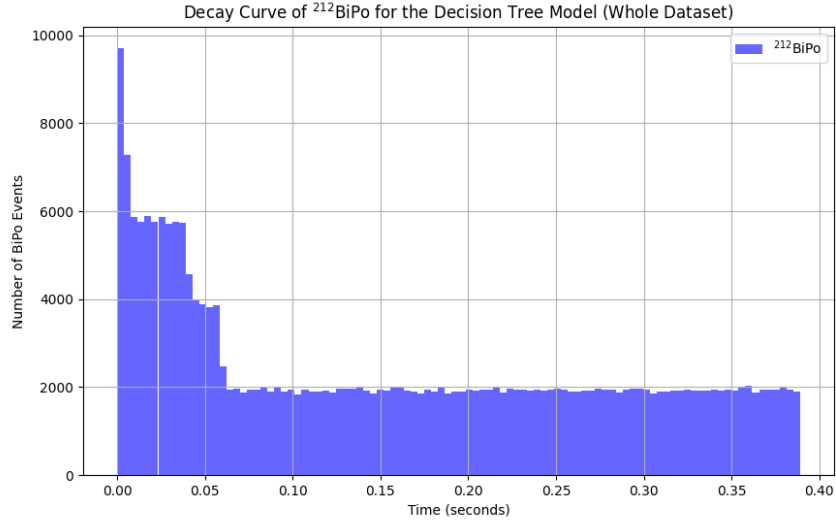


Figure 5.36: Decay Curve of the classified BiPo212 events predicted by the Decision Tree model for the whole dataset. The distribution shows a rapid initial decrease but fails to capture the short-lived exponential decay expected for ^{212}Po , instead flattening out due to misclassification of background or longer-lived events.

5.3.3 Separate Decision Trees for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$

Model Architecture and Training Strategy

To improve classification accuracy for the two decay chains individually, separate Decision Tree models were trained for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$. This approach allows the model to specialize in the characteristics of each decay chain, reducing misclassification due to overlapping energy and temporal features between chains.

The training and evaluation strategy follows the same methodology as the model trained on the full dataset. Initially, the Decision Tree models were trained without hyperparameter optimization to establish a baseline performance. Both models achieved an accuracy of 74%, which, while reasonable, indicated potential for improvement.

To enhance the model's performance, hyperparameter optimization was performed using Grid Search with Cross-Validation (`GridSearchCV`). The same hyperparameter ranges were selected for the Grid Search as for the model trained on the full dataset, ensuring consistency in the optimization strategy and enabling direct comparison of the models. This consistent approach ensures that the improvements observed can be attributed to the data separation rather than changes in the optimization strategy.

For the $^{214}\text{BiPo}$ model, the best parameters were:

- max_depth: 10
- min_samples_split: 5
- min_samples_leaf: 1

The optimal configuration achieved a cross-validation accuracy of 82.1%.

For the $^{212}\text{BiPo}$ model, the optimal parameters were:

- max_depth: 10
- min_samples_split: 5
- min_samples_leaf: 2

The best cross-validation accuracy was 82.2%.

In the following subsections, evaluation the model performance will be discussed in detail.

²¹⁴BiPo Model Evaluation

Performance Metrics The optimized Decision Tree model for the ²¹⁴BiPo chain achieved an accuracy of 82% on the test set. Table 5.7 shows the classification report summarizing precision, recall (TPR), and F1-score for each class.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.95	0.74	0.83
²¹⁴ Bi (1)	0.72	0.75	0.73
²¹⁴ Po (2)	0.92	0.99	0.95
External Background (3)	0.73	0.80	0.77

Table 5.7: Class-wise evaluation metrics of the Decision Tree model trained on the ²¹⁴BiPo chain. The model demonstrates high accuracy for Po events, while Bi events show improved performance compared to the model trained on the full dataset.

The model shows excellent classification performance for ²¹⁴Po, achieving a TPR of 99% and an efficiency of 92%. The classification for ²¹⁴Bi has improved compared to the full dataset model, with a TPR of 75% and an efficiency of 72%. Background classes also perform well, with internal background achieving high precision (95%) and external background showing balanced performance.

A breakdown of misclassifications reveals:

- ²¹⁴Bi is primarily misclassified as external background (14.04%) and internal background (13.82%).
- ²¹⁴Po is correctly classified in most cases, with minor confusion with ²¹⁴Bi (4.64%) and external background (2.79%).

The confusion matrix for the ²¹⁴BiPo Decision Tree model (Figure 5.37) further illustrates the classification performance. Ideally, a well-performing model would exhibit a bright diagonal, indicating that most predictions match the true labels. However, in this case, while the matrix shows a strong diagonal for ²¹⁴Po, indicating robust classification, the diagonal is less pronounced for ²¹⁴Bi, which shows considerable leakage into the background classes. The background classes (internal and external) also display some degree of misclassification into each other, though they generally maintain higher precision compared to the Bi events.

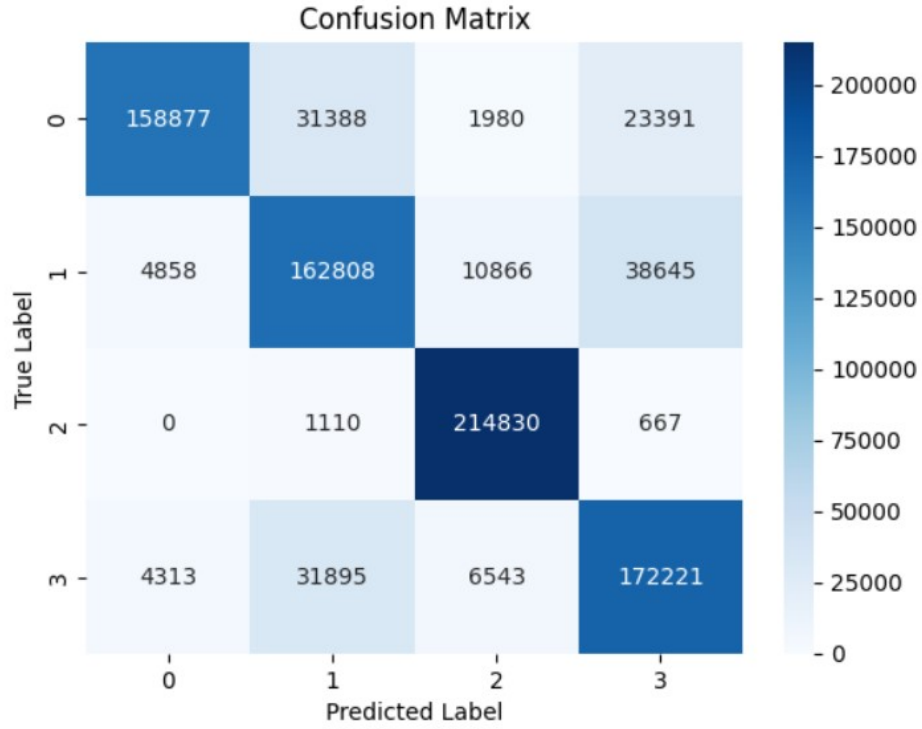


Figure 5.37: Confusion matrix of the optimized Decision Tree model trained on the $^{214}\text{BiPo}$ chain. The diagonal structure is prominent for ^{214}Po , indicating effective classification. In contrast, the diagonal for ^{214}Bi is less pronounced, showing significant misclassification into internal and external background classes. **This imbalance highlights the model's challenge in distinguishing ^{214}Bi events from background due to overlapping features.**

Energy Distributions The energy distributions of ^{214}Bi and ^{214}Po events predicted by the Decision Tree model for the BiPo214 chain are shown in Figures 5.38 and 5.39, respectively.

The ^{214}Bi energy spectrum (Figure 5.38) shows a broad distribution characteristic of beta decay, but with several irregularities when compared to the overlaid true distribution. The overall shape resembles a "chimney"-like structure, rising sharply to a peak near 1 MeV, followed by a plateau and a gradual decline. This pattern suggests that the Decision Tree model fails to capture the smooth, continuous nature of the true beta spectrum and instead clusters predictions in specific regions. A gap appears between approximately 0.7 MeV and 1.0 MeV, consistent with the behavior observed in the ANN model. This gap likely arises from misclassification as ^{214}Po or background, which dominate this overlapping region. As a result, the true spectral shape is only partially recovered.

In contrast, the ^{214}Po energy spectrum (Figure 5.39) displays a smooth monoenergetic peak around 0.78 MeV, in good agreement with the overlaid true distribution. The peak is symmetric and well-localized, reflecting the Decision Tree model's ability to reliably classify alpha decays when the energy signature is distinctive. The shape closely resembles a Gaussian, indicating that ^{214}Po is identified with high fidelity and minimal contamination.

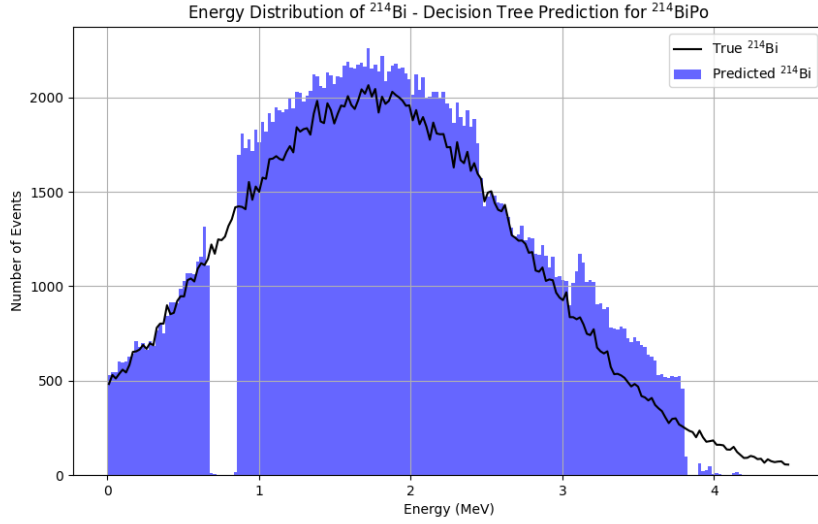


Figure 5.38: Energy spectrum of ^{214}Bi events classified by the Decision Tree model for the $^{214}\text{BiPo}$ chain. A chimney-like shape appears with a peak near 1 MeV. The overlaid true distribution shows discrepancies in the predicted distribution, especially the gap between 0.7-1.0 MeV caused by misclassification.

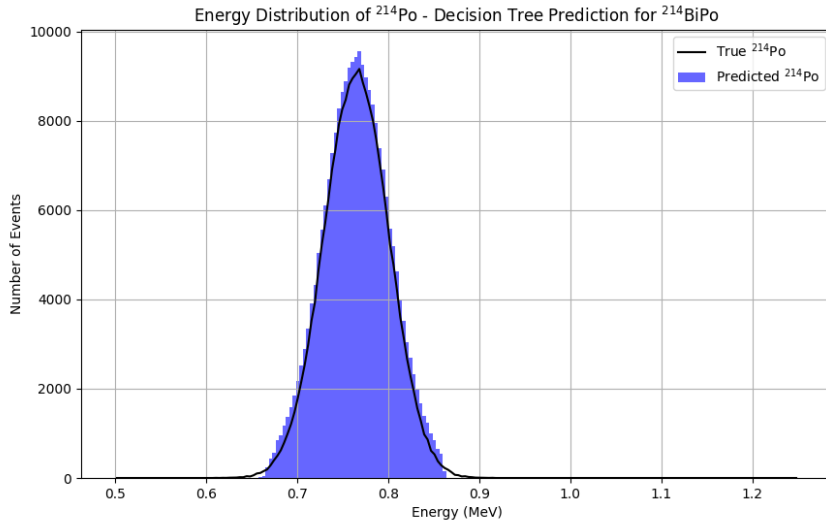


Figure 5.39: Energy spectrum of ^{214}Po events classified by the Decision Tree model for the $^{214}\text{BiPo}$ chain. The model captures a well-defined monoenergetic peak at 0.78 MeV, closely matching the true alpha decay distribution (line).

Decay Curve The decay curve predicted by the Decision Tree model for the BiPo214 chain exhibits similar issues to those observed with the ANN model for the same chain. The expected exponential decay shape is not clearly visible, and the distribution becomes nearly uniform after an initial drop. This pattern indicates significant misclassification of background events as $^{214}\text{BiPo}$ decays. Due to the striking resemblance to the ANN model's decay curve, the detailed plot is included in the Appendix E.2a for completeness.

$^{212}\text{BiPo}$ Model Evaluation

Performance Metrics The Decision Tree model for the $^{212}\text{BiPo}$ chain achieved an accuracy of 82% on the test set. Table 5.8 presents the performance metrics.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.96	0.78	0.86
External Background (3)	0.75	0.79	0.77
^{212}Bi (4)	0.73	0.74	0.73
^{212}Po (5)	0.87	0.98	0.92

Table 5.8: Class-wise evaluation metrics of the Decision Tree model trained on the $^{212}\text{BiPo}$ chain. The model shows strong performance for Po events, while Bi events are reasonably well-classified. Background classification is consistent with the ANN performance.

The model performs particularly well for ^{212}Po , achieving a TPR of 98% and an efficiency of 87%. Classification of ^{212}Bi shows reasonable performance with a TPR of 74% and an efficiency of 73%.

Class-wise misclassification patterns include:

- ^{212}Bi is often misclassified as external background (16.05%) or internal background (9.79%).
- ^{212}Po shows high accuracy, with minor confusion with ^{212}Bi (9.34%) and external background (2.88%).

The confusion matrix for the $^{212}\text{BiPo}$ Decision Tree model (Figure 5.40) demonstrates the classification performance of the optimized model. While the matrix shows a strong diagonal for ^{212}Po , indicating reliable classification, the diagonal for ^{212}Bi is notably less defined, reflecting substantial misclassification into background classes. The external background class also exhibits some overlap with ^{212}Bi , but overall, the precision for Po events is significantly higher. This pattern of classification is consistent with earlier observations that the model is more proficient at identifying Po due to its distinct spectral signature, while Bi events remain prone to confusion.

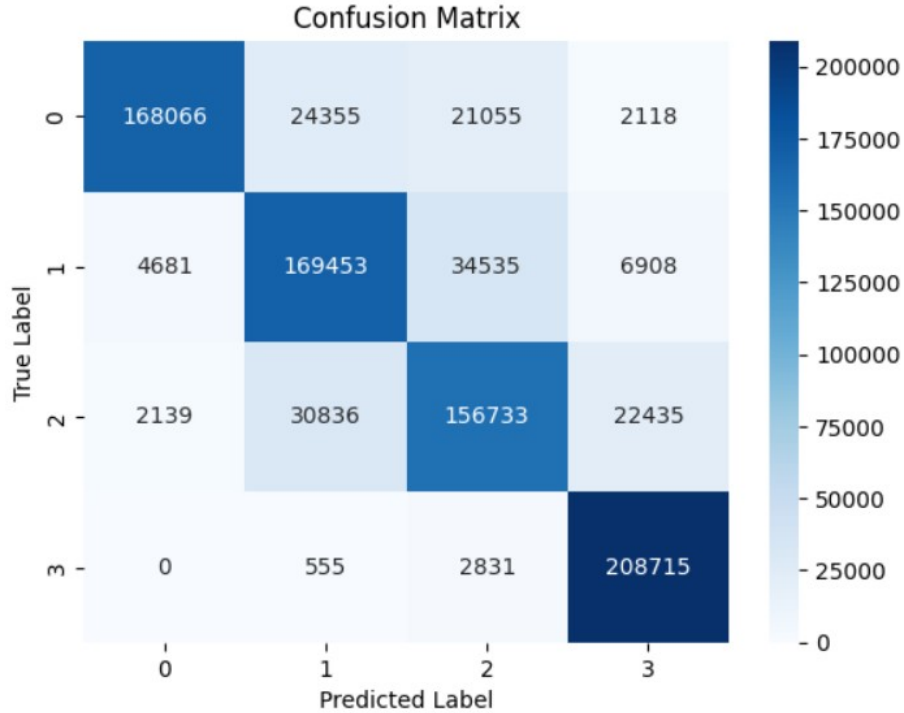


Figure 5.40: Confusion matrix of the optimized Decision Tree model trained on the $^{212}\text{BiPo}$ chain. The diagonal structure is well-defined for ^{212}Po , indicating effective classification, while ^{212}Bi exhibits significant misclassification into background classes, particularly external background. **This discrepancy highlights the challenge of accurately identifying ^{212}Bi events compared to the clearly distinguishable ^{212}Po events.**

Energy Distributions The energy distribution of classified ^{212}Bi events, shown in Figure 5.41, generally resembles the expected beta spectrum but diverges from the overlaid true distribution in some regions. A prominent gap between approximately 0.7 MeV and 1.0 MeV is observed, consistent with the behavior seen in the ANN model for the same chain. This gap likely results from misclassification

of true ^{212}Bi events as ^{212}Po or background, due to feature overlap in this energy region. Additionally, sharp spikes are visible at the edges of this missing interval, indicating that some background events are incorrectly classified as ^{212}Bi . **These artifacts highlight limitations in the model's ability to handle ambiguous energy regions, despite retaining the general beta shape.**

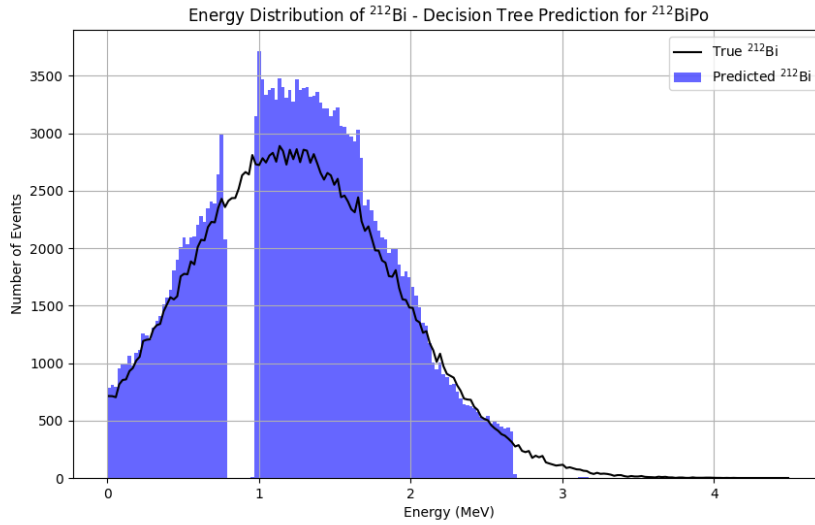


Figure 5.41: Energy spectrum of events classified as ^{212}Bi by the Decision Tree model for the $^{212}\text{BiPo}$ chain. The true distribution is overlaid. A clear gap is visible between 0.7 and 1.0 MeV, attributed to misclassification as Po or background. Peaks near the gap edges suggest contamination from other events.

In contrast, the ^{212}Po energy distribution (Figure 5.42) shows a well-defined monoenergetic peak near 0.85 MeV, closely matching the expected alpha decay signature as seen in the overlaid true distribution. The peak is smooth and symmetric, reflecting accurate classification of ^{212}Po events. This performance confirms that the Decision Tree model effectively identifies alpha decays when the energy signature is distinctive and well-separated from background.

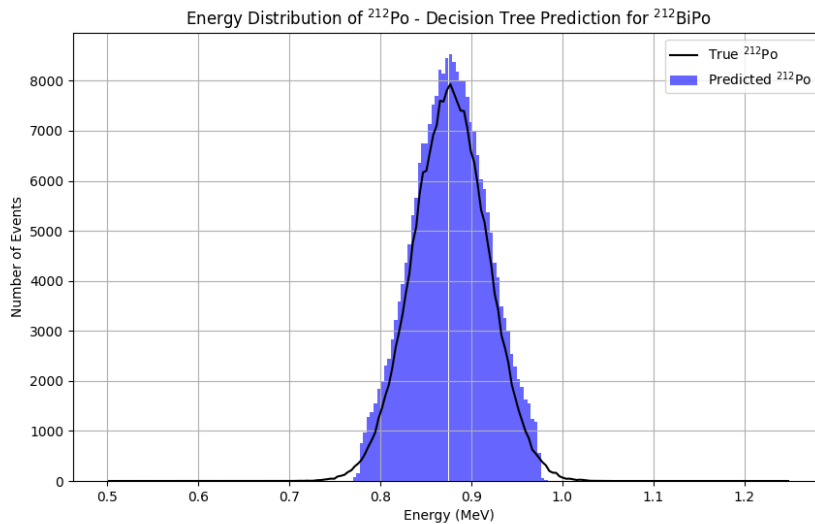


Figure 5.42: Energy spectrum of events classified as ^{212}Po by the Decision Tree model for the $^{212}\text{BiPo}$ chain. A narrow monoenergetic peak near 0.85 MeV matches the overlaid true alpha decay spectrum, indicating high classification accuracy.

Decay Curve The decay curve for the BiPo212 chain, similar to the BiPo214 chain, predicted by the Decision Tree model suffers from the same problems as the ANN model for the BiPo212 chain. The curve does not reflect the short decay time of ^{212}Po , instead showing a flattened distribution indicative

of background misclassification. As this curve is also analogous to the ANN model’s result, the plot is provided in the Appendix E.2b for reference.

5.3.4 Refined Classification with Higher Probability Selection

The final classification using the combined Decision Tree models follows the same combination strategy as applied in the ANN models. The purpose of this combined approach is to leverage the strengths of each specialized model and make a more robust final classification. However, like the ANN models, this approach proved to be highly ineffective.

The refined Decision Tree model performs significantly worse than the individual models trained on separate chains. The overall accuracy of the combined model is only 64%, which is substantially lower than the optimized Decision Tree trained on the full dataset (71%). In fact, this result is closer to the performance of the unoptimized model (61%), indicating that the combined approach fails to meaningfully enhance classification accuracy.

A key observation from the analysis is that some of the plots from the combined model exhibit patterns that appear almost random, especially in the case of ^{214}Bi . These plots can be found in the Appendix E.4 and E.5 for thorough inspection, while only the performance metrics are presented here.

Performance Metrics

The refined Decision Tree model achieved an overall accuracy of 64%, which is significantly lower than the optimized single model trained on the full dataset (71%). The detailed classification report is presented in Table 5.9, showing the precision, recall (TPR), and F1-score for each class.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.82	0.78	0.80
^{214}Bi (1)	0.41	0.12	0.19
^{214}Po (2)	0.81	0.81	0.81
External Background (3)	0.51	0.85	0.64
^{212}Bi (4)	0.48	0.58	0.53
^{212}Po (5)	0.82	0.72	0.77

Table 5.9: Class-wise evaluation metrics of the combined Decision Tree model. The model exhibits relatively strong classification for Po isotopes and external background but struggles significantly with the Bi isotopes, especially ^{214}Bi , which shows the lowest recall and F1-score.

The model performs best for ^{214}Po and ^{212}Po , achieving TPRs of 81% and 72%, and efficiencies of 81% and 82%, respectively. This is consistent with the trend seen in previous models, where Po events are more accurately classified due to their distinct energy and timing features.

In contrast, the classification performance for ^{214}Bi is exceptionally poor, with a TPR of only 12% and an efficiency of 41%. This indicates that most ^{214}Bi events are being misclassified into other classes, primarily as ^{212}Bi (25.97%) and ^{212}Po (18.89%). Such high misclassification highlights the model’s difficulty in distinguishing between Bi events from different chains.

The ^{212}Bi classification is slightly better, with a TPR of 58% and efficiency of 48%. However, significant confusion with ^{214}Bi (37.76%) indicates persistent challenges in separating the two beta-decaying isotopes.

The background classes also exhibit varying levels of accuracy. Internal background achieves a reasonable precision of 82% and TPR of 78%, while external background shows a TPR of 85% but relatively low precision of 51%.

A breakdown of misclassifications shows:

- ^{214}Bi is heavily misclassified as ^{212}Bi (25.97%) and ^{212}Po (18.89%), reflecting challenges with beta spectrum overlap.
- ^{214}Po remains relatively well classified, with some minor leakage into ^{212}Po (8.86%) and ^{212}Bi (5.53%).
- ^{212}Bi is frequently misclassified as ^{214}Bi (37.76%), again demonstrating significant overlap between beta decay events.
- ^{212}Po shows good classification but still experiences some confusion with ^{212}Bi (6.35%) and ^{214}Po (6.13%).

The confusion matrix (Figure 5.43) highlights the poor performance of the combined model. Unlike an ideal matrix, where a clear diagonal indicates strong classification, this matrix shows a diffuse pattern with no sharply defined diagonal. ^{214}Bi , in particular, displays significant off-diagonal activity, with a large portion of events being misclassified as ^{212}Bi or ^{212}Po . **This lack of clear separation between classes is a major limitation of the combined model, particularly when compared to the more specialized models.**

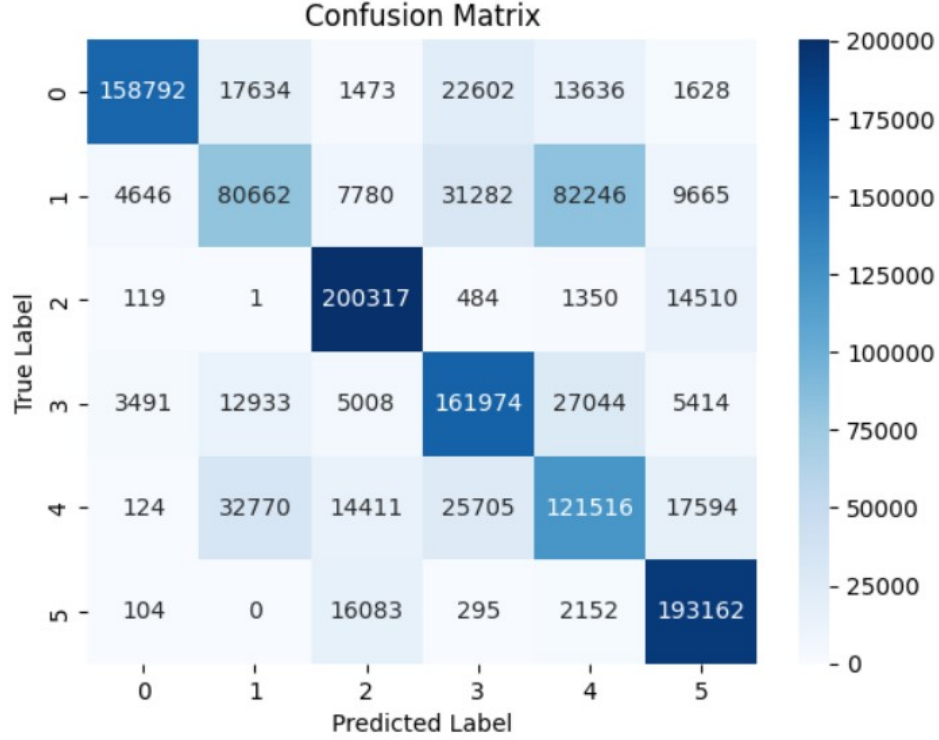


Figure 5.43: Confusion matrix of the combined Decision Tree model. The matrix lacks a clear diagonal, especially for ^{214}Bi , which shows considerable misclassification into ^{212}Bi and ^{212}Po . The relatively better performance for Po isotopes is visible through more concentrated diagonal values, but overall, the matrix structure is diffuse, indicating inconsistent classification accuracy.

5.4 Summary

In this chapter, ANNs and Decision Tree models were trained and evaluated for the full dataset, as well as separately for the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains.

The ANN models demonstrated superior classification performance for Po isotopes compared to Bi isotopes, with the full dataset model achieving a peak accuracy of 71%. The separate ANN models showed improved classification for Po events, while the classification of Bi events remained challenging, largely due to the inherent difficulty of distinguishing between beta decays.

Similarly, the Decision Tree models exhibited a comparable trend. The optimized single model trained on the full dataset reached an accuracy of 71%. The separate Decision Tree models demonstrated improved accuracy for Po isotopes but continued to face challenges with Bi classification, emphasizing the difficulty in correctly identifying beta-decay events.

Interestingly, combining specialized models using a higher probability selection strategy did not yield better results. Both for ANNs and Decision Trees, the combined approach proved less effective than the optimized single model, indicating that model fusion does not necessarily lead to performance improvements.

Overall, for both models, in regions where class overlap is severe, the model tends to drown in ambiguity, misclassifying events with similar energy or timing features.

A comprehensive comparison of the ANN and Decision Tree models will be presented after introducing the remaining models (SimpleRNN, LSTM, and BiLSTM with Attention) in the next chapter. This will allow for a holistic assessment of different machine learning techniques for the classification of BiPo events.

Chapter 6

Recurrent Neural Networks for Sequential Event Prediction

Recurrent Neural Networks (RNNs) are a class of neural networks designed to process sequential data by retaining information from previous time steps. This makes them particularly well-suited for tasks involving time series or ordered data. In the context of BiPo event classification, RNN-based models can potentially improve classification accuracy by capturing temporal correlations between events.

This chapter explores the use of RNN-based models for BiPo event classification. Three different architectures are presented here: SimpleRNN, Long Short-Term Memory (LSTM), and Bidirectional LSTM with Attention. Each model’s architecture, training strategy, and performance are evaluated after discussing the preprocessing steps needed to format the data for RNNs. Post-processing methods for pair classification are also covered, aiming to enhance model robustness.

By leveraging temporal patterns, RNN models may offer improvements over feedforward neural networks and decision trees discussed in the previous chapter. The methods explored hereafter aim to capture temporal dependencies and evolving event signatures that traditional models overlook.

6.1 Preprocessing for RNNs

RNNs are specifically designed to capture temporal dependencies by processing sequences of data. Unlike feedforward models, RNNs require input data to be structured as ordered sequences. Therefore, preprocessing for RNN models differs from the standard approach used for previous models. In particular, it is necessary to transform the event data into sequences, with each sequence representing a series of consecutive events. This format allows the RNN models to learn temporal patterns that may enhance classification performance.

The initial preprocessing steps are identical to those performed for previous models, including feature engineering and scaling. These steps involve transforming cylindrical coordinates (r, ϕ, z) and energy, as well as applying appropriate scaling to normalize the feature values. However, unlike in the previous models where time was treated as an absolute quantity, RNNs are designed to capture temporal patterns within sequences. Therefore, to provide the model with a meaningful notion of timing, the time difference between each event and the first event in its sequence is computed and used as a feature. This relative time difference helps the network focus on temporal intervals between events rather than their absolute timestamps, which are less relevant in the context of sequence modeling. The first event is chosen as the reference point because each input sequence spans a fixed number of events (10 in this case), and this anchor provides a consistent frame of reference for all subsequent events in the sequence. Note that this is not equivalent to using absolute time, as the computed differences encode only intra-sequence temporal relationships, independent of when the sequence begins in the dataset.

Additionally, a new feature, *distance*, is introduced to quantify the spatial separation of each event from the first event in the sequence. The distance is calculated using cylindrical coordinates as follows:

$$\text{distance} = \sqrt{(r \cos(\phi) - r_0 \cos(\phi_0))^2 + (r \sin(\phi) - r_0 \sin(\phi_0))^2 + (z - z_0)^2}$$

Here, r , ϕ , and z represent the cylindrical coordinates of the current event, while r_0 , ϕ_0 , and z_0 represent the coordinates of the first event in the sequence. This relative distance feature helps the RNN model understand spatial relationships between events.

Each sequence consists of 10 consecutive events, constructed using a sliding window approach. The first sequence includes events 1 to 10, the second includes events 2 to 11, the third spans events 3 to 12,

and so on. This overlapping strategy ensures that the temporal context is preserved while maximizing the amount of training data available to the model.

The choice of the number of events per sequence is primarily due to computational constraints, as longer sequences significantly increase memory usage and training time. Furthermore, using excessively long sequences could dilute temporal correlations, making it harder for the model to capture meaningful patterns. Additionally, previous studies have shown that while RNNs are capable of capturing long-term dependencies, they often perform adequately with moderately sized sequences, especially when temporal correlations do not extend over long periods [34].

After generating the sequences, the data is first split into training and test sets using an 80/20 ratio. The training set is then further divided into training and validation subsets using another 80/20 split. The random state is set to 0 for reproducibility. The same sequence generation process is applied for both the full dataset and the separately trained chains, with the only difference being the dataset used to generate the sequences.

6.2 Simple Recurrent Neural Networks (SimpleRNN)

RNNs are a type of neural network architecture designed to handle sequential data by maintaining a hidden state that is updated at each time step. Unlike traditional feedforward networks, RNNs can use their internal state (memory) to process sequences of inputs, making them particularly useful for tasks where the order of data points matters, such as time series analysis or natural language processing [30].

6.2.1 Structure of SimpleRNN

A SimpleRNN processes an input sequence one step at a time, updating its hidden state and producing an output. Each RNN cell takes the current input (x_t) and the hidden state from the previous time step (h_{t-1}) to generate the current hidden state (h_t). This process is described by the following equation:

$$h_t = \tanh(W \cdot x_t + U \cdot h_{t-1} + b)$$

Where:

- W and U are the weight matrices for the input and hidden state, respectively.
- b is the bias term.
- $\tanh$ is the activation function used in the RNN cell.

Figure 6.1 illustrates the internal structure of a SimpleRNN cell.

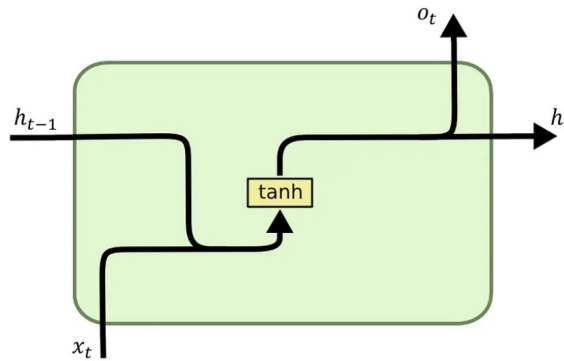


Figure 6.1: Internal structure of a SimpleRNN cell. The cell takes the current input x_t and the previous hidden state h_{t-1} and applies the $\tanh$ activation to update the current hidden state h_t . Image taken from [35].

6.2.2 Processing Sequences with SimpleRNN

A SimpleRNN network processes a sequence by iterating through each time step, as shown in Figure 6.2. In the given example, the input sequence "I am groot" is transformed into numerical vectors through an embedding layer. In the context of this work, the input sequence length is 10, and the embedding

dimension (features) is 6. The hidden state is passed from one time step to the next, allowing the model to retain memory of previous elements in the sequence.

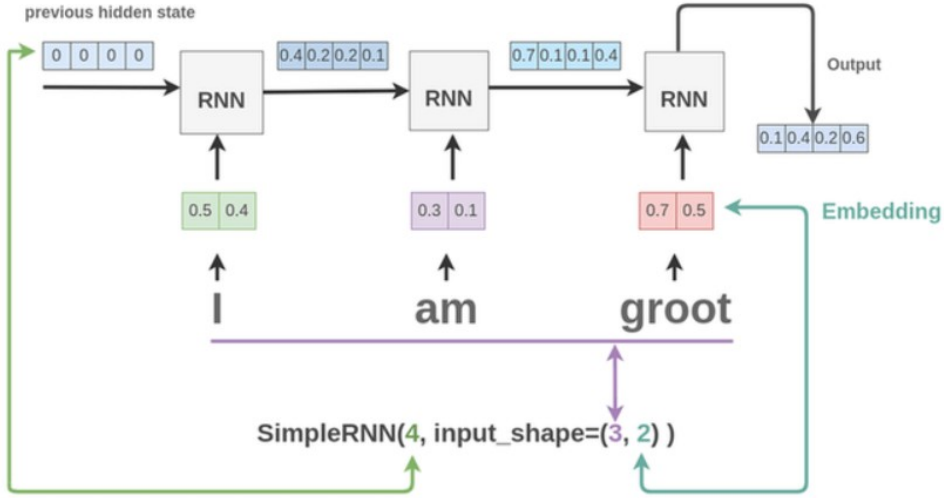


Figure 6.2: SimpleRNN processing a sequence. Here, "I am groot" represents a sequence processed by RNN cells, with each word being embedded into a numerical vector. The hidden state from each step is passed to the next RNN cell. In our case, the sequence length is 10, and the embedding size is 6. Image taken from [35].

6.2.3 Layered SimpleRNN

In the model developed for this work, a two-layer RNN architecture is used. The first layer has 64 units and returns sequences, which are then fed into the second layer of 32 units, also configured to return sequences. This setup allows for capturing more complex temporal dependencies. The output from the final RNN layer is passed through a Dense layer with 16 neurons (ReLU activation) before outputting the final softmax predictions.

The final output of this model is also a sequence of length 10, with each time step corresponding to one event and its predicted classification label. Each event in the input sequence is individually labeled, and the model performs classification for every event in the sequence. Figure 6.3 illustrates the multi-layer structure of SimpleRNN.

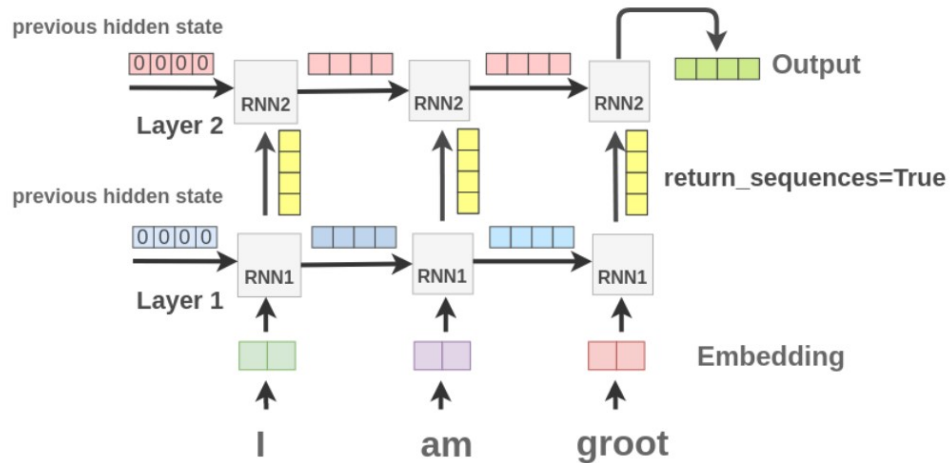


Figure 6.3: Layered SimpleRNN structure with two layers. The first layer (RNN1) has 64 units and outputs a sequence that is fed into the second layer (RNN2) with 32 units. Both layers are configured to return sequences. The final output is a softmax layer that provides predictions for each step in the input sequence. Image taken from [35].

6.2.4 SimpleRNN Model Trained on the Full Dataset

Model Architecture and Training Strategy

The baseline SimpleRNN model is a recurrent neural network designed to process sequential data. It is trained on the full dataset after applying the necessary preprocessing steps described in the previous section to generate sequences of events. The model is structured with two SimpleRNN layers followed by dense layers to output classification probabilities.

The architecture is as follows:

- **Input layer:** accepts sequences of 10 events, each with 6 engineered and scaled features (energy of the event, radial distance, z coordinate, azimuthal angle phi, time difference from the first event, and relative distance from the first event).
- **First RNN layer:** A SimpleRNN layer with 64 units using the tanh activation function:

$$h_t = \tanh(W \cdot x_t + U \cdot h_{t-1} + b)$$

This layer processes the input sequence and outputs another sequence of the same length, allowing the network to capture temporal dependencies across time steps.

- **Second RNN layer:** A SimpleRNN layer with 32 units, also using the tanh activation function. This layer continues to process the output of the first RNN layer, refining the temporal representation.
- **Dense layer:** A fully connected layer with 16 neurons and ReLU activation.
- **Output layer:** A softmax layer for multi-class classification. The softmax function ensures that output values form a valid probability distribution over the six possible event classes.

The model is compiled with the categorical cross-entropy loss function and the Adam optimizer, similar to the ANN model. The training was performed over a maximum of 200 epochs with a batch size of 256. However, early stopping halted the training at 97 epochs, as further training did not improve validation loss. The ReduceLROnPlateau callback was employed to decrease the learning rate if the loss plateaus, with a factor of 0.5 and a patience of 4 epochs. The initial learning rate is 0.0010. EarlyStopping was configured with a patience of 10 epochs to prevent overfitting by restoring the best weights after no improvement.

The loss and accuracy curves over the training and validation sets are included in Appendix F.1 and the ROC curve is shown in Appendix F.2.

Performance Metrics

The SimpleRNN model was trained using a dataset split of 80% for training and 20% for testing, with a `random_state` set to 0 to ensure reproducibility. Of the 80% training data, 80% was used for training and the remaining 20% for validation. This split strategy was consistently applied across all RNN models in this work. The data was organized into sequences as described in the preprocessing section.

The SimpleRNN model achieved an overall accuracy of 73% on the test set. Table 6.1 shows the classification report summarizing precision, recall (TPR), and F1-score for each class.

Table 6.1: Class-wise evaluation metrics of the SimpleRNN model trained on the full dataset. The model demonstrates strong classification performance for Po events due to their temporal and energetic distinctiveness. The classification of Bi isotopes, however, is less accurate, likely due to the overlapping beta spectra and temporal similarities with the background classes.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.94	0.72	0.82
²¹⁴ Bi (1)	0.54	0.39	0.45
²¹⁴ Po (2)	0.92	0.94	0.93
External Background (3)	0.64	0.78	0.70
²¹² Bi (4)	0.50	0.64	0.56
²¹² Po (5)	0.92	0.91	0.92

The model exhibits excellent performance for ²¹⁴Po and ²¹²Po, achieving high TPRs of 94% and 91%, respectively, and comparable efficiencies of 92%. These results are consistent with the distinct energy peaks and shorter time delays characteristic of alpha decays, which the RNN effectively captures.

In contrast, the model's performance for Bi isotopes is significantly lower. ^{214}Bi achieves a TPR of only 39% and an efficiency of 54%, while ^{212}Bi shows slightly improved performance with a TPR of 64% and efficiency of 50%. This is likely due to the inherent challenges of distinguishing beta decay events from the background and from each other, given their broad energy distributions and temporal overlaps.

Background classes show mixed results. Internal background (class 0) has a high precision of 94% but a lower TPR of 72%, indicating that while most predicted internal background events are accurate, some true internal events are misclassified. The external background (class 3) demonstrates a more balanced performance, with a TPR of 78% and a precision of 64%.

A detailed analysis of misclassifications reveals the following patterns:

- ^{214}Bi is primarily misclassified as ^{212}Bi (23.60%) and internal background (12.73%), reflecting the spectral and temporal overlaps between these classes.
- ^{214}Po is mostly correctly classified, but there is minor confusion with ^{212}Po (6.66%) and external background (0.43%).
- ^{212}Bi has a significant portion misclassified as ^{214}Bi (31.87%), demonstrating the challenge of distinguishing beta decays from different decay chains.
- ^{212}Po is well classified with a high TPR, though some events are mistakenly labeled as ^{214}Po (5.99%).

These trends are reflected in the confusion matrix (Figure 6.4), which highlights the distribution of correctly and incorrectly classified events. Ideally, a well-performing model would display a bright diagonal, indicating that most predictions align with true labels. However, while the Po classes exhibit a strong diagonal, indicating good classification performance, the diagonal for Bi classes is less prominent, suggesting persistent challenges in correctly identifying these events. Additionally, some background classes also exhibit off-diagonal elements, indicating partial confusion between internal and external background.

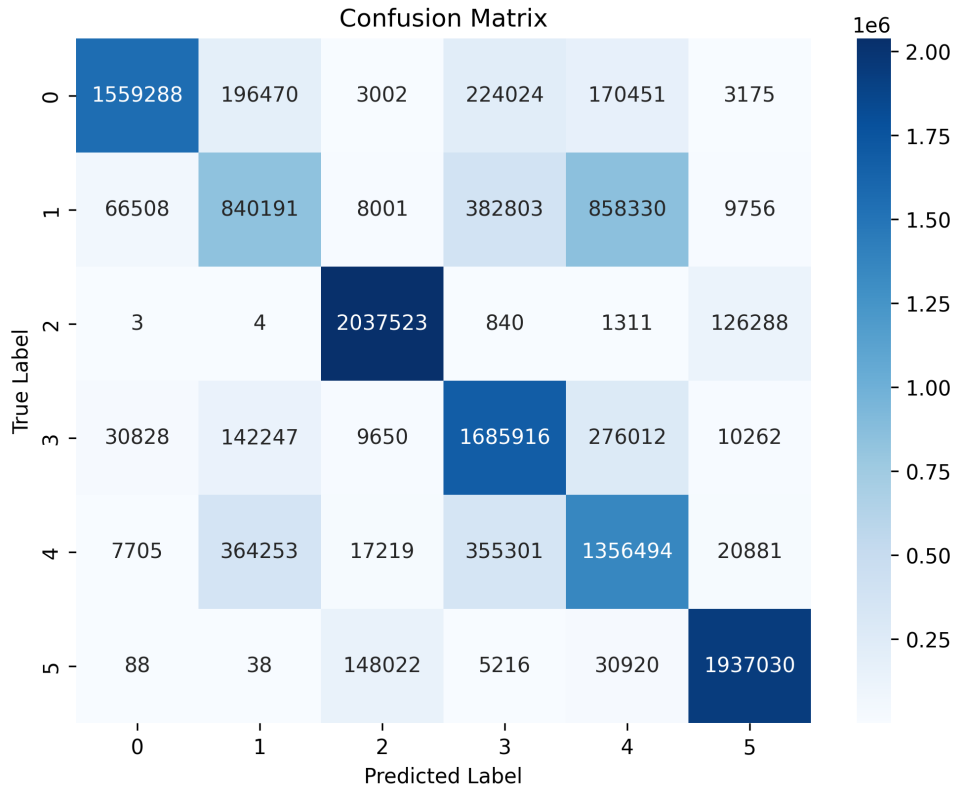


Figure 6.4: Confusion matrix of the SimpleRNN model trained on the full dataset. The matrix shows a strong diagonal for Po events, indicating successful classification. However, the diagonal structure for Bi events is less defined, indicating significant misclassification, particularly between ^{214}Bi and ^{212}Bi . Background classes also show some misclassification between internal and external events, reflecting the inherent challenges in distinguishing them based on the available features.

Energy Distributions

The ^{214}Bi spectrum (Figure 6.5) closely follows the expected high-energy beta distribution, beginning sharply at approximately 1.75 MeV and extending beyond 3.5 MeV. A minor background-like feature is again observed near 3.1 MeV, consistent with previous models and likely attributable to residual misclassification. While the high-energy region is well separated, the predicted spectrum does not closely match the overlaid true distribution, and several events are misclassified as ^{214}Bi .

The ^{212}Bi spectrum (Figure 6.6) complements this by capturing the lower energy region. A significant improvement over previous models is observed in the form of a continuous and uninterrupted energy spectrum, with no missing gap between 0.7 and 1.0 MeV. This suggests that the SimpleRNN architecture has learned to handle the overlapping feature space between ^{212}Bi and ^{212}Po without introducing discontinuities. However, the predicted spectrum still differs from the overlaid true distribution and includes many misclassified events.

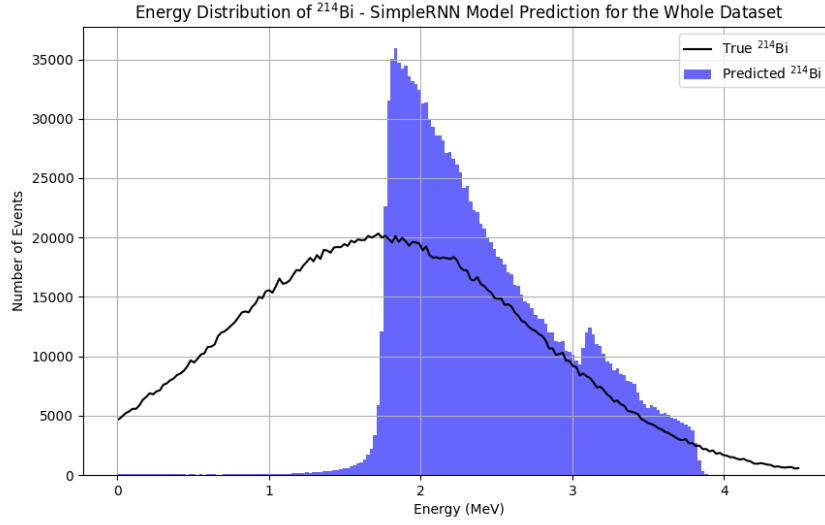


Figure 6.5: Energy spectrum of ^{214}Bi events using the SimpleRNN model trained on the full dataset. The predicted distribution (blue) covers the upper range of the beta spectrum starting around 1.75 MeV. The overlaid true distribution (line) highlights regions of misclassification and deviation.

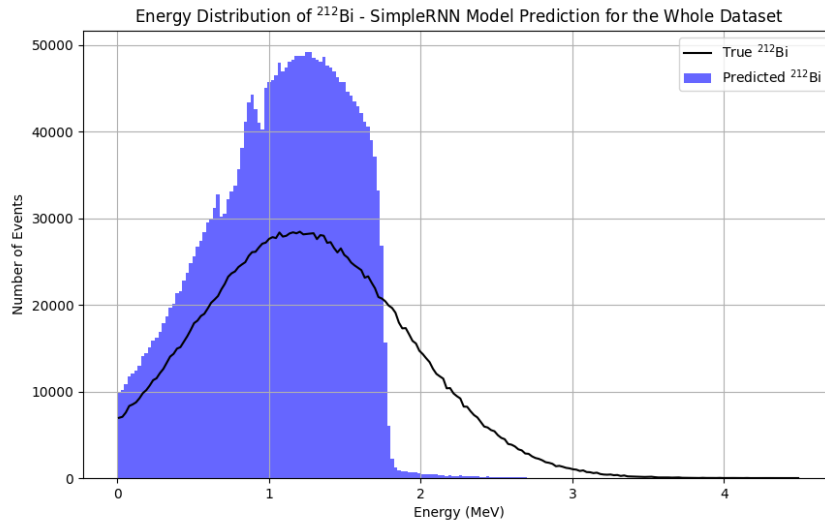


Figure 6.6: Energy spectrum of ^{212}Bi events using the SimpleRNN model. The predicted distribution (blue) spans the lower half of the beta spectrum and does not show a gap in the 0.7–1.0 MeV region. However, differences from the true distribution (line) indicate residual misclassification.

The ^{214}Po spectrum (Fig. 6.7) exhibits a well-defined and symmetric monoenergetic peak centered around 0.78 MeV, consistent with the expected α decay energy. Compared to the ANN model, the distribution here is noticeably smoother, especially near the rising edge, indicating improved model stability and reduced background contamination. The predicted spectrum aligns well with the overlaid true distribution, though it ends sharply around 0.82 MeV, marking the boundary with ^{212}Po events.

The ^{212}Po spectrum (Fig. 6.8) also presents a clean, bell-shaped monoenergetic peak around 0.89 MeV. The distribution is smooth and well-constrained, showing improvement over the ANN results where the spectrum appeared more diffuse. The predicted peak aligns well with the true distribution. However, as with previous models, a sharp boundary remains in the overlap region near 0.82 MeV. This separation results in a distinct threshold, with ^{214}Po dominating the lower energy range and ^{212}Po the upper, likely due to model uncertainty in this ambiguous feature space.

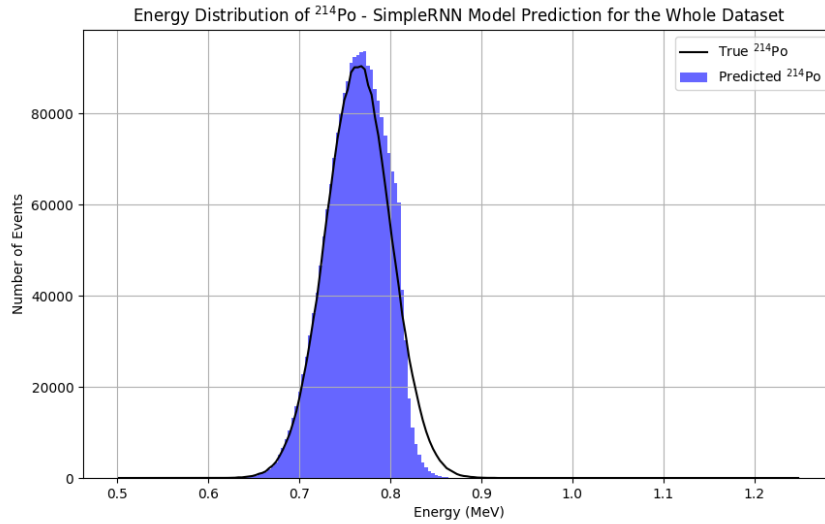


Figure 6.7: Energy spectrum of classified ^{214}Po events using the SimpleRNN model trained on the full dataset. The predicted distribution (blue) is smooth and symmetric, and matches the overlaid true distribution (line) well. A sharp cutoff appears near 0.82 MeV, where the ^{212}Po region begins.

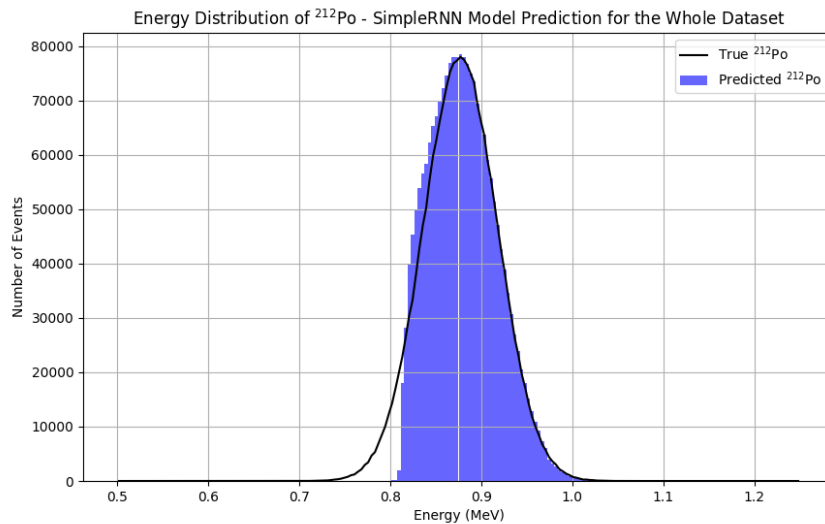


Figure 6.8: Energy spectrum of classified ^{212}Po events using the SimpleRNN model trained on the full dataset. The predicted peak (blue) near 0.89 MeV aligns closely with the true alpha energy (line), and a sharp threshold appears around 0.82 MeV, marking the transition from ^{214}Po events.

Decay Curves

The decay curves in Figures 6.9 and 6.10 display the time difference distributions between events classified as Bi and Po by the SimpleRNN model for the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ decay chains, respectively. Unlike the results from the ANN and Decision Tree models, a clear exponential decay trend is visible in both cases.

In Figure 6.9, the decay curve for $^{214}\text{BiPo}$ events shows an initial excess of events at short time differences, attributed to misclassified background or events from the $^{212}\text{BiPo}$ chain. However, after this initial peak, the curve quickly transitions into a smoothly decreasing distribution that closely resembles the expected exponential decay. This indicates that, despite some contamination, the model has successfully learned to identify the time structure associated with the true $^{214}\text{Bi} \rightarrow ^{214}\text{Po}$ sequence.

For $^{212}\text{BiPo}$ (Figure 6.10), the plot has been zoomed into the region of interest to better resolve the decay structure, since the initial bins are overwhelmingly populated by background and unrelated events. In the focused view, the curve displays a steep, clean exponential drop, characteristic of the short half-life of ^{212}Po . While the absolute event count in the early bins is still influenced by misclassification, the underlying decay trend remains physically consistent.

Although the classification accuracy of the SimpleRNN model is similar to the ANN and Decision Tree models, it captures the actual time structure of BiPo decays more clearly. This shows that RNNs are better at learning time-related patterns in the data.

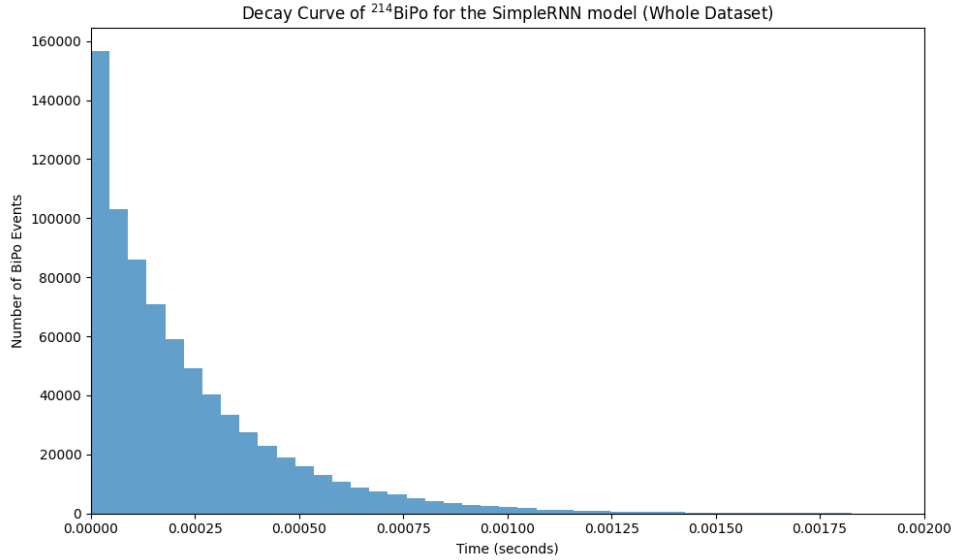


Figure 6.9: Decay curve of the classified $^{214}\text{BiPo}$ events using the SimpleRNN model. An initial rise due to misclassification is followed by a distinct exponential trend matching the known lifetime of ^{214}Po .

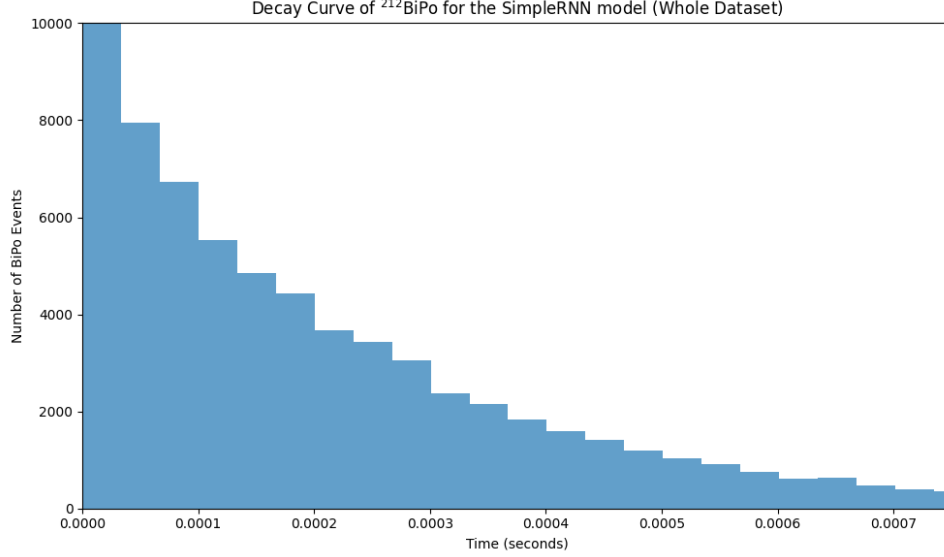


Figure 6.10: Decay curve of the classified $^{212}\text{BiPo}$ events using the SimpleRNN model, zoomed in to the region of interest. A clean exponential falloff is visible, consistent with the short lifetime of ^{212}Po .

For the SimpleRNN model, the r and z distributions of events classified as external background closely resemble those obtained with the ANN model. This similarity indicates that the SimpleRNN is also highly effective at learning and reproducing the spatial signature of background events. The model successfully identifies the radial clustering and vertical asymmetry associated with external backgrounds, suggesting a strong internal representation of these features. The corresponding r and z distributions for background-labeled events are shown in Appendix F.3, reinforcing the conclusion that the SimpleRNN model captures spatial correlations in the background data with high fidelity.

6.2.5 Separate SimpleRNN Models for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$

Model Architecture and Training Strategy

To mitigate the overlap in energy spectra and temporal features between the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains, two separate SimpleRNN models were trained, each specialized for one decay chain. This approach enables more targeted temporal learning and reduces misclassification caused by inter-chain similarities.

- The $^{214}\text{BiPo}$ model was trained on truth labels $\{0, 1, 2, 3\}$, corresponding to internal background, ^{214}Bi , ^{214}Po , and external background. Training stopped at 77 epochs.
- The $^{212}\text{BiPo}$ model was trained on truth labels $\{0, 3, 4, 5\}$, corresponding to internal background, external background, ^{212}Bi , and ^{212}Po . Training stopped at 87 epochs.

Both models use the same architecture as the full-dataset SimpleRNN model to ensure consistency in evaluation:

- **Input layer:** accepts sequences of 10 events, each with 6 engineered and scaled features (energy of the event, radial distance, z coordinate, azimuthal angle ϕ , time difference from the first event, and relative distance from the first event).
- **First RNN layer:** A SimpleRNN layer with 64 units using the tanh activation function.
- **Second RNN layer:** A SimpleRNN layer with 32 units, also using the tanh activation function.
- **Dense layer:** A fully connected layer with 16 neurons and ReLU activation.
- **Output layer:** A softmax layer for four-class classification. This produces a normalized probability distribution over the four output classes in each model.

Both datasets were split into 80% training and 20% testing sets using `random_state = 0` for reproducibility. The training set was further split into 80% training and 20% validation data. The models were trained for a maximum of 200 epochs with a batch size of 256. Early stopping and learning rate reduction callbacks were used to prevent overfitting and ensure optimal convergence. Training and validation curves for both models are included in Appendix F.4 and F.6, and the ROC curves can be found in Appendix F.5 and F.7.

²¹⁴BiPo Model Evaluation

Performance Metrics The model achieves an overall accuracy of 82% on the test set, matching the best-performing ANN models in this category. Table 6.2 presents the class-wise precision (efficiency), recall (TPR), and F1-score.

Table 6.2: Evaluation metrics for the SimpleRNN model trained on the ²¹⁴BiPo chain. The model demonstrates excellent classification performance for ²¹⁴Po, while ²¹⁴Bi classification remains challenging due to overlap with background classes.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.93	0.72	0.81
²¹⁴ Bi (1)	0.71	0.73	0.72
²¹⁴ Po (2)	0.99	1.00	0.99
External Background (3)	0.71	0.85	0.77

The strongest classification is observed for ²¹⁴Po, with near-perfect metrics, TPR of 100% and precision of 99%. This reflects its distinct monoenergetic alpha peak and short decay time, which the RNN can effectively identify within sequential patterns. On the other hand, ²¹⁴Bi achieves a TPR of 73% and an efficiency of 71%, which is an improvement over the full model but still limited by the ambiguity of beta decay features.

Misclassification analysis shows:

- ²¹⁴Bi is often confused with internal background (16.00%) and external background (13.02%), with only a negligible fraction misclassified as ²¹⁴Po (0.04%).
- ²¹⁴Po is extremely well-isolated, with only minor leakage into external background (0.54%), ²¹⁴Bi (0.48%), and internal background (0.18%).

The confusion matrix in Figure 6.11 illustrates these findings. A strong diagonal is clearly visible for ²¹⁴Po, confirming excellent classification. However, for ²¹⁴Bi, misclassification into background classes introduces a more diffuse pattern, suggesting remaining limitations in separating beta decays using temporal patterns alone.

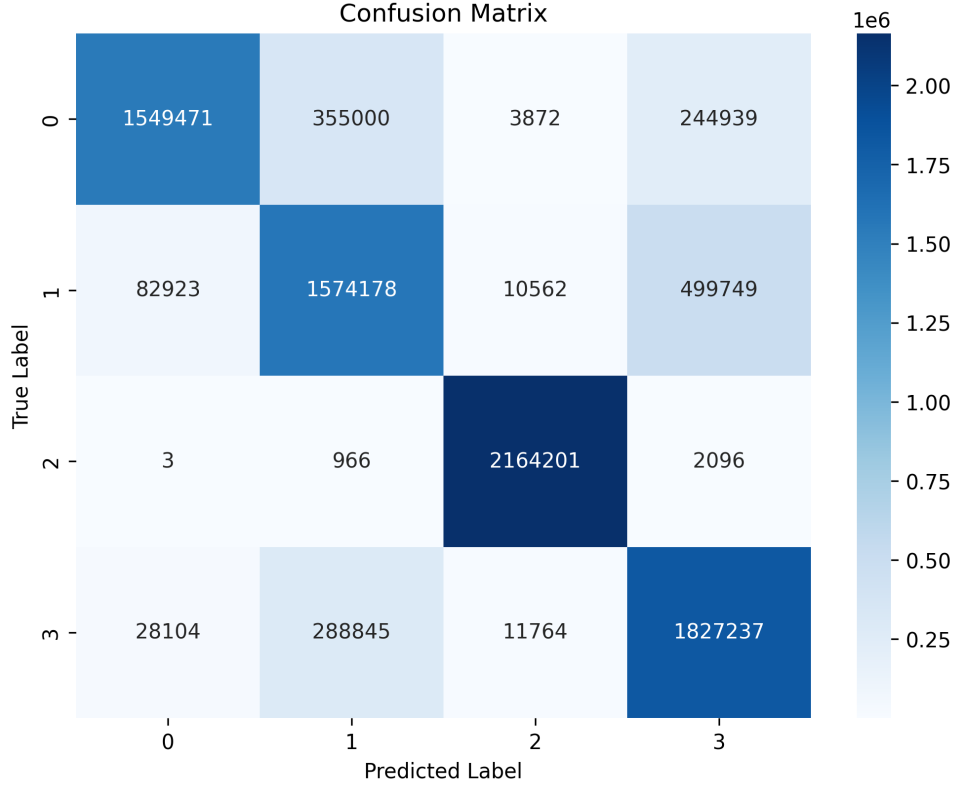


Figure 6.11: Confusion matrix of the $^{214}\text{BiPo}$ SimpleRNN model. The model achieves nearly perfect classification for ^{214}Po . In contrast, ^{214}Bi continues to show leakage into internal and external backgrounds due to overlapping spatial and energy features.

Energy Distributions Figures 6.12 and 6.13 present the energy distributions for ^{214}Bi and ^{214}Po events classified by the SimpleRNN model trained specifically on the $^{214}\text{BiPo}$ decay chain.

The ^{214}Bi spectrum (Figure 6.12) closely resembles the expected continuous beta distribution, spanning from approximately 0.2 MeV to nearly 4 MeV. Unlike the ANN model, there is no visible depletion between 0.6-0.9 MeV, suggesting that **the RNN-based model more effectively classifies events across the full beta spectrum**. A minor bump above 3 MeV persists, as seen in the ANN predictions, likely resulting from background contamination at higher energies. Overall, the spectrum exhibits smoother continuity and better coverage across the energy range, with strong agreement between the predicted and true distributions.

The ^{214}Po distribution (Figure 6.13) shows a clean, narrow monoenergetic peak centered around 0.785 MeV, consistent with the expected alpha decay energy and closely matching the true distribution. The peak is sharply defined and symmetric, indicating strong classification performance for alpha events. The smooth structure and lack of artifacts further support the model's ability to isolate this decay signature with high precision.

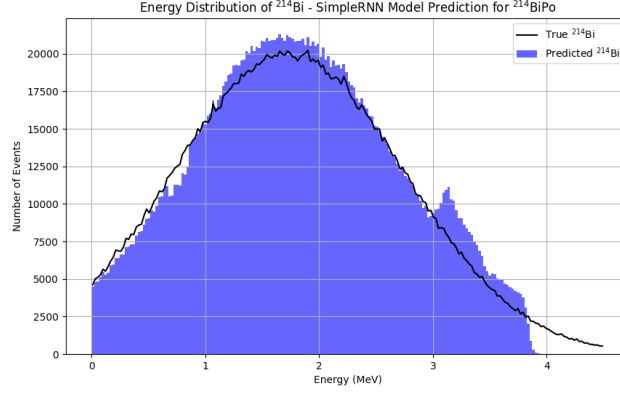


Figure 6.12: Energy distribution of events classified as ^{214}Bi by the SimpleRNN model trained on the $^{214}\text{BiPo}$ decay chain. The predicted spectrum (blue) closely follows the true distribution (line), with no gap between 0.6-0.9 MeV and a small excess above 3 MeV suggesting mild background leakage.

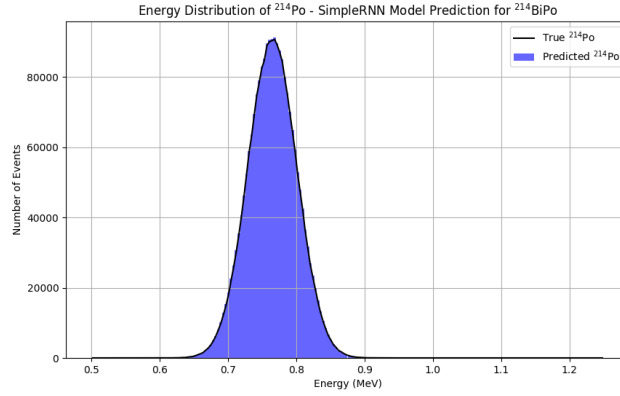


Figure 6.13: Energy distribution of events classified as ^{214}Po by the SimpleRNN model trained on the $^{214}\text{BiPo}$ decay chain. A sharp, symmetric peak appears near 0.785 MeV, closely aligned with the expected alpha decay energy and true distribution (line).

Decay Curve Figure 6.14 shows the decay time distribution for $^{214}\text{BiPo}$ events as predicted by the SimpleRNN model trained solely on the $^{214}\text{BiPo}$ chain and background events. In contrast to the full dataset model or the ANN counterpart, this decay curve exhibits a clean, uninterrupted exponential decline, closely aligned with the expected temporal signature of ^{214}Po decay.

The absence of an initial spike, which was previously observed in models trained on the full dataset, indicates a significant reduction in early-time misclassifications. This is likely due to the model's focused exposure to only relevant decay patterns during training, allowing it to better isolate true BiPo sequences from background or unrelated coincidences.

The overall shape of the distribution is smooth and consistent, beginning with a well-defined peak at short time intervals followed by a rapid and continuous decline. This profile matches the physical expectation for ^{214}Bi decaying into ^{214}Po , and suggests that the model has successfully internalized the temporal structure of the decay process. While some background contamination may still persist at longer time differences, it appears less significant than in previous models, further confirming the effectiveness of the model in capturing realistic decay behavior.

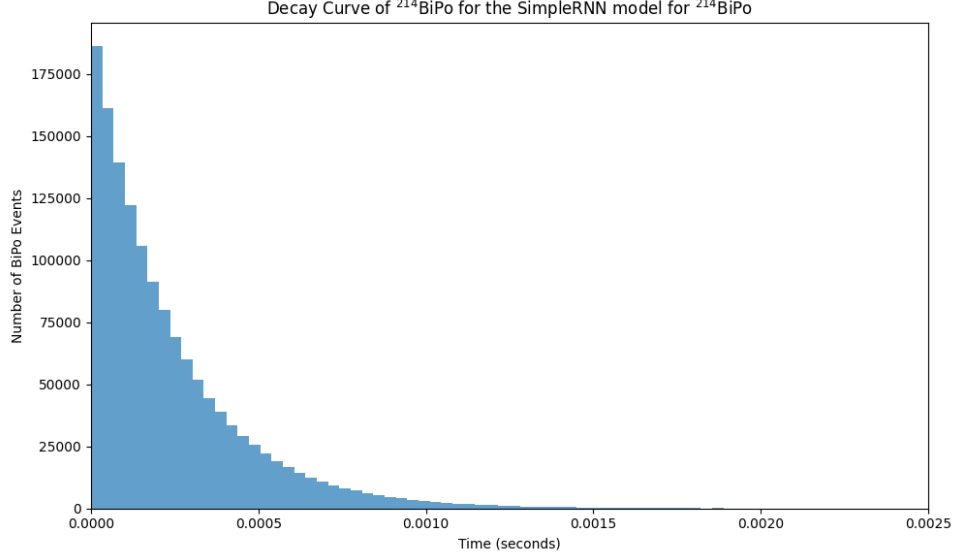


Figure 6.14: Decay curve for $^{214}\text{BiPo}$ events classified by the SimpleRNN model trained specifically for the $^{214}\text{BiPo}$ chain. The curve exhibits a smooth and clear exponential decline characteristic of the decay process, without the initial overpopulation of early bins seen in broader models.

$^{212}\text{BiPo}$ Model Evaluation

Performance Metrics The model achieves an overall accuracy of 83% on the test set. Table 6.3 shows the evaluation metrics for each class, including precision (efficiency), recall (TPR), and F1-score.

Table 6.3: Evaluation metrics for the SimpleRNN model trained on the $^{212}\text{BiPo}$ chain. The model achieves excellent classification for ^{212}Po , while ^{212}Bi classification is improved but still affected by confusion with background.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.96	0.76	0.85
External Background (3)	0.72	0.83	0.77
^{212}Bi (4)	0.73	0.77	0.75
^{212}Po (5)	0.98	0.98	0.98

The strongest classification is again observed for ^{212}Po , with a TPR of 98.19% and an efficiency of 97.95%. This reflects its distinct temporal and energetic characteristics, which the model captures reliably in sequential data. For ^{212}Bi , the model achieves a TPR of 76.91% and an efficiency of 72.66%, an improvement over previous models but still limited by overlap with background classes.

Misclassification analysis shows:

- ^{212}Bi is misclassified primarily as internal background (11.37%) and external background (14.57%), with a small fraction incorrectly classified as ^{212}Po (1.40%).
- ^{212}Po exhibits excellent separation, with minor leakage into ^{212}Bi (1.25%), external background (0.61%), and internal background (0.19%).

The confusion matrix in Figure 6.15 confirms these trends. ^{212}Po shows a sharply defined diagonal structure, indicating highly accurate classification. In contrast, ^{212}Bi displays broader off-diagonal spread due to confusion with background events. Nonetheless, the overall structure is more coherent than in the full dataset model, underscoring the benefit of specialized training.

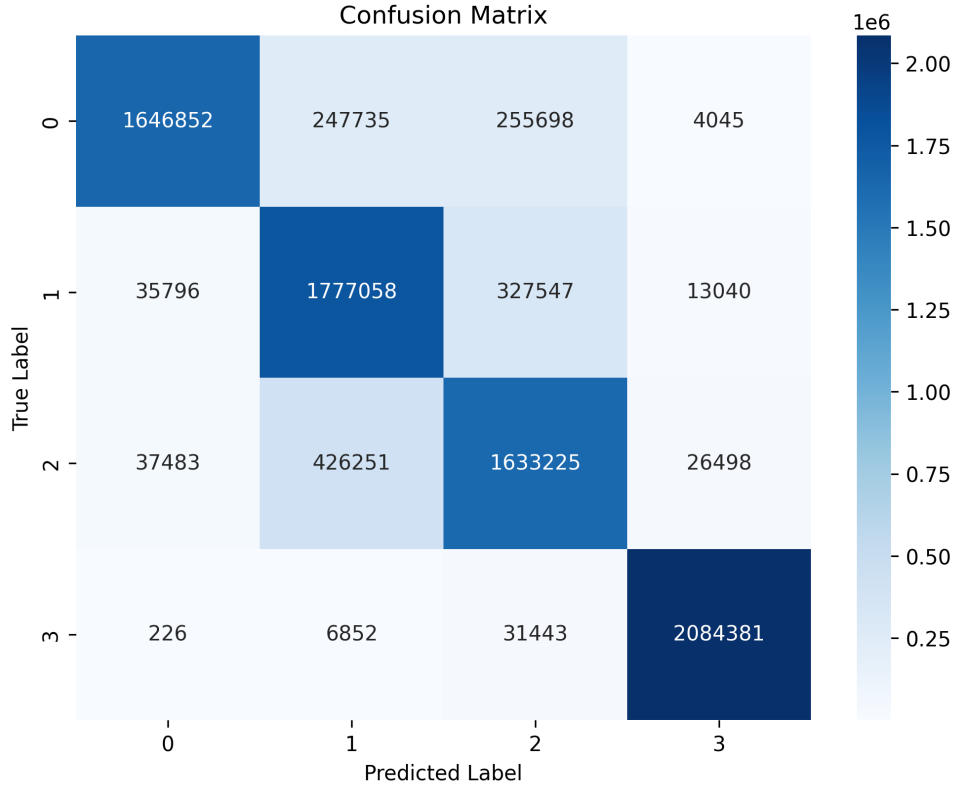


Figure 6.15: Confusion matrix of the $^{212}\text{BiPo}$ SimpleRNN model. ^{212}Po classification is near-perfect, while ^{212}Bi shows notable misclassification into background classes, primarily external background.

Energy Distributions Figures 6.16 and 6.17 show the energy spectra of the classified ^{212}Bi and ^{212}Po events, respectively, as predicted by the SimpleRNN model trained specifically on the $^{212}\text{BiPo}$ decay chain.

The ^{212}Bi energy spectrum (Figure 6.16) covers a broad, continuous range characteristic of beta decay. Unlike the ANN model, the SimpleRNN prediction does not exhibit a gap in the 0.75-1.0 MeV region. This suggests that the model has retained more events in this ambiguous range instead of misclassifying them as Po or background. While small fluctuations and uneven peaks persist throughout the spectrum, likely due to residual background contamination, the overall shape more closely follows the true beta distribution. Some misclassified events remain around the peak compared to the true distribution, but the improved continuity reflects better handling of overlapping features.

The ^{212}Po distribution (Figure 6.17) exhibits a sharp, symmetric monoenergetic peak centered near 0.895 MeV, consistent with the expected alpha decay signature and closely matching the true distribution. The peak is smooth and narrow, with minimal asymmetry or background leakage. This strong result demonstrates that the model effectively identifies ^{212}Po events based on their distinct energy signature.

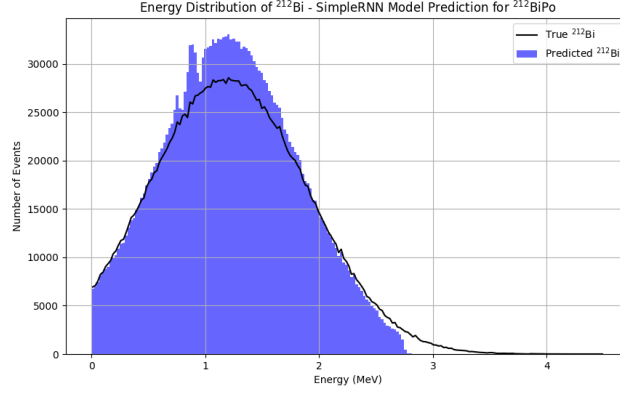


Figure 6.16: Energy distribution of events classified as ^{212}Bi by the SimpleRNN model trained on the $^{212}\text{BiPo}$ decay chain. The predicted beta spectrum (blue) is continuous and closely follows the true distribution (line), with no missing region near 0.75–1.0 MeV. Minor fluctuations suggest residual background misclassification.

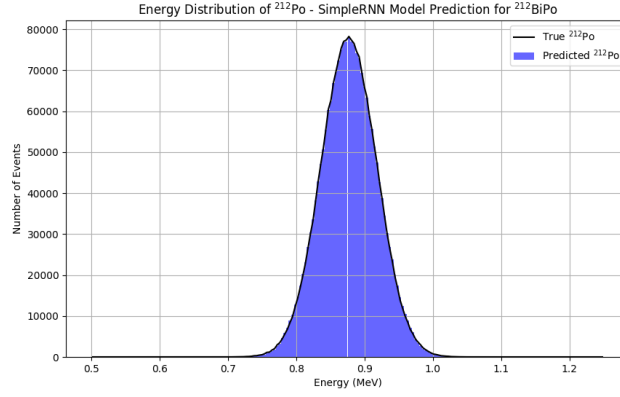


Figure 6.17: Energy distribution of events classified as ^{212}Po by the SimpleRNN model trained on the $^{212}\text{BiPo}$ decay chain. A narrow, symmetric monoenergetic peak appears around 0.895 MeV, consistent with the true alpha decay energy (line), indicating accurate classification.

Decay Curve Figure 6.18 displays the decay time distribution for $^{212}\text{BiPo}$ events classified by the SimpleRNN model trained exclusively on the BiPo212 chain and background. The curve closely follows the expected exponential decay behavior of ^{212}Po , particularly in the initial region, which reflects a clean classification of true BiPo coincidences.

Unlike the corresponding ANN model (Figure 5.19), this curve lacks the large irregularities and non-physical flattening at later time differences. Although some deviations due to misclassified events persist in the tail, the overall trend remains physical and non-random. Additionally, the initial region does not suffer from a sharp artificial spike as seen in the model trained on the entire dataset. This indicates that the SimpleRNN model is better at isolating true BiPo212 decays, learning the short-lived nature of ^{212}Po with improved temporal resolution.

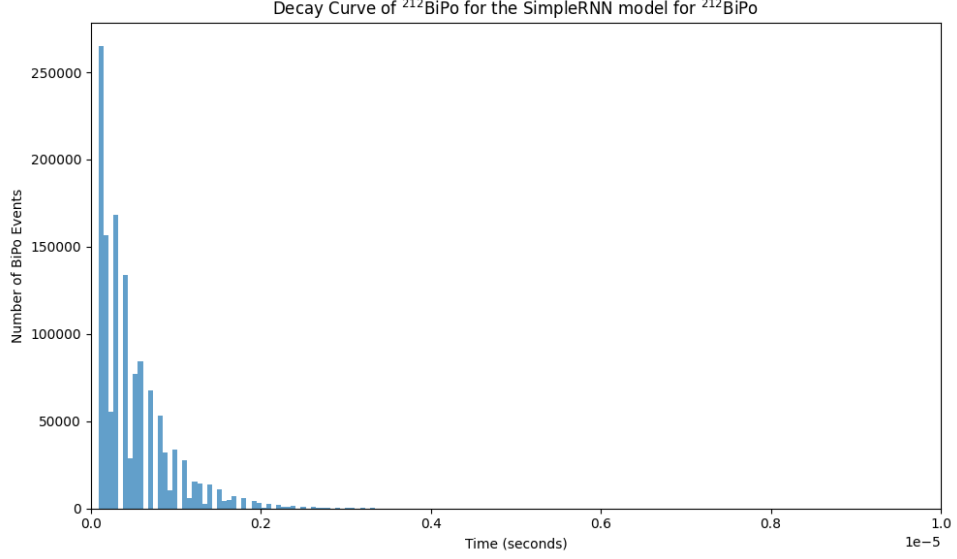


Figure 6.18: Decay curve of $^{212}\text{BiPo}$ events as predicted by the SimpleRNN model trained specifically on the BiPo212 chain. The trend closely follows the expected exponential shape without an initial spike, reflecting improved temporal learning compared to ANN and full-dataset models.

6.2.6 Final Classification with Higher Probability Selection

To leverage the specialized learning of the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ models, both trained separately on their respective decay chains, a final combined classification approach was implemented. This method mirrors the strategy used in the ANN framework: both SimpleRNN models were used to make predictions over the full dataset, and for each event, the class with the highest softmax probability across both models was selected as the final prediction.

The underlying motivation for this strategy was to retain the strengths of each specialized model, especially their performance on Po isotopes, while potentially compensating for weaknesses in Bi classification. However, in practice, this combined approach fails to outperform the full SimpleRNN model. It achieves an overall accuracy of only 66%, significantly lower than the 73% accuracy obtained with the full-dataset model. The combined model shows erratic misclassification behavior, especially for ^{214}Bi and ^{212}Bi events.

Due to this poor performance and the availability of a clearly superior model, a deeper investigation involving energy distributions or decay curve analysis is not pursued for the combined model. Instead, only the key performance metrics and the confusion matrix are reported for completeness.

Performance Metrics

The classification report is shown below:

Table 6.4: Classification report for the combined SimpleRNN model. The model performs relatively well for Po isotopes and background, but exhibits significantly degraded accuracy for Bi isotopes, particularly ^{214}Bi .

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.82	0.77	0.79
^{214}Bi (1)	0.49	0.23	0.32
^{214}Po (2)	0.72	0.94	0.82
External Background (3)	0.55	0.88	0.67
^{212}Bi (4)	0.52	0.52	0.52
^{212}Po (5)	0.93	0.60	0.73

Misclassification trends further illustrate the model's limitations:

- **^{214}Bi :** Only 23.34% of the true ^{214}Bi events are correctly classified. The majority are misclassified into ^{212}Bi (35.83%) and internal background (7.88%), with noticeable leakage into Po classes.

- ^{214}Po : This class retains high classification quality, with a TPR of 93.89% and efficiency of 72.42%. However, a large number of Po events (26.57%) are misclassified into ^{212}Po .
- ^{212}Bi : Exhibits high confusion with ^{214}Bi (34.09%), indicating that the fusion strategy worsens the separation between these two beta decayers.
- ^{212}Po : Maintains good performance, though its TPR drops compared to the individual models.

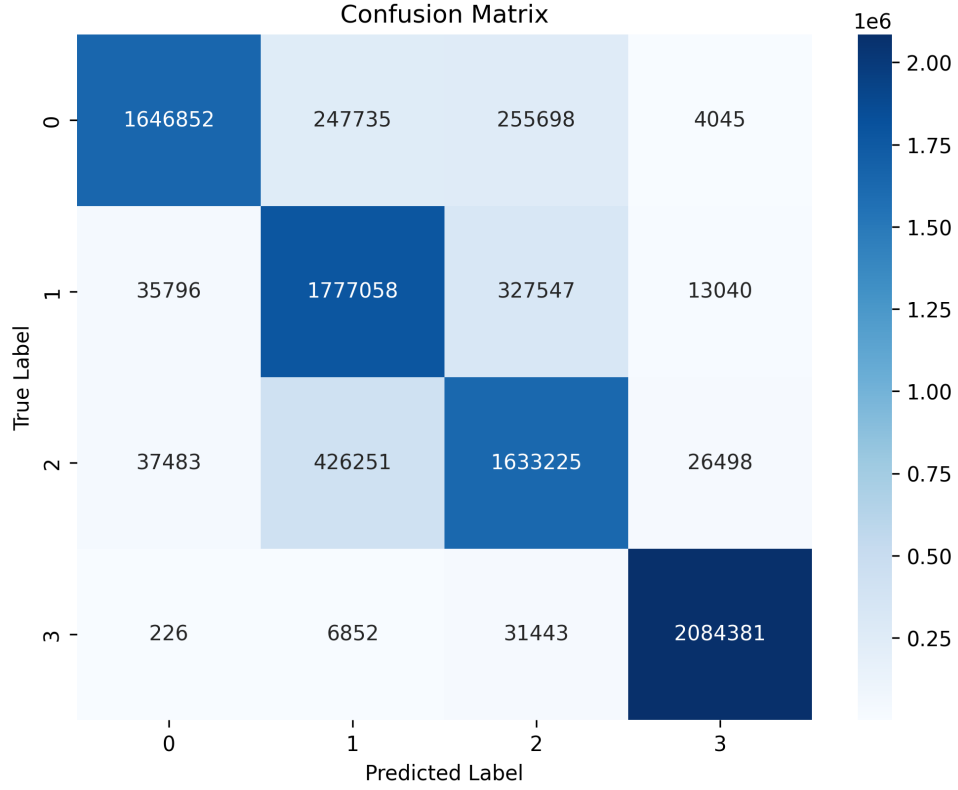


Figure 6.19: Confusion matrix of the combined SimpleRNN model. The structure lacks a clear diagonal, with particularly poor classification of ^{214}Bi and ^{212}Bi . Po classes retain relatively better performance, but **the overall dispersion highlights that combining two specialized models does not lead to any meaningful improvement in classification performance.**

6.3 Long Short-Term Memory Model

LSTM networks are an extension of RNNs that are specifically designed to capture long-range dependencies in sequential data. While SimpleRNNs can retain short-term memory through hidden state propagation, they often suffer from vanishing and exploding gradients when dealing with longer sequences [30]. LSTMs address this limitation through the introduction of a more complex memory structure and gated mechanisms that regulate the flow of information.

6.3.1 Structure of the LSTM Cell

An LSTM cell maintains two key components across time steps: the hidden state (h_t), which is analogous to that of a SimpleRNN, and the cell state (C_t), which acts as the long-term memory. The flow of information through the cell is controlled by three gates: the forget gate, input gate, and output gate. Each of these gates is composed of a sigmoid activation function that determines how much of each component should be remembered or discarded.

Figure 6.20 shows the internal architecture of a standard LSTM cell.

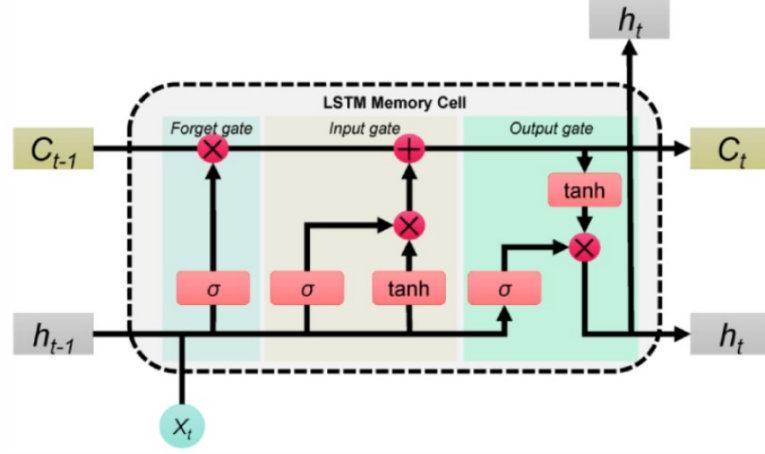


Figure 6.20: Internal structure of an LSTM memory cell. The cell incorporates three gates: forget, input, and output, each controlling different aspects of information flow through the cell. The long-term memory is preserved via the cell state C_t . Image taken from [36].

6.3.2 Mathematical Formulation

At each time step t , the LSTM performs the following operations:

- **Forget gate:** Decides which parts of the previous cell state C_{t-1} should be discarded.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

- **Input gate:** Determines what new information should be added to the cell state.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{and} \quad \tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

- **Cell state update:** Combines the above to update the current cell state.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

- **Output gate:** Computes the output hidden state based on the updated cell state.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{and} \quad h_t = o_t \cdot \tanh(C_t)$$

6.3.3 Comparison with SimpleRNN

Compared to SimpleRNNs, LSTMs provide a mechanism to learn which information to keep, forget, or expose to the next time step. This results in improved learning stability over long sequences, making LSTMs a natural progression in sequence modeling tasks. In this classification setup, LSTMs were employed in a two-layer configuration similar to the SimpleRNN, with both layers returning sequences to preserve per-event information. The LSTM outputs were further processed by a dense layer before producing the final classification results for each time step.

6.3.4 LSTM Model Trained on the Full Dataset

Model Architecture and Training Strategy

The model is trained on the full dataset using the same preprocessing steps as described previously, where sequences of 10 events are constructed, each containing six engineered and scaled features (energy of the event, radial distance, z coordinate, azimuthal angle ϕ , time difference from the first event, and relative distance from the first event). The architecture is as follows:

- **Input layer:** accepts sequences of 10 events, each represented by 6 input features.

- **First LSTM layer:** An LSTM layer with 32 units, using the tanh activation function. This layer processes the full input sequence and returns another sequence of the same length:

$$h_t = \text{LSTM}(x_t, h_{t-1}, c_{t-1})$$

where c_{t-1} is the cell state carried across time steps.

- **Second LSTM layer:** A subsequent LSTM layer with 16 units and the same activation, further processing the sequence.
- **Dense layer:** A fully connected dense layer with 8 neurons and ReLU activation, which maps the high-level features into a lower-dimensional representation.
- **Output layer:** A softmax-activated dense layer that outputs a probability distribution over the six possible event classes for each of the 10 time steps in the input sequence.

The model is compiled using the categorical cross-entropy loss function and the Adam optimizer. Training is performed for a maximum of 200 epochs with a batch size of 256. To avoid overfitting and stagnation, two callbacks are used: `ReduceLROnPlateau`, which halves the learning rate if the validation loss plateaus for four consecutive epochs, and `EarlyStopping`, which stops training after 10 epochs with no improvement and restores the best model weights. In this case, early stopping halted training at epoch 110, restoring weights from epoch 100.

The choice of architecture, particularly the relatively small number of LSTM units and two-layer structure, is primarily due to hardware limitations. More complex models with larger capacity could be explored with access to greater computational resources.

Despite LSTM’s theoretical advantage over SimpleRNN in capturing long-term dependencies, the performance of this model turns out to be nearly identical to the SimpleRNN model, as will be shown in the next section. Therefore, visualizations are not shown again here, and are instead included in Appendix F.4.

Performance Metrics

The LSTM model was trained using the same data split strategy as the SimpleRNN and ANN models: an 80%–20% training–testing split with `random_state = 0` for reproducibility, and an additional 80%–20% split of the training data into training and validation sets. The data was structured into sequences of 10 events, each consisting of six engineered and scaled features as described earlier.

The LSTM model achieved an overall accuracy of 73% on the test set, which is nearly identical to the SimpleRNN model. Table 6.5 provides the class-wise precision (efficiency), recall (TPR), and F1-score.

Table 6.5: Class-wise evaluation metrics of the LSTM model trained on the full dataset. Like the SimpleRNN model, this architecture performs best on Po events, while Bi events remain challenging to distinguish from background.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.94	0.72	0.82
²¹⁴ Bi (1)	0.55	0.38	0.45
²¹⁴ Po (2)	0.92	0.95	0.93
External Background (3)	0.63	0.78	0.70
²¹² Bi (4)	0.50	0.64	0.56
²¹² Po (5)	0.92	0.92	0.92

The LSTM model performs exceptionally well for the Po classes. ²¹⁴Po achieves a TPR of 95% and an efficiency of 92%, while ²¹²Po achieves 92% for both metrics. These results are consistent with the monoenergetic nature of alpha decays and their sharp timing features, which are readily learned by the recurrent architecture.

In contrast, classification performance is lower for Bi events. ²¹⁴Bi has a TPR of 38% and an efficiency of 55%, while ²¹²Bi is classified with a TPR of 64% and an efficiency of 50%. These low values reflect the difficulty of separating beta decays from background events and other signal classes due to their overlapping energy and temporal distributions.

The background classes perform moderately well. Internal background achieves a high precision of 94% but only a 72% TPR, suggesting that while predictions are usually correct when made, some internal events are missed. External background exhibits more balanced behavior with 78% recall and 63% precision.

Misclassification patterns further illustrate the model’s challenges:

- ^{214}Bi is most frequently confused with ^{212}Bi (23.15%), internal background (12.96%), and external background (9.19%), showing a high degree of overlap in feature space.
- ^{214}Po shows excellent separation, with minor leakage into ^{212}Po (5.51%) and background classes.
- ^{212}Bi is often misclassified as ^{214}Bi (31.96%), reinforcing the trend seen in earlier models that beta emitters from different chains are difficult to distinguish.
- ^{212}Po shows robust classification, with only small confusion primarily with ^{214}Po and background classes.

These findings are visualized in the confusion matrix in Figure 6.21, which shows a strong diagonal for Po classes but more diffuse patterns for Bi and background classes.

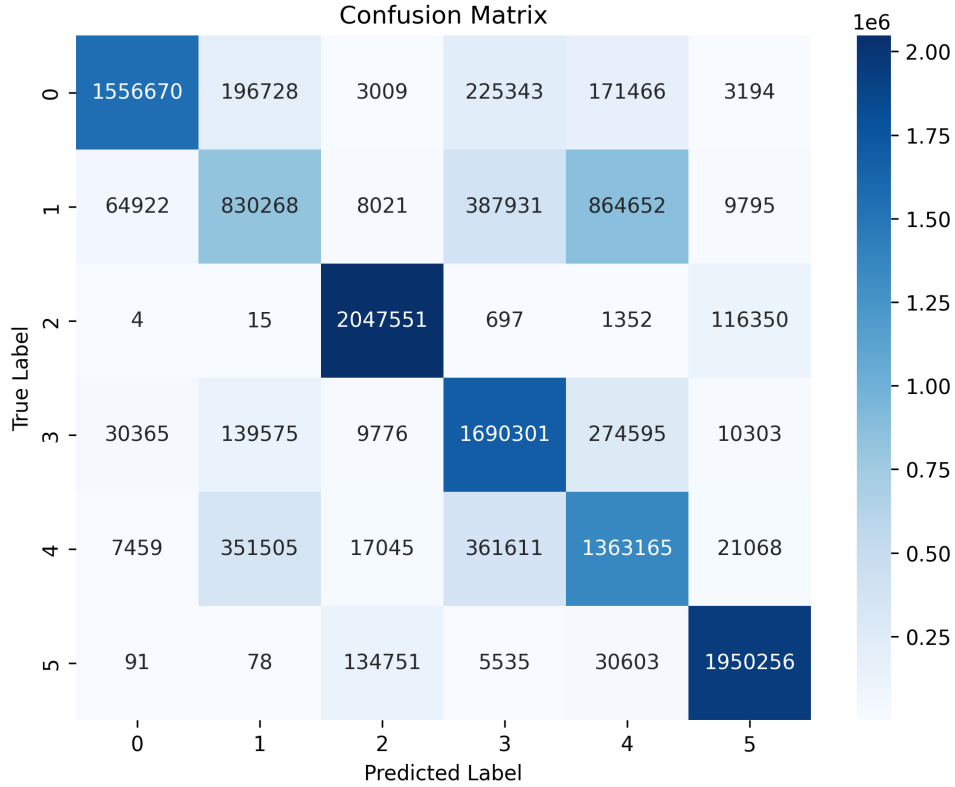


Figure 6.21: Confusion matrix of the LSTM model trained on the full dataset. The matrix shows strong diagonal entries for Po classes, indicating high classification accuracy. However, the Bi classes exhibit significant off-diagonal misclassifications, especially between each other and background signals, **highlighting the model's continued difficulty in resolving overlapping beta decay signals.**

The visualization patterns are nearly identical to those observed in the SimpleRNN model, so they are not shown again here and can be found in Appendix F.4.

6.3.5 Separate LSTM Models for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$

Model Architecture and Training Strategy

To reduce the inter-chain confusion arising from overlapping energy and timing features between the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains, two separate LSTM models were trained, each specialized for one decay chain. This separation allows for more focused learning and evaluation within the context of each chain's unique decay structure.

Compared to the full dataset model, a slightly deeper LSTM architecture was used for the separate models. However, this architecture matches the one used for the SimpleRNN models, to allow a fair comparison between the two approaches. The reduced number of events per chain also makes it computationally feasible to use larger LSTM layers without overloading system memory.

The architecture is as follows:

- **Input layer:** Accepts sequences of 10 events, each with 6 engineered and scaled features (energy, radial position, z , ϕ , time difference from the first event, and spatial distance from the first event).
- **First LSTM layer:** 64 units, tanh activation, returns full sequences.
- **Second LSTM layer:** 32 units, tanh activation, returns full sequences.
- **Dense layer:** 16 units, ReLU activation.
- **Output layer:** Softmax activation for 4-class classification.

Both datasets were split into 80% training and 20% testing sets using `random.state = 0` to ensure reproducibility. The training data was further split into 80% training and 20% validation subsets. Training was performed over a maximum of 200 epochs using a batch size of 256. EarlyStopping (patience = 10) and ReduceLROnPlateau (patience = 4, factor = 0.5) callbacks were used to prevent overfitting and improve convergence.

The results for both models closely mirror those of the SimpleRNN architecture. Therefore, in this section, only the performance metrics are presented for each chain.

²¹⁴BiPo Model Evaluation

Performance Metrics The LSTM model specialized for the ²¹⁴BiPo chain was trained on truth labels {0, 1, 2, 3}, representing internal background, ²¹⁴Bi, ²¹⁴Po, and external background.

The model achieves an overall accuracy of 82% on the test set. Table 6.6 presents the precision (efficiency), recall (TPR), and F1-score for each class.

Table 6.6: Evaluation metrics for the LSTM model trained on the ²¹⁴BiPo chain. The model achieves excellent performance for ²¹⁴Po, while classification for ²¹⁴Bi is improved compared to the full-dataset model.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.93	0.72	0.81
²¹⁴ Bi (1)	0.71	0.73	0.72
²¹⁴ Po (2)	0.99	1.00	0.99
External Background (3)	0.71	0.85	0.77

The classification of ²¹⁴Po is the most successful, with a recall of 100% and efficiency of 99%. The short time delays and sharp energy peak of alpha decays contribute to this separability. ²¹⁴Bi, in contrast, achieves a recall of 73% and efficiency of 71%, an improvement over the full dataset model, though it still suffers from confusion with background events.

Misclassification patterns include:

- ²¹⁴Bi is primarily misclassified as internal background (15.96%) and external background (12.99%), with minimal confusion into ²¹⁴Po (0.04%).
- ²¹⁴Po maintains high purity with only minor misclassification into external background (0.54%), ²¹⁴Bi (0.48%), and internal background (0.18%).

The confusion matrix for the LSTM model trained on the ²¹⁴BiPo chain is shown in Figure 6.22. A clearly defined diagonal is observed for the ²¹⁴Po class, indicating near-perfect classification. In contrast, ²¹⁴Bi exhibits noticeable off-diagonal entries, with considerable misclassification into both internal and external background classes.

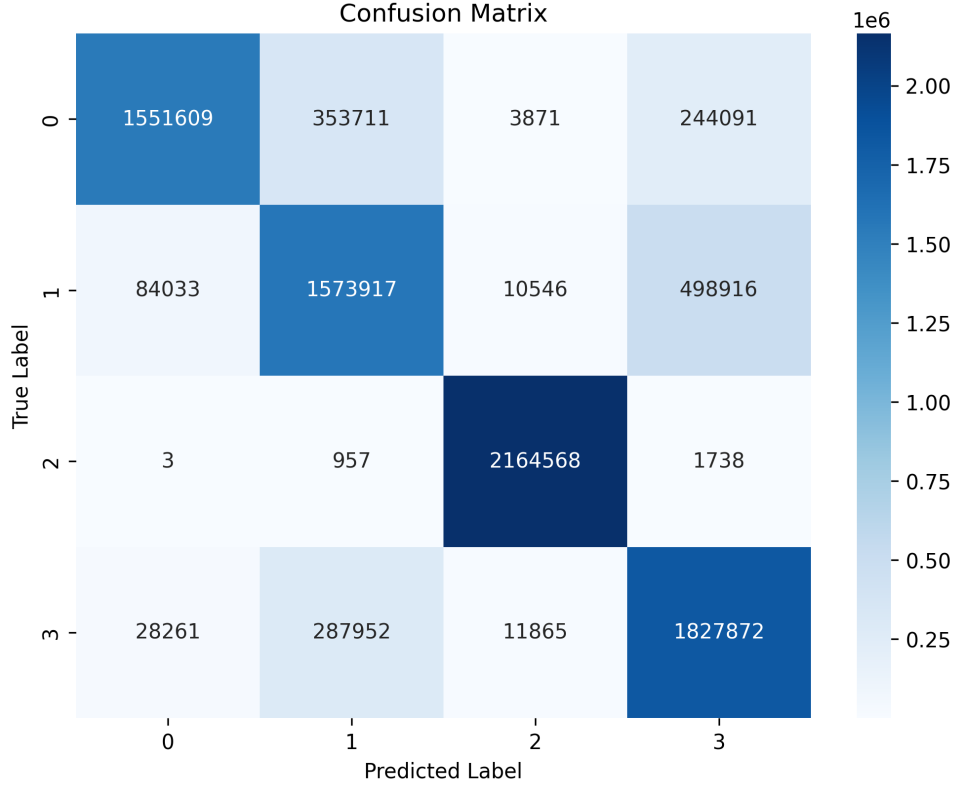


Figure 6.22: Confusion matrix of the LSTM model trained on the $^{214}\text{BiPo}$ chain. A strong diagonal is observed for ^{214}Po , confirming its robust classification. However, ^{214}Bi shows notable leakage into internal and external background classes, reducing classification purity.

Visualizations are omitted here as they are effectively identical to those shown for the SimpleRNN model. They are included in Appendix F.4 for reference.

$^{212}\text{BiPo}$ Model Evaluation

Performance Metrics The LSTM model trained on the $^{212}\text{BiPo}$ chain achieves a test accuracy of 83%. Table 6.7 summarizes the class-wise evaluation metrics for this model.

Table 6.7: Evaluation metrics for the LSTM model trained on the $^{212}\text{BiPo}$ chain. ^{212}Po classification remains highly accurate, while the ^{212}Bi classification continues to be challenged by overlaps with background events.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.96	0.76	0.85
External Background (3)	0.72	0.82	0.77
^{212}Bi (4)	0.73	0.77	0.75
^{212}Po (5)	0.98	0.98	0.98

The ^{212}Po class achieves near-perfect performance with 98% recall and efficiency. The distinct alpha decay signature again enables reliable detection. ^{212}Bi performance is moderate with a recall of 77% and efficiency of 73%, suggesting that some confusion with background and other classes persists.

Misclassification breakdown:

- ^{212}Bi is primarily misclassified as external background (14.65%) and internal background (11.35%), with minor leakage into ^{212}Po (1.40%).
- ^{212}Po is correctly classified with extremely high precision, with only small confusion into ^{212}Bi (1.24%), external background (0.61%), and internal background (0.19%).

Figure 6.23 presents the confusion matrix for the $^{212}\text{BiPo}$ LSTM model. The ^{212}Po class forms a sharply defined diagonal cluster, confirming highly accurate classification. The ^{212}Bi class, however,

continues to show misclassification into external background and internal background, though to a lesser extent than seen in full dataset models.

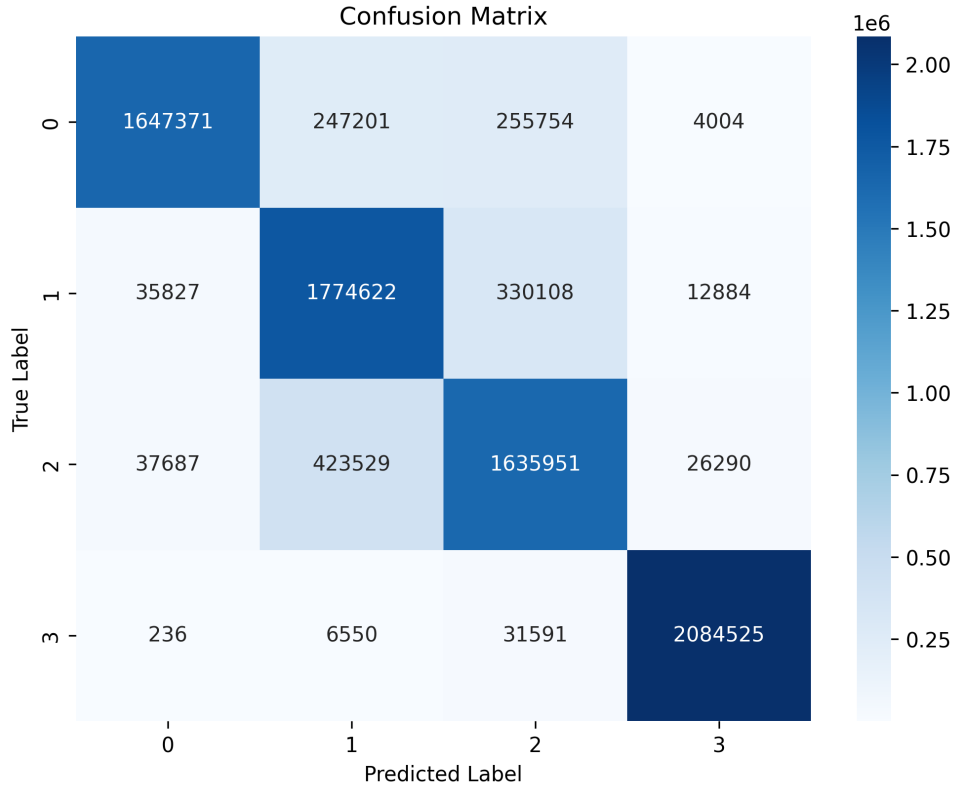


Figure 6.23: Confusion matrix of the LSTM model trained on the $^{212}\text{BiPo}$ chain. The diagonal dominance for ^{212}Po reflects successful classification. ^{212}Bi shows a broader distribution across external and internal background classes, indicating ongoing difficulty in isolating beta events from background despite improvements over full dataset models.

As with the $^{214}\text{BiPo}$ model, visualizations for this model are not repeated here to avoid redundancy, and can be found in Appendix F.4.

6.3.6 Final Classification with Higher Probability Selection

The method for combining the individual LSTM models follows the same approach used for the ANN and SimpleRNN models. Each event is passed through both specialized models, and the final prediction is selected based on the higher softmax probability between the two. While this strategy theoretically allows each model to specialize and make more confident predictions, it does not yield improved performance in practice.

Performance Metrics

The combined LSTM model performs significantly worse than the LSTM model trained directly on the full dataset. With an overall accuracy of only 62%, it falls far short of the 73% accuracy achieved by the full model. Given this considerable drop in performance, further detailed analysis, plots, or visualizations are not provided here. Instead, the classification report is included for completeness.

Table 6.8: Class-wise evaluation metrics of the combined LSTM model. Although ^{214}Po retains high recall, the performance of other classes, particularly ^{212}Po , suffers dramatically compared to all previous models.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.82	0.77	0.79
^{214}Bi (1)	0.49	0.24	0.32
^{214}Po (2)	0.62	0.95	0.75
External Background (3)	0.55	0.88	0.68
^{212}Bi (4)	0.52	0.52	0.52
^{212}Po (5)	0.92	0.37	0.53

The classification of Po isotopes, which consistently performed well in all prior models due to their distinctive alpha signatures, is notably degraded here, especially ^{212}Po , which achieves only 37% recall despite high precision. This sharp decline suggests that the model combination strategy compromises the confidence and specificity of the Po classification.

Misclassification analysis reveals significant class leakage:

- ^{214}Bi is heavily misclassified into ^{212}Bi (36.03%) and internal background (7.79%).
- ^{214}Po shows high recall (95.49%) but poor precision (62.13%) due to 36.95% of predictions being false positives from ^{212}Po .
- ^{212}Bi remains confused with ^{214}Bi (33.89%) and background classes.
- ^{212}Po , previously the best-performing class, now suffers a severe recall drop to 37.46%, with many misclassified as ^{214}Po .

These trends are reflected in the confusion matrix (Figure 6.24), which lacks a strong diagonal and displays widespread misclassification, especially between Bi classes and background. The overall structure indicates that the combined model lacks the class separation observed in more focused models.

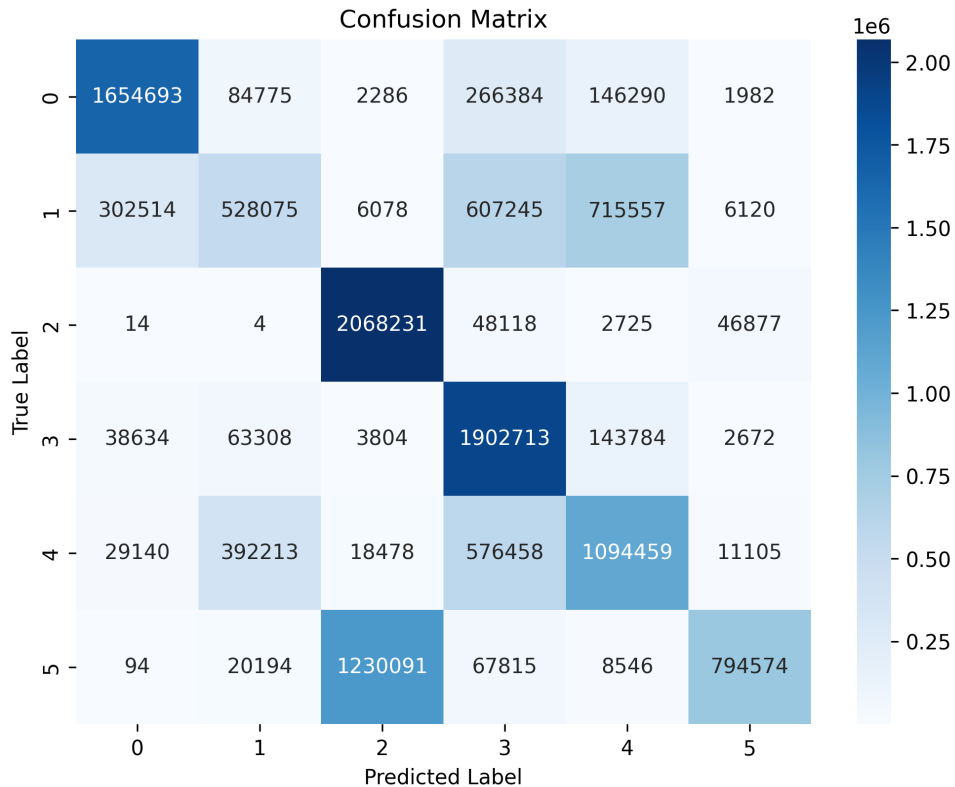


Figure 6.24: Confusion matrix of the combined LSTM model. The absence of a clearly defined diagonal and the presence of strong off-diagonal entries indicate extensive misclassification, particularly for Bi classes. ^{212}Po , previously classified with high accuracy, now suffers major confusion with ^{214}Po and background.

Despite the architectural advantages of LSTM networks, the results indicate that they offer no significant improvement over SimpleRNNs for this classification task. In particular, the combination of specialized LSTM models underperforms even the baseline full-dataset model. These observations suggest that unidirectional sequential modeling alone may not be sufficient to fully capture the complex temporal and spatial correlations present in BiPo events.

To address this limitation, the next section explores Bidirectional LSTM (BiLSTM) architectures. By processing sequences in both forward and backward directions, BiLSTMs offer the potential to leverage contextual information from both past and future time steps—an approach that may better suit the nature of coincident decay events in this dataset.

6.4 Bidirectional LSTM with Attention

In previous models, it was observed that ^{214}Po and ^{212}Po events were identified with high confidence due to their distinct monoenergetic alpha peaks. However, the corresponding ^{214}Bi and ^{212}Bi events, occurring just before the Po decays in the BiPo sequence, were more difficult to classify. Since these decays occur in tightly correlated pairs, leveraging only forward temporal information (as in unidirectional RNNs or LSTMs) limits the model’s ability to identify the earlier, less distinct Bi events based on the presence of a clearly identifiable Po event later in the sequence.

To address this limitation, BiLSTMs are introduced. By processing the sequence in both forward and backward directions, BiLSTMs allow the model to capture future context when predicting the current time step. This is particularly valuable in the context of BiPo event classification, where the temporal correlation between a Po decay and its preceding Bi event is a key signature of a true coincidence. Through backward processing, the model can use the strong features of Po events to inform the classification of earlier, ambiguous Bi events, potentially improving identification of full BiPo pairs. Moreover, BiLSTMs enhance learning of global sequence patterns, reduce information bottlenecks in early time steps, and offer a richer temporal representation, crucial for distinguishing overlapping backgrounds and subtle signal structures.

6.4.1 Overview of BiLSTM

LSTM networks are a powerful extension of RNNs, designed to capture long-range temporal dependencies through gated memory mechanisms [37]. While standard LSTMs process sequences in a single (typically forward) direction, BiLSTMs enhance this by learning from both past and future contexts simultaneously.

In a BiLSTM, two separate LSTM layers are used: one processes the sequence from $t = 1$ to $t = T$ (forward), and the other from $t = T$ to $t = 1$ (backward). The outputs from both directions are concatenated at each time step, enabling the model to incorporate information from the full temporal context. This is especially advantageous for tasks where the classification at a given time step depends not only on the history but also on future observations.

The structure of a BiLSTM is illustrated in Figure 6.25. Each input x_t is passed through both the forward and backward LSTM layers, producing hidden states $\vec{h}_t$ and $\overleftarrow{h}_t$, respectively. These are then combined to form the final output for time step t .

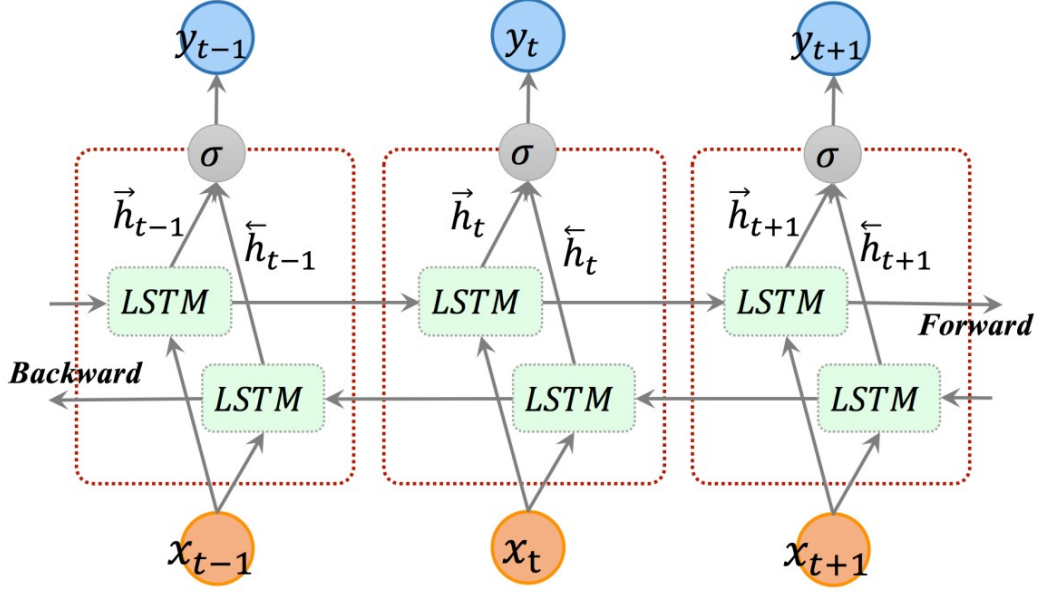


Figure 6.25: Schematic of a BiLSTM. The forward LSTM processes the sequence from past to future, while the backward LSTM processes it in reverse. At each time step, the hidden states from both directions are concatenated before being passed to subsequent layers. Image adapted from [38].

6.4.2 Attention Mechanism

While BiLSTMs enable the model to extract both past and future temporal features, they treat all time steps equally and lack a mechanism to emphasize particularly informative moments in the sequence. In real-world applications, particularly those involving temporally ordered physical processes such as BiPo decays, not all parts of the sequence carry equal relevance for classification. To address this, an attention mechanism is incorporated.

The attention mechanism allows the model to learn which time steps are most important for a given prediction by computing a weighted combination of all time steps in the sequence. These weights are learned based on how relevant other positions in the sequence are to the current position being processed. This enables the model to focus more on key features, such as distinct Po peaks or specific timing correlations between events.

In this work, a simplified version of *scaled dot-product attention* is employed [39], where the same input tensor serves as the query (Q), key (K), and value (V). Although mathematically identical in this implementation, conceptually:

- **Query (Q)** represents the current time step the model is focusing on.
- **Key (K)** encodes all time steps in the sequence against which the query is compared.
- **Value (V)** contains the actual features or information associated with each time step that should be retrieved or emphasized.

The attention mechanism computes how much attention the current time step (query) should pay to every other time step (keys), and uses this to combine the values accordingly.

The attention score between Q and K is calculated using scaled dot-product attention:

$$\text{score}(Q, K) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right)$$

Here, d_k is the dimensionality of the key vectors. Scaling by $\sqrt{d_k}$ helps mitigate the risk of large dot products pushing the softmax function into regions with very small gradients, which would otherwise make learning inefficient.

The final attention output is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V$$

This operation results in a context-aware representation of the input sequence, where each time step is influenced not only by its own features but also by a weighted combination of all other time steps, based on learned relevance. This enables the network to dynamically prioritize and combine temporal information in a more interpretable and adaptive way than standard sequence models. In the context of this work, such context-aware representations allow the model to better capture interactions between events within a BiPo decay sequence, ultimately enhancing the classification of temporally correlated decays.

Scaled Dot-Product Attention

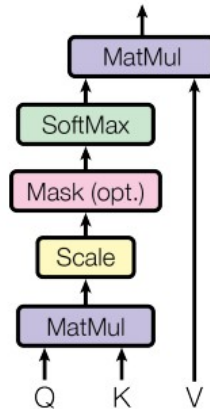


Figure 6.26: Flow diagram of scaled dot-product attention. The query (Q) and key (K) vectors are multiplied and scaled before passing through a softmax, which produces the attention weights. These are then applied to the value (V) vectors to generate the final output. Image adapted from [39].

6.4.3 Combining BiLSTM with Attention

The combination of BiLSTMs and attention mechanisms creates a highly expressive model architecture capable of both understanding bidirectional context and selectively focusing on the most important parts of a sequence.

In the architecture implemented for this work, the input is first passed through a BiLSTM layer with 64 units and `return_sequences=True`, producing a sequence of context-enriched representations. These are then passed through the custom attention layer described above, which computes a weighted combination of all time steps based on their relevance to each other. This mechanism enables the network to focus more on time steps that carry strong decay signatures or unique spatial-temporal characteristics.

The result of the attention layer is passed to a `TimeDistributed` dense layer with ReLU activation, followed by a final softmax output layer for classification at each time step.

The `TimeDistributed` wrapper in Keras applies the same dense layer independently to each time step in the sequence. This is essential in models like BiLSTM with attention, where the output of the attention layer retains the temporal dimension (i.e., the sequence length). By wrapping the dense layer with `TimeDistributed`, the model ensures that each time step is processed individually while sharing the same learned weights across all time steps.

This differs from the other models used in this work, such as the SimpleRNN or ANN, where the temporal structure is either reduced earlier in the pipeline or flattened entirely before classification. In the BiLSTM with attention model, however, retaining and individually classifying each time step allows the model to make context-aware predictions for every point in the sequence, which is critical when identifying specific Bi or Po events within a temporally correlated event window.

6.4.4 BiLSTM Model with Attention Trained on the Full Dataset

Model Architecture and Training Strategy

While BiLSTMs can access both past and future context, they treat all time steps equally. The addition of an attention mechanism allows the model to dynamically focus on the most informative portions of each

sequence, enabling better discrimination between event classes, particularly in scenarios with complex temporal correlations, such as BiPo decays.

The architecture is composed of the following components:

- **Input layer:** Receives sequences of 10 events, each represented by 6 engineered and scaled features: energy, radial distance, z coordinate, azimuthal angle ϕ , time difference (relative to the first event in the sequence), and spatial distance (also relative to the first event).
- **BiLSTM layer:** A BiLSTM layer with 64 units that processes the input sequence in both forward and backward directions. This enables the model to access full temporal context for each event.
- **Attention layer:** A custom scaled dot-product attention mechanism is applied to the output of the BiLSTM layer. The same sequence is used as the query, key, and value tensors. This mechanism learns the importance of each time step by computing attention scores and generating a context-aware representation of the sequence.
- **TimeDistributed Dense layer:** A fully connected layer with 64 ReLU-activated neurons is applied to each time step to refine the extracted representations.
- **Output layer:** A final softmax-activated dense layer outputs class probabilities for each time step, ensuring a probabilistic interpretation over the six event classes.

The model was compiled using the categorical cross-entropy loss function and the Adam optimizer. Training was performed for a maximum of 200 epochs with a batch size of 256. To avoid overfitting and optimize convergence, two callbacks were employed:

- **ReduceLROnPlateau:** Monitors validation loss and reduces the learning rate by a factor of 0.5 if no improvement is seen over four consecutive epochs.
- **EarlyStopping:** Halts training if no improvement is observed over 10 epochs and restores the best model weights.

Training stopped at epoch 86 due to early stopping, with the best model found at epoch 76. At that point, the model achieved a training and validation accuracy of 90.7%, with a loss of 0.231. A detailed evaluation of classification performance will follow in the next section. Training and validation curves are presented in Appendix F.16, and ROC curves are included in Appendix F.17.

Performance Metrics

The BiLSTM model with attention was trained using the same data splitting strategy applied across all sequence-based models: 80% of the data was allocated for training and 20% for testing, with a `random.state` set to 0 to ensure reproducibility. Within the training set, an additional 80/20 split was applied to separate training and validation data. The input consisted of sequences of 10 events, each containing 6 engineered features.

The model achieved a test accuracy of 91%, significantly outperforming the previous RNN-based architectures. Table 6.9 summarizes the classification performance per class.

Table 6.9: Class-wise evaluation metrics of the BiLSTM model with attention trained on the full dataset. The model demonstrates strong and balanced performance across all classes, including challenging beta emitters, benefiting from the context-aware sequence modeling introduced by the attention mechanism.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.96	0.83	0.89
^{214}Bi (1)	0.90	0.89	0.90
^{214}Po (2)	0.93	0.94	0.94
External Background (3)	0.84	0.94	0.89
^{212}Bi (4)	0.88	0.90	0.89
^{212}Po (5)	0.94	0.93	0.93

The attention-enhanced BiLSTM shows excellent performance for ^{214}Po and ^{212}Po , achieving TPRs of 94% and 93% respectively, with matching precision values. This reflects the model’s capacity to emphasize distinct and temporally coherent signal patterns, such as those associated with monoenergetic alpha decays.

Classification performance for the more challenging beta decays has also improved significantly. ^{214}Bi achieved a TPR of 89% and efficiency of 90%, while ^{212}Bi followed closely with a TPR of 90% and

precision of 88%. This improvement is attributed to the attention mechanism’s ability to focus on critical temporal and energetic dependencies that standard sequence models may overlook.

Background classes also benefit from the enhanced modeling: internal background (class 0) maintains a high precision of 96% and a TPR of 83%, while external background (class 3) reaches a balanced performance with 94% TPR and 84% precision.

A breakdown of misclassification reveals:

- ^{214}Bi events are most often confused with ^{212}Bi (7.96%), with minor leakage into background classes.
- ^{214}Po is nearly perfectly classified, with less than 1% confusion across all classes.
- ^{212}Bi occasionally overlaps with ^{214}Bi (9.32%) but is otherwise well distinguished.
- ^{212}Po also maintains very high classification accuracy, with only minor confusion into ^{214}Po (5.66%) and background classes.

These trends are further confirmed by the confusion matrix in Figure 6.27. The matrix shows strong diagonal dominance across all classes, reflecting the model’s robust ability to separate event types. Off-diagonal values are minimal, particularly for Po isotopes, while beta emitters and background classes now exhibit much less overlap compared to previous models.

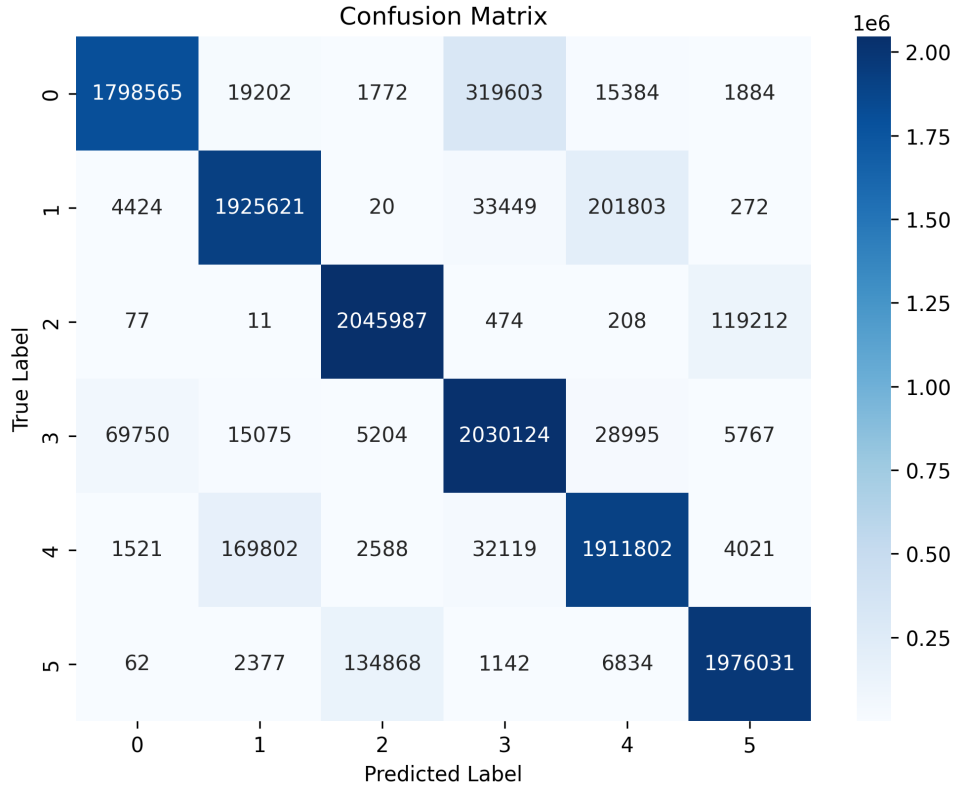


Figure 6.27: Confusion matrix of the BiLSTM model with attention trained on the full dataset. The matrix displays clear diagonal dominance across all classes, indicating accurate classification performance. The reduction of off-diagonal confusion, particularly for the Bi isotopes and background classes, **highlights the model’s ability to leverage temporal context effectively via attention.**

Energy Distributions

Figures 6.28 and 6.29 display the energy spectra of ^{214}Bi and ^{212}Bi events classified by the BiLSTM model with an attention mechanism, trained on the full dataset.

The ^{214}Bi energy distribution (Figure 6.28) exhibits the general shape of a continuous beta spectrum, extending beyond 3 MeV. However, a pronounced peak appears around 2 MeV, which is not expected in the true physical spectrum. This anomaly likely reflects background contamination or misclassifications clustering around that energy range. Despite this artifact, the overall trend of the ^{214}Bi distribution maintains continuity across the spectrum, and no clear threshold-based separation from ^{212}Bi is observed,

indicating an improved ability to distinguish events based on features beyond energy alone. Aside from the anomalous peak, the predicted distribution aligns well with the true spectrum.

In contrast, the ^{212}Bi spectrum (Figure 6.29) captures the expected broad shape of a beta spectrum remarkably well. Unlike previous models, there is no sharp cutoff or missing region, and the spectrum transitions smoothly across the energy range. The absence of a hard boundary between the two Bi isotopes highlights the strength of the BiLSTM with attention in utilizing contextual information from the entire sequence to resolve overlapping feature spaces. The predicted distribution shows strong agreement with the true one across the entire energy range.

This performance suggests that the combination of bidirectional recurrence and attention enables the model to extract more nuanced patterns that aid in disambiguating difficult cases, particularly where beta spectra overlap significantly.

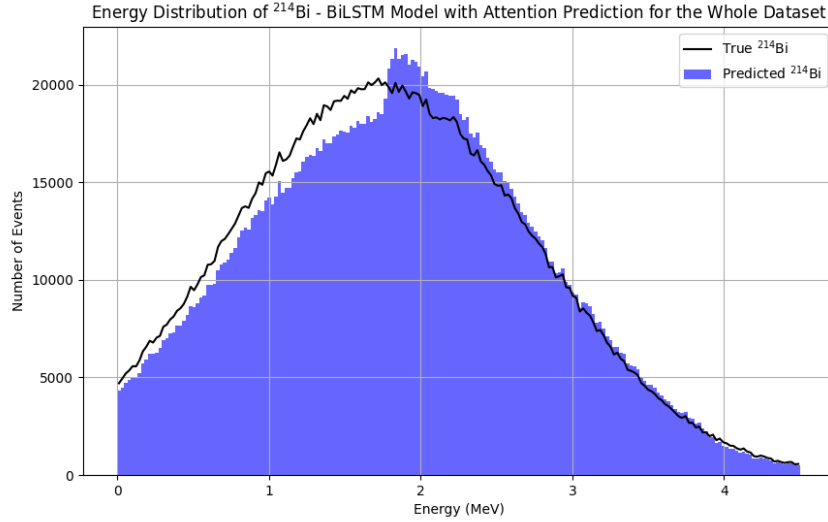


Figure 6.28: Energy distribution of events classified as ^{214}Bi by the BiLSTM model with attention, trained on the full dataset. The predicted spectrum (blue) is mostly consistent with the true distribution (line), except for an unexpected peak around 2 MeV likely caused by background misclassification.

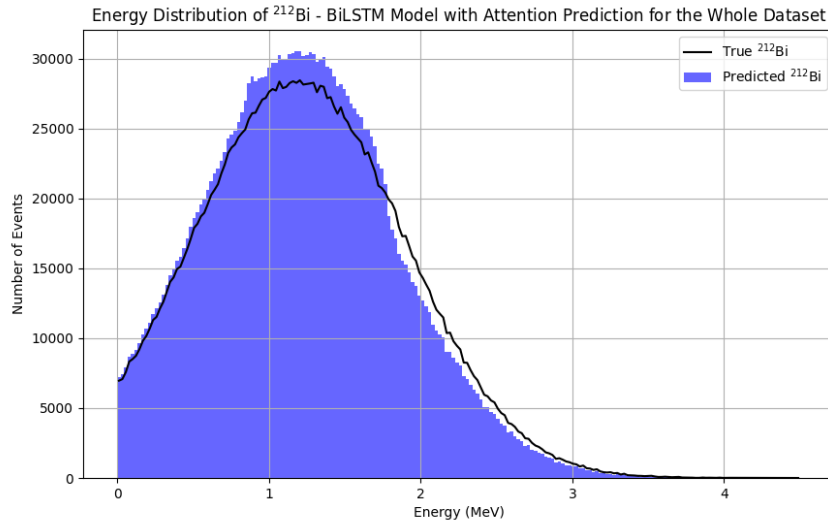


Figure 6.29: Energy distribution of events classified as ^{212}Bi by the BiLSTM model with attention. The predicted beta spectrum (blue) closely matches the true distribution (line), with no missing regions or abrupt boundaries.

Figures 6.30 and 6.31 show the energy distributions of ^{214}Po and ^{212}Po events, respectively, as predicted by the BiLSTM model with an attention mechanism on the full dataset.

The ^{214}Po spectrum (Figure 6.30) features a sharp, monoenergetic peak centered around 0.78 MeV, consistent with the expected alpha decay energy. Compared to earlier models, the shape is cleaner and better defined, without the sharp rise or cutoffs seen in the SimpleRNN or ANN models. Interestingly, although the dominant peak remains well-preserved, there is still some ambiguity in the overlap region. However, a few events are still classified in this overlap range, in contrast to previous models that imposed a much stricter separation. This suggests that the model begins to learn finer distinctions without enforcing a hard threshold. The predicted distribution aligns closely with the true spectrum.

Similarly, the ^{212}Po spectrum (Figure 6.31) displays a well-defined monoenergetic peak near 0.87 MeV, characteristic of alpha decay. The peak is symmetric and cleanly captured. As with ^{214}Po , some events from the overlapping region are still classified as ^{212}Po , even though the majority remain correctly identified. This slight diffusion across the boundary contrasts with the harder separations of earlier models and indicates the BiLSTM with attention can tolerate and learn from ambiguity in overlapping energy regions, potentially improving generalization. The overall shape matches the true distribution very well.

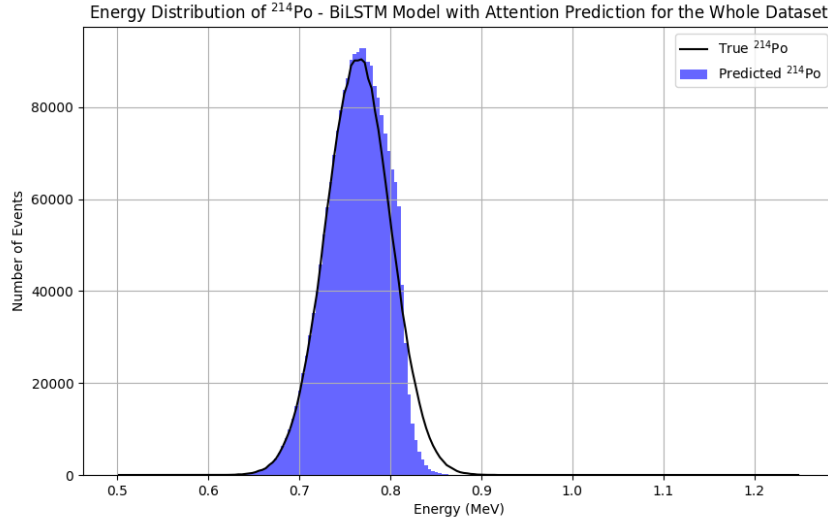


Figure 6.30: Energy distribution of ^{214}Po events predicted by the BiLSTM with Attention model on the full dataset. A sharp monoenergetic peak appears around 0.78 MeV, consistent with the true alpha decay energy. Some diffusion into the ^{212}Po overlap region indicates softer separation boundaries than in previous models.

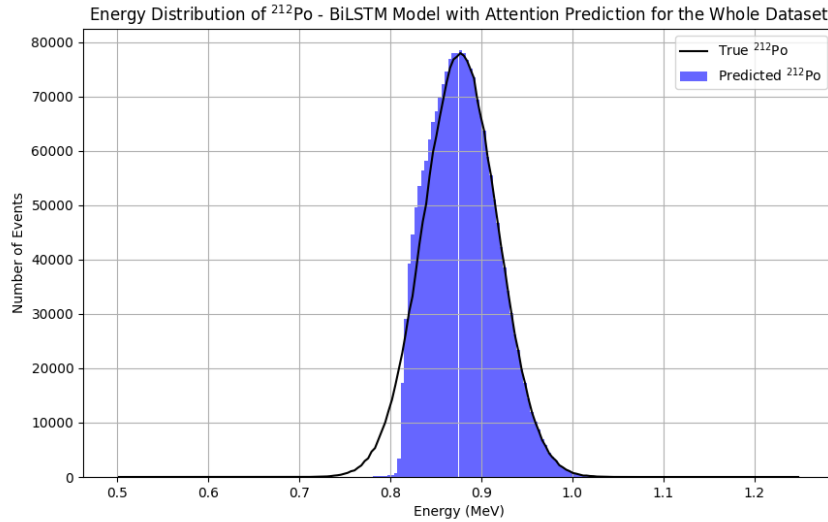


Figure 6.31: Energy distribution of ^{212}Po events predicted by the BiLSTM with Attention model on the full dataset. The clean, symmetric peak around 0.87 MeV aligns closely with the true distribution. Minor classification into the ^{214}Po overlap region suggests improved generalization in ambiguous areas.

Decay Curves

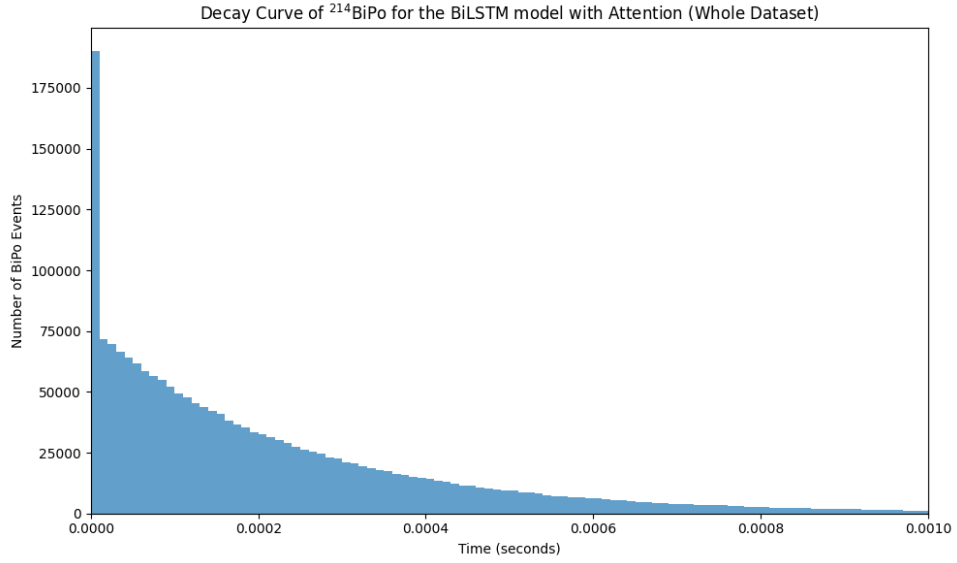
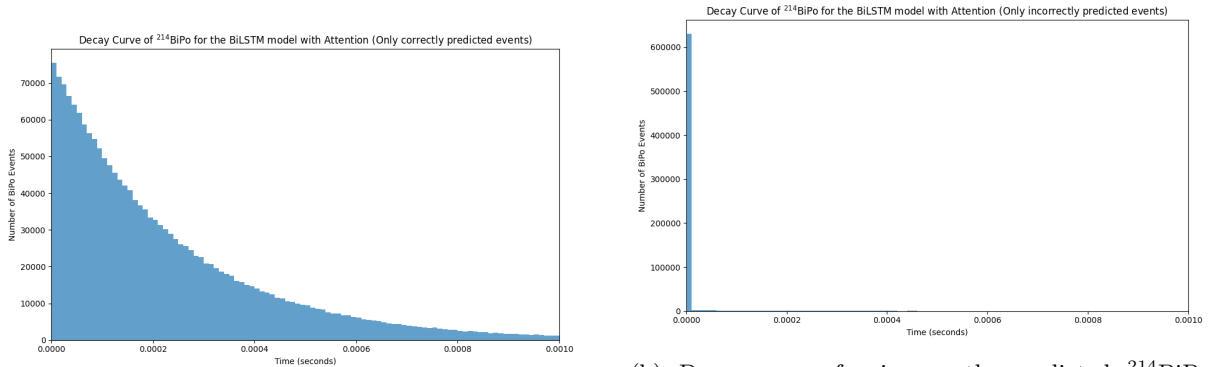


Figure 6.32: Decay curve of $^{214}\text{BiPo}$ events predicted by the BiLSTM model with Attention, using the full dataset. A sharp spike appears at the shortest time differences, followed by an exponential decline. This peak suggests the presence of false event pairs, likely due to background or misclassification.

To understand how misclassification affects the decay curve, both correctly and incorrectly classified BiPo pairs are plotted separately, as shown below.



(a) Decay curve for correctly predicted $^{214}\text{BiPo}$ events. The distribution exhibits a clean exponential trend, as expected from the short-lived nature of ^{214}Po .

(b) Decay curve for incorrectly predicted $^{214}\text{BiPo}$ events. The steep peak at the start, followed by a flat tail, suggests that many background coincidences are misidentified as true BiPo pairs.

Figure 6.33: Decay curves of $^{214}\text{BiPo}$ events predicted by the BiLSTM with Attention model, split by classification outcome. The correctly predicted events align well with physical expectations, while the incorrectly predicted events account for the unphysical early spike observed in the total distribution.

Figure 6.32 displays the decay time distribution for BiPo214 events predicted by the BiLSTM with Attention model using the entire dataset. While a generally exponential trend is visible, the curve begins with a steep spike in event counts at very short time differences, similar to what was observed in previous models.

To diagnose this behavior, the total distribution is decomposed into two subsets: correctly predicted BiPo events (Figure 6.33a) and incorrectly predicted BiPo events (Figure 6.33b). The decay curve of the correctly identified events follows a clean exponential shape, which aligns well with the expected decay characteristics of ^{214}Po . This confirms that when the model identifies the Bi and Po pair correctly, it also captures the physical decay behavior with high precision.

In contrast, the curve for incorrectly predicted BiPo pairs displays a sharp concentration of events at the lowest time differences, with little to no meaningful exponential structure beyond the first bin. This

strongly suggests that the initial spike in the total distribution originates from misclassified background or unrelated coincidences.

This breakdown demonstrates that the overall decay curve predicted by the BiLSTM with Attention model retains a physically meaningful exponential shape, except for the pronounced initial spike. By separately analyzing the correctly and incorrectly predicted events, it becomes evident that this spike arises primarily from misclassified background pairs. Once these are excluded, the remaining distribution aligns well with the expected decay behavior of true BiPo events. In comparison to the ANN and SimpleRNN models where the full decay curves were significantly distorted or flattened due to widespread misclassification, this result highlights a distinct advantage of the BiLSTM with Attention, which is its ability to preserve clean, temporally consistent decay patterns even when operating on noisy, overlapping data.

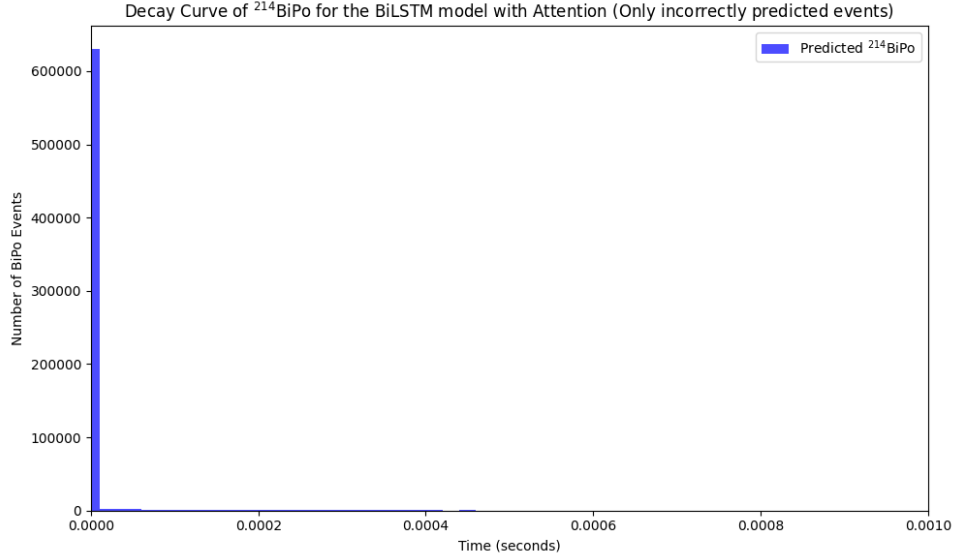


Figure 6.34: Decay time distribution of $^{212}\text{BiPo}$ events predicted by the BiLSTM model with Attention on the whole dataset. While the overall curve appears dominated by an initial spike, the physically meaningful structure is obscured due to a high number of misclassified events.

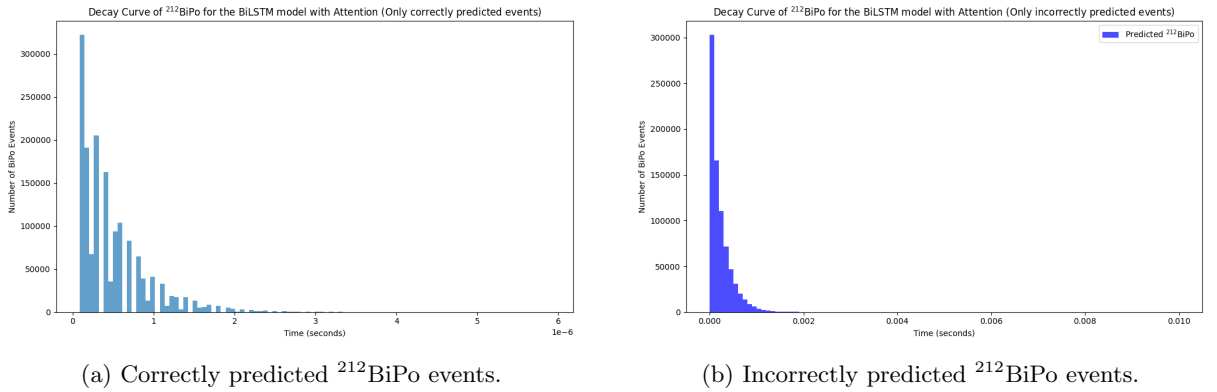


Figure 6.35: Decay curves for $^{212}\text{BiPo}$ events predicted by the BiLSTM with Attention model, separated by classification accuracy.

The decay curve for the entire $^{212}\text{BiPo}$ dataset (Figure 6.34) is overwhelmingly dominated by a sharp spike at very short time differences, which obscures the true underlying decay structure. This behavior is in stark contrast to the physical expectation of an exponential decay and is largely caused by the misclassification of unrelated or background events as BiPo212.

When correctly and incorrectly classified events are plotted separately (Figure 6.35), the underlying dynamics become more apparent. The correctly classified events (Figure 6.35a) exhibit a clear exponential decay trend that closely follows the expected behavior of true $^{212}\text{BiPo}$ coincidences. This indicates that the model has successfully learned to identify a significant portion of the physical signal, even within the challenging context of overlapping chains.

In contrast, the decay curve of incorrectly predicted events (Figure 6.35b) shows a separate and unrelated exponential structure. While this decay pattern is not consistent with the ^{212}Po half-life, it suggests that the model may also be detecting other temporal correlations, possibly from the $^{214}\text{BiPo}$ chain or background, that are not relevant to $^{212}\text{BiPo}$.

Although these misclassified events are numerous, they primarily cluster at larger decay times and could be effectively removed by applying a cut on the maximum allowed BiPo time difference. This filtering would enhance signal purity by eliminating spurious coincidences, thereby improving the utility of the BiLSTM with Attention model for decay-based event selection.

Overall, the ability of the model to recover a clean exponential decay from the correctly predicted subset, despite significant background interference, highlights its robustness and effectiveness compared to previous models like the ANN and SimpleRNN, which struggled to produce physically interpretable results under similar conditions.

6.4.5 Separate BiLSTM Models for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$

To further improve classification performance and reduce inter-chain confusion, two separate BiLSTM models with attention were trained independently on the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ subsets. This approach mirrors the strategy adopted in earlier models (ANN, SimpleRNN, LSTM), where each model learns to specialize in identifying event classes from a specific decay chain.

Model Architecture and Training Strategy

The architecture used for both separate-chain BiLSTM models is identical to the one used for the full dataset. This consistency ensures a fair comparison between models trained on isolated chains and those trained jointly.

- **Input:** sequences of 10 events, each with 6 engineered features (energy, r , z , ϕ , time difference, and relative distance).
- **BiLSTM layers:** A bidirectional LSTM layer with 64 units followed by an attention mechanism and a second bidirectional LSTM layer with 32 units, both using the tanh activation.
- **Dense layers:** A `TimeDistributed` dense layer with 64 ReLU-activated neurons and a final softmax layer for multi-class classification.

The models were trained for a maximum of 200 epochs with a batch size of 256. Early stopping and `ReduceLROnPlateau` callbacks were used to optimize convergence and prevent overfitting. The training-validation split followed the same 80/20 strategy applied to all prior models, with `random_state = 0` to ensure reproducibility.

The $^{214}\text{BiPo}$ model stopped training at epoch 92, restoring the best weights from epoch 82. The $^{212}\text{BiPo}$ model stopped at epoch 82, with the best performance at epoch 72.

Given that the model architecture and training strategy remain consistent with the full dataset model, and the training curves follow a similar trajectory, loss and accuracy plots are not shown again here but are included in Appendix F.19 and F.21 for completeness.

$^{214}\text{BiPo}$ Model Evaluation

Performance Metrics The BiLSTM model with attention trained specifically on the $^{214}\text{BiPo}$ chain achieved an overall accuracy of 94% on the test set, making it the most accurate model among all those presented thus far for this subset. Table 6.10 presents the class-wise evaluation metrics including precision (efficiency), recall (TPR), and F1-score.

Table 6.10: Evaluation metrics for the BiLSTM model trained on the $^{214}\text{BiPo}$ chain. The model demonstrates excellent performance across all classes, particularly for ^{214}Po , with near-perfect classification.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.96	0.83	0.89
^{214}Bi (1)	0.97	0.98	0.97
^{214}Po (2)	1.00	1.00	1.00
External Background (3)	0.85	0.95	0.90

Classification of ^{214}Po is nearly flawless, with a TPR of 100% and efficiency of 100%, indicating that the model almost never misses a true Po event nor mislabels another event as Po. The performance

for ^{214}Bi is also remarkably strong, with a TPR of 98% and efficiency of 97%, a significant improvement compared to earlier models. Internal and external backgrounds are also well captured, with high precision and recall values.

Breakdown of misclassifications shows:

- ^{214}Bi events are occasionally misclassified as internal (1.48%) or external background (1.42%), though these rates are minor.
- ^{214}Po shows minimal leakage into background or Bi classes, with a false classification rate of under 0.5%.

Figure 6.36 shows the confusion matrix for the model. The matrix exhibits a strong diagonal across all classes, especially for ^{214}Po , indicating high classification confidence. The classification of Bi events also shows a well-defined concentration along the diagonal, indicating significantly reduced misclassification compared to previous models.

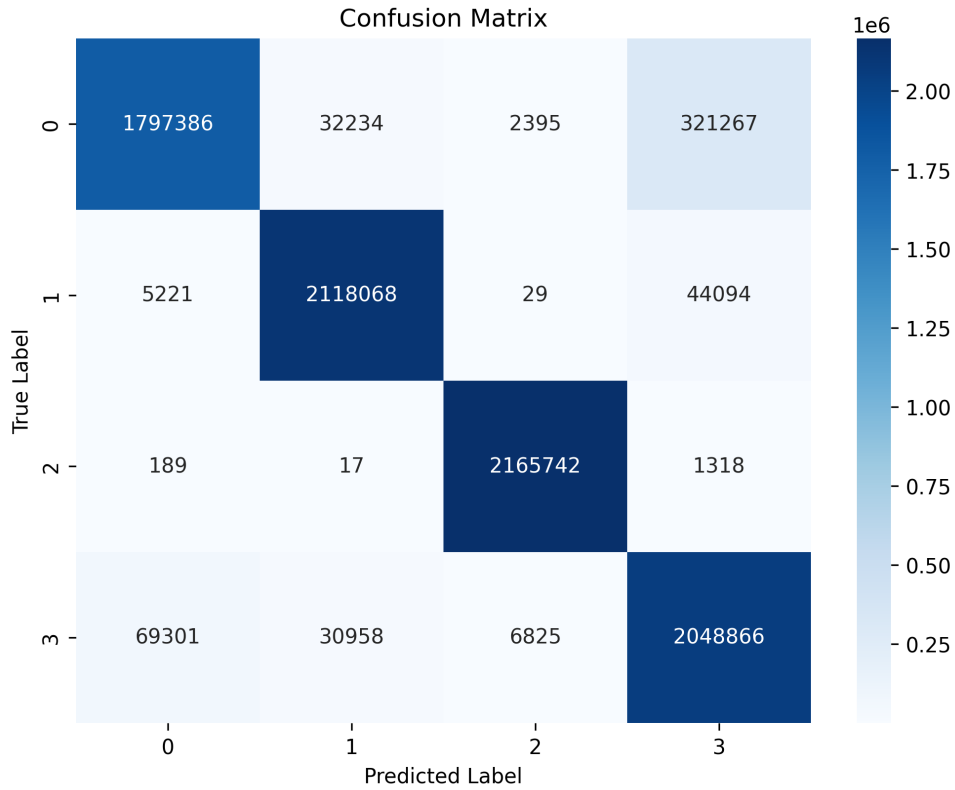


Figure 6.36: Confusion matrix of the BiLSTM model with attention trained on the $^{214}\text{BiPo}$ chain. The matrix shows highly focused diagonal structures for all classes, particularly ^{214}Po and ^{214}Bi . **This indicates that the model has effectively learned to separate signal from background with minimal confusion.**

Energy Distributions and Decay Curve The BiLSTM model with attention trained specifically on the $^{214}\text{BiPo}$ chain displays the expected physical behavior across both energy and timing domains. As seen in Figure 6.37, the ^{214}Bi energy spectrum exhibits the continuous shape typical of a beta decay, with a smooth rise and fall over the energy range. There are no abrupt discontinuities or gaps, suggesting the model captures the full spread of beta emission energies associated with ^{214}Bi . The predicted distribution aligns almost perfectly with the true distribution.

In Figure 6.38, the ^{214}Po energy distribution forms a sharp, monoenergetic peak centered near 0.78 MeV, consistent with the alpha decay signature. The symmetry and narrowness of the peak reinforce the model's precision in identifying ^{214}Po events. The predicted distribution matches the true distribution with excellent agreement.

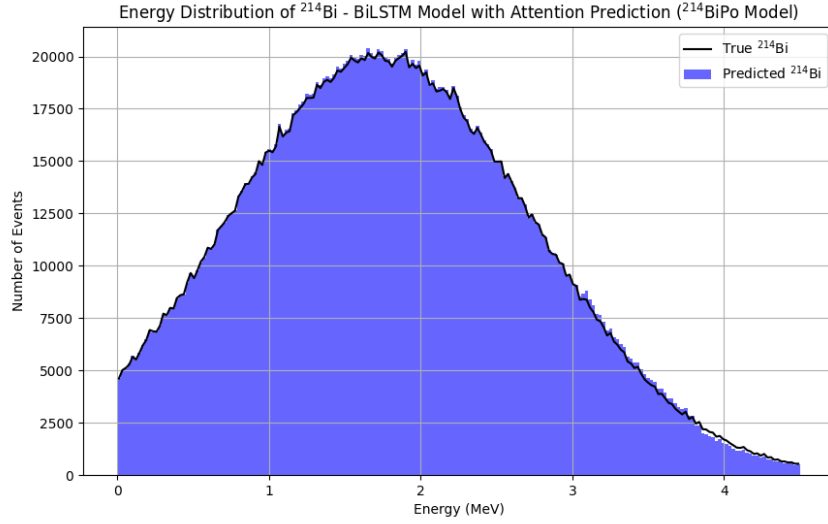


Figure 6.37: Energy distribution of events classified as ^{214}Bi by the BiLSTM model with attention, trained specifically on the $^{214}\text{BiPo}$ chain. The predicted spectrum closely follows the true beta decay shape, covering the full energy range without discontinuities.

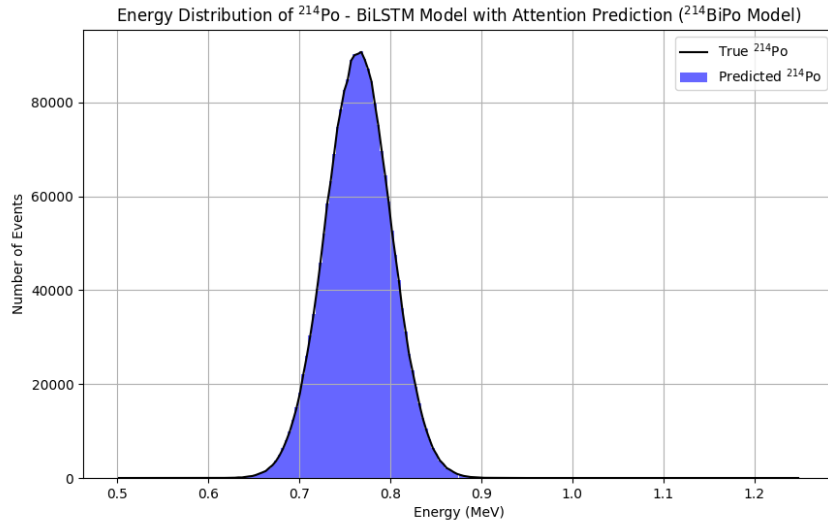


Figure 6.38: Energy distribution of events classified as ^{214}Po by the BiLSTM model with attention, trained specifically on the $^{214}\text{BiPo}$ chain. A sharp monoenergetic peak appears around 0.78 MeV, closely matching the true alpha spectrum and demonstrating high classification accuracy.

The decay time distribution in Figure 6.39 further confirms that the model has effectively learned the temporal pattern of true BiPo decays. The exponential drop in event counts with increasing time delay follows the theoretical expectation for $^{214}\text{BiPo}$ sequences. The absence of an initial spike or non-physical tail supports the model's robustness in distinguishing signal from background.

Together, these results demonstrate that the BiLSTM with attention not only correctly classifies ^{214}Bi and ^{214}Po based on energy signatures but also captures their sequential timing structure, leading to a consistent and physically meaningful event reconstruction.

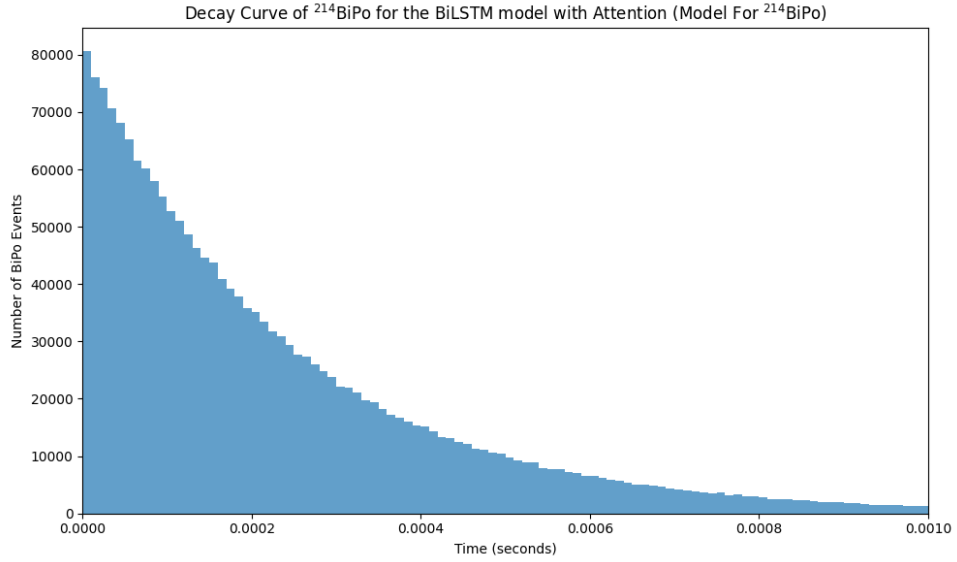


Figure 6.39: Decay time distribution for $^{214}\text{BiPo}$ events predicted by the BiLSTM model with attention. The curve shows a clean exponential trend without artifacts, indicating the model has learned the correct temporal relationship between decays in the BiPo214 chain.

$^{212}\text{BiPo}$ Model Evaluation

Performance Metrics The BiLSTM model with attention trained on the $^{212}\text{BiPo}$ chain achieved outstanding classification performance, reaching an overall accuracy of 94% on the test set. Table 6.11 summarizes the key performance metrics across all classes.

Table 6.11: Class-wise evaluation metrics for the BiLSTM model with attention trained on the $^{212}\text{BiPo}$ chain. The model achieves strong precision and recall across all classes, with particularly high performance for the alpha-emitting Po class.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.96	0.84	0.90
External Background (3)	0.85	0.95	0.90
^{212}Bi (4)	0.97	0.98	0.97
^{212}Po (5)	0.99	1.00	0.99

As with the $^{214}\text{BiPo}$ model, the classification of Po events is nearly perfect. The model identifies ^{212}Po events with a recall of 100% and a precision of 99%, while ^{212}Bi achieves 98% recall and 97% precision. These results reflect the model's capacity to leverage both temporal and contextual dependencies through attention to sharply distinguish alpha and beta decays.

Background classes also show substantial improvements. Internal background (class 0) maintains high precision (96%) and a strong TPR (84%), while external background (class 3) achieves a TPR of 95% and precision of 85%, suggesting robust generalization across all classes.

A breakdown of misclassifications indicates:

- ^{212}Bi is primarily misclassified as external background (1.64%) and internal background (1.06%), with minimal leakage into ^{212}Po (0.37%).
- ^{212}Po is highly separable, with minor misclassifications into external background (0.35%) and ^{212}Bi (0.25%).

The confusion matrix (Figure 6.40) reflects this performance. The diagonal is sharp and well-defined for all classes, particularly for ^{212}Po , indicating minimal misclassification. The Bi and background classes also exhibit significantly tighter clustering than in previous models, confirming the attention-enhanced model's superior ability to isolate decay signatures.

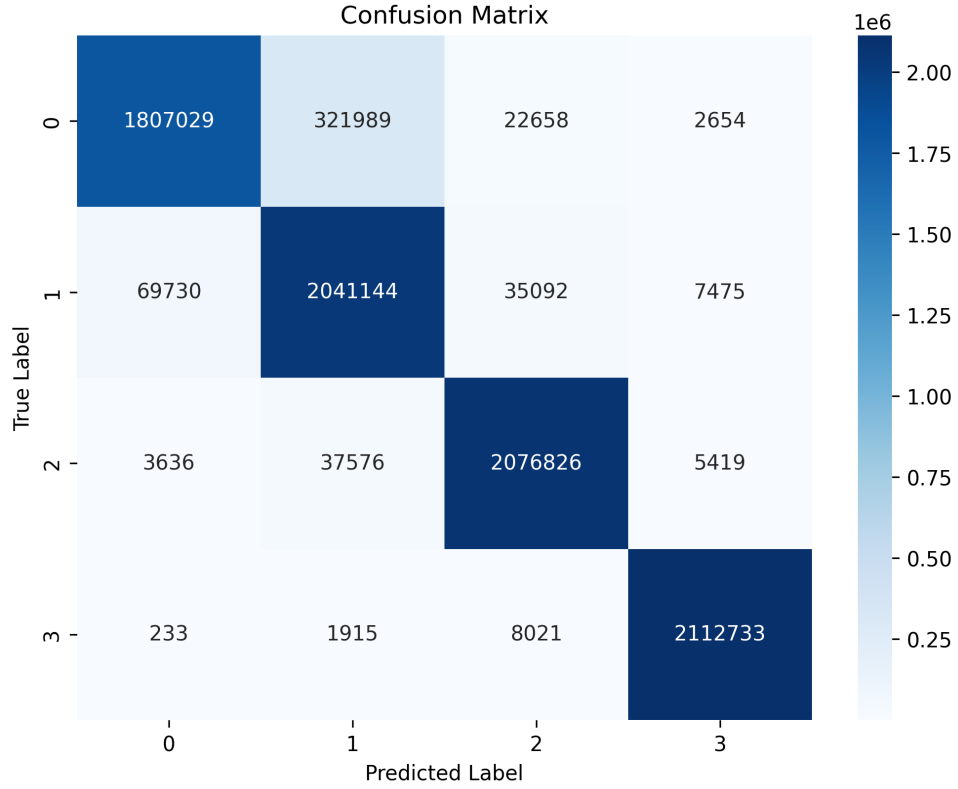


Figure 6.40: Confusion matrix of the BiLSTM model with attention trained on the $^{212}\text{BiPo}$ chain. The matrix shows exceptionally strong diagonal dominance, especially for ^{212}Po . Bi and background events also exhibit well-contained clustering, indicating precise classification and minimal overlap between classes.

Energy Distributions and Decay Curve The beta spectrum of ^{212}Bi in Figure 6.41 exhibits a continuous shape that aligns with the theoretical beta decay profile, with no abrupt cutoffs or missing regions. The predicted distribution matches the true distribution closely across the full energy range. Likewise, the alpha decay of ^{212}Po produces a sharp and symmetric monoenergetic peak in Figure 6.42, centered around 0.9 MeV. This peak matches the known energy of ^{212}Po alpha emissions, and the predicted distribution aligns very well with the true spectrum, indicating that the model learns the energy signature very effectively.

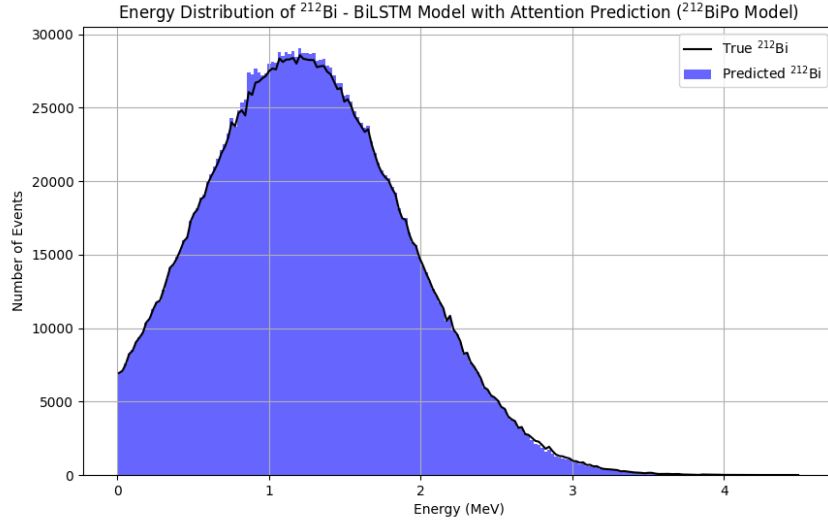


Figure 6.41: Energy distribution of events classified as ^{212}Bi by the BiLSTM model trained for the $^{212}\text{BiPo}$ chain. The predicted spectrum smoothly follows the expected beta decay shape and shows excellent agreement with the true distribution.

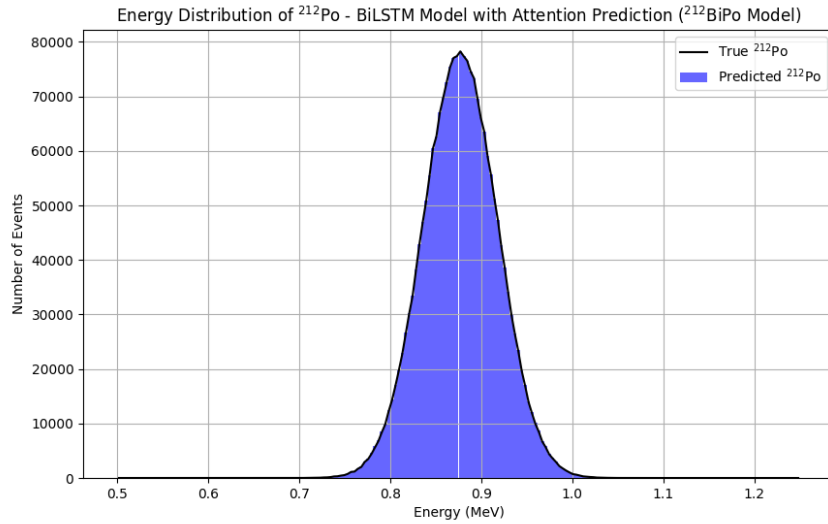


Figure 6.42: Energy distribution of events classified as ^{212}Po by the BiLSTM model trained for the $^{212}\text{BiPo}$ chain. A sharp monoenergetic peak around 0.9 MeV closely matches the true alpha spectrum of ^{212}Po , demonstrating the model's strong classification performance.

The decay time distribution in Figure 6.43 displays a clear exponential pattern without noise or unphysical features. Compared to other models, this decay curve is the most well-resolved, showing strong agreement with the expected short half-life of ^{212}Po decays. The absence of background contamination or misclassified events in this zoomed decay window highlights that the model captures the temporal dynamics of true BiPo212 events precisely.

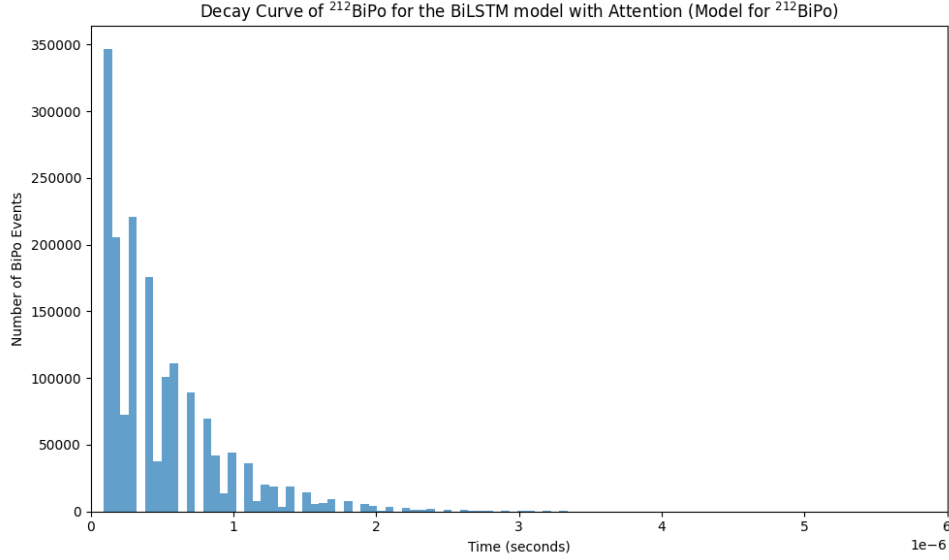


Figure 6.43: Decay curve of $^{212}\text{BiPo}$ events predicted by the BiLSTM model trained for BiPo212. The exponential shape is consistent with the short half-life of ^{212}Po . The absence of distortion or misclassified background at longer time scales reflects the model’s ability to learn the underlying temporal structure of true BiPo decays.

6.4.6 Final Classification with Higher Probability Selection

The model combination strategy applied here follows the same principle used in previous architectures: each event is passed through both specialized BiLSTM models (one for $^{214}\text{BiPo}$ and one for $^{212}\text{BiPo}$), and the final classification is determined by selecting the class with the highest softmax probability. This approach aims to leverage the chain-specific expertise of each model to improve classification fidelity.

Performance Metrics

Despite achieving a moderately high overall accuracy of 78%, this performance is still significantly lower than the model trained on the full dataset, which attained a test accuracy of 90% with the BiLSTM model and attention mechanism. Although the combined model performs better than the ANN, SimpleRNN, and LSTM combination strategies, it fails to retain the sharp class separability exhibited by the full dataset model. This discrepancy suggests that critical temporal correlations may be disrupted when models are specialized separately, and the integration via probability selection fails to restore the original context.

In addition, the combining mechanism introduces ambiguity in handling overlapping event types, particularly in cases where both models produce confident but conflicting predictions. This limitation, combined with the lower accuracy and increased complexity, renders further investigation into this approach unjustified within the scope of this work.

Table 6.12 shows the class-wise performance metrics.

Table 6.12: Classification metrics for the combined BiLSTM model with attention using higher-probability selection. Despite improved performance relative to other combination strategies, the model still falls significantly short of the full-model accuracy.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.89	0.84	0.86
^{214}Bi (1)	0.69	0.79	0.74
^{214}Po (2)	0.75	0.81	0.78
External Background (3)	0.77	0.95	0.85
^{212}Bi (4)	0.77	0.59	0.67
^{212}Po (5)	0.82	0.66	0.73

An analysis of misclassifications supports this conclusion:

- ^{214}Bi retains a recall of 79%, but 30% of its predictions are misclassified as ^{212}Bi .

- ^{214}Po achieves a recall of 81%, yet 24% of its predictions are incorrectly labeled as ^{212}Po .
- ^{212}Po , which is typically well-classified in all full-model architectures, experiences a performance drop with only 66% recall and substantial leakage into ^{214}Po .

The confusion matrix, shown in Figure 6.44, further illustrates this loss of resolution, with visible dispersion along off-diagonal elements—particularly among the Bi and Po classes.

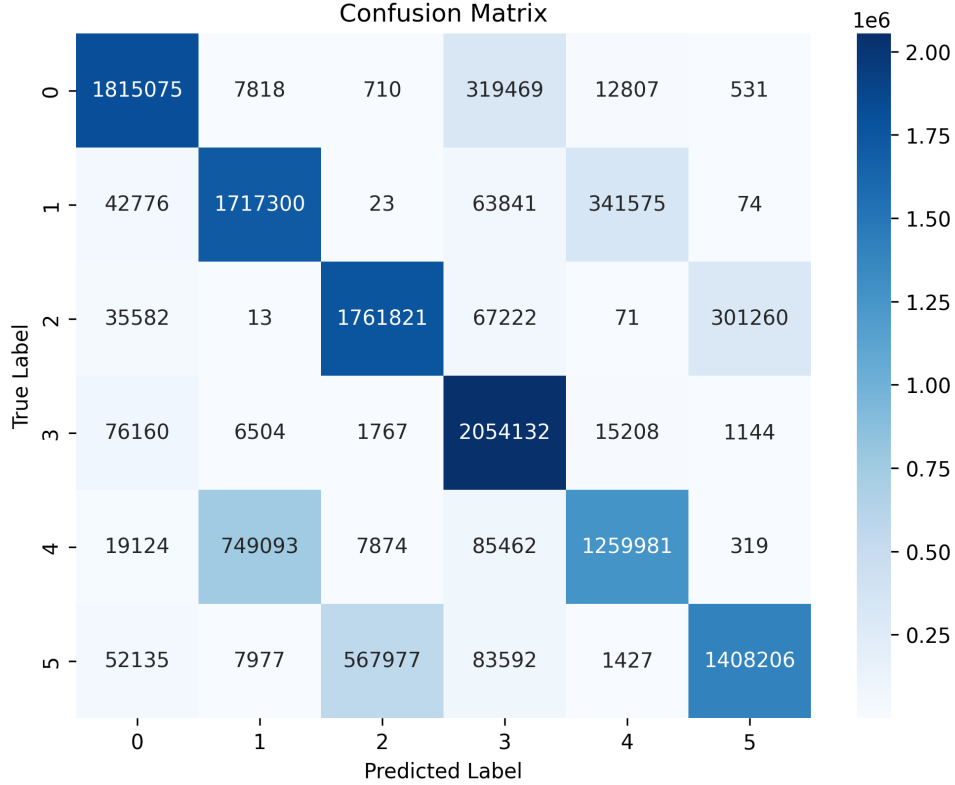


Figure 6.44: Confusion matrix for the combined BiLSTM model with attention. The off-diagonal diffusion, especially between $^{214}\text{Bi}/^{212}\text{Bi}$ and $^{214}\text{Po}/^{212}\text{Po}$, reflects the inability of the combined model to recover the precision achieved by the unified full-dataset model.

6.5 Post-Processing for Pair Classification

While the preceding sections focused on classifying individual events using sequence-based models, the ultimate objective of this analysis is the identification of BiPo pairs, specifically, the detection of a β decay (Bi) followed by a correlated α decay (Po) within a short time window. These BiPo coincidences represent the true physical signatures of radioactive decay chains and are essential for background suppression in OSIRIS.

Pair classification is a crucial post-processing step that transforms per-event predictions into physics-relevant decay signatures. By leveraging the outputs of the classification model, we can scan through the predicted labels and construct BiPo pairs based on temporal proximity and decay type.

This step is carried out exclusively using the BiLSTM model with attention trained on the full dataset. Among all the models, only this one achieves sufficiently high classification performance and yields a balanced and statistically meaningful number of both Bi and Po events. This makes it the only realistic candidate for reconstructing BiPo coincidences in a mixed-chain environment, which is representative of real experimental data where both $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains are present.

The following sections detail the methodology used to identify candidate BiPo pairs and evaluate their physical consistency through energy spectra and decay time distributions.

6.5.1 BiPo Pair Enforcement Strategy

The output of the BiLSTM with attention model consists of sequence-level classifications for each event, identifying them as either background or signal classes. However, these predictions are generated independently at each time step and do not inherently account for the temporal structure of radioactive

decay chains. Specifically, the model may predict isolated ^{214}Bi or ^{212}Po events without a corresponding partner, which violates the expected BiPo coincidence structure.

To ensure physical consistency in post-processing, a pair enforcement algorithm is applied to the predicted labels. This algorithm traverses the predicted label sequences and enforces only valid BiPo pairs: ^{214}Bi followed by ^{214}Po (labels $1 \rightarrow 2$), and ^{212}Bi followed by ^{212}Po (labels $4 \rightarrow 5$). Any unpaired or out-of-order predictions are reassigned to the background class (label 0). This step acts as a temporal filter that strengthens the integrity of the classification results by aligning them with the physics of decay sequences.

Algorithm Overview:

- Iterate through each predicted sequence.
- If a Bi candidate (label 1 or 4) is immediately followed by a Po candidate (label 2 or 5), the pair is retained.
- If a Po candidate appears without a preceding Bi, it is reassigned to background.
- Similarly, Bi candidates without a Po following are also reassigned to background.
- All other labels (e.g., internal or external background) are left unchanged.

This correction mechanism ensures that only time-coherent decay sequences contribute to the final BiPo tagging, significantly reducing false positives and improving the fidelity of event-level predictions. It also aligns the output more closely with the kind of event patterns expected in real detector environments.

Performance Metrics

After enforcing the BiPo pair structure on the predicted labels, the classification performance is re-evaluated. As described in the previous section, only valid Bi $\rightarrow$ Po pairs ($1\rightarrow 2$ for $^{214}\text{BiPo}$ and $4\rightarrow 5$ for $^{212}\text{BiPo}$) are retained, while unpaired Bi or Po candidates are reassigned to the background class (label 0). This approach ensures physical consistency but comes with trade-offs in classification accuracy.

The resulting model achieves an overall accuracy of 85%, which is slightly lower than the original BiLSTM model trained on the full dataset (91%), but still significantly better than all other combined classification models. Table 6.13 presents the class-wise precision, recall, and F1-score after pair enforcement.

Table 6.13: Performance metrics after applying BiPo pair enforcement. Precision and recall values for Bi and Po classes now reflect valid pair classification rather than independent per-event tagging.

Class	Precision (Efficiency)	Recall (TPR)	F1-score
Internal Background (0)	0.64	0.85	0.73
^{214}Bi (1)	0.94	0.85	0.89
^{214}Po (2)	0.94	0.85	0.89
External Background (3)	0.84	0.94	0.89
^{212}Bi (4)	0.94	0.82	0.88
^{212}Po (5)	0.94	0.82	0.88

The pair enforcement procedure results in perfectly balanced counts for each Bi and its corresponding Po class, such that the number of ^{214}Bi events (class 1) is identical to the number of ^{214}Po events (class 2), and similarly for ^{212}Bi and ^{212}Po (classes 4 and 5). This symmetry validates the transformation from event-wise to pair-wise classification and is a critical feature for physically meaningful decay tagging.

Despite the strong performance for Bi and Po classes, internal background (class 0) shows degraded performance, with a low precision of 64%. This is expected, as all unmatched or invalidly ordered Bi and Po predictions are reassigned to this class, inflating the background population with previously signal-like events. Nevertheless, this reclassification is essential to uphold the physics constraints of true BiPo decays.

A breakdown of misclassification behavior shows:

- ^{214}Bi events are predicted with 94% precision and 85% recall, with most confusion occurring with ^{212}Bi (5.94%).
- ^{214}Po shows nearly identical metrics and confusions, mirroring its Bi counterpart due to the enforced pairing.

- ^{212}Bi and ^{212}Po also exhibit tight agreement, with precision values above 94% and misclassification primarily into each other.
- **Internal background** absorbs the remaining unpaired or improperly ordered events, leading to a drop in its classification reliability.

The confusion matrix in Figure 6.45 reflects these trends. While not as sharp and diagonally dominant as the original BiLSTM model, it still shows a strong diagonal for all Bi and Po classes, confirming the preservation of classification strength in valid pairs. Off-diagonal spread is more pronounced for the background class, which absorbs misaligned predictions during the enforcement step.

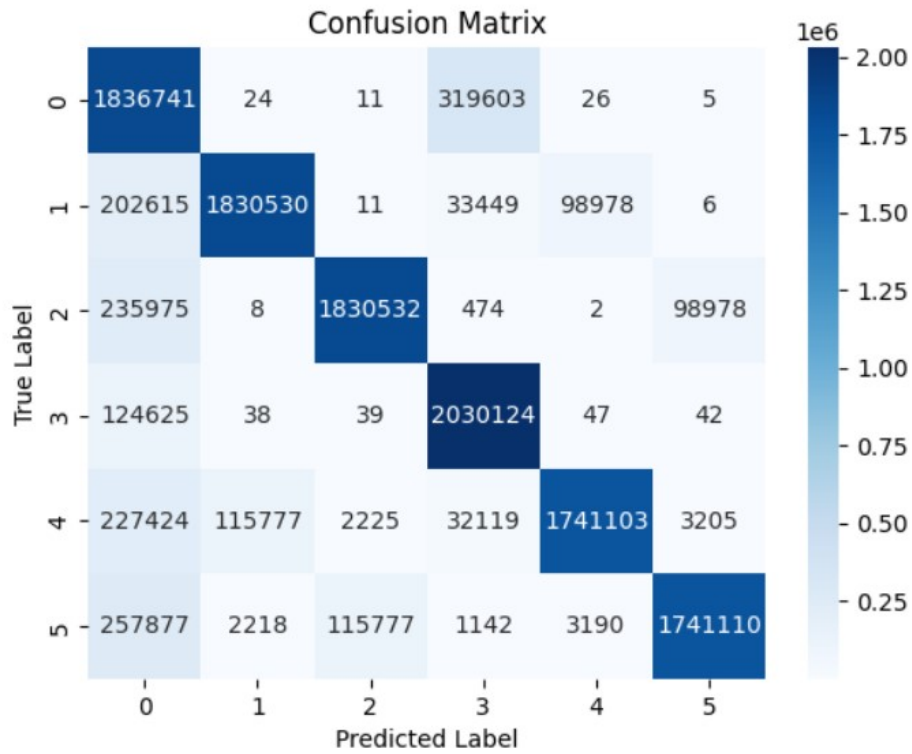


Figure 6.45: Confusion matrix after BiPo pair enforcement. While the matrix remains strongly diagonal for valid BiPo pairs, the background class (0) exhibits greater dispersion due to reassignment of unpaired or improperly ordered signal events. **Nevertheless, the structure confirms the model's ability to preserve physical decay relationships under pair classification.**

6.5.2 Energy Distributions

The Bi energy distributions post pair-enforcement (Figures 6.47 and 6.46) remain consistent with the expected shapes for β decays. Both ^{212}Bi and ^{214}Bi exhibit continuous spectra with asymmetric peaks and long high-energy tails, which are characteristic features of allowed β transitions.

Crucially, these distributions are preserved even after enforcing strict BiPo pairing, indicating that the pair enforcement algorithm does not distort the model's physical predictions. In fact, the previously observed anomalous structure in the peak of the ^{214}Bi distribution (Figure 6.28) has disappeared, resulting in a smoother and more natural spectrum. This suggests that the enforcement step not only preserves meaningful physics but may also improve it by eliminating ambiguous or misclassified events. In general, the predicted spectra follow the true beta shapes closely, although they contain slightly fewer events than the true distributions.

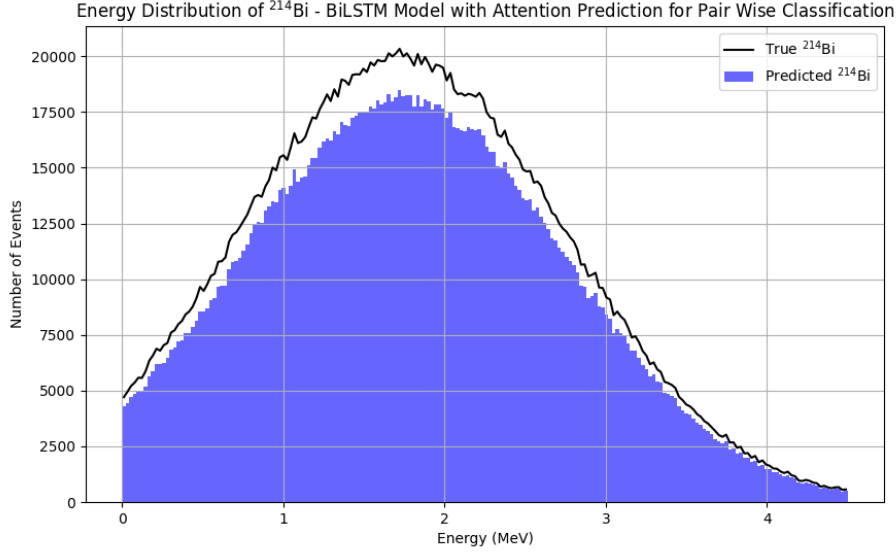


Figure 6.46: Energy distribution of ^{214}Bi after BiPo pair enforcement. The smooth, bell-like structure reflects the expected β energy spectrum and aligns well with the true distribution. The artificial peak previously seen in the full-model classification (Figure 6.28) is no longer present, indicating a more physically accurate classification.

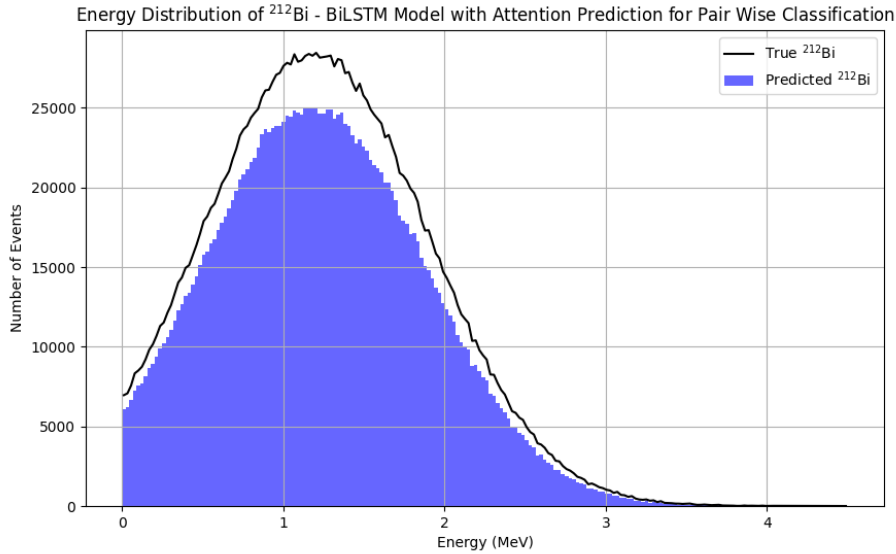


Figure 6.47: Energy distribution of ^{212}Bi after BiPo pair enforcement. The predicted spectrum retains its continuous β -like shape and agrees well with the true distribution, though with slightly fewer events.

To assess whether the BiPo pair enforcement procedure preserves the underlying physical characteristics of alpha decays, the energy spectra of ^{214}Po and ^{212}Po are examined and fitted with Gaussian distributions. Since alpha decays are monoenergetic, their energy spectra are expected to resemble narrow Gaussian peaks, in contrast to the broader, continuous beta spectra of Bi isotopes, which are left unfitted due to their intrinsic shape.

The goodness of the fits is quantified using the coefficient of determination (R^2), a statistical measure that evaluates how well the Gaussian model describes the observed energy distributions. It is calculated as:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \quad (6.1)$$

where $SS_{\text{res}} = \sum_i (y_i - f(x_i))^2$ is the sum of squared residuals between the observed data points y_i and the fitted model predictions $f(x_i)$, and $SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$ is the total sum of squares relative to the

mean of the observed data $\bar{y}$. An R^2 value close to 1 indicates that the model explains most of the variability in the data, reflecting a strong agreement between the fitted curve and the measurements.

In the context of alpha decays, where the true physical expectation is a narrow, monoenergetic peak, a Gaussian fit with a high R^2 value confirms that the observed energy distribution retains the expected sharp spectral shape. Therefore, a high R^2 not only quantifies the fit quality but also validates that the post-processing and BiPo pair enforcement procedures preserve the fundamental physical characteristics of the Po decays without introducing significant distortions.

Figure 6.48 displays the Gaussian fit for ^{214}Po . The fitted parameters are:

$$\mu_{\text{fit}} = 0.766 \text{ MeV}, \quad \sigma_{\text{fit}} = 0.034 \text{ MeV}, \quad R^2 = 0.984.$$

These are compared against the ground truth values obtained directly from the labeled dataset:

$$\mu_{\text{true}} = 0.765 \text{ MeV}, \quad \sigma_{\text{true}} = 0.036 \text{ MeV}.$$

The absolute deviations are:

$$|\mu_{\text{fit}} - \mu_{\text{true}}| = 0.001 \text{ MeV}, \quad |\sigma_{\text{fit}} - \sigma_{\text{true}}| = 0.002 \text{ MeV},$$

indicating excellent agreement with the expected distribution.

Similarly, the energy spectrum for ^{212}Po is shown in Figure 6.49, with fitted parameters:

$$\mu_{\text{fit}} = 0.876 \text{ MeV}, \quad \sigma_{\text{fit}} = 0.039 \text{ MeV}, \quad R^2 = 0.975,$$

compared to the true values:

$$\mu_{\text{true}} = 0.877 \text{ MeV}, \quad \sigma_{\text{true}} = 0.041 \text{ MeV},$$

yielding deviations of:

$$|\mu_{\text{fit}} - \mu_{\text{true}}| = 0.001 \text{ MeV}, \quad |\sigma_{\text{fit}} - \sigma_{\text{true}}| = 0.002 \text{ MeV}.$$

These results confirm that the model preserves the monoenergetic structure of Po decays with high precision even after pair enforcement. The preservation of peak positions and widths within ~ 2 keV error margins, together with R^2 values above 0.97, highlights the model's capacity to retain fine-grained physical details and ensures the viability of BiPo tagging based on energy signatures. Furthermore, the lack of distortion in the fitted distributions demonstrates that the post-processing step does not degrade the spectral integrity of the classified events.

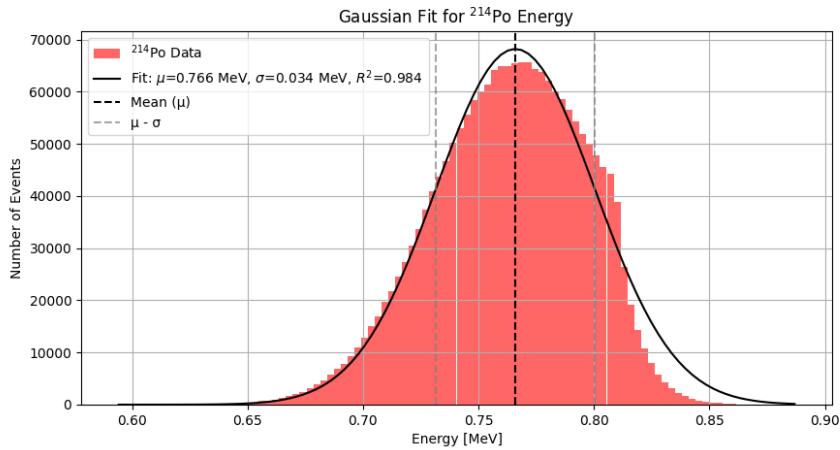


Figure 6.48: Gaussian fit for ^{214}Po energy distribution after BiPo pair enforcement. The model reproduces the monoenergetic alpha peak at 0.766 MeV with $\sigma = 0.034$ MeV and $R^2 = 0.984$, matching closely with the reference fit ($\mu = 0.765$ MeV, $\sigma = 0.036$ MeV).

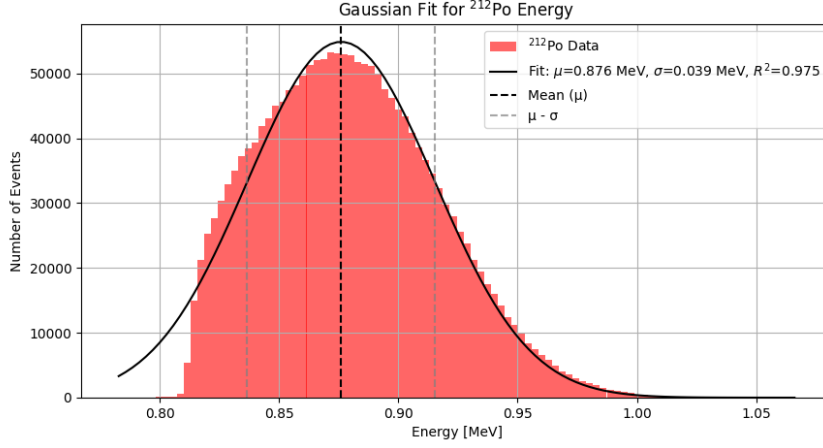


Figure 6.49: Gaussian fit for ^{212}Po energy distribution after BiPo pair enforcement. The fitted peak at 0.876 MeV with $\sigma = 0.039$ MeV and $R^2 = 0.975$ is in excellent agreement with the expected values of 0.877 MeV and 0.041 MeV from the dataset.

6.5.3 Decay Curve

Decay Curve Fitting

To investigate whether the BiPo pair identification preserves the expected temporal behavior of decaying isotopes, the decay time distributions of $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ events are fitted with exponential functions:

$$N(t) = N_0 e^{-t/\tau}$$

These fits allow extraction of the mean lifetimes τ , which are compared to the true dataset values. The quality of the fits is quantified by the coefficient of determination (R^2), providing a measure of how well the exponential model describes the data.

Figure 6.50 shows the fitted decay curve for $^{214}\text{BiPo}$ events. For physical consistency, the first bin is ignored due to unphysical pile-up, and a time threshold of 1 ms is applied to exclude extreme outliers. The extracted mean lifetime is:

$$\tau_{\text{fit}} = 2.372 \times 10^{-4} \text{ s},$$

which corresponds to a half-life of:

$$T_{1/2} = \tau \ln(2) \approx 164 \text{ } \mu\text{s}.$$

The dataset reference value for ^{214}Po is $\tau_{\text{true}} = 2.37 \times 10^{-4}$ s. The relative error between the fit and the true value is:

$$\frac{|\tau_{\text{fit}} - \tau_{\text{true}}|}{\tau_{\text{true}}} = \frac{|2.372 \times 10^{-4} - 2.370 \times 10^{-4}|}{2.370 \times 10^{-4}} \approx 0.08\%.$$

The R^2 value for the fit is 0.797, indicating a moderate goodness of fit. Although slight deviations exist, the exponential trend is well captured, confirming that the BiPo pairing procedure retains the expected temporal behavior for ^{214}Po decay.

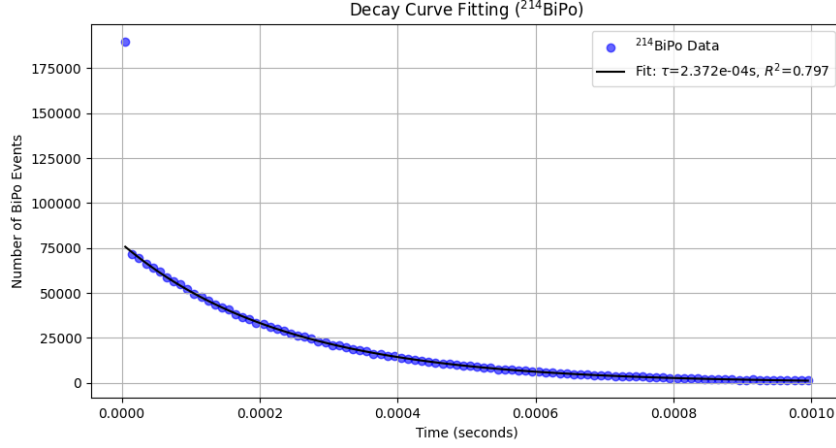


Figure 6.50: Decay curve fitting for $^{214}\text{BiPo}$ events after pair classification. The fit yields a mean lifetime $\tau = 2.372 \times 10^{-4}$ s with $R^2 = 0.797$. Data points beyond 1 ms were excluded, and the first bin was ignored to mitigate unphysical distortions.

In contrast, Figure 6.51 shows the fitted decay curve for $^{212}\text{BiPo}$ events. A stricter time threshold of 10^{-5} s is applied, cutting off a larger fraction of events to prevent distortion from misclassified long-lived outliers. The extracted lifetime is:

$$\tau_{\text{fit}} = 4.350 \times 10^{-7} \text{ s},$$

compared to the dataset value:

$$\tau_{\text{true}} = 4.288 \times 10^{-7} \text{ s}.$$

The corresponding relative error is:

$$\frac{|\tau_{\text{fit}} - \tau_{\text{true}}|}{\tau_{\text{true}}} = \frac{|4.350 \times 10^{-7} - 4.288 \times 10^{-7}|}{4.288 \times 10^{-7}} \approx 1.45\%.$$

The R^2 value is 1.000, suggesting an almost perfect fit. While an exact $R^2 = 1$ is unusual in experimental data, it is plausible here because the distortions mainly occur at larger timescales that have been excluded by the applied threshold. In the short time window relevant for ^{212}Po decay (microsecond scale), the data closely follows the expected exponential profile without significant background contamination.

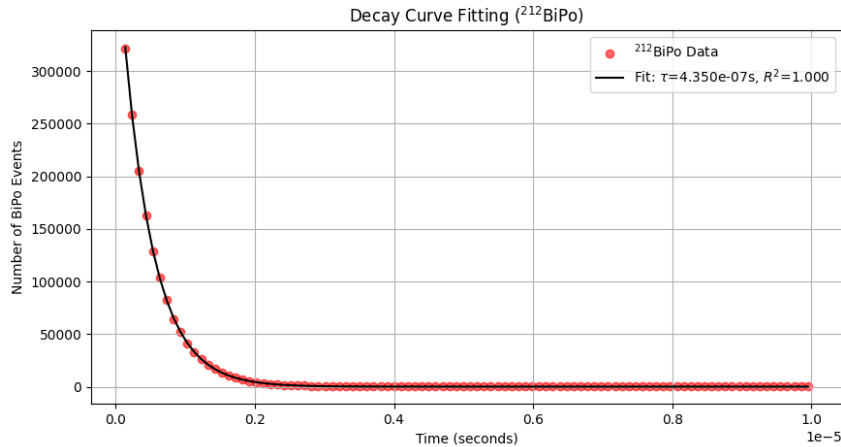


Figure 6.51: Decay curve fitting for $^{212}\text{BiPo}$ events after pair classification. The fit yields a mean lifetime $\tau = 4.350 \times 10^{-7}$ s with $R^2 = 1.000$. Data points beyond 10^{-5} s were excluded to remove distortions from misclassified long-lived events.

In summary, the decay time fits confirm that the BiPo pairing procedure successfully preserves the

correct temporal decay signatures. Minor deviations for $^{214}\text{BiPo}$ are observed, but $^{212}\text{BiPo}$ decay is captured with exceptional fidelity after applying appropriate time selection cuts.

6.6 Summary

This chapter presented a suite of RNN models for classifying time-ordered detector events, with a focus on identifying BiPo decays in the OSIRIS experiment. Beginning with SimpleRNNs and LSTMs and culminating in a BiLSTM enhanced with an attention mechanism, the progression of architectures demonstrated increasing ability to capture temporal dependencies in the data.

The BiLSTM with Attention achieved the highest classification accuracy (91%) and delivered strong performance across all event classes, particularly for Po isotopes. Separate models trained on individual decay chains provided useful insight but underperformed when combined, reinforcing the importance of unified models in realistic scenarios.

A final post-processing step translated per-event predictions into BiPo decay pairings, yielding symmetric Bi and Po distributions and decay curves that match expected physics. Though this step reduced overall accuracy, primarily due to reassigning unpaired events as background, it enabled the recovery of interpretable, time-correlated decay signatures.

A detailed comparison of all models developed in this thesis, including ANN, Decision Tree, and RNN-based approaches, will be provided in the following chapter.

Chapter 7

Model Comparison and Discussion

Having developed and evaluated a variety of machine learning models for BiPo event classification including ANN, RNNs, and Decision Trees, this chapter serves to synthesize their outcomes into a unified discussion. While the previous chapters examined each model independently, here we draw comparisons across architectures in terms of classification accuracy, performance on Bi and Po classes, and behavior on energy and decay signatures.

This chapter aims to assess how well each approach captures the underlying physics of the problem and to identify the strengths and limitations of different strategies. Additionally, traditional cut-based methods are revisited and contrasted with machine learning outcomes using the best-performing model as reference.

The chapter concludes with a discussion of interpretability, computational trade-offs, and broader applicability of these models, followed by limitations of the current approach and potential directions for future work.

7.1 Comparison of the Machine Learning Models

This section provides a high-level comparison of the machine learning models developed in this thesis, including ANN, Decision Trees, SimpleRNN, LSTM, and BiLSTM with Attention. The focus lies on evaluating classification accuracy and class-wise performance across models, particularly for the key signal isotopes (^{214}Bi , ^{214}Po , ^{212}Bi , and ^{212}Po).

The comparison includes models trained on the full dataset as well as those trained separately on $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains. Models that perform poorly, especially those based on combined predictions, are briefly discussed and ruled out as viable approaches.

A detailed discussion of model behavior and final conclusions, including a comparison to the traditional cut-based method, follows in subsequent sections.

7.1.1 Performance Summary

Table 7.1 presents a comparative summary of the models trained on the full dataset. These include the ANN, Decision Tree, SimpleRNN, LSTM, and the BiLSTM with Attention. Only full-dataset models are included here, as they represent the most realistic scenario in which both $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ decay chains are present simultaneously, consistent with the conditions in real-world detector data.

Models trained separately on individual decay chains were valuable for understanding classification behavior and class-specific learning. However, their combination via higher probability selection did not result in meaningful performance improvements and even led to reduced classification accuracy (e.g., 62-66%), making them unsuitable for practical application. As such, neither the separate-chain models nor their combined outputs are considered in this summary table.

Table 7.1: Classification performance summary for models trained on the full dataset. Metrics are reported separately for ^{214}Bi , ^{214}Po , ^{212}Bi , and ^{212}Po , including recall (R) and precision (P). The BiLSTM with Attention achieves the highest accuracy and most balanced class-wise performance. The pair-classified variant ensures consistent Bi–Po matching but shows slightly reduced overall accuracy due to stricter pair enforcement.

Model	Acc.	Bi214 R	Po214 R	Bi212 R	Po212 R	Bi214 P	Po214 P	Bi212 P	Po212 P
ANN (Full)	0.73	0.39	0.94	0.64	0.91	0.54	0.92	0.50	0.92
Decision Tree (Optimized)	0.71	0.39	0.94	0.63	0.90	0.54	0.92	0.49	0.91
SimpleRNN	0.73	0.38	0.94	0.64	0.91	0.55	0.92	0.50	0.92
LSTM	0.73	0.38	0.95	0.64	0.92	0.55	0.92	0.50	0.92
BiLSTM + Attention	0.91	0.89	0.94	0.90	0.93	0.90	0.93	0.88	0.93
BiLSTM + Attention (Pairs)	0.85	0.85	0.85	0.82	0.82	0.94	0.94	0.94	0.94

Among all evaluated models, the BiLSTM with Attention stands out with the highest overall accuracy of 91% and excellent recall and precision values for both Bi and Po events. This architecture’s ability to model bidirectional temporal dependencies and focus on salient features through attention proves especially effective in capturing the temporal patterns associated with BiPo decays.

In contrast, the ANN, Decision Tree, SimpleRNN, and LSTM models exhibit similar performance, with overall accuracies of 71–73%. These models succeed in capturing Po events due to their sharper energy and time distributions, but struggle with Bi classification, largely due to the broader beta spectrum and feature overlap with backgrounds.

The pair-classified BiLSTM model, which enforces Bi–Po temporal pairings during post-processing, maintains strong precision and recall for signal events, though the overall accuracy drops slightly to 85%. This is due in part to the reassignment of unmatched events (e.g., lone Bi or Po events) to the internal background class, which slightly distorts the overall distribution but leads to more physically interpretable results.

For completeness, it is worth noting that the models trained separately on $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains showed strong performance when evaluated on their respective decay sequences, achieving accuracies of 82–94% and particularly high recall for Po isotopes. However, due to their narrow scope and lack of realism for mixed-chain datasets, they serve mainly as supportive experiments rather than viable classification models for practical application.

7.1.2 Physics Interpretability and Model Suitability

Beyond raw classification accuracy, a key requirement for model deployment in a physics experiment like OSIRIS is the ability to reproduce physically interpretable observables—namely, clean energy spectra and meaningful decay time distributions for BiPo sequences. This section evaluates the trained models not only by how well they classify individual events, but by their success in reconstructing these underlying physics signatures.

Decay Time Distributions:

The decay curves for BiPo sequences offer a direct probe of whether the model has truly learned the temporal correlation between Bi and Po events. Both the ANN and Decision Tree models, despite achieving moderate classification accuracies, fail to produce exponential decay curves. Their outputs show an initial excess followed by a nearly flat or random distribution, indicative of widespread misclassification of background as Bi, diluting any true decay pattern. As such, these models are unable to capture the time structure of BiPo decays and are therefore unsuitable for applications requiring temporal consistency.

In contrast, the RNN-based models (SimpleRNN and LSTM) produce decay curves that align more closely with physical expectations. While these models share similar classification accuracy with the ANN and Decision Tree, they succeed in recovering the exponential decay of Po isotopes, particularly after zooming into the relevant time region for ^{212}Po . This demonstrates that recurrent architectures are better suited for tasks requiring temporal modeling, even when their class-level accuracy does not vastly exceed simpler models.

The BiLSTM with Attention model further improves upon this trend. Not only does it achieve the highest classification accuracy, but it also reproduces clean exponential decay curves for both $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ events. When the predictions are split into correctly and incorrectly classified pairs, the exponential behavior is almost perfectly recovered for the former, while the latter explain the initial spike and flattening seen in the total distributions. This highlights the model’s robustness. It captures both the correct time ordering and decay scale when its predictions are accurate, and its misclassifications are structurally distinct and identifiable.

Energy Spectra:

The ANN and Decision Tree models show sharp class boundaries in the energy spectra, especially for the Po isotopes. However, their Bi spectra often display artificial gaps or oversegmentation, particularly in the range between 0.7-1.0 MeV. This suggests reliance on hard energy thresholds, which fails to generalize in overlapping regions.

RNN-based models like the SimpleRNN and LSTM demonstrate smoother energy spectra, especially for the Bi classes. The Po peaks also have similar width as the original distribution, indicating more precise identification of alpha decays. However, Bi misclassification remains a challenge for these models as well, limiting their physics fidelity despite improved time modeling.

The BiLSTM with Attention model offers the most realistic energy distributions. Its Bi spectra are continuous and span the expected beta ranges without introducing artificial gaps or sharp boundaries. The Po spectra also show well-centered monoenergetic peaks with minimal background contamination. Importantly, the model does not impose hard boundaries in overlapping regions but rather learns to tolerate and resolve ambiguity, improving generalization.

Post-Processing and Final Model Choice for Deployment:

While the BiLSTM with Attention model achieves the best standalone performance, it still operates at the per-event level. In real-world applications, BiPo decays manifest as time-correlated event pairs. Hence, post-processing is essential to recover pair-level signatures.

The final post-processed model, based on BiLSTM with Attention and augmented with pair enforcement, yields slightly lower classification accuracy but preserves the integrity of energy spectra and decay time distributions. Moreover, it guarantees one-to-one Bi-Po matching, critical for reliable tagging in OSIRIS. This trade-off makes it the most suitable machine learning model for deployment.

On Separate Chain Models:

For completeness, separate models trained on the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains are briefly noted. While these models excel within their isolated subsets, they are unrealistic in practice. OSIRIS observes both chains simultaneously, and the attempt to merge predictions using higher probability selection significantly degraded overall accuracy and interpretability. As such, neither separate models nor their combined output are viable for operational use.

Conclusion:

In summary, while ANN and Decision Tree models provide a useful benchmark, they fail to capture temporal correlations. The RNN-based models add temporal depth but still struggle with Bi classification. The BiLSTM with Attention model stands out as the most capable, and after post-processing, emerges as the optimal candidate for BiPo pair identification in OSIRIS. This final model will be directly compared with the traditional cut-based method in the following section to assess the best overall strategy for BiPo tagging.

A qualitative summary of these visual distributions is provided in Table 7.2.

Table 7.2: Qualitative comparison of predicted distributions (energy spectra and decay curves) for different models trained on the full dataset. Evaluations are based on visual inspection of how closely each model reproduces the expected physical shapes.

Feature	ANN	Decision Tree	SimpleRNN	LSTM	BiLSTM + Attention	BiLSTM + Attention (Pairs)
Energy of ^{214}Bi	Poor	Poor	Poor	Poor	Good	Excellent
Energy of ^{214}Po	Good	Good	Excellent	Excellent	Excellent	Excellent
Decay Curve of $^{214}\text{BiPo}$	Very Poor	Very Poor	Good	Good	Excellent	Excellent
Energy of ^{212}Bi	Poor	Poor	Poor	Poor	Excellent	Excellent
Energy of ^{212}Po	Good	Good	Excellent	Excellent	Excellent	Excellent
Decay Curve of $^{212}\text{BiPo}$	Very Poor	Very Poor	Moderate	Moderate	Good	Good

7.2 Comparison with Cut-Based Methods

7.2.1 Numerical Comparison of Classification Performance

This subsection presents a quantitative comparison between the two cut-based approaches, classical and optimized, and the best-performing machine learning model: the BiLSTM with Attention, post-processed for BiPo pair classification. The comparison is based on two key evaluation metrics: TPR and Efficiency, as defined in Section 4.1.1 (Recall and Precision, respectively).

Table 7.3: Comparison of TPR and Efficiency for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ decay chains across different classification methods. While cut-based approaches are limited by fixed selection thresholds, the BiLSTM + Attention model with pair post-processing achieves a better balance between sensitivity and signal retention.

Method	$^{214}\text{BiPo}$		$^{212}\text{BiPo}$	
	TPR	Efficiency	TPR	Efficiency
Classical Cuts	68.66%	47.34%	99.99%	14.54%
Optimized Cuts	61.95%	46.81%	91.86%	51.52%
BiLSTM + Attention (Pairs)	84.53%	93.94%	82.08%	94.45%

The table highlights several key differences:

- For the $^{214}\text{BiPo}$ chain, the BiLSTM + Attention model outperforms both classical and optimized cuts in both TPR and efficiency. This indicates that the model successfully captures subtle temporal and energetic correlations necessary for identifying these longer-lived decays.
- For the $^{212}\text{BiPo}$ chain, the classical cuts achieve an almost perfect TPR of 99.99%, but at the cost of extremely low efficiency (14.54%). The optimized cuts significantly improve efficiency (51.52%) while maintaining a high TPR (91.86%), suggesting a better trade-off.
- In contrast, the BiLSTM + Attention model maintains strong performance across both chains, with balanced TPR and very high efficiency for both. This demonstrates the model’s ability to retain a larger portion of true signal events without being overly inclusive of background, a key advantage in realistic low-statistics scenarios like OSIRIS.

These numerical results confirm that, while cut-based methods can be fine-tuned to prioritize either purity or completeness, machine learning, when combined with a robust post-processing strategy, offers a compelling solution that maintains high performance across multiple objectives and decay chains.

7.2.2 Qualitative Comparison of Physical Signatures

Although the quantitative metrics give a broad view of model performance, they do not fully reflect how well each method captures the physical characteristics of BiPo events. This subsection compares the energy spectra and decay curves obtained using the three approaches.

Energy Spectra: Classical cuts often eliminate useful portions of the energy spectrum, particularly in the ^{212}Po region. Optimized cuts improve signal retention, especially for ^{212}Bi and ^{212}Po , without significantly increasing background. In contrast, the BiLSTM model, with its learned representations, accurately reconstructs the full energy distributions of all decay products, even in overlapping regions.

Decay Curves: The classical method yields a good exponential decay for ^{214}Po but performs poorly for ^{212}Po due to overly restrictive energy cuts. Optimized cuts recover the ^{212}Po signal, but some background remains. The BiLSTM model shows clean exponential decay curves for both chains, with even better temporal resolution after misclassified events are filtered via pair post-processing.

7.2.3 Practical Recommendation for BiPo Event Tagging

Despite the simplicity and interpretability of cut-based methods, they suffer from inflexibility, chain-specific optimization, and hard-threshold-based information loss. The classical cuts are overly strict for ^{212}Po , leading to very low signal retention. Optimized cuts improve this but still require extensive tuning and compromise between purity and efficiency.

In contrast, the BiLSTM with Attention model, combined with pair post-processing, strikes a strong balance. It captures both temporal and energetic features of BiPo decays, produces physically consistent decay curves and spectra, and retains performance across both decay chains simultaneously. Therefore, the BiLSTM with Attention (with pair post-processing) is recommended as the most robust and scalable approach for BiPo event classification in OSIRIS.

7.3 Limitations

Despite the promising performance of the machine learning models developed in this work, several limitations must be acknowledged:

- **Unrealistic class distribution:** The dataset used here includes equal proportions of all six classes (background, Bi, and Po for both $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$), which is not representative of real OSIRIS data. In reality, more than 99% of events are background, and even among signal events, $^{214}\text{BiPo}$ is more common than $^{212}\text{BiPo}$.

While the theoretical goal of a classifier is to generalize independently of class frequencies, in practice, the distribution used during training can bias both learning and performance evaluation. For example, a model trained on a balanced dataset may overestimate its ability to detect rare events when evaluated on similarly balanced test data. In deployment, however, the overwhelming presence of background can lead to a surge in false positives unless the model is explicitly trained to handle imbalance.

Thus, although balancing was helpful for stable training and learning class boundaries, future versions of the model must address the real-world imbalance using resampling, class-weighted loss functions, or post-training calibration to avoid bias and ensure reliable performance in an operational setting.

- **Simplified simulation:** The Monte Carlo simulation used is a toy model that approximates key features of BiPo decays but does not include all physical effects and detector-specific characteristics. The official OSIRIS simulation, once available, will provide a more realistic dataset and should be used to retrain and validate models.
- **Computational constraints:** Due to limited hardware resources, models were trained using a short sequence length (10) and relatively simple architectures. This restricted the exploration of more complex temporal patterns and deeper networks. With more computational power, architectures like deeper BiLSTMs or transformers could potentially improve performance.
- **No validation on real data:** All results were obtained on simulated data only. The domain gap between simulation and real detector data (e.g., due to noise or calibration effects) may affect model performance. Bridging this gap through domain adaptation or hybrid simulations is essential for future application.
- **No uncertainty estimation:** The models do not provide uncertainty estimates for their predictions. In low-background experiments like OSIRIS, knowing how confident the model is in its classification is crucial. Future work should explore Bayesian methods or uncertainty calibration techniques.

7.4 Outlook and Future Work

Building on the promising results of this thesis, several directions can be pursued to improve realism, robustness, and applicability of BiPo event classification in the OSIRIS detector:

- **Use of the official OSIRIS simulation:** The most immediate step forward is to retrain and validate the models using the full, physics-informed OSIRIS simulation once it becomes available. This simulation will include detailed detector responses, background processes, and realistic noise models, essential for transitioning from proof-of-concept to deployment.
- **Handling extreme class imbalance:** The models must be adapted to handle the real-world distribution, where background events vastly outnumber signal events. Techniques such as dynamic resampling, class-weighted loss functions, anomaly detection approaches, or semi-supervised learning may help maintain performance under these conditions.
- **Exploration of more advanced architectures:** With access to better computational resources, more complex models such as Transformer-based architectures or attention-rich models (e.g., temporal attention, hierarchical attention networks) should be explored to capture long-range temporal dependencies and context more effectively than current RNN-based approaches.
- **Integration of uncertainty estimation:** Future models should quantify the confidence of their predictions. Probabilistic frameworks such as Monte Carlo dropout, Bayesian neural networks, or ensemble methods can help flag uncertain classifications and improve reliability in high-stakes applications.
- **Real-time or online inference:** Investigating whether trained models can be adapted for online classification of streaming data would be an important step toward real-time monitoring in OSIRIS or similar detectors.

- **Explainability and interpretability:** Further development of tools to interpret model decisions, such as attention heatmaps or feature attribution, would support transparency, especially in experimental physics applications where model trustworthiness is essential.

7.5 Closing Remarks

This chapter presented a comprehensive comparison of classical cut-based techniques and various machine learning models for BiPo event classification in OSIRIS. While simpler models such as ANNs and Decision Trees provide useful benchmarks, they fall short in capturing the temporal dynamics of decay chains. In contrast, RNN-based architectures, particularly the BiLSTM with Attention and pair post-processing, demonstrate strong performance across both classification metrics and physical consistency of output spectra and decay curves.

The final section compared this best-performing machine learning approach against traditional cuts, highlighting its superior balance between efficiency and sensitivity. These findings support the potential of sequence-based deep learning models as powerful tools for automated decay identification in next-generation neutrino experiments.

A comprehensive summary of the thesis findings is provided in the final conclusion chapter.

Chapter 8

Conclusion

This work has explored the classification of BiPo events in OSIRIS through a combination of physics-informed strategies and machine learning techniques. At the heart of this work lies the challenge of distinguishing fast-coincidence signals from sea of background. While classical methods grounded in physical thresholds have long been the standard, their rigidity leaves much to be desired when navigating high-dimensional, overlapping feature spaces. It never ends, the need to refine, retrain, and reimagine how signal can be separated from noise. In this process, machine learning becomes a catalyst, enabling more dynamic and responsive approaches to event classification.

This thesis explored whether machine learning can offer a more flexible solution. The story began with foundational principles in neutrino physics, oscillations, and mass hierarchy, leading to the scientific motivations behind JUNO and its pre-detector, OSIRIS (Chapter 2). The simulated dataset used throughout this study was then introduced (Chapter 3), followed by the establishment of baselines using classical and optimized cut-based approaches (Chapter 4).

Chapter 5 marked the entry into machine learning, where artificial neural networks and decision trees were trained on engineered features. While these models already outperformed cut-based methods in several metrics, it wasn't until Chapter 6 that the full potential of temporal modeling was revealed. RNNs, especially the BiLSTM architecture with attention and pairwise post-processing, demonstrated a significant leap in performance.

This performance, however, must be viewed in the context of the limitations discussed in Chapter 7. The simulated dataset used here, with its balanced class distribution and simplified physics, does not yet reflect the complexity of real OSIRIS data. Furthermore, the computational constraints imposed by hardware limited the scope of the architectures explored. While the results are promising, the path to deployment will require the use of realistic simulations, advanced handling of extreme class imbalance, and better uncertainty estimation.

Even within these constraints, the findings offer a glimpse of what is possible, bringing the approach one step closer to a model that moves beyond fixed rules to uncover underlying patterns.

Looking ahead, the implications extend beyond OSIRIS. As JUNO prepares to probe the neutrino mass ordering, the ability to identify and reject background with precision becomes increasingly vital. Machine learning, if carefully validated and integrated, may become a core tool not only for BiPo rejection but for the broader goals of neutrino physics, from spectrum unfolding to rare event searches.

More broadly, this thesis reflects the growing convergence of disciplines. The intersection of experimental particle physics and machine learning is no longer a novelty, it is a necessity. By combining physical insight with computational power, we expand the toolkit with which we explore the universe.

In the end, as the field advances, the hope is that methods like those explored here will not only support experiments like OSIRIS and JUNO, but also inspire further collaboration between physicists and computer scientists. In this evolving landscape, perhaps the most exciting developments are yet to come, the ones we can't yet model, but are just beginning to learn how to listen for.

Appendix A

Code and Repository Access

All code, model architectures, training notebooks, and additional resources used in this thesis are available on GitHub at:

<https://github.com/anuragxorma/BiPoClassification>

The repository includes scripts for data preprocessing, model training and evaluation, as well as documentation and instructions for reproducing the results.

Appendix B

Supplementary Figures for Data Description

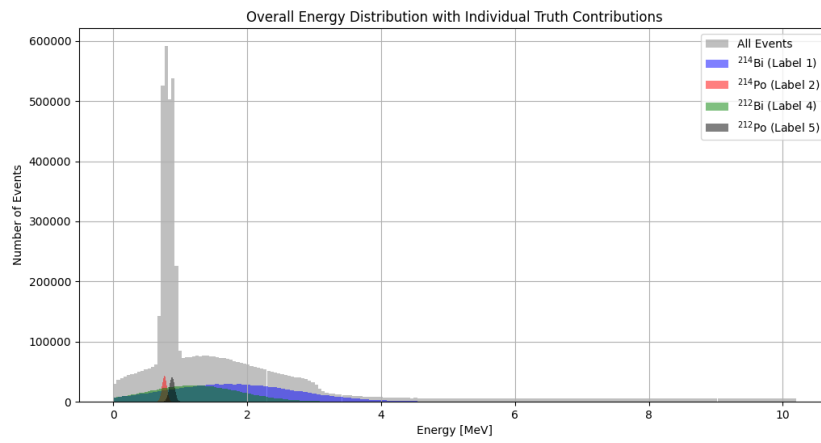


Figure B.1: All energy distributions of the dataset

Appendix C

Supplementary Figures for Classical Cuts and Optimized Cuts

C.1 Decay Curves

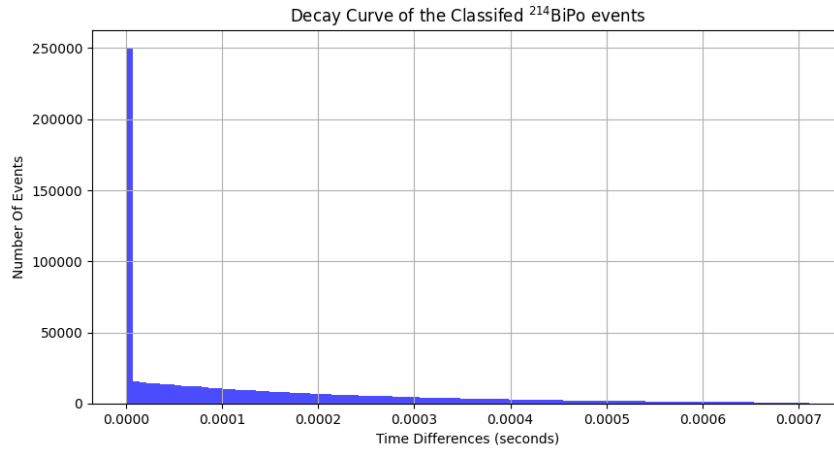


Figure C.1: Decay curve of ^{214}Bi without zooming after applying the classical cuts

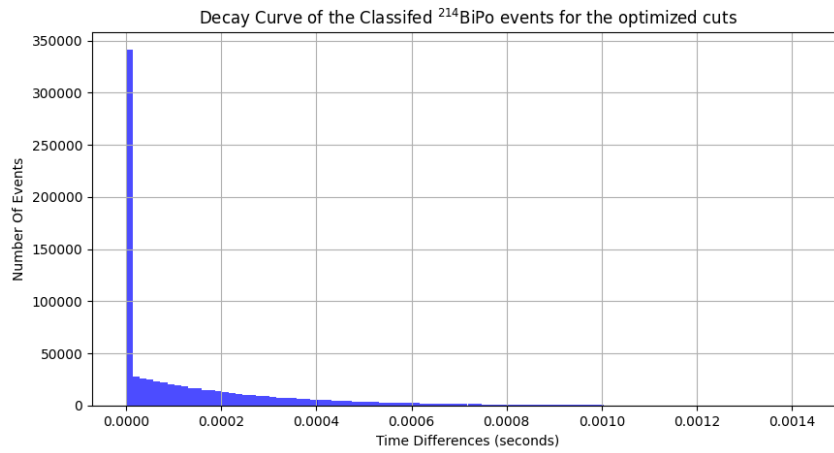


Figure C.2: Decay curve of ^{214}Bi without zooming after applying the optimized cuts

C.2 Optuna Optimization

To analyze the optimization progress, Figures C.4 and C.4 present the objective values obtained across multiple trials for both the $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains. The x-axis represents the number of trials, while the y-axis corresponds to the objective function value (SNR).

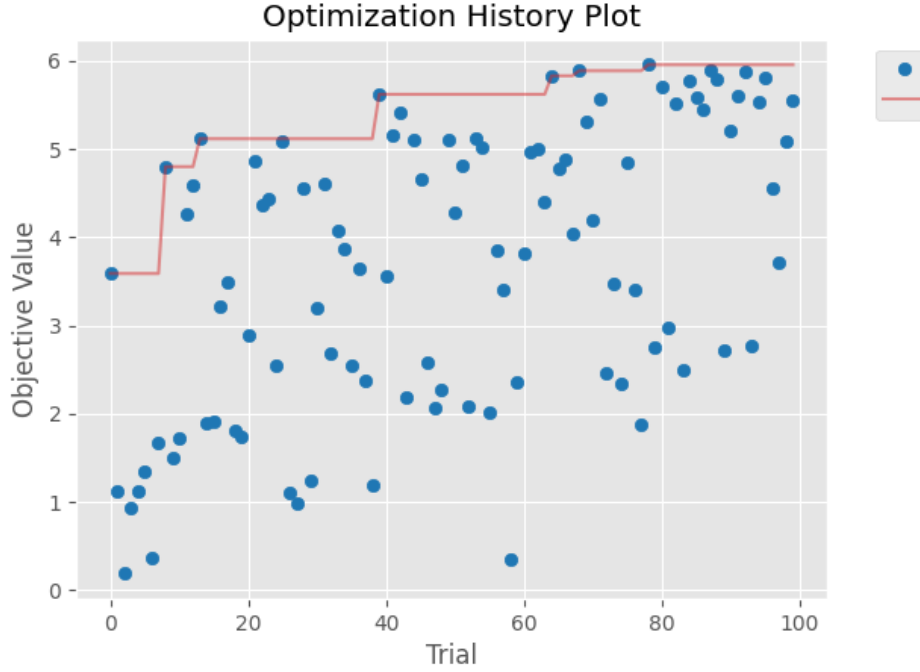


Figure C.3: Optimization training curves for the $^{214}\text{BiPo}$ chain. The objective function (SNR) is plotted as a function of the number of trials. The curves illustrate how the optimizer refines the cut values over time, leading to improved SNR.

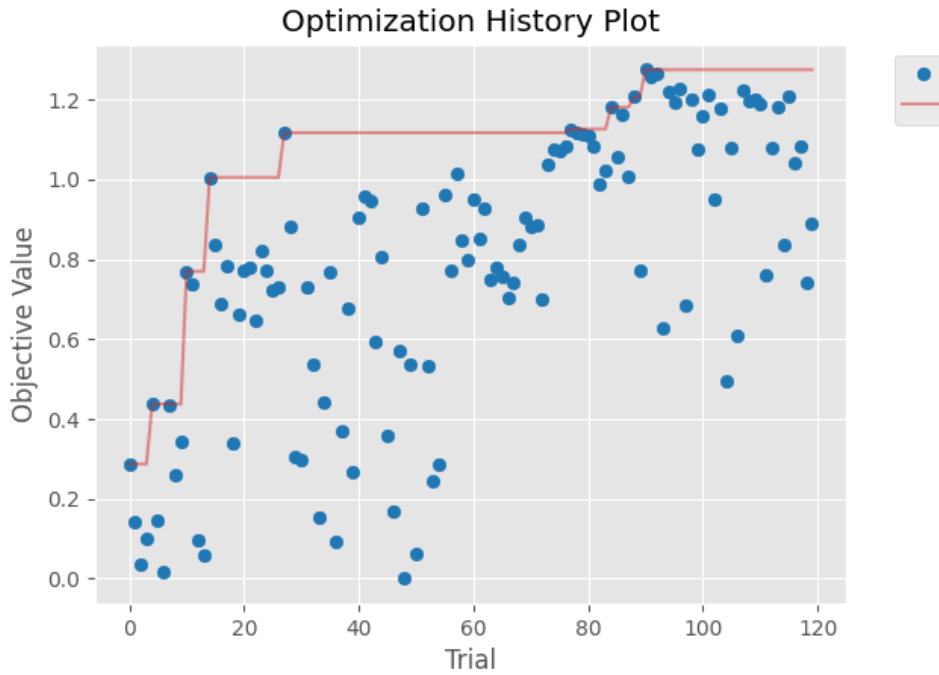


Figure C.4: Optimization training curves for the $^{212}\text{BiPo}$ chain.

The early phase of optimization exhibits large variations in the objective function, as the optimizer

explores different regions of the hyperparameter space. As trials progress, the optimizer gradually focuses on higher-performing parameter configurations, leading to an increase in the best observed SNR.

Since the objective function is designed to maximize SNR, the curve shows an increasing trend over the trials. The optimization process for the $^{214}\text{BiPo}$ chain converged after 100 trials, whereas the $^{212}\text{BiPo}$ chain required 120 trials before reaching an optimal solution. The difference in trial count arises due to the complexity of the parameter space and how quickly the optimizer identifies the best-performing cut values.

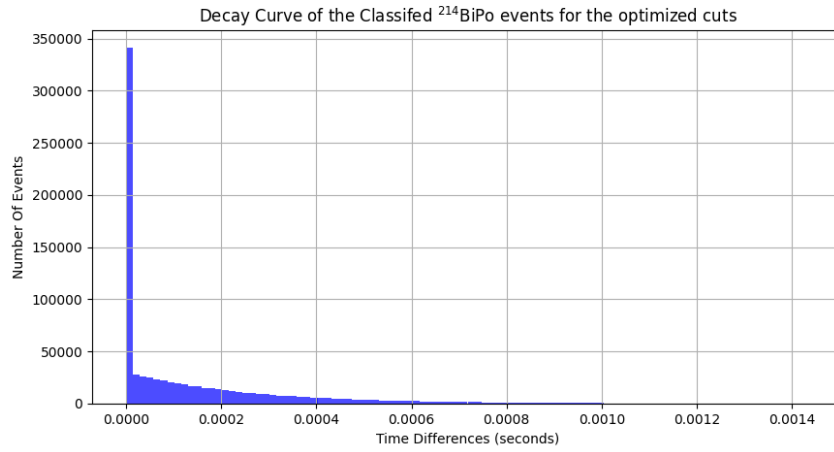


Figure C.5: Time difference distribution of classified $^{214}\text{BiPo}$ events after applying optimized cuts.

Appendix D

Supplementary Figures for Artificial Neural Network

D.1 Artificial Neural Networks

D.1.1 Model for Both Chains

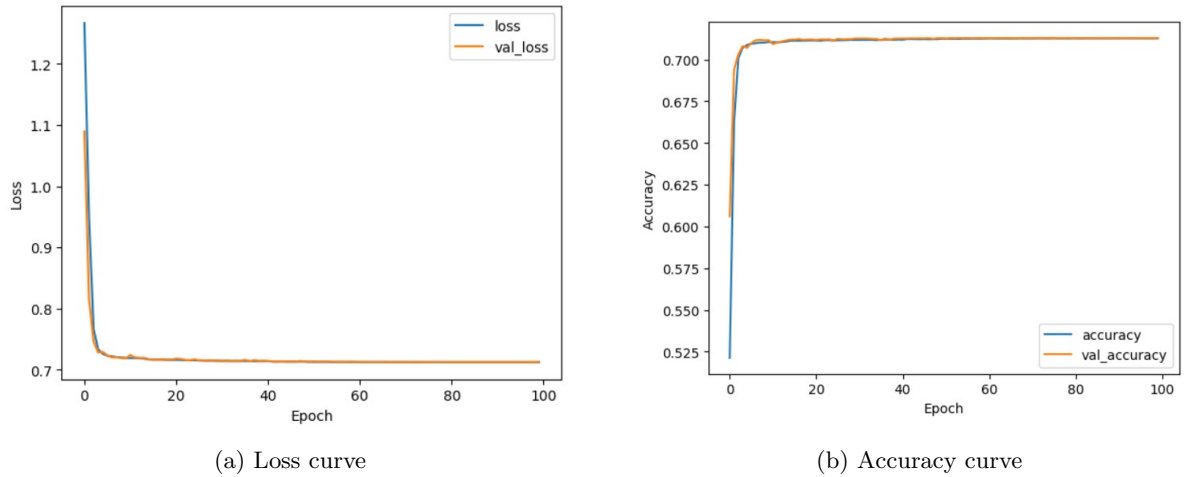


Figure D.1: Training curves (loss and accuracy) for the whole ANN model

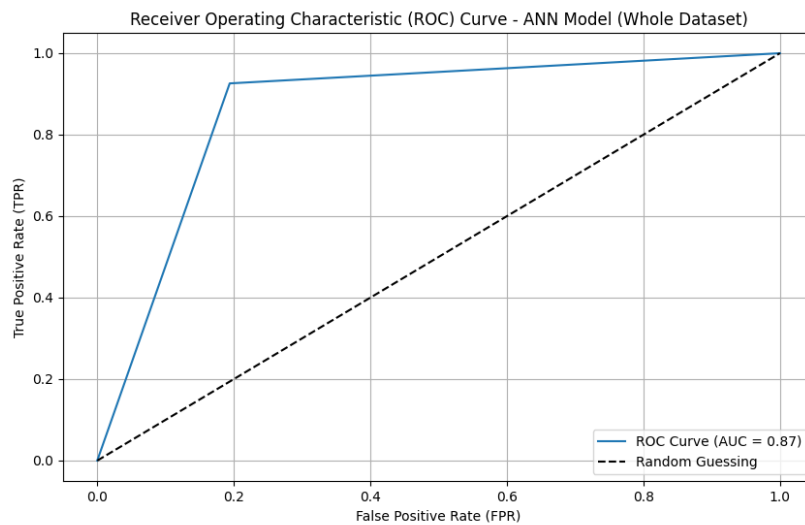


Figure D.2: ROC curve for the whole ANN model

D.2 Model for $^{214}\text{BiPo}$ Chain

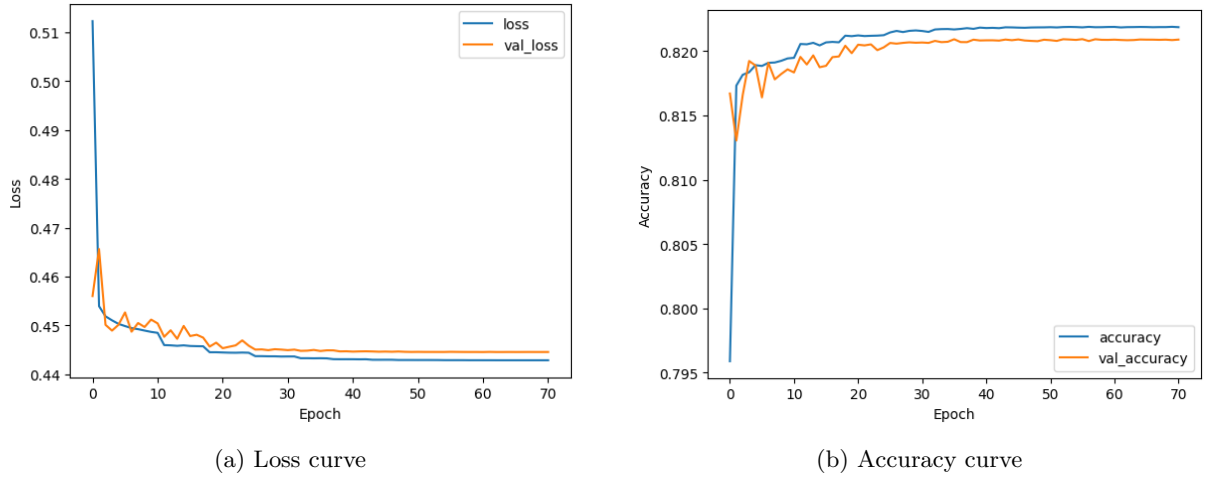


Figure D.3: Training curves (loss and accuracy) for the specialized ANN model for $^{214}\text{BiPo}$

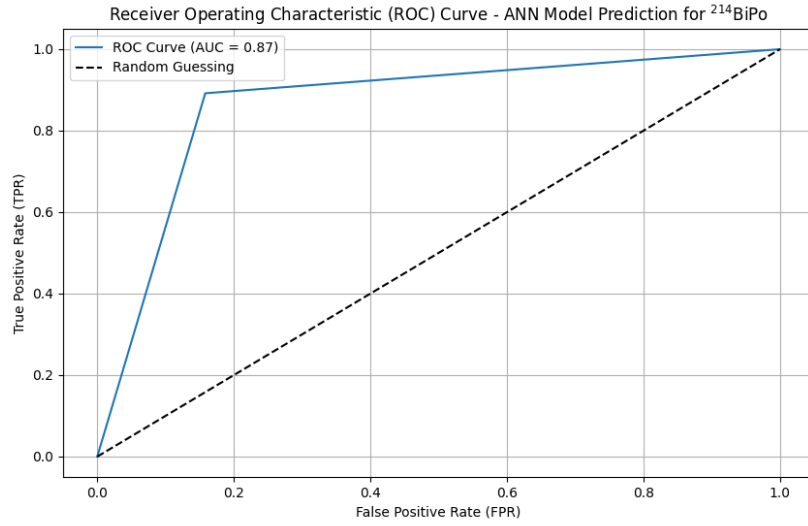


Figure D.4: ROC curve for the specialized ANN model for $^{214}\text{BiPo}$

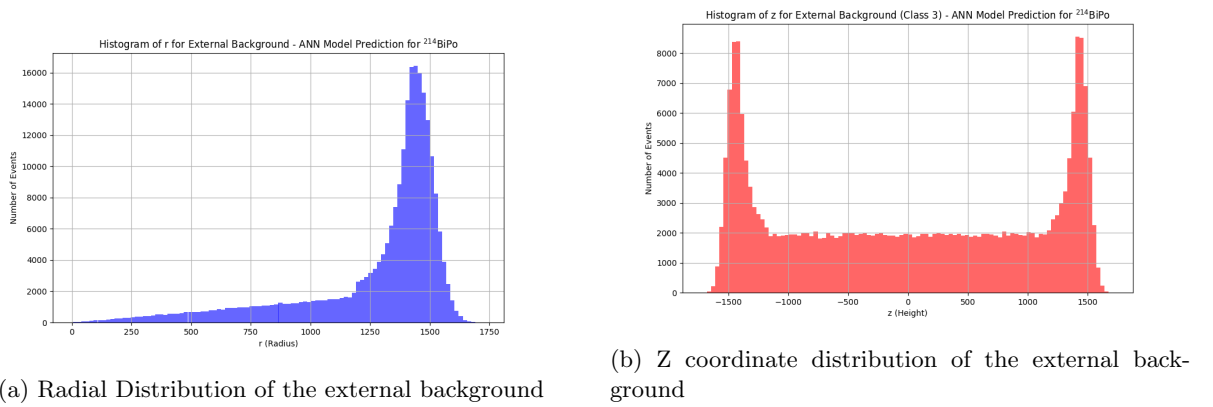


Figure D.5: Distribution of the external background predicted by the specialized ANN model for $^{214}\text{BiPo}$

D.3 Model for $^{212}\text{BiPo}$ Chain

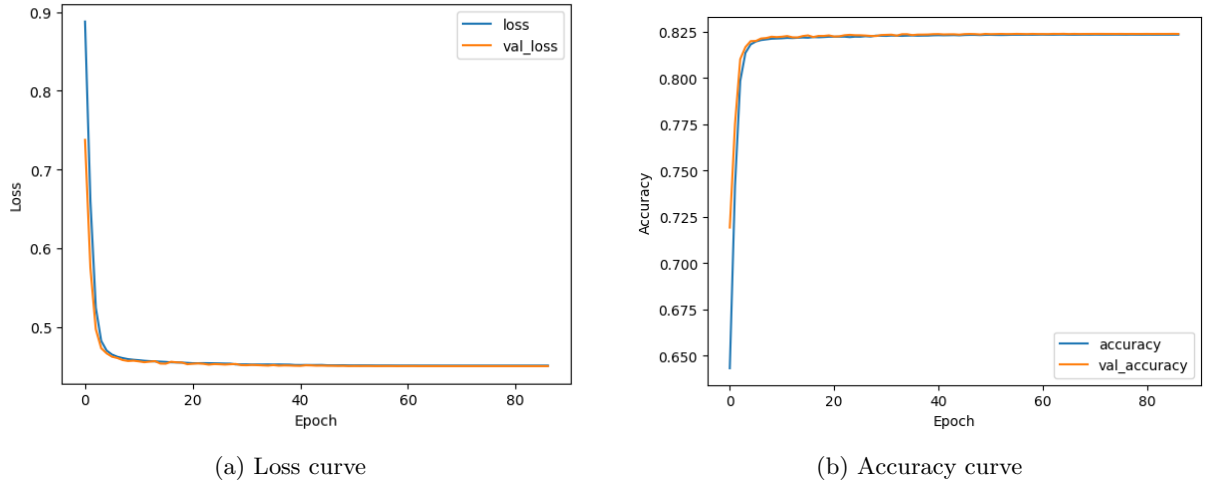


Figure D.6: Training curves (loss and accuracy) for the specialized ANN model for $^{212}\text{BiPo}$

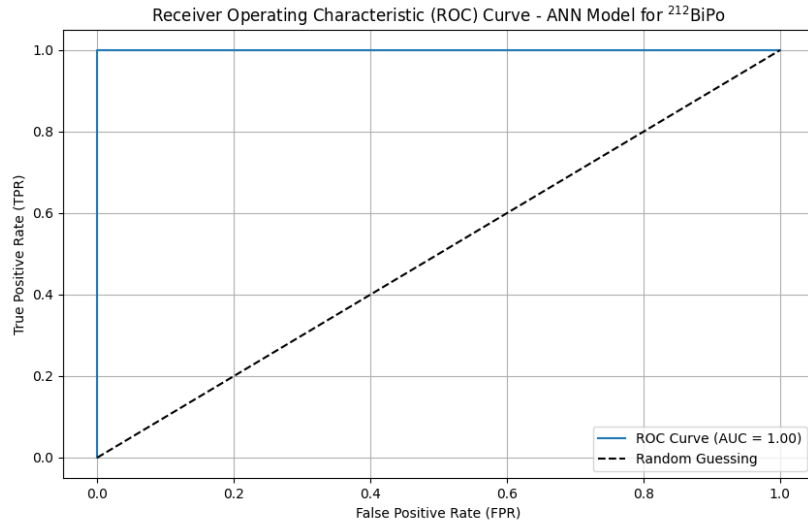


Figure D.7: ROC curve for the specialized ANN model for $^{212}\text{BiPo}$

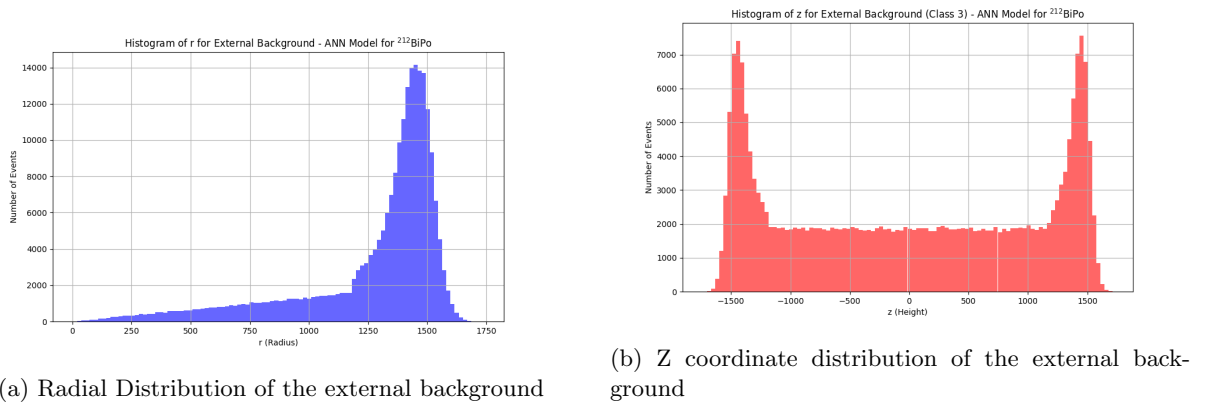
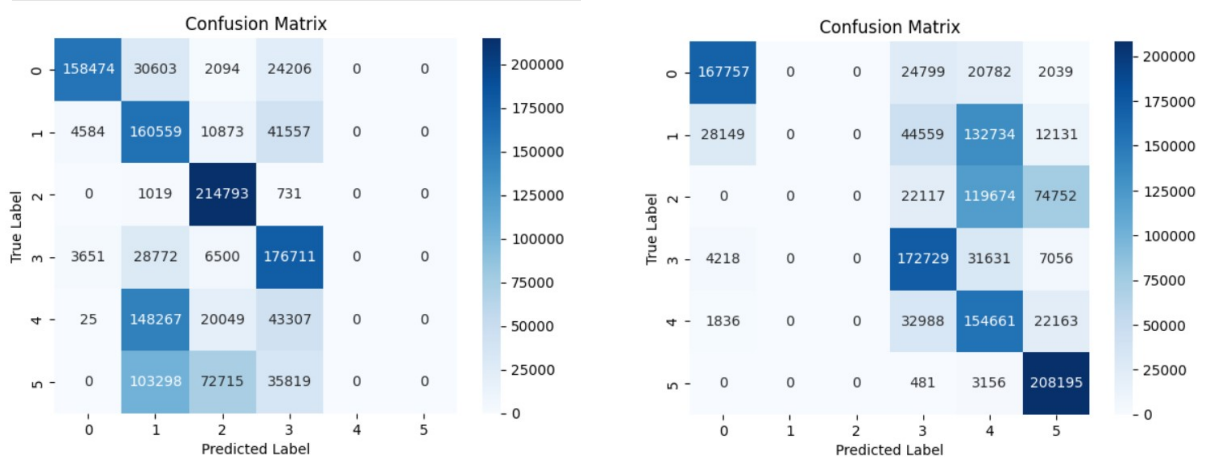


Figure D.8: Distribution of the external background predicted by the specialized ANN model for $^{212}\text{BiPo}$

D.4 Additional Diagrams



(a) Confusion matrix from the $^{214}\text{BiPo}$ -specific ANN model applied to the full dataset.

(b) Confusion matrix from the $^{212}\text{BiPo}$ -specific ANN model applied to the full dataset.

Figure D.9: Confusion matrices showing how each specialised ANN model performs when applied to the full dataset containing both chains.

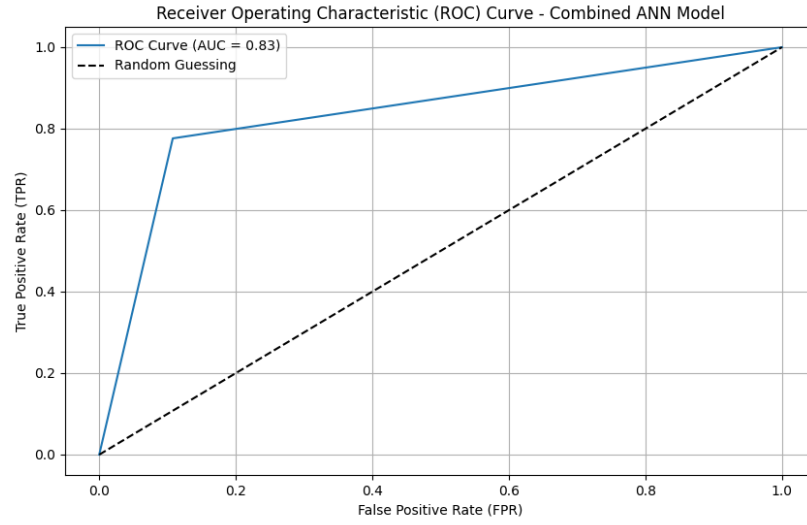


Figure D.10: ROC curve for the refined ANN model after combining both the specialized models

Appendix E

Supplementary Figures for Decision Tree Model

E.1 Decision Tree Model for the whole dataset

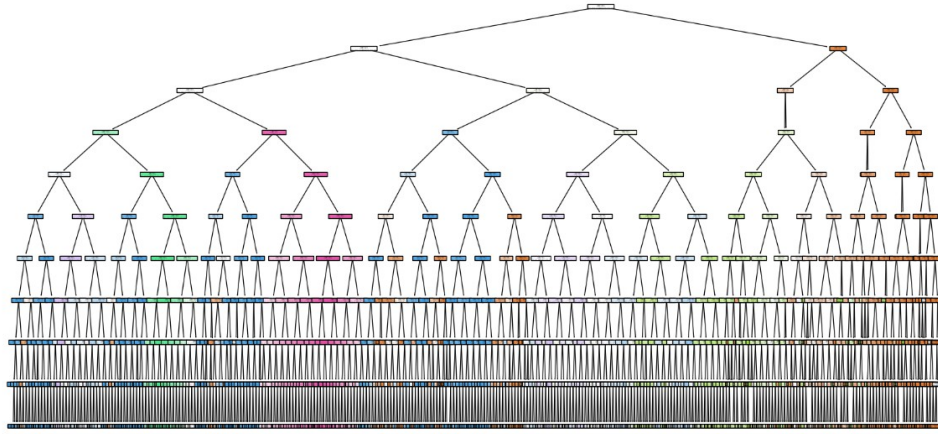
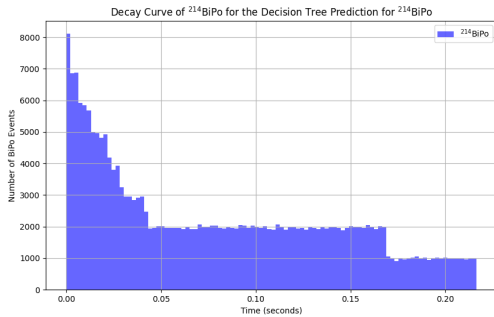
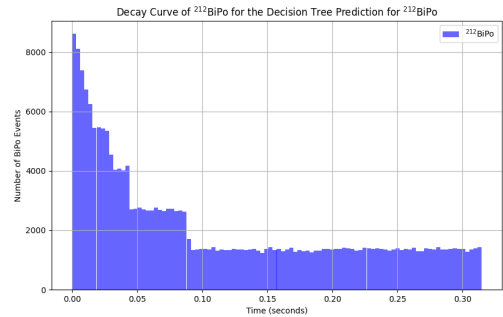


Figure E.1: Visualization of the trained decision tree model used to classify all event types in the dataset.

E.2 Specialized Models for each chain



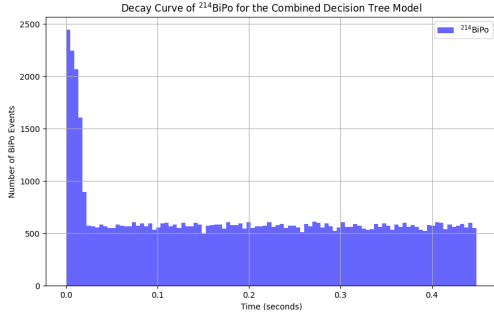
(a) Predicted decay curve for $^{214}\text{BiPo}$ using the decision tree model trained on $^{214}\text{BiPo}$ events.



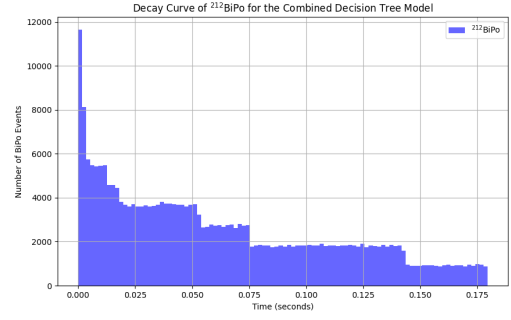
(b) Predicted decay curve for $^{212}\text{BiPo}$ using the decision tree model trained on $^{212}\text{BiPo}$ events.

Figure E.2: Decay curves predicted by the specialized decision tree models for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains.

E.3 Refined Classification with Higher Probability Selection

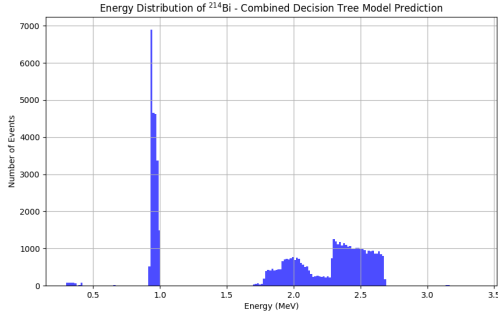


(a) Predicted decay curve for $^{214}\text{BiPo}$ using the refined decision tree model, obtained by combining the specialized models.

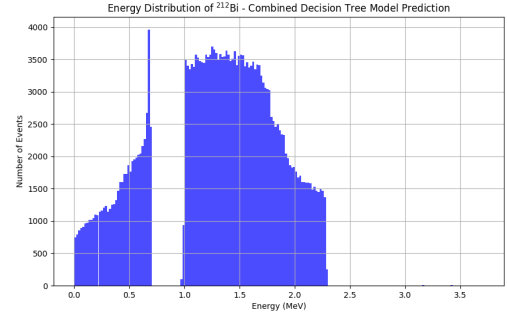


(b) Predicted decay curve for $^{212}\text{BiPo}$ using the refined decision tree model, obtained by combining the specialized models.

Figure E.3: Decay curves for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ predicted by the refined decision tree model, which integrates both specialized models.

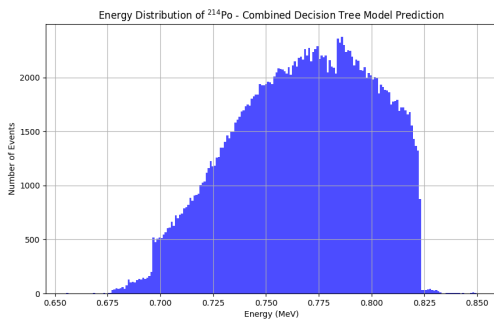


(a) Predicted energy distribution of ^{214}Bi as classified by the refined decision tree model.

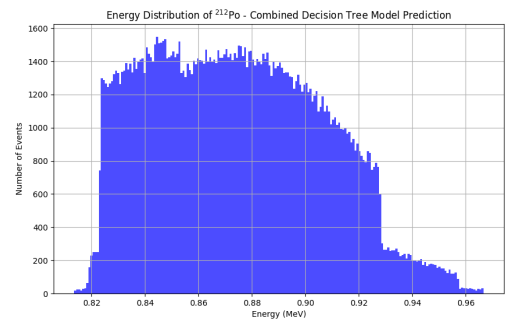


(b) Predicted energy distribution of ^{212}Bi as classified by the refined decision tree model.

Figure E.4: Energy distributions of ^{214}Bi and ^{212}Bi events predicted by the refined decision tree model that combines both specialized models.



(a) Predicted energy distribution of ^{214}Po as classified by the refined decision tree model.



(b) Predicted energy distribution of ^{212}Po as classified by the refined decision tree model.

Figure E.5: Energy distributions of ^{214}Po and ^{212}Po events predicted by the refined decision tree model that combines both specialized models.

Appendix F

Supplementary Figures for Recurrent Neural Networks

F.1 Simple Recurrent Neural Network Figures

F.1.1 Model for Both Chains

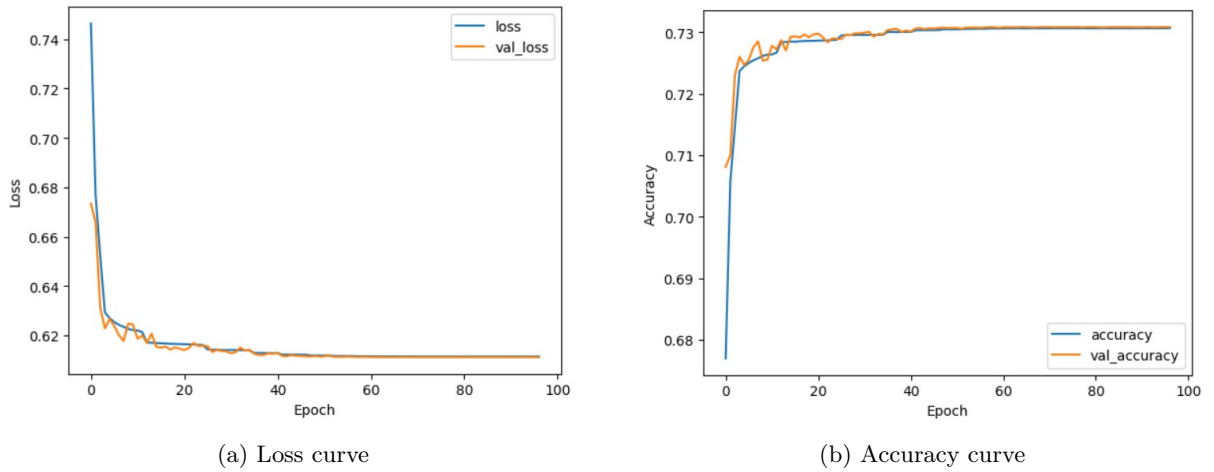


Figure F.1: Training curves (loss and accuracy) for the whole SimpleRNN model

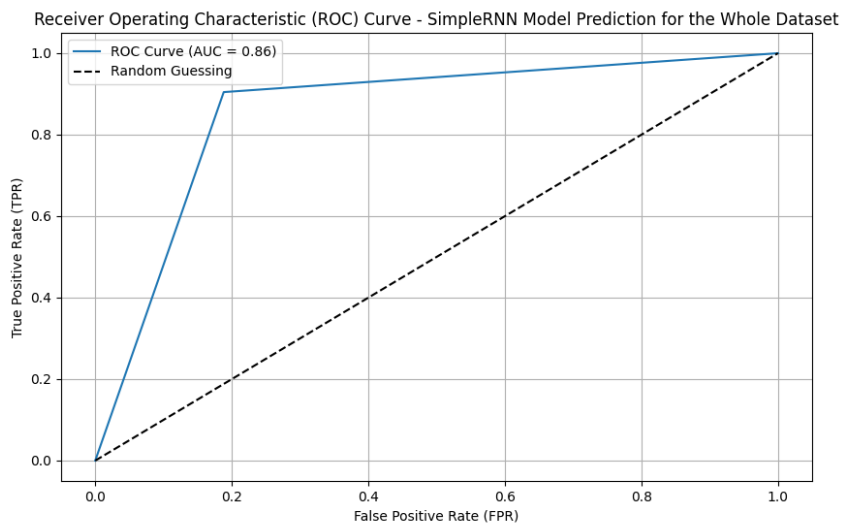
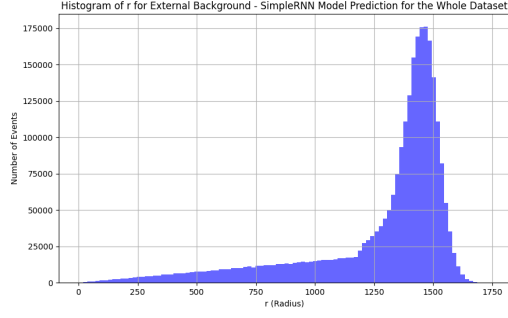
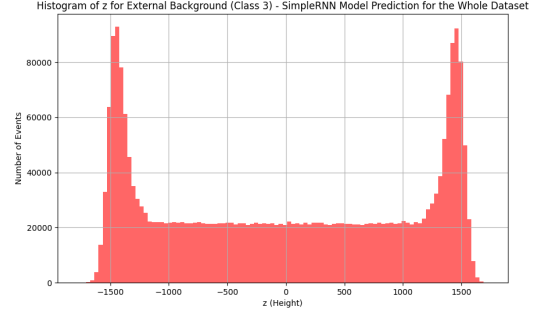


Figure F.2: ROC curve for the whole SimpleRNN model



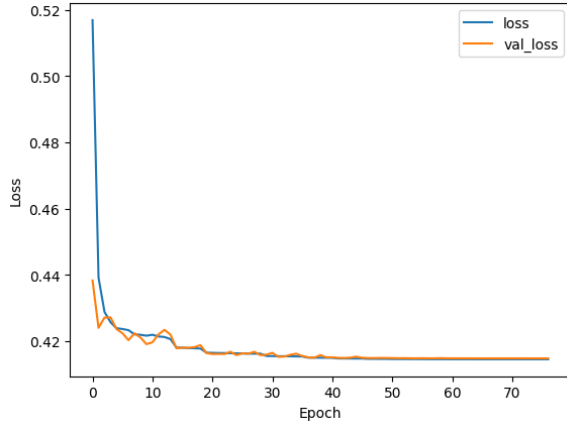
(a) Radial Distribution of the external background



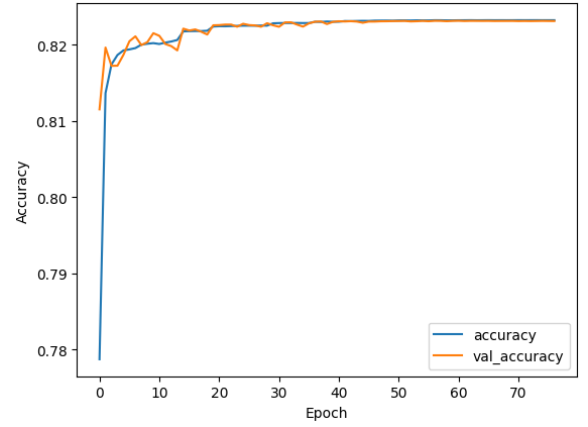
(b) Z coordinate distribution of the external background

Figure F.3: Distribution of the external background predicted by the SimpleRNN model

F.2 Model for $^{214}\text{BiPo}$ Chain



(a) Loss curve



(b) Accuracy curve

Figure F.4: Training curves (loss and accuracy) for the specialized SimpleRNN model for $^{214}\text{BiPo}$

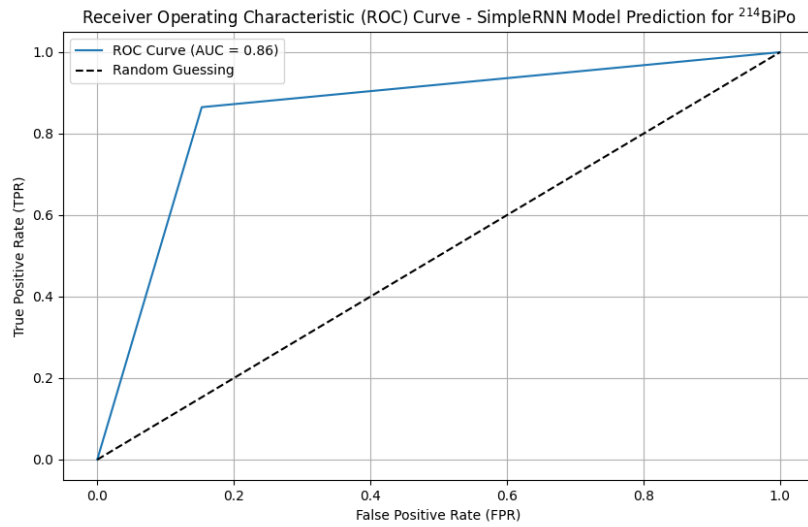


Figure F.5: ROC curve for the specialized SimpleRNN model for $^{214}\text{BiPo}$

F.3 Model for $^{212}\text{BiPo}$ Chain

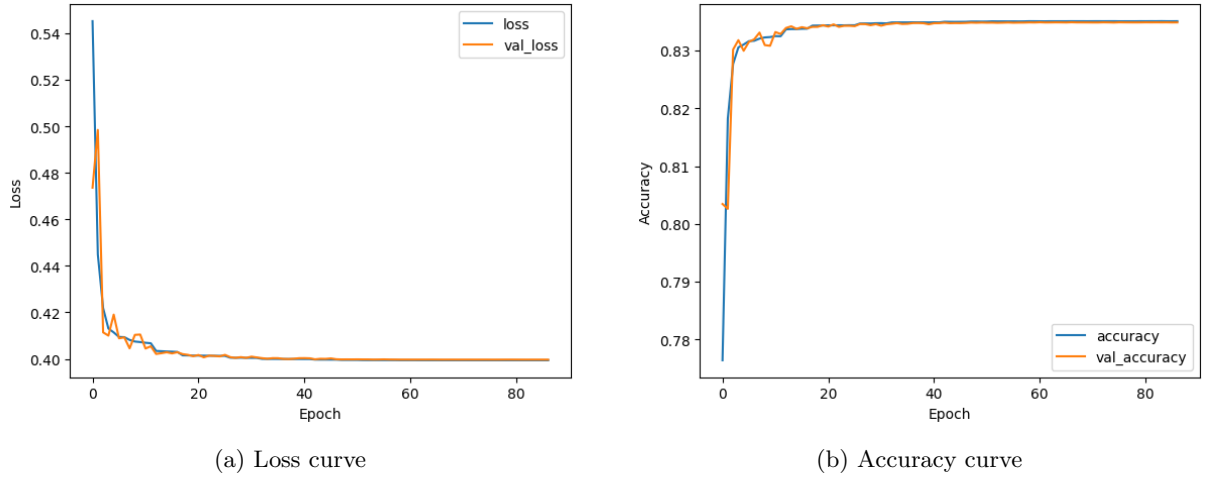


Figure F.6: Training curves (loss and accuracy) for the specialized SimpleRNN model for $^{212}\text{BiPo}$

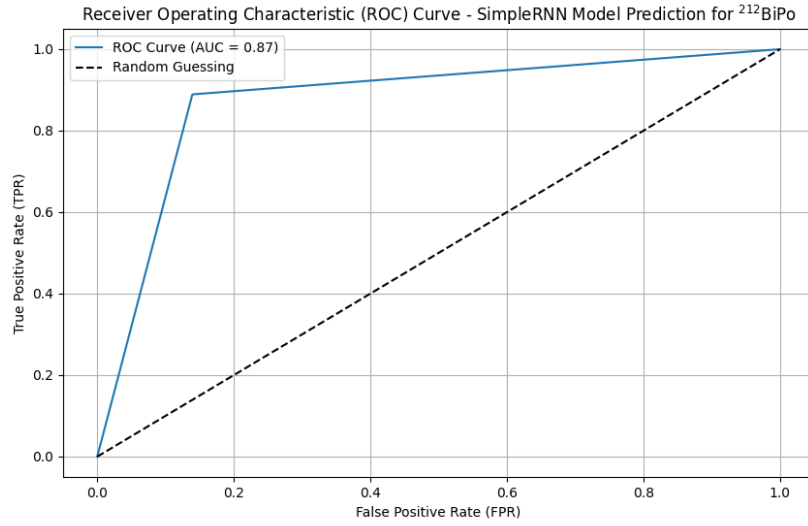


Figure F.7: ROC curve for the specialized SimpleRNN model for $^{212}\text{BiPo}$

F.4 LSTM Figures

F.4.1 Model for Both Chains

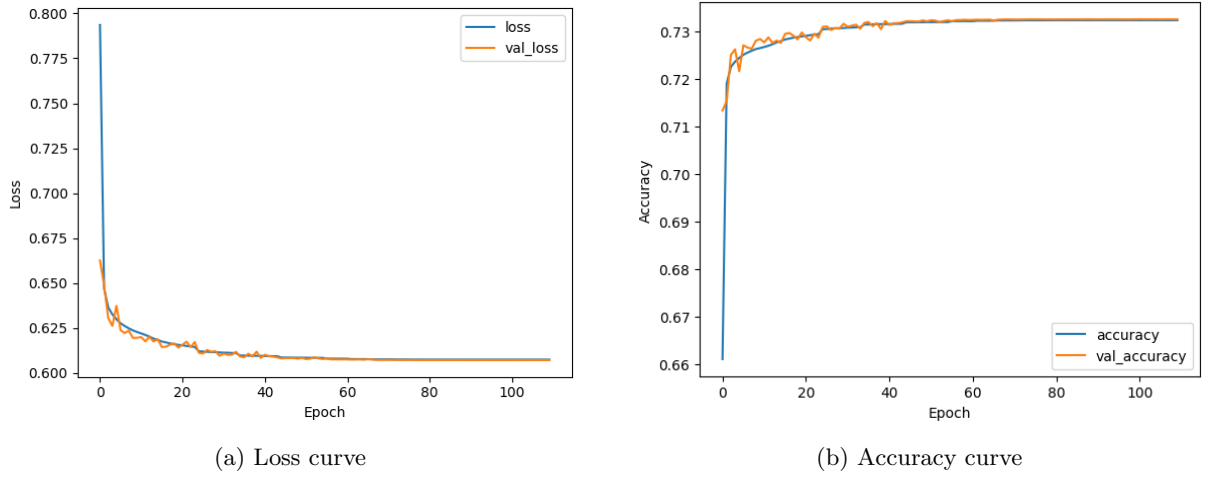


Figure F.8: Training curves (loss and accuracy) for the whole LSTM model

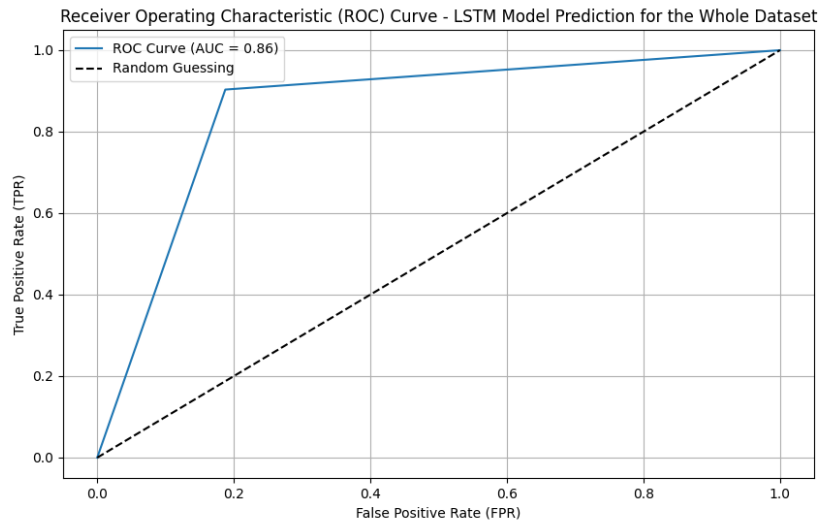
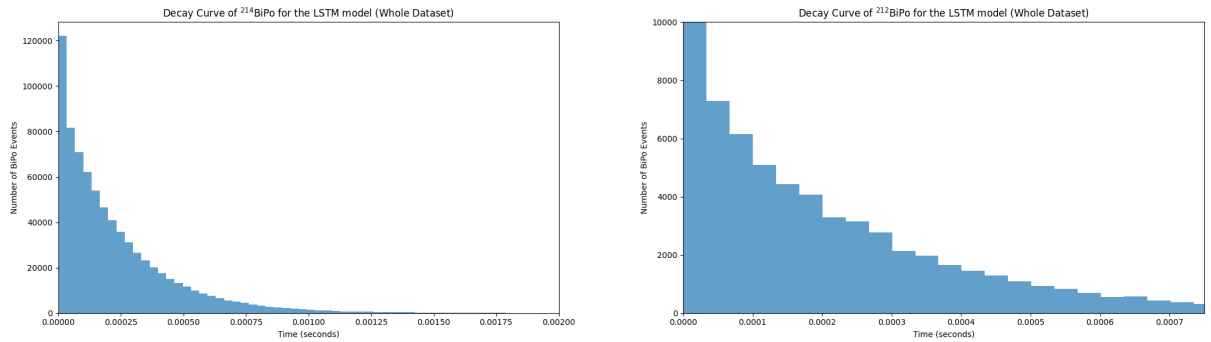


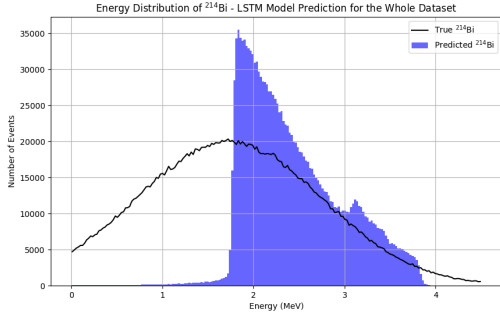
Figure F.9: ROC curve for the whole LSTM model



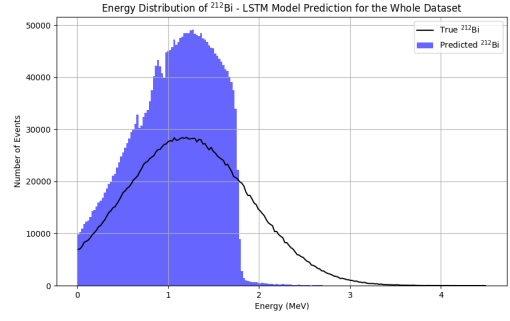
(a) Predicted decay curve for $^{214}\text{BiPo}$ using the full LSTM model.

(b) Zoomed-in predicted decay curve for $^{212}\text{BiPo}$ using the full LSTM model.

Figure F.10: Predicted decay curves for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ from the LSTM model trained on the full dataset.

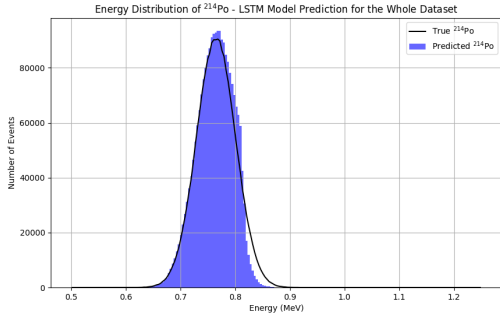


(a) Predicted energy distribution of ^{214}Bi as classified by the whole LSTM model.

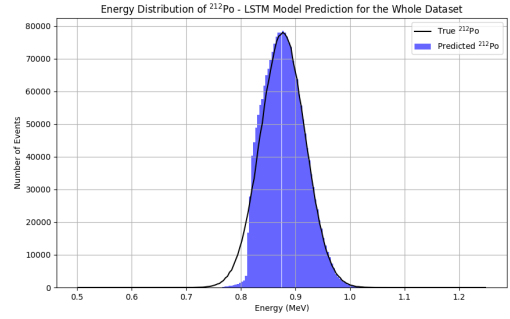


(b) Predicted energy distribution of ^{212}Bi as classified by the whole LSTM model.

Figure F.11: Energy distributions of ^{214}Bi and ^{212}Bi events predicted by the LSTM model trained on the whole dataset.



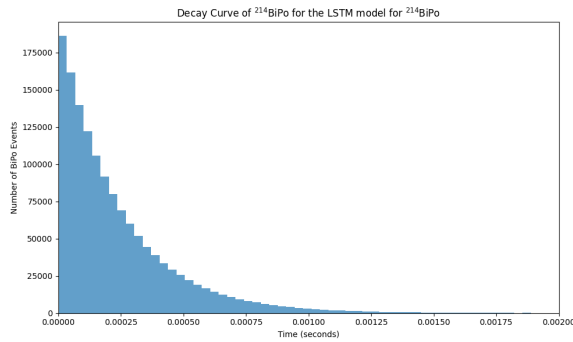
(a) Predicted energy distribution of ^{214}Po as classified by the whole LSTM model.



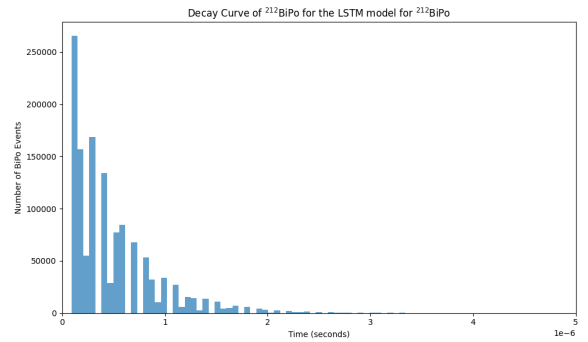
(b) Predicted energy distribution of ^{212}Po as classified by the whole LSTM model.

Figure F.12: Energy distributions of ^{214}Po and ^{212}Po events predicted by the LSTM model trained on the whole dataset.

F.5 Specialized Models for each chain

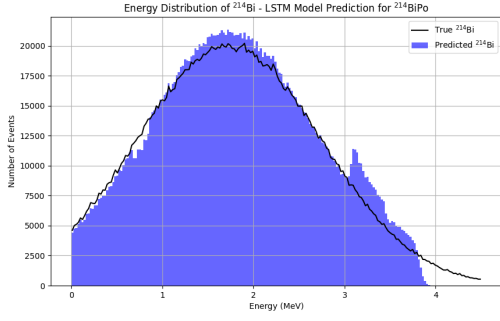


(a) Predicted decay curve for $^{214}\text{BiPo}$ using the LSTM model trained on $^{214}\text{BiPo}$ events.

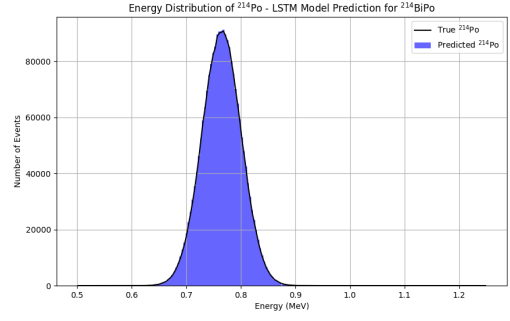


(b) Predicted decay curve for $^{212}\text{BiPo}$ using the LSTM model trained on $^{212}\text{BiPo}$ events.

Figure F.13: Decay curves predicted by the specialized LSTM models for $^{214}\text{BiPo}$ and $^{212}\text{BiPo}$ chains.

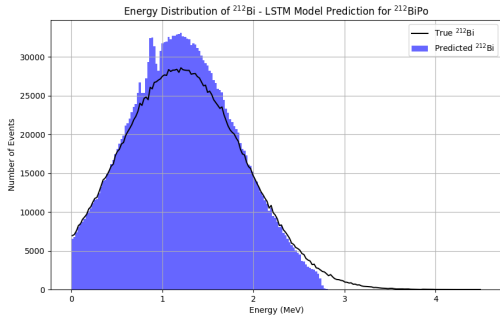


(a) Predicted energy distribution for ^{214}Bi using the LSTM model trained on $^{214}\text{BiPo}$ events.

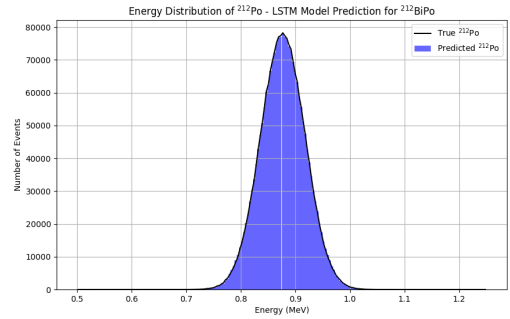


(b) Predicted energy distribution for ^{214}Po using the LSTM model trained on $^{214}\text{BiPo}$ events.

Figure F.14: Predicted energy distributions for ^{214}Bi and ^{214}Po from the LSTM model specialized for the $^{214}\text{BiPo}$ chain.



(a) Predicted energy distribution for ^{212}Bi using the LSTM model trained on $^{212}\text{BiPo}$ events.

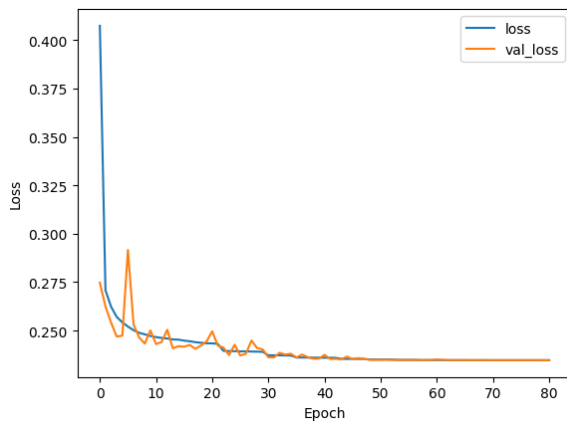


(b) Predicted energy distribution for ^{212}Po using the LSTM model trained on $^{212}\text{BiPo}$ events.

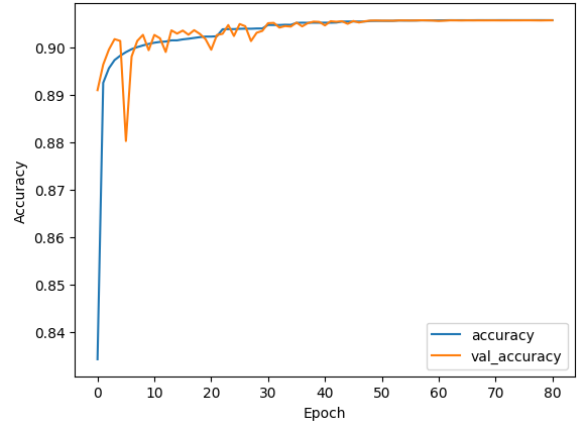
Figure F.15: Predicted energy distributions for ^{212}Bi and ^{212}Po from the LSTM model specialized for the $^{212}\text{BiPo}$ chain.

F.6 BiLSTM with Attention Figures

F.6.1 Model for Both Chains



(a) Loss curve



(b) Accuracy curve

Figure F.16: Training curves (loss and accuracy) for the whole BiLSTM model with Attention.

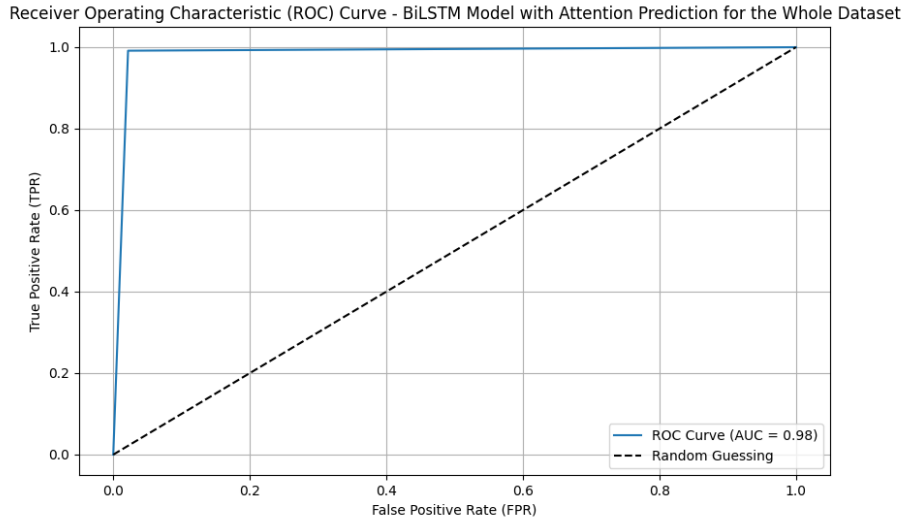
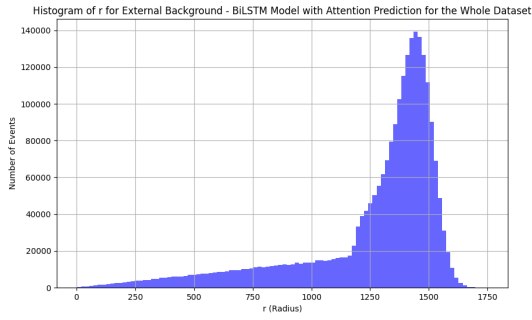
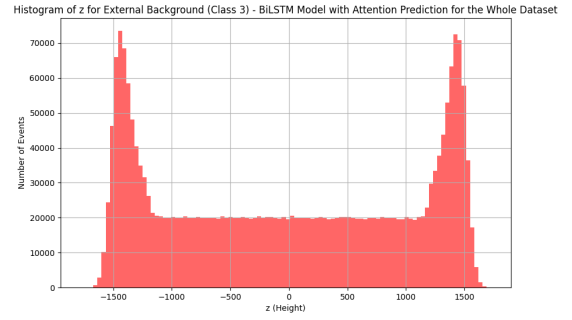


Figure F.17: ROC curve for the whole BiLSTM model with Attention.



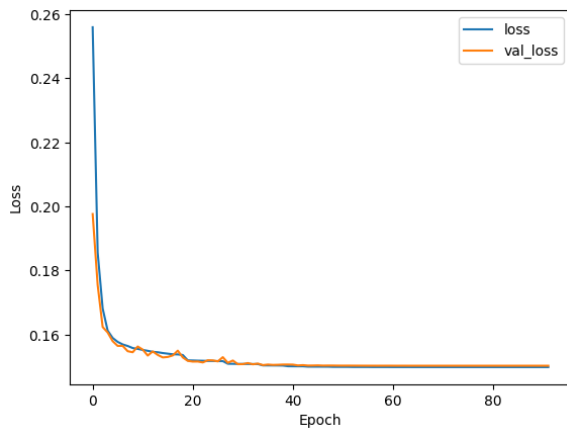
(a) Radial Distribution of the external background



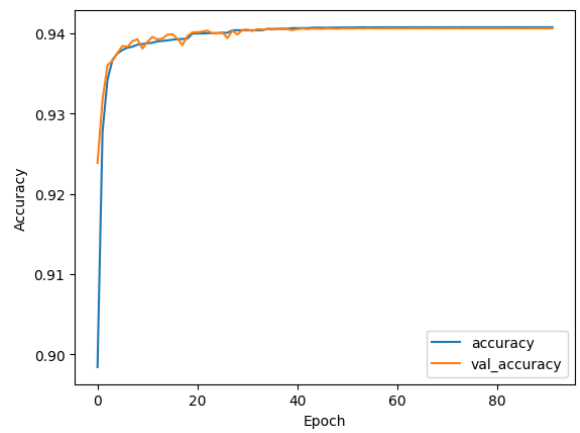
(b) Z coordinate distribution of the external background

Figure F.18: Distribution of the external background predicted by the BiLSTM model with Attention

F.7 Model for $^{214}\text{BiPo}$ Chain



(a) Loss curve



(b) Accuracy curve

Figure F.19: Training curves (loss and accuracy) for the specialized BiLSTM model with Attention for $^{214}\text{BiPo}$

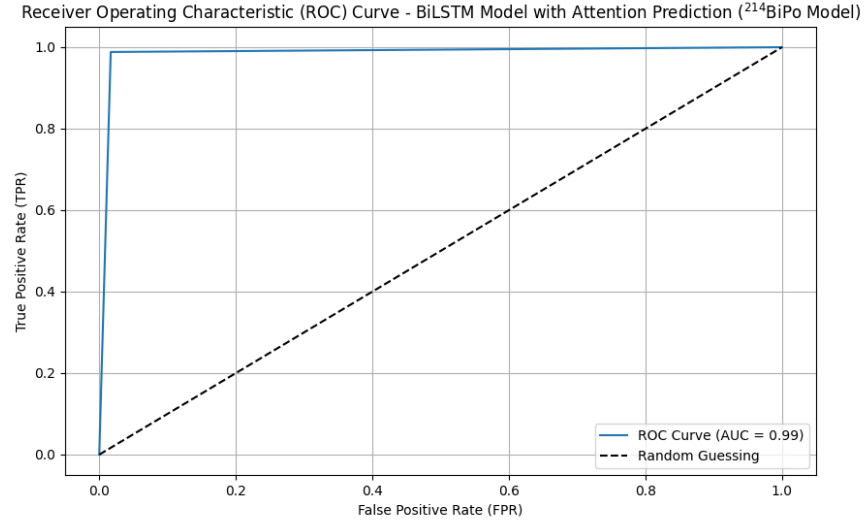
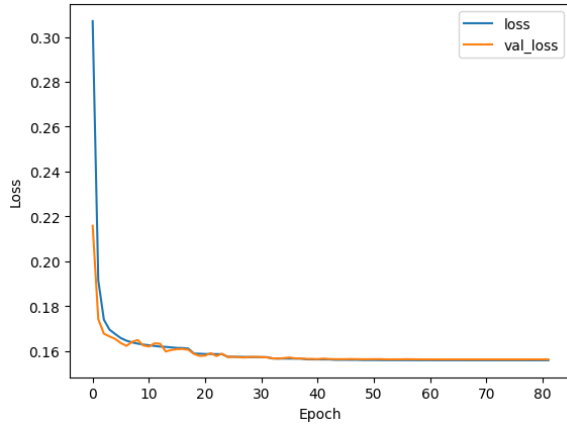
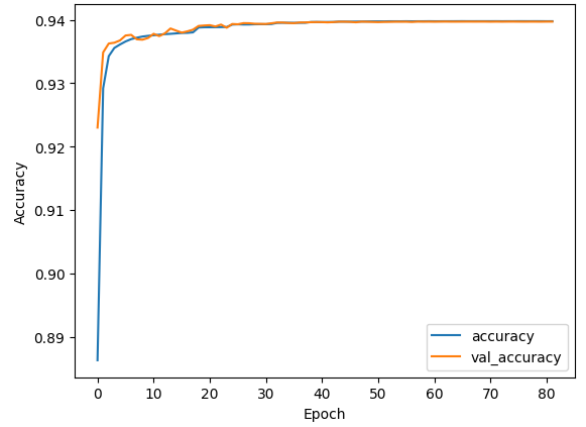


Figure F.20: ROC curve for the specialized BiLSTM model with Attention for $^{214}\text{BiPo}$

F.8 Model for $^{212}\text{BiPo}$ Chain



(a) Loss curve



(b) Accuracy curve

Figure F.21: Training curves (loss and accuracy) for the specialized BiLSTM model with Attention for $^{212}\text{BiPo}$

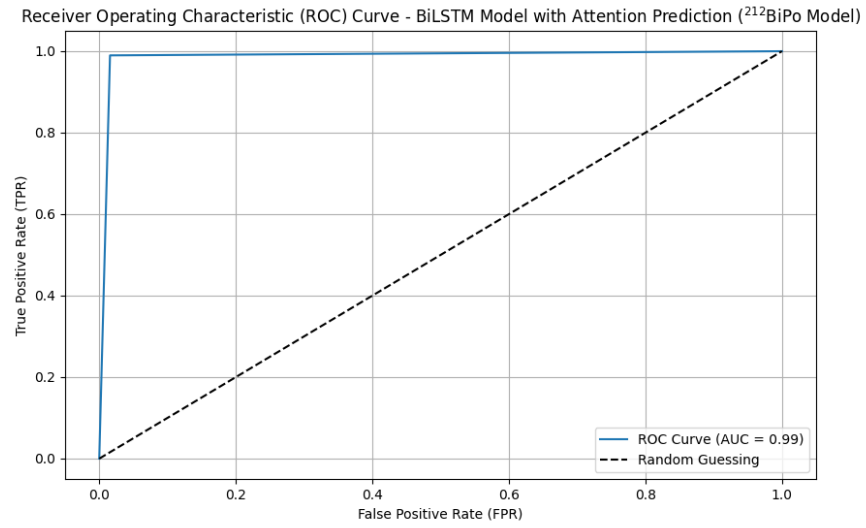


Figure F.22: ROC curve for the specialized BiLSTM model with Attention for $^{212}\text{BiPo}$

F.9 Pair-Wise Classification

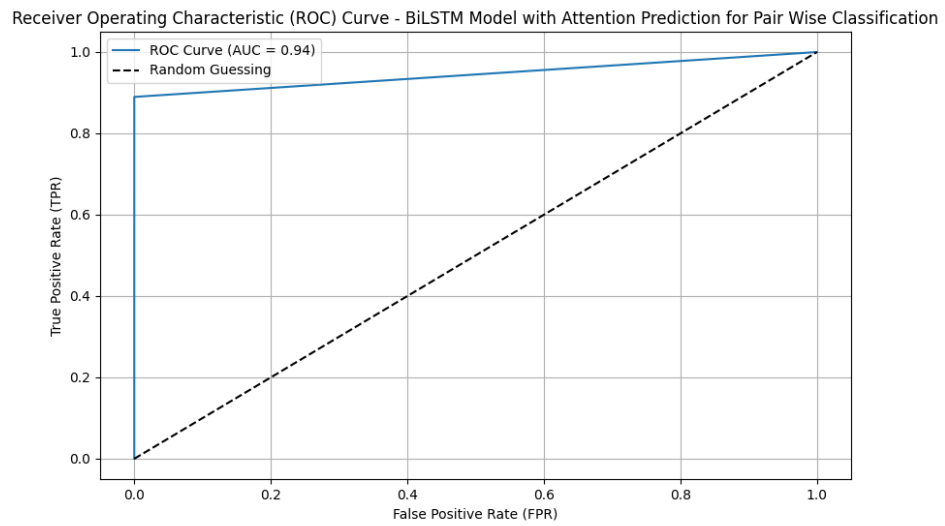
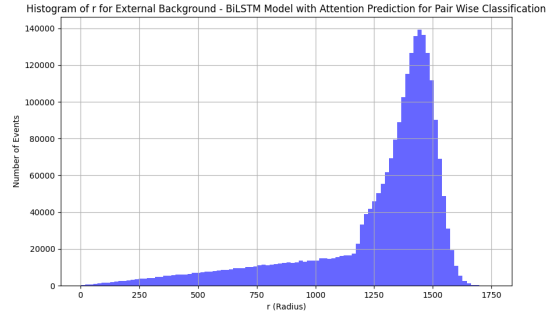
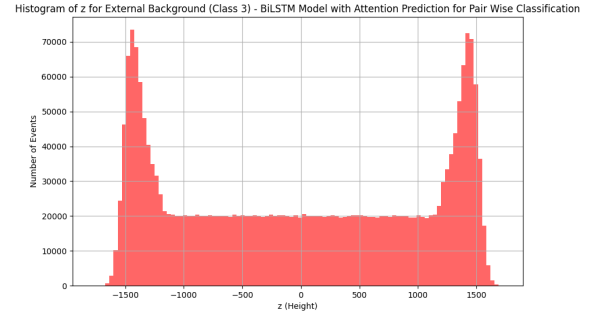


Figure F.23: ROC curve for the BiLSTM model with attention after post-processing, used for classifying BiPo pairs.



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(b) Z-coordinate distribution of the external background events.

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¹Differences in energy peak positions compared to Chapter 3 can be attributed to differences in the underlying simulation and detector modeling.

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Acronyms

ACU Automated Calibration Unit.

ANN Artificial Neural Network.

ASHA Asynchronous Successive Halving Algorithm.

AV Acrylic Vessel.

BiLSTM Bidirectional Long Short-Term Memory.

CD Central Detector.

CP Charge Parity.

DAQ Data Acquisition.

FC Feldman-Cousins.

FV Fiducial Volume.

HV High Voltage.

IBD Inverse Beta Decay.

IO Inverted Ordering.

JUNO Jiangmen Underground Neutrino Observatory.

LCU Laser Calibration System.

LHS Liquid Handling System.

LS Liquid Scintillator.

LSTM Long Short-Term Memory.

MC Monte Carlo.

MCP Microchannel Plate.

ML Machine Learning.

NMH Neutrino Mass Hierarchy.

NO Normal Ordering.

OSIRIS Online Scintillator Internal Radioactivity Investigation System.

p.e. Photoelectrons.

PMT Photomultiplier Tube.

ReLU Rectified Linear Unit.

RNN Recurrent Neural Network.

ROC Receiver Operating Characteristic.

SiPM Silicon Photomultiplier.

SNO Sudbury Neutrino Observatory.

SNR Signal-to-Noise Ratio.

TPE Tree-structured Parzen Estimator.

TPR True Positive Rate.

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